Final research report

Repository: georgeredo, branch `claude/lucid-heisenberg-bpyq7k`. **Built:** 2026-10-23 by `research_notes/FINAL REPORT/build/` (`build.py`, then `render.py` for the PDF). **Inventory of every study:** `INVENTORY.md`, the file beside this one.

This report is read-only with respect to the research. It adds no test, variant or fetch, opens no holdout and changes no rule. Every metric is recomputed from a study's committed trade-level output, or from the scratchpad copy that INVENTORY.md names, with one exception: two studies (#31 and #40) were rerun deterministically from their committed scripts at their own commits, on the same stored inputs. Step 6's rule membership (#17) is recomputed by its committed `step6/trade.py` `choose()`, on the scratchpad population it was built from. Where a figure cannot be produced, the report writes "not available — ". GeorgeFi's own recorded card returns appear only in sections marked **vendor-recorded, not re-marked** (#1, and the grade check in #29f).

Units. Return per trade is given as:

- % of premium paid, for long options;
- % of max risk, for spreads and butterflies;
- % of the credit received, for naked short straddles and strangles;
- % of cash set aside, for cash-secured puts;
- % of notional, for stock trades. Their "$" column is $ per 100 shares, because every stock trade here is 100 shares.

$ per trade is one lot, after every fee the study charged.

Page 1. Every strategy, sorted by mean trade

One row per strategy, on the period that carries its verdict, sorted by mean trade (% in each row's own unit). The table pools all names when a study judged them together, and shows one row per name or variant when it did not. The interval is the 20-session moving-block bootstrap of the mean % a trade (`audit_lib.session_stats`, 2,000 draws). It is "n/a" when the period has fewer than 60 sessions. "Biggest win/loss" are $ for one trade, with its entry date. The verdict is the study's own, as pre-registered.

#	strategy	name	period	trades	win %	avg win	avg loss	mean trade	biggest win	biggest loss	20-session block 95% (mean %)	verdict
6e	H15 carry lane: the 14:05 SPXW short strangle at ±0.5 expected move, held to …	SPXW	TEST	407	70.8	+86.5% / +646	−147.6% / −1,047	**+18.02 % credit** / +151.17	+4,578 (2025-04-10)	−7,131 (2026-03-09)	[+3.54, +29.89]	Open, NOT promotable
1	GeorgeFi vendor cards (the vendor's own recorded P&L)	all names	Epoch 1	546	78.8	+22.8% / +143	−14.8% / −156	**+14.82 % prem** / +79.21	+4,020 (2026-02-06)	−1,375 (2026-02-18)	n/a (< 60 sessions)	Not evidence
41a	EMA system v0: PREVIEW on 2026-10-12 → 16 (5 sessions)	Config A X1	preview 2026-10-12→16	57	61.4	+41.4% / +135	−38.2% / −103	**+10.67 % prem** / +43.03	+717 (2026-10-14)	−773 (2026-10-15)	n/a (< 60 sessions)	PREVIEW
6d	T-E: the 13:00 SPXW ATM straddle held to PM settlement, ex FOMC/CPI/NFP (H13)	SPXW	TEST	388	67.0	+52.8% / +998	−80.7% / −1,768	**+8.75 % credit** / +85.56	+4,100 (2025-03-06)	−34,022 (2025-04-09)	[+1.86, +14.77]	FAIL
27a	After-hours exercise study: fly5 iron ±5, case (b) the post-market decision	all names	ii	987	55.7	+95.6% / +325	−106.2% / −358	**+6.25 % risk** / +22.52	+2,196 (2026-06-22)	−2,196 (2026-04-07)	[−5.84, +15.27]	DROP
6c	T-C: the 13:00 SPXW short straddle bought back 15:55 (H11)	SPXW	2025-06→2026-10	339	63.4	+46.5% / +833	−66.7% / −1,246	**+5.11 % credit** / +72.43	+3,371 (2025-09-17)	−8,499 (2026-06-11)	[−0.06, +9.01]	FAIL
26c	Paper tracker: condor_d20 (SPY 2-wide 20-delta iron condor, 13:00–15:00)	SPY	ii	2,090	82.1	+17.0% / +28	−51.5% / −84	**+4.68 % risk** / +7.67	+60 (2025-04-10)	−190 (2026-04-13)	[+2.43, +6.59]	Positive in ii

#	strategy	name	period	trades	win %	avg win	avg loss	mean trade	biggest win	biggest loss	20-session block 95% (mean %)	verdict
21a	Phase 2 hyp C: 15:00 SPY/QQQ P1 iron ±2 butterfly, held to settlement	all names	OOS 2025–26	878	52.3	+69.1% / +54	−66.3% / −52	**+4.49 % risk** / +3.41	+153 (2026-01-28)	−133 (2026-01-07)	[−2.96, +12.33]	Not confirmed
20a	Phase 2 hyp B: boundary touch — SPY 2-wide 25-delta credit spread, held (the …	SPY	OOS 2025–26	2,600	88.1	+11.9% / +21	−55.2% / −96	**+3.91 % risk** / +6.84	+45 (2025-11-14)	−191 (2026-04-13)	[+1.56, +5.68]	No edge
25	Phase 3 hyp F: the 15:30 SPY/QQQ long call ±5 butterfly (cash account)	all names	OOS 2025–26	879	62.2	+19.0% / +63	−21.1% / −77	**+3.85 % risk** / +9.89	+259 (2026-04-29)	−335 (2025-03-04)	[+1.49, +6.20]	Passes all four, but the cell was named after C’s OOS was printed
26b	Paper tracker: fly5 (SPY/QQQ ±5 butterflies at 15:00 and 15:30: iron, call, put)	all names	ii	5,271	60.5	+21.8% / +70	−24.2% / −82	**+3.62 % risk** / +9.64	+280 (2026-04-29)	−471 (2026-05-12)	[+1.77, +5.47]	Iron clear of zero in i and ii
20b	Phase 2 hyp B: the midday SPY iron condor (em1, mid-VIX), the midday pick	SPY	OOS 2025–26	1,202	81.6	+15.5% / +26	−50.9% / −85	**+3.25 % risk** / +5.55	+56 (2026-04-29)	−179 (2026-07-24)	[+0.21, +6.04]	“Needs X”
1	GeorgeFi vendor cards (the vendor’s own recorded P&L)	all names	Epoch 2	603	48.1	+32.6% / +191	−24.2% / −189	**+3.13 % prem** / −6.23	+3,313 (2026-07-13)	−3,400 (2026-07-13)	[−0.74, +9.76]	Not evidence
26a	Paper tracker: condor_em1 (SPY 2-wide iron condor at ± the ATM straddle, …	SPY	ii	2,068	80.4	+17.5% / +28	−59.1% / −96	**+2.46 % risk** / +4.10	+63 (2026-07-29)	−188 (2025-04-09)	[−0.54, +5.02]	Historical periods through zero
27b	After-hours exercise study: condor_d20, case (b)	SPY	ii	1,176	83.3	+37.0% / +60	−170.6% / −277	**+2.37 % risk** / +3.84	+867 (2026-09-16)	−1,799 (2026-04-07)	[−12.63, +14.57]	DROP
41b	EMA system v1: PREVIEW on three drawn weeks (up, flat, down)	Config A X1	three weeks pooled	64	54.7	+39.2% / +167	−42.4% / −195	**+2.23 % prem** / +2.98	+1,357 (2026-02-03)	−873 (2026-02-03)	[−14.91, +28.63]	PREVIEW
21c	Phase 2 hyp C: 15:00 SPY/QQQ P3 call ±5 butterfly, held to settlement	all names	OOS 2025–26	878	58.5	+23.1% / +73	−27.4% / −92	**+2.15 % risk** / +4.52	+274 (2026-04-29)	−471 (2026-05-12)	[−0.59, +4.78]	Not confirmed
21b	Phase 2 hyp C: 15:00 SPY/QQQ P2 call ±2 butterfly, held to settlement	all names	OOS 2025–26	880	51.2	+66.7% / +53	−65.8% / −53	**+2.09 % risk** / +1.34	+151 (2026-01-28)	−134 (2026-01-07)	[−5.35, +10.06]	Not confirmed
28b	SPXW analogue B: 20-delta iron condor, wings 2×ratio, 13:00–15:00 (condor_d20)	SPXW	selection	2,157	76.8	+17.5% / +289	−51.4% / −849	**+1.49 % risk** / +24.69	+605 (2026-07-29)	−1,850 (2026-04-13)	[−1.12, +3.46]	FAIL
21d	Phase 2 hyp C: debit vertical 0.70/0.30, universe B, held (the vertical pick)	all names	OOS 2025–26	1,761	51.6	+70.7% / +348	−72.9% / −361	**+1.25 % prem** / +4.95	+6,049 (2025-04-11)	−7,091 (2025-04-10)	[−2.84, +4.65]	Loses less than the naked contract but still loses
7j-atm_put	S17 Index put-writing: monthly ATM cash-secured put	all names	2012-06→2026-10	334	81.4	+1.7% / +511	−4.0% / −1,168	**+0.65 % capital** / +199.27	+2,117 (2026-07-17)	−9,871 (2020-02-21)	[+0.16, +1.07]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …
6a	T-A: the 13:00 short ATM 0DTE straddle, SPY/QQQ/IWM, bought back 15:55 (H9)	all names	2025-06→2026-10	1,017	62.2	+38.6% / +70	−62.0% / −108	**+0.62 % credit** / +2.57	+343 (2026-06-12)	−1,187 (2026-06-11)	[−4.33, +4.41]	FAIL
7j-put_30d	S17 Index put-writing: monthly 0.30-delta cash-secured put	all names	2012-06→2026-10	334	85.6	+1.1% / +327	−3.5% / −963	**+0.46 % capital** / +141.65	+1,427 (2026-05-15)	−9,371 (2020-02-21)	[+0.02, +0.79]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …

#	strategy	name	period	trades	win %	avg win	avg loss	mean trade	biggest win	biggest loss	20-session block 95% (mean %)	verdict
37b	Edge H8: turn of the month (SPY, QQQ stock)	QQQ	single test	172	59.3	+1.8% / +504	−1.7% / −463	**+0.387 % notional** / +110.22	+3,373 (2026-07-30)	−2,370 (2026-01-29)	[+0.006, +0.694]	SPY FAIL; QQQ PASS at 1× only
37b	Edge H8: turn of the month (SPY, QQQ stock)	SPY	single test	117	60.7	+1.5% / +608	−1.5% / −597	**+0.311 % notional** / +134.00	+2,808 (2026-07-30)	−2,464 (2024-07-30)	[−0.052, +0.642]	SPY FAIL; QQQ PASS at 1× only
7j-put_15d	S17 Index put-writing: monthly 0.15-delta cash-secured put	all names	2012-06→2026-10	334	94.0	+0.5% / +155	−4.2% / −1,056	**+0.26 % capital** / +82.52	+719 (2026-05-15)	−8,446 (2020-02-21)	[−0.10, +0.47]	PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 …
37a	Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED	QQQ	single test	108	61.1	+0.6% / +195	−0.5% / −128	**+0.210 % notional** / +69.59	+1,161 (2024-07-30)	−542 (2026-07-28)	[+0.049, +0.355]	PASS both names
37a	Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED	SPY	single test	77	55.8	+0.6% / +225	−0.3% / −125	**+0.190 % notional** / +70.43	+823 (2024-07-30)	−620 (2021-01-26)	[+0.059, +0.355]	PASS both names
32	Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)	SPY	single test	2,177	50.8	+0.5% / +181	−0.4% / −165	**+0.032 % notional** / +11.09	+3,027 (2020-03-13)	−1,750 (2025-04-07)	[−0.013, +0.084]	FAIL
32	Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)	QQQ	single test	3,299	50.5	+0.5% / +134	−0.5% / −117	**+0.027 % notional** / +9.42	+2,401 (2024-08-02)	−1,449 (2025-04-07)	[−0.006, +0.064]	FAIL
35	Edge H4: market intraday momentum (SPY, QQQ stock)	QQQ	test 2025-10-13→2026-10-22	257	49.8	+0.2% / +120	−0.2% / −99	**+0.014 % notional** / +10.32	+902 (2026-07-29)	−378 (2025-11-20)	[−0.016, +0.048]	FAIL
9a	Lifecycle P1: Quick Scalps → cash-secured put sale (portfolio)	all names	hold-out 2025-10→2026-10	122	63.1	+0.1% / +51	−0.2% / −66	**+0.01 % risk** / +8.09	+136 (2026-08-03)	−198 (2025-10-14)	[−0.02, +0.03]	Rejected
35	Edge H4: market intraday momentum (SPY, QQQ stock)	SPY	test 2025-10-13→2026-10-22	257	45.5	+0.1% / +89	−0.1% / −80	**−0.004 % notional** / −2.95	+573 (2026-07-29)	−373 (2026-09-30)	[−0.028, +0.020]	FAIL
6b	T-B: gamma-conditioned last-30-minute momentum in SPY shares (H10)	SPY	negative-gamma days 2024-05→2026-10	299	46.2	+0.1% / +77	−0.2% / −102	**−0.031 % notional** / −19.59	+522 (2025-04-04)	−699 (2025-03-04)	[−0.056, −0.003]	FAIL
7c	S03 Quick Scalps with weekly expirations (1–7 days)	all names	2017-06→2026-10	818	37.7	+26.6% / +59	−16.2% / −34	**−0.08 % prem** / +1.27	+463 (2020-05-15)	−227 (2020-03-10)	[−1.87, +1.70]	Nothing promoted
4b	GeorgeFi entries with the trail engine (reconstruction C, G2)	all names	ledger E2	516	39.0	+18.8% / +93	−12.2% / −56	**−0.13 % prem** / +2.27	+1,753 (2026-07-06)	−777 (2026-03-16)	[−2.08, +3.55]	No robust entry timing
28a	SPXW analogue A: iron fly ±5×ratio at 15:00 and 15:30 (fly5 iron)	SPXW	selection	882	58.8	+15.0% / +534	−22.5% / −812	**−0.41 % risk** / −19.82	+2,569 (2026-09-16)	−3,550 (2026-03-09)	[−2.28, +1.41]	FAIL
43	Compass ICT: sweep → displacement/MSS → FVG on SPXW 0DTE	v0	exploration 2022-05→2026-10-09	142	37.3	+35.5% / +955	−22.4% / −492	**−0.76 % prem** / +48.26	+9,735 (2026-07-27)	−3,285 (2025-05-05)	[−7.77, +7.54]	Exploration only
19	Phase 2 hyp A: the low-moc_mag closing pin (SPXW short call spread, 15:30, held)	SPXW	OOS 2025–26	177	87.6	+2.6% / +63	−29.5% / −709	**−1.39 % risk** / −33.03	+174 (2025-04-14)	−2,311 (2025-04-30)	[−3.53, +0.30]	NO EDGE
15	Step 4: turn-up buy (1.5× the low) + 20% trailing stop, near the money	all names	check 2025–26	26,258	29.0	+40.8% / +386	−19.5% / −161	**−2.02 % prem** / −2.26	+21,609 (2025-04-07)	−39,941 (2025-04-09)	[−3.63, −0.05]	A trailing stop beats holding but still loses.

#	strategy	name	period	trades	win %	avg win	avg loss	mean trade	biggest win	biggest loss	20-session block 95% (mean %)	verdict
4	GeorgeFi's own trades re-priced at the NBBO (G1)	all roots	ledger E2	594	45.1	+26.4% / +156	−26.3% / −205	**−2.54 % prem** / −42.07	+3,019 (2026-09-21)	−3,462 (2026-07-13)	[−6.22, +2.54]	Not supported after execution
40b	Compass EMA ladder: "take profits are the nearest EMAs" PREVIEW (E1 ATM, E2 …	E1	preview 2025-02 (fresh)	1,761	30.8	+15.1% / +16	−10.7% / −11	**−2.75 % prem** / −2.49	+176 (2025-02-03)	−86 (2025-02-07)	n/a (< 60 sessions)	PREVIEW
2b	George rebuild: the best of 144 variants (sweep / ict+gex / morning / …	all names	out-of-sample 2026-03→10	385	40.3	+12.2% / +20	−13.3% / −16	**−3.04 % prem** / −1.78	+122 (2026-06-25)	−103 (2026-06-29)	[−4.57, −1.42]	Rejected
5	Deep research round 1: flow rule F1 on SPY (H2 family)	SPY	2025-07→2026-10	638	35.9	+13.9% / +22	−13.2% / −19	**−3.45 % prem** / −4.12	+176 (2025-12-18)	−84 (2026-07-29)	[−4.53, −2.12]	Rejected
7g	S12b 10:00 long option in the first-half-hour direction	all names	2022-11→2026-10	2,499	33.4	+145.6% / +171	−78.3% / −93	**−3.60 % prem** / −4.87	+2,478 (2025-04-09)	−413 (2026-06-25)	[−9.45, +2.93]	Nothing promoted
7e	S10 Short premium: fade the Quick Scalps signal (single contract, 2× stop, to …	all names	2022-11→2026-10	2,886	55.4	+74.4% / +71	−100.7% / −87	**−3.63 % credit** / +0.52	+456 (2025-04-09)	−426 (2026-06-09)	[−9.79, +2.83]	Nothing promoted
9b	Lifecycle P2: Quick Scalps → credit put spread to expiry (portfolio)	all names	hold-out 2025-10→2026-10	196	87.8	+12.5% / +112	−119.5% / −1,084	**−3.65 % risk** / −34.72	+1,945 (2025-10-14)	−5,195 (2026-01-16)	[−16.19, +4.15]	Rejected
7d	S09 Quick Scalps on SPXW 0DTE	SPXW	2022-05→2026-10	1,229	33.8	+37.8% / +318	−25.0% / −232	**−3.81 % prem** / −46.18	+1,484 (2025-04-09)	−1,001 (2026-03-09)	[−5.48, −2.15]	REJECTED
2	George rebuild: the GeorgeFi-like reference (sweep / ict+gex / all / none / …	all names	out-of-sample 2026-03→10	1,399	37.2	+12.8% / +17	−13.7% / −15	**−3.84 % prem** / −2.99	+141 (2026-06-26)	−128 (2026-06-25)	[−5.02, −3.01]	Rejected
17	Step 6: the six pre-registered runner rules × three exits	B1a	check 2025→2026-10-09	6,350	41.6	+31.8% / +686	−29.5% / −643	**−3.97 % prem** / −89.48	+1,957 (2026-06-17)	−2,891 (2025-07-30)	[−4.79, −3.41]	FAIL
29c	Picker, S2: "5 minutes" as a hard exit	all names	whole span (verdict)	2,943	38.3	+14.1% / +67	−15.6% / −63	**−4.18 % prem** / −13.04	+907 (2025-04-09)	−993 (2026-02-05)	[−5.32, −3.14]	Labelled
7a	S01 Quick Scalps (owner baseline) on index 0DTE	all names	2022-11→2026-10	2,886	34.3	+36.2% / +33	−26.4% / −23	**−4.92 % prem** / −4.19	+346 (2026-06-09)	−157 (2025-04-07)	[−6.31, −3.69]	REJECTED
29d	Picker, S3: SPXW rides to the official close when in the money at 15:58	SPXW	whole span (verdict)	908	37.0	+21.9% / +269	−20.7% / −253	**−4.96 % prem** / −59.43	+1,387 (2025-04-09)	−2,953 (2025-04-07)	[−6.73, −3.49]	Labelled
7b	S02 Quick Scalps: 41-variant grid, walk-forward OOS (lower-bound selection)	all names	walk-forward OOS 2024-01→2026-10	3,200	34.8	+35.2% / +31	−26.9% / −22	**−5.31 % prem** / −3.81	+346 (2026-06-09)	−163 (2025-04-07)	[−6.87, −3.98]	Nothing promoted
7h	S15 VRP-conditioned 10:00 short ATM 0DTE straddle (unconditional row)	all names	2022-11→2026-10	1,941	58.0	+43.6% / +133	−73.2% / −220	**−5.47 % credit** / −15.46	+660 (2024-08-05)	−3,324 (2025-04-09)	[−9.93, −1.42]	Nothing promoted
29e	Picker, IMPROVED (picked on the cards): PDH sweep → put, 10:00–12:00, +25% or …	all names	whole span (verdict)	516	38.8	+16.3% / +96	−19.3% / −91	**−5.52 % prem** / −18.73	+477 (2024-08-07)	−1,173 (2026-02-20)	[−8.38, −2.95]	FAIL on every name
29a	Picker (GeorgeFi rule), PRIMARY: opening-range fade, stop 0.68, target 1.25, …	all names	whole span (verdict)	2,943	35.9	+21.9% / +102	−21.7% / −92	**−6.05 % prem** / −22.62	+1,387 (2025-04-09)	−2,953 (2025-04-07)	[−7.41, −4.82]	FAIL on every name
29b	Picker, SECONDARY: the narrative's +6% breakeven lock and VIX trail	all names	whole span (verdict)	2,943	39.8	+16.8% / +81	−21.4% / −88	**−6.20 % prem** / −21.27	+1,387 (2025-04-09)	−1,183 (2025-04-23)	[−7.38, −5.02]	Labelled

#	strategy	name	period	trades	win %	avg win	avg loss	mean trade	biggest win	biggest loss	20-session block 95% (mean %)	verdict
30d	Compass features BV1b: the V1 veto at 3% (PASSED, Phase B)	0DTE→60 DTE, 15 names	holdout (opened)	251,808	31.9	+134.9% / +692	−73.0% / −363	**−6.73 % prem** / −26.77	+45,477 (2026-03-30)	−26,113 (2026-03-30)	[−9.38, −3.86]	PASS
7i	S16b Prior-day fade with a weekly contract (time exit)	all names	2017-06→2026-10	1,041	37.8	+51.7% / +118	−42.3% / −87	**−6.78 % prem** / −9.91	+1,275 (2022-10-13)	−445 (2026-06-12)	[−11.05, −3.10]	Nothing promoted
17	Step 6: the six pre-registered runner rules × three exits	A1a	check 2025→2026-10-09	1,035	36.6	+37.2% / +430	−33.4% / −287	**−7.54 % prem** / −24.69	+4,161 (2025-04-09)	−3,533 (2025-04-04)	[−15.50, −0.10]	FAIL
30b	Compass features V1: veto a spread over 10% of mid (PASSED)	0DTE→60 DTE, 15 names	holdout (opened)	474,526	29.5	+153.1% / +676	−76.1% / −322	**−8.43 % prem** / −27.16	+45,477 (2026-03-30)	−29,001 (2026-10-19)	[−11.25, −5.25]	PASS
8e	V8: month-end 13:00 long straddle on unread chains	all names	2017-06→2022 (unread chains)	105	39.0	+63.1% / +503	−55.4% / −463	**−9.14 % prem** / −86.06	+3,222 (2022-03-31)	−3,552 (2020-02-28)	[−23.88, +10.99]	Rejected
40a	Compass EMA ladder: PREVIEW 2025-03, 13 versions (rerun)	V1	preview 2025-03	2,781	15.5	+37.5% / +14	−17.7% / −6	**−9.20 % prem** / −2.76	+148 (2025-03-28)	−31 (2025-03-05)	n/a (< 60 sessions)	PREVIEW
8d	V3a: single-name 0DTE short ATM straddle 10:00 → 15:50	all names	hold-out	1,622	56.4	+42.1% / +200	−76.1% / −315	**−9.47 % prem** / −24.84	+2,452 (2026-06-12)	−7,240 (2026-07-02)	[−19.41, −2.80]	Rejected
30c	Compass features X3: sell a same-day long option at the 14:00 bid (PASSED)	0DTE→60 DTE, 15 names	holdout (opened)	233,454	29.0	+122.0% / +271	−63.3% / −128	**−9.51 % prem** / −11.72	+19,167 (2025-11-20)	−4,253 (2025-11-21)	[−13.87, −7.42]	PASS
30e	Compass features BX3e: the 14:00 exit on the physically settled names only ...	0DTE→60 DTE, 15 names	holdout (opened)	203,289	28.4	+122.1% / +152	−63.8% / −80	**−11.03 % prem** / −14.09	+7,779 (2026-07-02)	−3,458 (2021-12-17)	[−15.23, −9.12]	PASS
30a	Compass features: the base trades (every feature is judged against them)	0DTE→60 DTE, 15 names	holdout (opened)	580,234	28.2	+158.6% / +672	−77.5% / −307	**−11.03 % prem** / −31.26	+45,477 (2026-03-30)	−32,491 (2026-10-19)	[−13.95, −7.83]	Descriptive
7f	S12 Intraday momentum: first-half-hour direction → 10:00–15:00 long option	all names	2022-11→2026-10	2,214	30.4	+100.1% / +62	−65.3% / −33	**−15.07 % prem** / −4.36	+1,188 (2026-07-29)	−222 (2025-04-08)	[−20.45, −9.22]	Nothing promoted
13	Step 2: buy near the money, sell at +100% or hold (within 0.5% of the price)	all names	check 2025–26	227,851	40.1	+80.8% / +621	−85.2% / −506	**−18.66 % prem** / −54.46	+9,039 (2025-04-09)	−18,091 (2025-11-28)	[−20.55, −16.99]	Every group loses before and after fees; near the money 29% double, ...
31	Compass direction: post-earnings drift in 31–60 DTE options (rerun)	single names	exploration (holdout unopened)	12,114	26.1	+79.5% / +471	−60.9% / −314	**−24.29 % prem** / −108.97	+18,029 (2023-10-30)	−13,866 (2026-06-26)	[−28.19, −19.82]	FAIL
8a	V1 T0: short every monthly ATM straddle across the optionable universe	all names	hold-out	12,484	44.4	+43.1% / +698	−78.3% / −994	**−24.41 % prem** / −243.08	+30,841 (2026-07-20)	−65,572 (2026-01-20)	[−32.05, −16.96]	Rejected
34	Edge H3: earnings IV-crush short iron condor (single names)	single names	exploration (holdout unopened)	2,209	30.3	+21.0% / +118	−52.5% / −283	**−30.24 % risk** / −161.54	+1,560 (2022-04-26)	−6,540 (2022-07-26)	[−32.38, −27.80]	FAIL
8c	V2b: long ATM straddle into the earnings report	single names	hold-out	3,543	10.1	+23.8% / +486	−37.2% / −436	**−31.05 % prem** / −342.34	+19,712 (2026-05-21)	−11,373 (2026-02-10)	[−34.85, −25.85]	Rejected
8b	V2a: short ATM straddle across the earnings report	single names	hold-out	3,682	25.6	+23.6% / +345	−56.5% / −630	**−35.95 % prem** / −380.58	+5,017 (2026-08-05)	−22,303 (2026-05-04)	[−38.94, −30.24]	Rejected

Studies with no trade-level result (each has its own short section below):

#	study	why no metrics	verdict
3	George engine audit: the phase 1 reproduction audit of #1–#2	the audit produced no trades of its own; its trade files are #2's, byte for byte (see #2).	Reproduced
9c	Lifecycle Tier I, daily tier, L-A3, H-FLO (by configuration)	committed as configuration-level cells (grid_by_setup.csv, cells.csv); the trade rows are in the scratchpad (lifecycle/archive, 281 MB) and were not re-run for	0 promoted
10	ATM straddle samples and the independent short-premium prereg (branch compass-research-2026-10-14)	price samples only (no trade with an exit); the pre-registered study never ran.	No result
11	Price-action catalogue and atlas (43 names)	descriptive tables, no trades.	Descriptive; V6 tested its levels
12	Step 1: every 0DTE contract broken down	a contract inventory, no trades.	Descriptive.
14	Step 3: 108 exits × 9 conditions	committed as per-rule summaries (exits.csv, exits_best.csv); the per-trade paths are committed but the exit grid was not re-run here.	Nothing made money in a way that held up; a −50% stop shrank …
16	Step 5: the runner model (model A/B) and its top 1% picks	the picks are committed as daily sums (top1_daily.csv), not trades.	Runners partly separable
18	Compass engine audit (six module groups)	the audit committed row-level paths and events (paths_*.parquet, events_*.parquet) and per-cell tables; its exit stacks were scored by cell, and no single trade	No directional rule survives after fees. What holds
22	Phase 2 hyp D: volatility breakout (straddles on the model's top decile)	committed as cells by scan minute and decile (cells_*.csv, minutes_*.parquet); no trade list of the pre-registered cell.	No edge
24	Phase 3 hyp E: short-premium lift of the vol model's quiet deciles	committed as by-scan cells and pooled tables; the candidate was set after the OOS read (see #26 condor_d20 for its trades).	No edge established — not pre-registrable
26d	Paper tracker: fly_spxw (SPX ±5 / ±25 flies)	a one-week sanity run outside the repository; no rows were kept.	Run only as a one-week sanity run outside the repository; replaced …
29f	Picker: flow gate and grade check	one session of flow (5 trades) and the vendor's own recorded returns (see #1, vendor-recorded).	Flow gate
30f	Compass features VN1 (venue SPY vs SPXW) and BC3 (V1 + X3 + D6)	VN1 pairs venues and BC3 needs the D6 direction feature; both are computed inside the study's screen code, and their per-trade arms are not on disk.	PASS
36	Edge H5 (SPY − SPXW late straddle) and H6 (monthly put spreads), forward only	forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted	No verdict.
38	Edge H9: overnight drift (forward only, withdrawn)	forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is "not available" because no forward session has been counted	No verdict.
39	Compass scanner (score ≥ 80, SPY/QQQ/IWM 0DTE)	signal-only: the gate failed before any option was priced, so trades, win rate, average win/loss, biggest win/loss, profit factor, drawdown, top-5 share and by-	FAIL at the gate
42	George A+ entry + EMA targets	the run was stopped before its prereg was committed; no file exists on disk or in history, so every metric is not available.	Never run.
44	Compass EMA: the owner's three EMA / ORB models	the run was stopped before computing: a partial engine and a session list exist in the scratchpad (compass_ema_partial/), no signal, trade or P&L.	Never computed.
45	DR-GCE gates 1 and 2 (Mode 1 Gamma Vacuum Squeeze)	signal-only: touch outcomes (signals.csv, gate2_pairs.csv), never traded to an exit; no P&L metric exists.	Gate 1 PASS
46	Trend-day test	never run (the trend-day share is only described in step 6).	Never registered, never run.

The rules that passed, with their numbers

Seven rules passed a pre-registered holdout or single test. None of them makes a losing base set profitable, except H7, which is a stock trade.

rule	what it does	the test	result (Δ = with − without, points of the debit a trade; H7 = $ a trade)	forward N (committed)
V1 spread veto	skip a base trade whose (ask − bid) ÷ mid > 10% at entry	Compass features Phase A holdout (a2e14e4, 2a49049)	Δ **+2.61 [+1.76, +3.46]**; −8.43% with vs −11.03% without	23,960 sessions (out of reach)

rule	what it does	the test	result (Δ = with − without, points of the debit a trade; H7 = $ a trade)	forward N (committed)
X3 14:00 exit	sell a 0DTE long entered before 14:00 at the 14:00 bid	same	Δ **+7.78 [+4.14, +10.95]**; −9.51% vs −17.29%	369 sessions
VN1 venue	keep the SPXW/SPY venue with the lower ((ask − mid) + fee ÷ 100) ÷ mid	same (failed on the first, duplicated, read)	Δ **+1.18 [+0.50, +2.20]**	587
BV1b	V1 at 3%	Phase B holdout (b6737c7, e5dd8c6), not virgin	Δ **+4.30 [+2.66, +5.85]**; −6.73% vs −11.03%	481
BX3e	X3 on the physically settled names only	same	Δ **+7.67 [+4.23, +10.69]**; −11.03% vs −18.71%	265
BC3	V1 veto + X3 exit + D6 direction, against all base trades	same	Δ **+6.77 [+4.82, +8.19]**; −4.26% vs −11.03%	341
H7 pre-FOMC drift	100 shares from the close before a 2:00 pm FOMC statement to the 13:58 bar	single test (edca472, 6c94bb8)	SPY **+$70.43 [+14.64, +129.31]**, QQQ **+$69.59 [+23.10, +122.14]**; QQQ clears Bonferroni at m = 13	366 SPY / 483 QQQ trades (fcc1f4d)

These rules are frozen as written:

- Compass's decision layer uses V1, X3 and VN1 exactly as frozen (`FORWARD_RULE.md` b101ae1, `PREREG_PICKER.md` §4 fees).
- The daily forward job counts every rule here against its N. No verdict is read before a rule reaches its N.

Also positive on a single historical read, but not passed:

- S17's index put-writing passes its carry-lane rule (#7j). It is a carry with alpha 0, not an edge.
- H8 passes for QQQ at 1× only.
- Phase 3 hyp F (#25) passes its four-part rule on the OOS that chose it.

The holdout ledger (the report's copy)

This is a copy of `research_notes/HOLDOUT_LEDGER.md` (9af93fc) with three rows added after it. The committed ledger is unchanged: nothing in it is wrong, it only predates DR-GCE.

Holdout A = index direction data 2017-06-01 → 2022-05-12. **Holdout B** = the 28 drawn months.

study (commit)	data and span	rules read	A	B
S04 (`b037899`)	SPY/QQQ/IWM bars 2017-06-01 → 2026-10-09	Quick Scalps signal, forward returns (+ ablations, not counted)	1	1
S06 (`4aba507`)	same	7 triggers × 4 filter sets	28	28
S12 (`be991de`)	bars 2017-06 → 2026-10-09	first-half-hour momentum, 3 windows	3	3
S12b (`be991de`)	0DTE 2022-11-11 → 2026-10-09	2 option expressions	0	2
S16a	bars 2017-06 → 2026-10-09	108 cells ÷ 3 names	36	36
S03, S16b (`3c8d6cb`)	weekly chains, sample 2017-06 → 2026-10-09	Quick Scalps weekly; prior-day fade 2 cells × 2 exits	5	5
H10 (registry)	SPY bars 2017 → 2026-10-09	last-30-minute momentum	1	1
H17 (registry)	SPX 2017 → 2026	4 day filters	4	4
S17	SPY/QQQ monthly chains 2012-06 → 2026-10	5 structures	5	5

study (commit)	data and span	rules read	A	B
Lifecycle daily tier (`50de62a`)	SPY/QQQ/IWM option EOD 2012-06 → 2026-10	≥ 19 walk-forward finalists	≥ 19	≥ 19
V6 / atlas (`0d706b6`, `1660366`)	43 names incl. SPY/QQQ/IWM bars 2017-06 → 2026-10	PDH/PDL, round numbers	2	2
V8 (registry)	SPXW 2017-06 → 2022-04, SPY 2017-06 → 2022-10 month-ends	month-end 13:00 straddle	1	1
Compass H1 (`4a02144`)	SPY 2017-01 → 2025-10-10, QQQ 2012-06 →	overnight reversal	1	1
Edge H4 (`dfa575d`)	SPY/QQQ 2012/2017 → 2026-10-22 (exploration context + test)	intraday momentum	1	1
Edge H7, H8 (`6c94bb8`)	SPY 2017 → 2026-09, QQQ 2012/2013 →	pre-FOMC drift; turn of month	2	2
Compass features (`a2e14e4`, `2a49049`, `e5dd8c6`)	ETFs 2012-06 →; SPXW/SPY/QQQ 0DTE 2022 → 2026-10-21; its own 43-month holdout read (twice)	23 features screened	23	23
Compass scanner signal check (`142fcf4`, `9e3888e`)	SPY/QQQ/IWM bars, Pick 2017-07-05 → 2022-11-18 (its non-held-out months; the 28 never opened)	the scanner score (RSI/VWAP/RVOL/structure family)	1	0
S01, S02 (`5e90201`, `6bf5934`)	SPY/QQQ/IWM 0DTE 2022-11-11 → 2026-10-09	Quick Scalps + 41 variants	0	42
S09, S10 (`7167541`, `3c8d6cb`)	SPXW 0DTE 2022-05 →; SPY/QQQ/IWM	QS on SPXW; QS fade (≥ 2 cells)	0	≥ 3
S15 (registry)	SPY/QQQ 0DTE 2022-11-11 → 2026-10-09	10:00 straddle, 3 conditions	0	3
H9, H11, H13, H15, H16 (registry)	SPY/QQQ/SPXW 0DTE 2022-05 / 2025-06 → 2026-10	straddles/strangles; grids of 1,890 + 2,736 configs	0	≥ 4,630
Steps 1–5 (`2185eb9` → `491a0c7`)	every 0DTE contract 2022-05-13 → 2026-10-09	buy/sell-and-hold, times, exits, turn-ups, volume (≥ 1 a step)	0	≥ 5
Lifecycle Tier I (`425c7f4`)	0DTE SPY/QQQ/IWM/SPXW 2022-05 → 2026-10-09	12 setups × 8 structures × 116 policies	0	11,136
V5, V7 (registry)	same-day chains 2022-05/11 →	calibration; event afternoons	0	2
Engine audit, phase 2 (4), phase 3 (E, F), backtest audit, SPXW analogues (`2820bb4`)	0DTE 2022 → 2026-10-09	≥ 1 each	0	≥ 9
Picker (`c502122`, `1e10053`)	SPY/QQQ/IWM/SPXW 0DTE 2022 → 2026-10-21	opening-range fade + improved	0	2
EMA system v1 preview	bars of 60 sessions before each week (2025-11, 2026-04 bars as warm-up only, no outcome)	—	0	0
Compass EMA ladder (previews 2025-02/03)	not held out; holdout not read	—	0	0
Step 6 (`30babaa`, `44b5547`)	0DTE 2022-05-13 → 2026-10-09, the 28 months never opened	18 rules	0	0
Compass ICT (`9d6c37e`, `0943030`)	exploration only; holdouts never opened	—	0	0
Compass EMA (this study, prereg)	not yet read	—	0	0
Compass EMA (`9af93fc`)	prereg only; the run was stopped before computing	—	0	0
DR-GCE gates 1, 2 (`01d5c8e`, `041baba`, `b20eb9c`; `294d4f2`, `00eb3d5`)	SPXW 0DTE exploration 2023-12-01 → 2026-10-09, the 28 months never opened (guard selftest 5/5)	Mode 1 / Mode 2 touch gate; direction vs volatility	0	0

study (commit)	data and span	rules read	A	B
This final report (2026-10-23)	reads only spans already read; reruns #31 (exploration X only) and #40 (2025-02/03 only)	—	0	0

Running count before Compass EMA's read: holdout A ≥ 133 rules; holdout B ≥ 15,966 rules (11,136 of them Lifecycle Tier I and ≥ 4,630 the short-premium grids, all in B's 13 option-era months; the 15 pre-2022-05 months carry the A-span studies' 133 less the scanner's 1 = 132).

Single names, not counted: V3 / V4 (11 singles' same-day chains 2017-06 → 2026-10), S06-TierA (34 names' bars 2017-06 →), Compass features' singles (2017-06 →, AVGO/MU holdout read), H3 (earnings condors 2019-04 → 2022-12), Compass direction (X names 2019-04 →; its holdout names never opened).

What has read each holdout, and what remains unopened

holdout	read by	unopened for
A (2017-06 → 2022-05-12)	≥ 133 rules (S04, S06, S12, S16a, S03/S16b, H10, H17, S17, the daily tier, V6, V8, Compass H1, H4, H7/H8, Compass features, the scanner's signal check)	every rule from the scanner on: Compass ICT (#43), Compass EMA (#44), the EMA ladder (#40), DR-GCE (#45), step 6 (#17)
B (28 months)	≥ 15,966 rules, before the holdout was defined (steps 1–5, S01/S02, S09/S10, S15, the short-premium grids, Tier I, V5/V7, the engine audit, phases 2–3, the picker)	the rules defined after it: #17, #39, #40, #43, #44, #45 ("unopened for this rule only")
Compass features' own (43 months + AVGO, MU, QQQ)	opened: Phase A twice (the first on duplicated rows), Phase B once	—
Compass direction: 1,600 names 2019-01 → 2022-12	—	**unopened** (the screen failed)
Edge H3: 1,600 names	—	**unopened** (not fetched)
The scanner's held-out tickers AMD, AVGO	—	**unopened**
Lifecycle hold-out 2025-10-01 → 2026-10-09	opened once per group (Tier I, L-A3, H-FLO, the daily tier, P1/P2)	—
V1–V8 hold-outs	each opened once	—
ThetaData lanes TEST (2025 → 2026-10)	read (T-A, T-B, T-C, T-E, H15)	—
Phase 2/3 OOS 2025-01 → 2026-10-09	read once each; this is the tracker's and the SPXW test's "selection" period	—
Forward from 2026-10-12	the paper tracker (9 sessions to 10-22)	features (0 sessions counted), H5 (none), H6 (from 2026-11-02), H7 (from 2026-10-28)

The data inventory

All market data is stored files. Nothing was fetched for this report.

data	span	where
Same-day option chains, 1-minute NBBO (every contract)	SPY 989 sessions 2022-11-11 → 2026-10-22; QQQ 989, same span; IWM 629, 2024-04-19 → 2026-10-21; SPXW 1,115, 2022-05-13 → 2026-10-22	scratchpad `theta/data_full/chains/` (forward sessions 2026-10-12 → 21 also committed in `data/theta_forward/`)
Underlying 1-minute bars	SPY 3,687 sessions 2012-06-01 → 2026-10-22; QQQ 3,619; IWM 3,618	scratchpad `theta/data/underlying/`
SPX prints and VIX, 1-minute	SPX 2,362 sessions, VIX 2,395, 2017-06-01 → 2026-10-22	scratchpad `theta/data/index/`
Official closes	Cboe's SPX history 1975 → 2026-10-21; UW daily bars	`data/official_closes/`
Option EOD and open interest, every contract	the account's bulk files 2026-10-06 → 12; index-ETF option EOD 2012-06 → 2026-10	`data_lake/thetadata/flat/`, scratchpad `lifecycle/eod_by_exp`
Stock daily bars, every symbol	2012-06 → 2026-10-13 (26,855 symbols listed)	`data_lake/thetadata/stock_eod/`

data	span	where
Unusual Whales per-ticker histories	daily bars, greek exposure, IV rank, realised volatility, VRP, earnings; about one year to 2026-10-13	`data_lake/unusualwhales/`
Unusual Whales recorder (flow, dark pool, net-prem ticks, minute bars, OI by strike)	from 2026-10-21	`data_recorder/` (git-ignored)
Post-market 1-minute bars (UW)	16 name-sessions fetched once for the after-hours study, plus the backtest cache 2025-05 → 2026-10	`data/postmarket_1m/`, scratchpad `bt/cache`
Macro calendar (FOMC, CPI, payrolls)	2022 → 2027	`data/macro_events_2022_2027.csv`
GeorgeFi evidence (cards, books, API captures)	2026-01-20 → 2026-10-13	`evidence/` at 7dfea57
Trade-level outputs used here and not committed (kept in the scratchpad)	tracker `data_recorder/paper_backtest_official` (316 MB); step 6 population `step6/` (39 MB); features `features_work/{expl,hold}`; direction rerun `dir/work/expl_trades.parquet`; EMA ladder rerun `ema_ladder_out/`	as listed

Forward tracking

The daily job. `scripts/daily_forward.py` runs after each close, in this order, stopping at the first failure:

1. fetch the session's seven files: the Theta Terminal on the owner's machine, or one SDK login inside the owner's window;
2. verify them;
3. merge them;
4. read the official close: UW's daily bar, or Cboe for SPX;
5. read the post-market decision price;
6. replay the session through the live paper tracker (condor_em1, fly5, condor_d20), settled on the official close;
7. append it to `research_notes/Compass phase 3/forward_cumulative.md`;
8. run the Compass features forward test (V1, X3, VN1, BV1b, BX3e, BC3 under `FORWARD_RULE.md` b101ae1);
9. run the edge-search forward tests (H5, H6 and H7).

A session is appended only whole. No verdict is read before a model reaches its committed N. Interim figures are logged without early stopping.

Where it stands (2026-10-23).

- **The paper tracker's period iii:** 9 sessions, 2026-10-12 → 22. Results in that window:
  - condor_em1: 44 trades, +$9.99 a trade.
  - fly5 iron: 36 trades, +$52.93.
  - condor_d20: 41 trades, +$23.38.

  This is an execution and data-quality record. The thresholds are 60 → 120 sessions for condor_em1, 150 for condor_d20's D1 re-test, and 250 a name for the flies.
- **The Compass features:** 0 forward sessions counted (from 2026-10-22).
- **H5:** forward from 2026-10-23.
- **H6:** from 2026-11-02.

H7 live. The pre-FOMC drift is wired into step 8 of the job (712b0ad). It trades 100 SPY / 100 QQQ from the official close before each scheduled 2:00 pm statement to the 13:58 bar's close. It runs on the official closes and minute bars the job already keeps, with the macro calendar and H7's dividend table. The first forward trade is the statement of 2026-10-28. N is 366 trades for SPY and 483 for QQQ (`FORWARD_RULE_H7.md` fcc1f4d), about 46 and 60 years. The verdict is read at N only.

The H9 removal. H9 (overnight drift) was registered forward-only (edca472) and wired in 712b0ad. The owner withdrew it on 2026-10-23, and it was removed from the job (c68f389) before any H9 forward row was written. The registered rule is unchanged.

Method (every section)

- **Fills.** As each study ran them: ask in, bid out, at the 1-minute NBBO unless the section says otherwise. Older studies (#2, #4b, #5) used Unusual Whales bars with a modelled spread. Each section states its fee schedule.
- **Win and loss.** A win is a trade whose P&L after fees is above zero. A breakeven counts as a loss.
- **Session span.** The calendar is every stored SPY minute-file session in the period's span. A session with no trade counts as zero.
- **Interval.** The interval is `audit_lib.session_stats` with blocks of 20 sessions (2,000 draws, its own seed), for the mean $ and the mean % a trade. A period under 60 sessions gets no interval. Some studies used other intervals, such as iid session or trade bootstraps, so the numbers here can differ from a study's own; the verdict is always the study's.
- **Drawdown.** Max drawdown is taken on the cumulative $ of session sums. The longest drawdown is the longest run of sessions spent below a previous peak.
- **Top-5 share.** This is the five best sessions' share of a positive total. Where the total is ≤ 0, the five best sessions' $ and the total without them are printed instead.
- **Random baselines.** Each is quoted from the study that computed it. For S10, S12, S12b and S16b, the mean of the study's own matched-control rows (seeds 11–15) is taken from the same trade file. None was re-simulated.
- **By year.** All periods together, grouped by entry year.
- **Charts.** Cumulative $ per name and pooled, with each period shaded and labelled.

Part A. Studies before the fresh start (75c29c9), read from 7dfea57

#1 GeorgeFi vendor cards (the vendor’s own recorded P&L)

Folder: `evidence/public_api/, research/georgefi_full_history/normalized/ (@7dfea57)`. **Commits:** — / errata 5ed66df / cf555c9. **Dates:** 2026-10-13.

Hypothesis. GeorgeFi’s published 0DTE and swing calls carry a positive return.

Rule. The vendor’s own cards: ticker, contract, entry and exit premium, size and P&L as GeorgeFi recorded them (books v2.0.0-ict, trade_now_v1, singles_today_v1, spx_plan_v1).

Fills and costs. The vendor’s recorded fills; no fee or spread applied by us.

Data. 1,149 closed cards, 2026-01-20 → 2026-10-13 (Epoch 1 2026-01-20 → 02-27; Epoch 2 from 2026-03-02).

Periods and holdouts. Epoch 1 / Epoch 2, descriptive. No holdout.

Pass rule (as pre-registered). None (descriptive). The grade check of the picker test (#29) reads the same cards.

Verdict. Not evidence: re-priced at the NBBO (#4) Epoch 1 falls from +$43,248 recorded to +$12,039 executable and Epoch 2 from −$3,757 to −$24,988 (G1).

> **Vendor-recorded, not re-marked.** Every figure below is the vendor’s own recorded card P&L, as published; none was re-priced on market quotes.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
v2.0.0-ict 0DTE	Epoch 1	373	25 / 29	82.6	+25.5, +161.46	−23.6, −195.44	+14.5, +39.00	**+16.92**, +99.27	n/a (< 60 sessions); n/a (< 60 sessions)
v2.0.0-ict SWING	Epoch 1	173	10 / 11	70.5	+16.1, +95.18	−3.6, −105.67	+10.7, +29.00	**+10.29**, +35.97	n/a (< 60 sessions); n/a (< 60 sessions)
all names	Epoch 1	546	28 / 29	78.8	+22.8, +142.65	−14.8, −155.97	+13.0, +36.00	**+14.82**, +79.21	n/a (< 60 sessions); n/a (< 60 sessions)
singles_today_v1 0DTE	Epoch 2	29	6 / 7	20.7	+43.3, +46.33	−45.8, −61.30	−50.0, −24.00	**−27.35**, −39.03	n/a (< 60 sessions); n/a (< 60 sessions)
singles_today_v1 SWING	Epoch 2	19	4 / 7	57.9	+20.3, +40.18	−16.2, −19.13	+8.2, +4.00	**+4.94**, +15.21	n/a (< 60 sessions); n/a (< 60 sessions)
spx_plan_v1 0DTE	Epoch 2	11	7 / 7	54.5	+97.4, +902.50	−79.0, −480.00	+54.1, +200.00	**+17.23**, +274.09	n/a (< 60 sessions); n/a (< 60 sessions)
trade_now_v1 0DTE	Epoch 2	55	35 / 69	25.5	+90.5, +1,247.32	−35.0, −667.44	−13.9, −130.00	**−3.06**, −180.05	[−32.03, +25.54]; [−540.38, +201.06]
v2.0.0-ict 0DTE	Epoch 2	429	113 / 159	51.0	+28.5, +129.99	−19.3, −120.22	+0.9, +3.00	**+5.13**, +7.51	[−0.81, +12.76]; [−17.29, +30.78]
v2.0.0-ict SWING	Epoch 2	60	28 / 121	56.7	+25.9, +94.03	−19.9, −94.02	+6.1, +22.00	**+6.06**, +12.54	[−8.91, +14.87]; [−57.80, +24.30]
all names	Epoch 2	603	132 / 162	48.1	+32.6, +190.56	−24.2, −188.56	+0.0, +0.00	**+3.13**, −6.23	[−0.74, +9.76]; [−62.87, +13.75]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
v2.0.0-ict 0DTE	Epoch 1	+4,020.00, +1,057.9%, 2026-02-06, O:SPX260206C06930000	−1,375.00, −78.6%, 2026-02-18, O:SPXW260218P06860000	3.91	310	1	60%
v2.0.0-ict SWING	Epoch 1	+910.00, +59.6%, 2026-01-29, O:USAR260130P00023000	−810.00, −21.4%, 2026-01-21, O:COIN260123P00230000	2.15	579	3	109%
all names	Epoch 1	+4,020.00, +1,057.9%, 2026-02-06, O:SPX260206C06930000	−1,375.00, −78.6%, 2026-02-18, O:SPXW260218P06860000	3.39	310	1	62%
singles_today_v1 0DTE	Epoch 2	+104.00, +50.5%, 2026-10-07, O:MU261007C01085000	−216.00, −63.5%, 2026-10-09, O:MU261009P01005000	0.20	1,132	7	total ≤ 0 (best 5 days −671; without them −461)
singles_today_v1 SWING	Epoch 2	+144.00, +38.9%, 2026-10-08, O:GOOGL261009P00347500	−70.00, −53.8%, 2026-10-08, O:TQQQ261009C00084000	2.89	0	0	100%
spx_plan_v1 0DTE	Epoch 2	+1,750.00, +140.0%, 2026-10-09, O:SPXW261009C07800000	−935.00, −94.4%, 2026-10-06, O:SPXW261006P07815000	2.26	2,140	5	162%
trade_now_v1 0DTE	Epoch 2	+3,312.50, +160.6%, 2026-07-13, O:SPXW260713P07505000	−3,400.00, −60.7%, 2026-07-13, O:SPXW260713C07570000	0.64	15,740	67	total ≤ 0 (best 5 days +8,318; without them −18,220)
v2.0.0-ict 0DTE	Epoch 2	+2,055.00, +24.9%, 2026-03-04, O:SPXW260304C06815000	−2,015.00, −24.7%, 2026-03-05, O:SPXW260305P06840000	1.13	4,066	63	227%
v2.0.0-ict SWING	Epoch 2	+501.00, +68.1%, 2026-03-04, O:IWM260313P00261000	−475.00, −19.2%, 2026-03-17, O:COIN260320C00210000	1.31	1,445	112	320%
all names	Epoch 2	+3,312.50, +160.6%, 2026-07-13, O:SPXW260713P07505000	−3,400.00, −60.7%, 2026-07-13, O:SPXW260713C07570000	0.94	14,299	101	total ≤ 0 (best 5 days +13,434; without them −17,191)

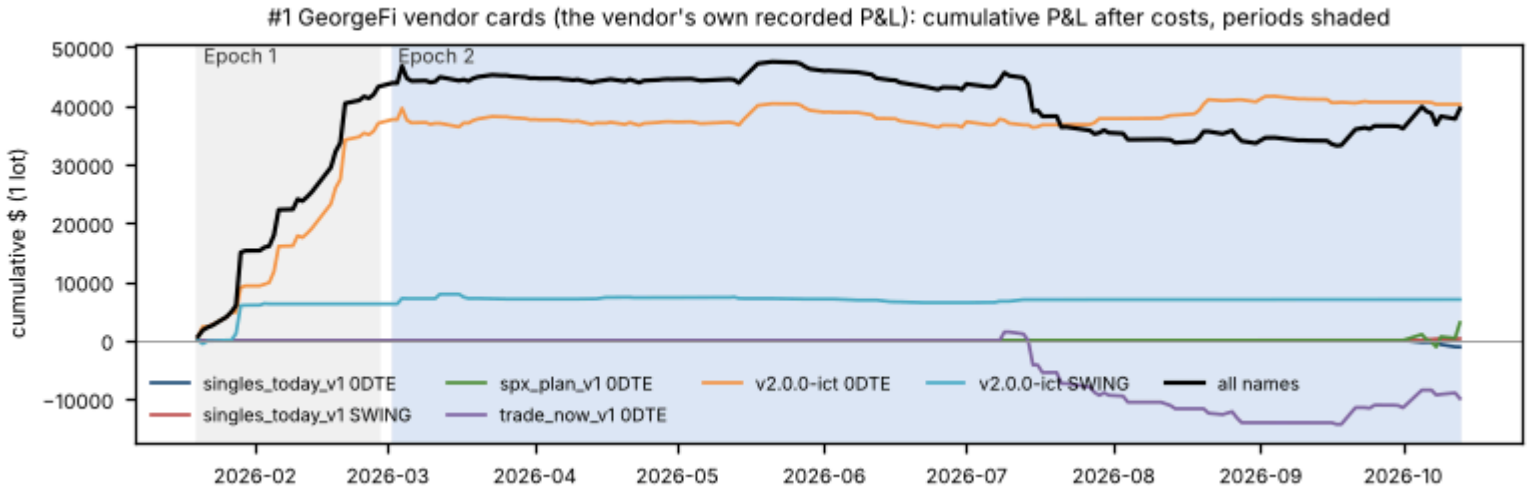
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- Epoch 1: ICT: Take profit at MAJOR liquidity pool 6.8%; TIER A +10% (1/5) 4.8%; EOD AUTO-CLOSE (3:58 PM) 4.6%; ICT TIER 1 PROFIT +11% (PARTIAL) 3.7%; ICT TIER 1 PROFIT +12% (PARTIAL) 3.3%; TIER A +10% (1/4) 2.9%; ICT TIER 1 PROFIT +13% (PARTIAL) 2.9%; TIER A +10% (2/8) 2.6%
- Epoch 2: ICT: Take profit at MAJOR liquidity pool 5.6%; SINGLES_TODAY_TRAIL 4.3%; STOP LOSS (-12%) 4.1%; GEX EXIT: Accelerator reversed against B 3.8%; SINGLES_TODAY_STOP 2.3%; STOP LOSS (0%) 2.0%; GEX REGIME SHIFT: NEG→POS while losing (1.8%; SWING PROFIT_TARGET PARTIAL EXIT (25%): 1.8%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	singles_today_v1 0DTE	29	20.7	−27.35	−39.03	−1,132
2026	singles_today_v1 SWING	19	57.9	+4.94	+15.21	+289
2026	spx_plan_v1 0DTE	11	54.5	+17.23	+274.09	+3,015
2026	trade_now_v1 0DTE	55	25.5	−3.06	−180.05	−9,902
2026	v2.0.0-ict 0DTE	802	65.7	+10.62	+50.18	+40,247
2026	v2.0.0-ict SWING	233	67.0	+9.20	+29.93	+6,975
2026	all names	1,149	62.7	+8.68	+34.37	+39,492



Amendments, deviations and integrity checks. pnl_reported is used as recorded; 24 cards carry a data_error flag and are kept, as the vendor kept them.

What it showed. The vendor’s record is positive in Epoch 1 and near zero in Epoch 2; the same rows re-priced at the market’s quotes are a loss in Epoch 2 (#4).

#2 George rebuild: the GeorgeFi-like reference (sweep / ict+gex / all / none / ladder)

Folder: `backtest/, BACKTEST_RESULTS.md (@7dfea57)`. **Commits:** — / — / 8f5b067. **Dates:** 2026-10-13.

Hypothesis. A level-sweep reversal into the nearest-strike 0DTE contract, managed by GeorgeFi's ladder, makes money after costs.

Rule. Levels PDH/PDL, PWH/PWL, PMH/PML, ORH/ORL and the prior-day call/put walls; a sweep within the last 3 bars closed back inside by max(0.25 ATR14, 0.02%) → buy the nearest-strike 0DTE call (low swept) or put (high swept) at the next bar's VWAP; ladder exit: stop −25%, T1 +25% half, T2 +45%, breakeven lock at +6%, trail at 65% of the high-water mark from +15%, 30-minute time stop, flat 15:55.

Fills and costs. The "spread" model: a half-spread of 1% (SPY, QQQ) or 2% (IWM) of premium each side plus $0.70 a contract each side; one contract. (Unusual Whales 1-minute bars, no NBBO.)

Data. SPY, QQQ, IWM 0DTE, 345 sessions 2025-06-02 → 2026-10-09 (SPX excluded: no index bars on the plan).

Periods and holdouts. In-sample < 2026-03-02, out-of-sample 2026-03-02 → 2026-10-09 (read). No holdout.

Pass rule (as pre-registered). None pre-registered (an exploratory grid of 144 variants).

Verdict. Rejected: 0 of 144 variants positive after costs (registry H1).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	in-sample 2025-06→2026-02	504	175 / 192	35.1	+11.7, +7.50	−18.7, −11.21	−6.2, −3.60	**−8.06**, −4.64	[−9.87, −6.89]; [−5.99, −3.78]
QQQ	in-sample 2025-06→2026-02	526	165 / 192	35.0	+14.3, +18.93	−12.8, −16.36	−3.1, −3.44	**−3.32**, −4.02	[−4.73, −2.65]; [−5.98, −2.72]
SPY	in-sample 2025-06→2026-02	550	172 / 192	37.6	+13.6, +16.40	−12.7, −14.62	−3.0, −3.21	**−2.79**, −2.95	[−4.25, −1.51]; [−4.84, −1.36]
all names	in-sample 2025-06→2026-02	1,580	184 / 192	35.9	+13.2, +14.45	−14.7, −14.11	−3.3, −3.40	**−4.65**, −3.84	[−5.64, −4.20]; [−4.95, −3.16]
IWM	out-of-sample 2026-03→10	483	143 / 160	41.6	+11.2, +8.79	−17.1, −11.13	−5.6, −3.16	**−5.30**, −2.84	[−7.54, −3.82]; [−4.49, −1.88]
QQQ	out-of-sample 2026-03→10	461	142 / 160	36.9	+13.6, +25.28	−11.5, −18.63	−2.7, −4.00	**−2.21**, −2.44	[−4.23, −0.59]; [−4.68, −0.12]
SPY	out-of-sample 2026-03→10	455	144 / 160	32.7	+13.9, +20.02	−12.6, −15.26	−3.1, −3.70	**−3.94**, −3.71	[−4.92, −3.04]; [−4.74, −2.58]
all names	out-of-sample 2026-03→10	1,399	155 / 160	37.2	+12.8, +17.40	−13.7, −15.05	−3.0, −3.68	**−3.84**, −2.99	[−5.02, −3.01]; [−3.83, −2.31]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	in-sample 2025-06→2026-02	+27.99, +28.9%, 2026-01-14, IWM260114C00260000	−62.96, −29.1%, 2025-09-17, IWM250917P00240000	0.36	2,350	192	total ≤ 0 (best 5 days +173; without them −2,510)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	in-sample 2025-06→2026-02	+100.79, +32.2%, 2025-11-20, QQQ251120P00600000	−104.39, −27.1%, 2025-11-20, QQQ251120C00593000	0.62	2,263	177	total ≤ 0 (best 5 days +432; without them −2,544)
SPY	in-sample 2025-06→2026-02	+89.69, +32.1%, 2025-10-17, SPY251017C00658000	−84.59, −27.2%, 2025-11-20, SPY251120C00660000	0.68	1,700	180	total ≤ 0 (best 5 days +340; without them −1,961)
all names	in-sample 2025-06→2026-02	+100.79, +32.2%, 2025-11-20, QQQ251120P00600000	−104.39, −27.1%, 2025-11-20, QQQ251120C00593000	0.57	6,127	185	total ≤ 0 (best 5 days +513; without them −6,584)
IWM	out-of-sample 2026-03→10	+49.81, +29.5%, 2026-03-11, IWM260311C00251000	−44.72, −29.4%, 2026-06-11, IWM260611P00289000	0.56	1,437	151	total ≤ 0 (best 5 days +223; without them −1,595)
QQQ	out-of-sample 2026-03→10	+141.28, +32.3%, 2026-06-26, QQQ260626C00705000	−127.93, −27.0%, 2026-06-25, QQQ260625C00711000	0.79	1,213	159	total ≤ 0 (best 5 days +676; without them −1,801)
SPY	out-of-sample 2026-03→10	+104.71, +32.2%, 2026-06-25, SPY260625C00730000	−81.11, −27.2%, 2026-04-02, SPY260402P00655000	0.64	1,720	160	total ≤ 0 (best 5 days +299; without them −1,985)
all names	out-of-sample 2026-03→10	+141.28, +32.3%, 2026-06-26, QQQ260626C00705000	−127.93, −27.0%, 2026-06-25, QQQ260625C00711000	0.68	4,276	159	total ≤ 0 (best 5 days +634; without them −4,817)

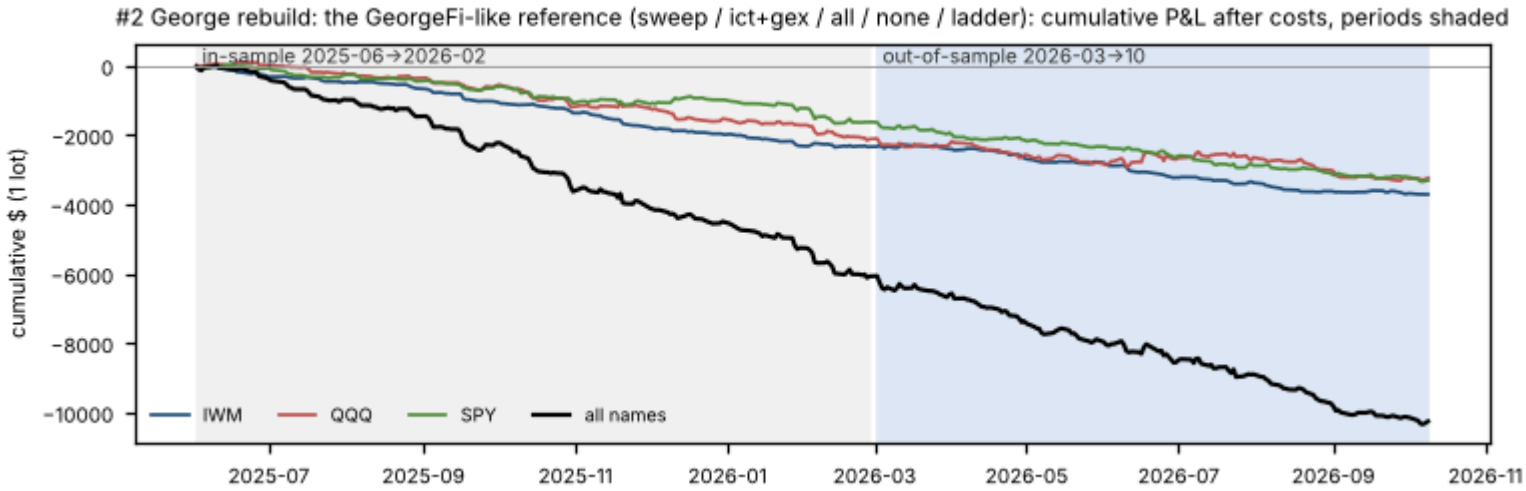
Random baseline: random-time entries, same contract choice and ladder (9,896 trades): −$3.20 a trade [−3.70, −2.70] under the spread model; the reference is −$3.44 [−4.23, −2.70]. No percentile was computed.

Exit reasons (share of trades):

- in-sample 2025-06→2026-02: BREAKEVEN 36.8%; STOP 26.1%; TRAIL 22.0%; T1+TRAIL 9.9%; T1+T2 4.2%; TIME_STOP 0.9%
- out-of-sample 2026-03→10: BREAKEVEN 38.5%; STOP 23.7%; TRAIL 23.6%; T1+TRAIL 10.4%; T1+T2 3.4%; TIME_STOP 0.6%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	399	34.1	−8.71	−4.98	−1,988
2026	IWM	588	41.2	−5.36	−2.93	−1,722
2025	QQQ	425	35.5	−3.40	−3.62	−1,537
2026	QQQ	562	36.1	−2.34	−3.02	−1,700
2025	SPY	444	38.3	−2.29	−2.22	−986
2026	SPY	561	33.2	−4.11	−4.14	−2,321
2025	all names	1,268	36.0	−4.68	−3.56	−4,511
2026	all names	1,711	36.9	−3.96	−3.36	−5,743



Amendments, deviations and integrity checks. Reproduced byte for byte by the phase 1 audit (#3): trades.csv sha256 0e8c41c819e74eb0… on both runs.

What it showed. The reference variant loses $3.44 a trade, no better than random entry times with the same exits.

#2b George rebuild: the best of 144 variants (sweep / ict+gex / morning / posgamma_fade / trail)

Folder: `backtest/ (@7dfea57)`. **Commits:** — / — / 8f5b067. **Dates:** 2026-10-13.

Hypothesis. The best-ranked of the 144 variants (selected in sample) holds out of sample.

Rule. As #2 but 10:00–12:00 only, sweep reversals only when spot is above the prior-day gamma flip, and the trail engine (no profit targets).

Fills and costs. As #2 (spread model).

Data. As #2.

Periods and holdouts. As #2.

Pass rule (as pre-registered). None pre-registered.

Verdict. Rejected: −$2.41 a trade [−4.10, −0.47]; out of sample −$1.8 [−4.5, +0.9].

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	in-sample 2025-06→2026-02	134	91 / 187	29.9	+12.6, +8.43	−16.1, −11.12	−6.1, −4.06	**−7.51**, −5.29	[−10.05, −5.20]; [−6.90, −3.75]
QQQ	in-sample 2025-06→2026-02	161	98 / 189	32.3	+16.5, +25.71	−12.4, −16.02	−3.1, −3.64	**−3.11**, −2.54	[−7.01, −0.26]; [−7.17, +2.10]
SPY	in-sample 2025-06→2026-02	154	96 / 186	36.4	+16.8, +23.80	−12.6, −15.68	−3.1, −3.46	**−1.91**, −1.33	[−4.22, +0.67]; [−4.90, +3.61]
all names	in-sample 2025-06→2026-02	449	145 / 191	33.0	+15.6, +20.32	−13.6, −14.38	−3.3, −3.60	**−4.01**, −2.94	[−5.87, −2.66]; [−5.23, −0.81]
IWM	out-of-sample 2026-03→10	131	79 / 160	39.7	+10.4, +9.25	−17.4, −14.23	−5.7, −3.92	**−6.41**, −4.91	[−8.92, −4.38]; [−6.62, −3.47]
QQQ	out-of-sample 2026-03→10	122	76 / 157	45.9	+12.4, +28.48	−10.2, −18.51	−2.6, −4.07	**+0.17**, +3.06	[−1.94, +2.98]; [−1.14, +8.89]
SPY	out-of-sample 2026-03→10	132	76 / 160	35.6	+14.0, +21.29	−11.8, −16.66	−2.9, −3.94	**−2.65**, −3.15	[−5.03, −0.65]; [−6.96, −0.25]
all names	out-of-sample 2026-03→10	385	116 / 160	40.3	+12.2, +19.85	−13.3, −16.36	−2.8, −3.98	**−3.04**, −1.78	[−4.57, −1.42]; [−3.61, +0.21]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	in-sample 2025-06→2026-02	+37.21, +53.2%, 2026-01-02, IWM260102P00248000	−35.88, −29.7%, 2025-11-21, IWM251121P00231000	0.32	709	187	total ≤ 0 (best 5 days +111; without them −819)
QQQ	in-sample 2025-06→2026-02	+264.54, +84.5%, 2025-11-20, QQQ251120P00600000	−88.87, −27.2%, 2025-11-21, QQQ251121C00588000	0.77	608	182	total ≤ 0 (best 5 days +647; without them −1,056)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	in-sample 2025-06→2026-02	+199.58, +71.5%, 2025-10-17, SPY251017C00658000	−83.26, −27.2%, 2025-11-21, SPY251121C00655000	0.87	329	84	total ≤ 0 (best 5 days +468; without them −673)
all names	in-sample 2025-06→2026-02	+264.54, +84.5%, 2025-11-20, QQQ251120P00600000	−88.87, −27.2%, 2025-11-21, QQQ251121C00588000	0.69	1,485	185	total ≤ 0 (best 5 days +774; without them −2,095)
IWM	out-of-sample 2026-03→10	+45.99, +33.6%, 2026-03-25, IWM260325C00250000	−43.58, −29.4%, 2026-06-25, IWM260625P00300000	0.43	665	140	total ≤ 0 (best 5 days +136; without them −780)
QQQ	out-of-sample 2026-03→10	+121.71, +29.8%, 2026-06-25, QQQ260625C00711000	−103.32, −27.1%, 2026-06-29, QQQ260629C00712000	1.31	211	40	146%
SPY	out-of-sample 2026-03→10	+76.74, +42.9%, 2026-04-06, SPY260406P00658000	−81.11, −27.2%, 2026-04-02, SPY260402P00655000	0.71	517	143	total ≤ 0 (best 5 days +242; without them −657)
all names	out-of-sample 2026-03→10	+121.71, +29.8%, 2026-06-25, QQQ260625C00711000	−103.32, −27.1%, 2026-06-29, QQQ260629C00712000	0.82	872	133	total ≤ 0 (best 5 days +607; without them −1,293)

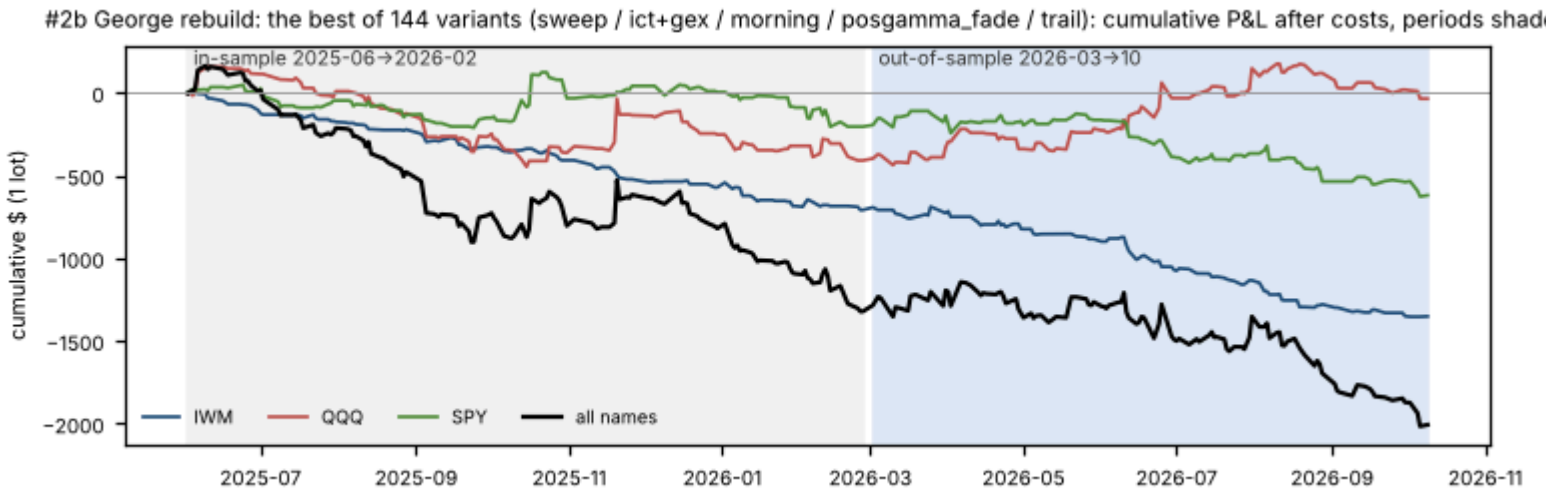
Random baseline: random-time baseline −$3.20 a trade (as #2); the best variant beats it by about $0.8, inside its interval.

Exit reasons (share of trades):

- in-sample 2025-06→2026-02: BREAKEVEN 40.5%; TRAIL 33.0%; STOP 25.4%; TIME_STOP 1.1%
- out-of-sample 2026-03→10: TRAIL 40.3%; BREAKEVEN 36.9%; STOP 22.3%; TIME_STOP 0.5%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	98	30.6	−8.38	−5.84	−572
2026	IWM	167	37.1	−6.13	−4.67	−780
2025	QQQ	136	30.9	−2.85	−1.86	−252
2026	QQQ	147	44.9	−0.62	+1.48	+217
2025	SPY	124	37.1	−1.39	+0.03	+4
2026	SPY	162	35.2	−2.92	−3.85	−624
2025	all names	358	33.0	−3.86	−2.29	−820
2026	all names	476	38.9	−3.34	−2.49	−1,187



Amendments, deviations and integrity checks. Selected as the best of 144 on the full sample (a selection effect the numbers include).

What it showed. Even the best of 144 variants loses after costs.

#3 George engine audit: the phase 1 reproduction audit of #1–#2

Folder: `audit/phase1/ (@7dfea57)` . **Commits:** — / — / 5ed66df (+ e95177c deterministic T5 rerun). **Dates:** 2026-10-13.

Hypothesis. The rebuild (#2) and the GeorgeFi validation tests T1–T7 reproduce from their artifacts.

Rule. Rerun of backtest/run.py and the validation scripts; byte comparison of outputs.

Fills and costs. —

Data. As #2.

Periods and holdouts. —

Pass rule (as pre-registered). Reproduction (no trading rule).

Verdict. Reproduced: the regenerated trades and baseline files are byte-identical once decompressed.

Metrics. not available — the audit produced no trades of its own; its trade files are #2's, byte for byte (see #2).

Amendments, deviations and integrity checks. Fixes the audit motivated were added, never overwritten (trading-days path, signals.json, deterministic T5 iteration).

What it showed. The rebuild is deterministic and reproducible; no new strategy was tested.

#4 GeorgeFi’s own trades re-priced at the NBBO (G1)

Folder: `research/georgefi_full_history/results/F_nbbo_\*.csv (@7dfea57)`. **Commits:** — / — / c27ff28. **Dates:** 2026-10-13.

Hypothesis. The vendor’s recorded trades were executable at the prices it recorded.

Rule. Each card’s own entry and exit second, filled at ThetaData’s NBBO (ask in, bid out), the card’s own size.

Fills and costs. NBBO at the reported second; no fee added.

Data. 1,220 cards with both sides quoted, 2026-01-20 → 2026-10-13, every root the vendor traded.

Periods and holdouts. ledger Epoch 1, ledger Epoch 2, paper book. Descriptive.

Pass rule (as pre-registered). None (an execution check).

Verdict. Not supported after execution: Epoch 1 +$22.2 a trade [day CI −11.2, +66.2], Epoch 2 −$42.1 [−78.8, −9.6], paper −$73.9 (registry G1); 58% of Epoch 1 rows filled better than the market.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
all roots	ledger E1	492	27 / 29	64.4	+15.2, +117.84	−23.1, −144.69	+6.7, +20.10	**+1.55**, +24.46	n/a (< 60 sessions); n/a (< 60 sessions)
all roots	ledger E2	594	132 / 162	45.1	+26.4, +156.24	−26.3, −205.09	−2.8, −7.90	**−2.54**, −42.07	[−6.22, +2.54]; [−121.60, −8.30]
all roots	paper book	84	38 / 61	36.9	+36.3, +132.66	−31.7, −194.74	−26.4, −77.80	**−6.58**, −73.92	[−12.81, −0.49]; [−96.33, −55.14]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
all roots	ledger E1	+1,988.80, +38.2%, 2026-02-20, SPXW PUT y3bxx2s50sc0nh8	−1,660.00, −88.5%, 2026-02-18, SPXW PUT 2k5f36h32m9rqf7	1.48	5,114	12	118%
all roots	ledger E2	+3,018.60, +392.0%, 2026-09-21, SPXW CALL dni7it4jw9zjisg	−3,462.40, −61.8%, 2026-07-13, SPXW CALL vkmuy6n1j94hoee	0.63	31,551	159	total ≤ 0 (best 5 days +10,353; without them −35,340)
all roots	paper book	+598.60, +89.3%, 2026-10-01, SPXW PUT 4oduouxg1tp6xd2	−462.80, −36.2%, 2026-08-06, SPXW PUT qoxj9ry0ux3n0cz	0.40	6,665	61	total ≤ 0 (best 5 days +1,532; without them −7,741)

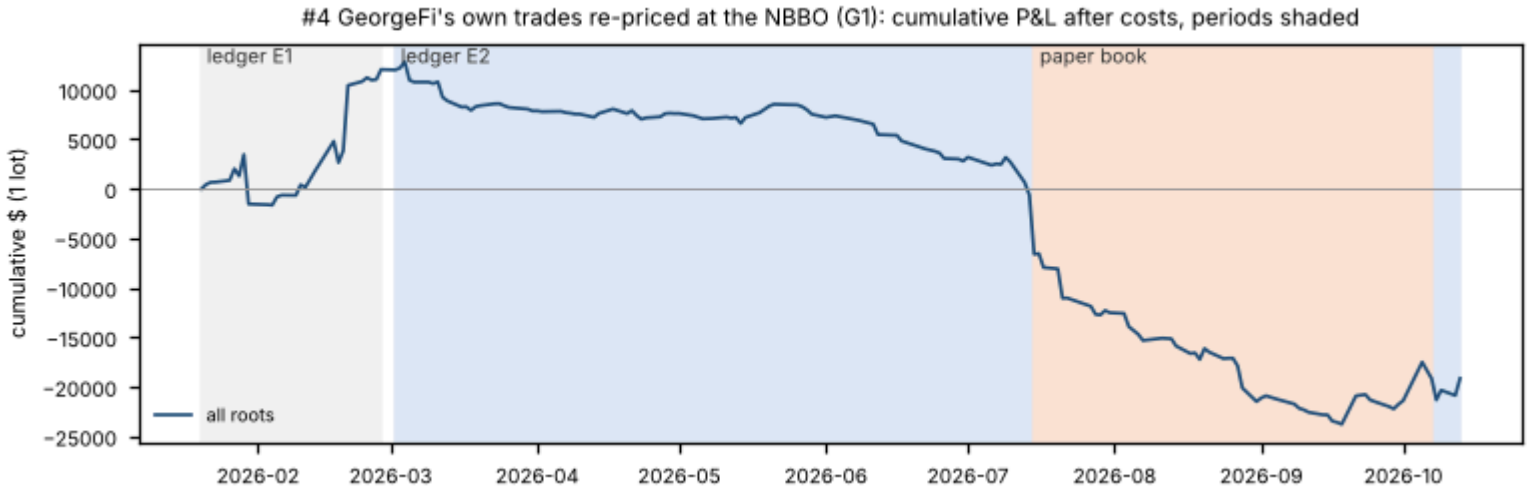
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ledger E1: vendor exit time, filled at the NBBO 100.0%
- ledger E2: vendor exit time, filled at the NBBO 100.0%
- paper book: vendor exit time, filled at the NBBO 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	all roots	1,170	52.6	−1.11	−16.38	−19,161



Amendments, deviations and integrity checks. Cards with a side unquoted, and the stock rows, are left out (as F_nbbo_trades.csv flags them).

What it showed. At the market’s own quotes the vendor’s live epoch loses $42 a trade.

#4b GeorgeFi entries with the trail engine (reconstruction C, G2)

Folder: `research/georgefi_full_history/results/C_trades.csv (@7dfea57)`. **Commits:** — / — / 5ed66df. **Dates:** 2026-10-13.

Hypothesis. GeorgeFi's entry minutes carry timing information when managed by a fixed trail engine.

Rule. The vendor's entry minute, the reconstructed contract, entry at the bar VWAP, the paper-lane trail engine.

Fills and costs. Spread model of #2 (half-spread + $0.70 a side).

Data. 972 reconstructed trades, 2026-01 → 2026-10.

Periods and holdouts. ledger Epoch 1 / Epoch 2 / paper.

Pass rule (as pre-registered). None (descriptive).

Verdict. No robust entry timing: Epoch 1 +$5.8 a contract [−5.7, +16.3], Epoch 2 +$2.3 [−10.5, +14.9]; MFE percentile against time-matched random minutes 53 and 48 (G2).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AAPL	ledger E1	1	1 / 1	100.0	+6.5, +1.38	—, —	+6.5, +1.38	**+6.49**, +1.38	n/a (< 60 sessions); n/a (< 60 sessions)
COIN	ledger E1	5	2 / 6	40.0	+10.7, +5.12	−27.9, −46.34	−27.9, −28.76	**−12.50**, −25.75	n/a (< 60 sessions); n/a (< 60 sessions)
CRCL	ledger E1	12	1 / 1	0.0	—, —	−27.9, −47.57	−27.9, −47.57	**−27.94**, −47.57	n/a (< 60 sessions); n/a (< 60 sessions)
CRM	ledger E1	1	1 / 1	100.0	+62.4, +49.54	—, —	+62.4, +49.54	**+62.43**, +49.54	n/a (< 60 sessions); n/a (< 60 sessions)
CRML	ledger E1	14	1 / 1	78.6	+14.8, +1.62	−27.9, −8.72	+14.8, +1.62	**+5.65**, −0.59	n/a (< 60 sessions); n/a (< 60 sessions)
CVNA	ledger E1	1	1 / 1	0.0	—, —	−27.9, −205.75	−27.9, −205.75	**−27.94**, −205.75	n/a (< 60 sessions); n/a (< 60 sessions)
GOOGL	ledger E1	3	1 / 1	0.0	—, —	−27.9, −21.64	−27.9, −21.64	**−27.94**, −21.64	n/a (< 60 sessions); n/a (< 60 sessions)
HOOD	ledger E1	3	1 / 1	66.7	+11.5, +6.67	−27.9, −42.15	+11.5, +6.67	**−1.67**, −9.61	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	ledger E1	65	15 / 28	64.6	+26.2, +56.03	−9.1, −15.44	+7.1, +11.06	**+13.74**, +30.74	n/a (< 60 sessions); n/a (< 60 sessions)
NVDA	ledger E1	14	2 / 3	57.1	+28.7, +24.94	−3.9, −5.48	+28.7, +24.94	**+14.71**, +11.90	n/a (< 60 sessions); n/a (< 60 sessions)
OKLO	ledger E1	1	1 / 1	0.0	—, —	−27.9, −50.70	−27.9, −50.70	**−27.94**, −50.70	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ	ledger E1	63	11 / 29	30.2	+13.2, +23.46	−6.5, −16.15	−2.0, −5.22	**−0.60**, −4.21	n/a (< 60 sessions); n/a (< 60 sessions)
RDDT	ledger E1	2	1 / 1	50.0	+55.6, +83.69	−27.9, −38.16	+13.8, +22.76	**+13.84**, +22.76	n/a (< 60 sessions); n/a (< 60 sessions)
RKLB	ledger E1	28	2 / 6	64.3	+20.0, +18.09	−3.9, −6.92	+6.2, +8.33	**+11.45**, +9.16	n/a (< 60 sessions); n/a (< 60 sessions)
RUTW	ledger E1	8	1 / 1	87.5	+8.3, +21.82	−3.0, −8.00	+8.2, +20.99	**+6.86**, +18.09	n/a (< 60 sessions); n/a (< 60 sessions)
SLV	ledger E1	3	1 / 1	0.0	—, —	−3.9, −6.96	−3.9, −6.96	**−3.92**, −6.96	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	ledger E1	59	11 / 25	55.9	+11.3, +140.45	−14.4, −150.45	+6.5, +6.48	**−0.02**, +12.25	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	ledger E1	89	16 / 24	42.7	+13.7, +33.69	−8.0, −22.63	−2.0, −4.08	**+1.29**, +1.41	n/a (< 60 sessions); n/a (< 60 sessions)
all names	ledger E1	372	25 / 29	49.2	+17.6, +52.28	−10.9, −39.16	−2.0, −2.72	**+3.15**, +5.83	n/a (< 60 sessions); n/a (< 60 sessions)
AAPL	ledger E2	2	2 / 8	50.0	+5.7, +4.28	−3.9, −5.04	+0.9, −0.38	**+0.91**, −0.38	n/a (< 60 sessions); n/a (< 60 sessions)
AMD	ledger E2	1	1 / 1	0.0	—, —	−27.9, −37.02	−27.9, −37.02	**−27.94**, −37.02	n/a (< 60 sessions); n/a (< 60 sessions)
AMZN	ledger E2	6	5 / 42	16.7	+15.7, +9.81	−8.7, −11.14	−3.9, −6.88	**−4.65**, −7.65	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AVGO	ledger E2	5	5 / 152	40.0	+7.4, +13.37	−19.9, −14.65	−3.9, −3.52	**−9.00**, −3.44	[−12.65, +5.61]; [−8.54, +16.97]
BAC	ledger E2	1	1 / 1	0.0	—, —	−27.9, −4.54	−27.9, −4.54	**−27.94**, −4.54	n/a (< 60 sessions); n/a (< 60 sessions)
COIN	ledger E2	1	1 / 1	100.0	+7.9, +21.60	—, —	+7.9, +21.60	**+7.91**, +21.60	n/a (< 60 sessions); n/a (< 60 sessions)
GLD	ledger E2	17	12 / 150	47.1	+29.3, +35.65	−11.9, −21.32	−3.9, −2.28	**+7.49**, +5.49	[+0.00, +32.98]; [−2.26, +35.60]
GOOGL	ledger E2	10	8 / 142	20.0	+13.8, +13.85	−10.5, −12.76	−3.9, −6.88	**−5.62**, −7.44	[−8.57, +8.89]; [−11.85, +8.27]
HOOD	ledger E2	1	1 / 1	0.0	—, —	−27.9, −9.95	−27.9, −9.95	**−27.94**, −9.95	n/a (< 60 sessions); n/a (< 60 sessions)
IBIT	ledger E2	2	2 / 6	0.0	—, —	−7.9, −4.73	−7.9, −4.73	**−7.92**, −4.73	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	ledger E2	19	10 / 68	36.8	+12.7, +29.26	−13.9, −30.69	−3.9, −7.44	**−4.12**, −8.60	[−8.20, +13.22]; [−17.07, +28.02]
META	ledger E2	17	6 / 39	35.3	+6.7, +14.51	−10.5, −24.59	−3.9, −17.12	**−4.40**, −10.79	n/a (< 60 sessions); n/a (< 60 sessions)
MSFT	ledger E2	1	1 / 1	100.0	+9.5, +37.38	—, —	+9.5, +37.38	**+9.51**, +37.38	n/a (< 60 sessions); n/a (< 60 sessions)
MU	ledger E2	4	2 / 3	50.0	+16.6, +30.31	−15.9, −24.27	+3.2, +10.43	**+0.31**, +3.02	n/a (< 60 sessions); n/a (< 60 sessions)
NDXP	ledger E2	16	14 / 44	56.2	+49.0, +445.57	−23.8, −230.46	+15.9, +98.25	**+17.18**, +149.80	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ	ledger E2	103	56 / 158	36.9	+15.6, +28.47	−12.4, −23.47	−2.0, −4.08	**−2.11**, −4.31	[−5.66, +2.78]; [−11.15, +5.33]
RUTW	ledger E2	23	17 / 77	56.5	+25.2, +157.01	−15.1, −91.68	+9.1, +53.44	**+7.67**, +48.88	[−1.20, +14.07]; [+5.86, +67.82]
SLV	ledger E2	3	3 / 29	66.7	+15.0, +5.40	−3.9, −2.52	+10.7, +3.73	**+8.70**, +2.76	n/a (< 60 sessions); n/a (< 60 sessions)
SMCI	ledger E2	1	1 / 1	0.0	—, —	−27.9, −18.78	−27.9, −18.78	**−27.94**, −18.78	n/a (< 60 sessions); n/a (< 60 sessions)
SMH	ledger E2	1	1 / 1	0.0	—, —	−3.9, −5.80	−3.9, −5.80	**−3.92**, −5.80	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	ledger E2	206	90 / 160	40.3	+20.1, +124.82	−11.8, −90.46	−3.0, −12.41	**+1.03**, −3.72	[−1.75, +4.20]; [−23.91, +9.77]
SPY	ledger E2	65	31 / 158	29.2	+10.3, +20.36	−10.0, −24.76	−2.0, −4.72	**−4.05**, −11.57	[−5.86, −1.98]; [−14.63, −4.11]
TSLA	ledger E2	8	6 / 126	62.5	+5.7, +6.64	−3.9, −11.07	+5.5, +2.69	**+2.11**, +0.00	[−3.92, +1.72]; [−12.08, +0.00]
UBER	ledger E2	1	1 / 1	0.0	—, —	−27.9, −7.67	−27.9, −7.67	**−27.94**, −7.67	n/a (< 60 sessions); n/a (< 60 sessions)
UCO	ledger E2	1	1 / 1	0.0	—, —	−27.9, −20.49	−27.9, −20.49	**−27.94**, −20.49	n/a (< 60 sessions); n/a (< 60 sessions)
USO	ledger E2	1	1 / 1	100.0	+27.5, +39.30	—, —	+27.5, +39.30	**+27.52**, +39.30	n/a (< 60 sessions); n/a (< 60 sessions)
all names	ledger E2	516	131 / 162	39.0	+18.8, +93.18	−12.2, −55.73	−2.0, −4.98	**−0.13**, +2.27	[−2.08, +3.55]; [−9.81, +20.03]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AAPL	ledger E1	+1.38, +6.5%, 2026-02-06, AAPL PUT 955xs5861je0sz7	+1.38, +6.5%, 2026-02-06, AAPL PUT 955xs5861je0sz7	∞	0	0	100%
COIN	ledger E1	+5.12, +10.7%, 2026-01-23, COIN PUT j55nf3xhhgeu4hi	−81.49, −27.9%, 2026-01-30, COIN PUT 09x52w4963s144b	0.07	129	6	total ≤ 0 (best 5 days −129; without them +0)
CRCL	ledger E1	−47.57, −27.9%, 2026-01-30, CRCL PUT 0g311hmh7608g99	−47.57, −27.9%, 2026-01-30, CRCL PUT 0g311hmh7608g99	0.00	571	1	total ≤ 0 (best 5 days −571; without them +0)
CRM	ledger E1	+49.54, +62.4%, 2026-01-30, CRM PUT 78506oa5o61r701	+49.54, +62.4%, 2026-01-30, CRM PUT 78506oa5o61r701	∞	0	0	100%
CRML	ledger E1	+1.62, +14.8%, 2026-01-30, CRML PUT 163cd9t5w354tz2	−11.09, −27.9%, 2026-01-30, CRML PUT btik9xf95i8b0pt	0.68	8	1	total ≤ 0 (best 5 days −8; without them +0)
CVNA	ledger E1	−205.75, −27.9%, 2026-01-30, CVNA PUT 3gris210t7m2b5w	−205.75, −27.9%, 2026-01-30, CVNA PUT 3gris210t7m2b5w	0.00	206	1	total ≤ 0 (best 5 days −206; without them +0)
GOOGL	ledger E1	−21.64, −27.9%, 2026-01-30, GOOGL PUT 3afgh5f7vn4ui47	−21.64, −27.9%, 2026-01-30, GOOGL PUT 3afgh5f7vn4ui47	0.00	65	1	total ≤ 0 (best 5 days −65; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
HOOD	ledger E1	+6.67, +11.5%, 2026-01-30, HOOD PUT 2co12g52z4bgs33	−42.15, −27.9%, 2026-01-30, HOOD PUT 774ne079d7stl33	0.32	29	1	total ≤ 0 (best 5 days −29; without them +0)
IWM	ledger E1	+369.33, +163.0%, 2026-02-12, IWM PUT 3591zk0339o8xj8	−80.06, −27.9%, 2026-02-13, IWM PUT 960e22lsdv24ai6	6.63	183	11	112%
NVDA	ledger E1	+24.94, +28.7%, 2026-02-06, NVDA CALL 10qr749120v26gj	−5.48, −3.9%, 2026-02-04, NVDA PUT 1848jkd063c3950	6.07	33	2	100%
OKLO	ledger E1	−50.70, −27.9%, 2026-01-30, OKLO PUT 9xy7jas7kw9524i	−50.70, −27.9%, 2026-01-30, OKLO PUT 9xy7jas7kw9524i	0.00	51	1	total ≤ 0 (best 5 days −51; without them +0)
QQQ	ledger E1	+42.32, +19.1%, 2026-02-20, QQQ CALL 72421cl1l5e521n	−121.77, −26.5%, 2026-02-18, QQQ CALL 402z467697wj3ie	0.63	470	29	total ≤ 0 (best 5 days +299; without them −564)
RDDT	ledger E1	+83.69, +55.6%, 2026-01-30, RDDT PUT 5725tnis6i49604	−38.16, −27.9%, 2026-01-30, RDDT PUT 485pwz3e339pbdj	2.19	0	0	100%
RKLB	ledger E1	+32.58, +37.0%, 2026-01-30, RKLB PUT 0gi7xw49q6uthdz	−6.92, −3.9%, 2026-01-30, RKLB PUT 02rl903c51x3m7l	4.70	0	0	100%
RUTW	ledger E1	+26.82, +8.2%, 2026-01-27, RUTW CALL 1891h5gtbs11ue7	−8.00, −3.0%, 2026-01-27, RUTW CALL 592b74h07fy9i2j	19.09	0	0	100%
SLV	ledger E1	−6.96, −3.9%, 2026-02-06, SLV PUT ar9594rv59q1lpa	−6.96, −3.9%, 2026-02-06, SLV PUT ar9594rv59q1lpa	0.00	21	1	total ≤ 0 (best 5 days −21; without them +0)
SPXW	ledger E1	+640.20, +21.7%, 2026-02-17, SPXW PUT 0vsr61tk6e355f0	−853.91, −27.2%, 2026-02-19, SPXW PUT u3a2f8rxkb60t4x	1.18	740	7	287%
SPY	ledger E1	+158.38, +63.8%, 2026-02-24, SPY CALL 894j813el751a5q	−130.56, −17.9%, 2026-02-10, SPY CALL 9zf937w5328f71i	1.11	459	17	620%
all names	ledger E1	+640.20, +21.7%, 2026-02-17, SPXW PUT 0vsr61tk6e355f0	−853.91, −27.2%, 2026-02-19, SPXW PUT u3a2f8rxkb60t4x	1.29	814	9	173%
AAPL	ledger E2	+4.28, +5.7%, 2026-03-18, AAPL PUT yco9879mvrod1qa	−5.04, −3.9%, 2026-03-27, AAPL PUT lf35l8q98xq5m4i	0.85	5	1	total ≤ 0 (best 5 days −1; without them +0)
AMD	ledger E2	−37.02, −27.9%, 2026-10-09, AMD PUT i3b03ewpkryh99v	−37.02, −27.9%, 2026-10-09, AMD PUT i3b03ewpkryh99v	0.00	37	1	total ≤ 0 (best 5 days −37; without them +0)
AMZN	ledger E2	+9.81, +15.7%, 2026-09-23, AMZN PUT t5exm42pv39sbea	−29.90, −27.9%, 2026-08-17, AMZN PUT 9jcbom24ks2h5nl	0.18	46	42	total ≤ 0 (best 5 days −46; without them +0)
AVGO	ledger E2	+16.97, +5.6%, 2026-03-13, AVGO PUT rvd99u8867d6qku	−29.04, −27.9%, 2026-10-12, AVGO PUT ekylizkrfb48uzw	0.61	40	2	total ≤ 0 (best 5 days −17; without them +0)
BAC	ledger E2	−4.54, −27.9%, 2026-10-09, BAC CALL x0z4k4fwb8gos4x	−4.54, −27.9%, 2026-10-09, BAC CALL x0z4k4fwb8gos4x	0.00	5	1	total ≤ 0 (best 5 days −5; without them +0)
COIN	ledger E2	+21.60, +7.9%, 2026-10-09, COIN CALL 9oyzh2v0xe3d9bs	+21.60, +7.9%, 2026-10-09, COIN CALL 9oyzh2v0xe3d9bs	∞	0	0	100%
GLD	ledger E2	+141.62, +92.9%, 2026-09-14, GLD CALL kw39iz49kt101o8	−78.64, −27.9%, 2026-03-18, GLD CALL 4hc062it0yxcvu4	1.49	79	128	277%
GOOGL	ledger E2	+24.50, +21.7%, 2026-08-21, GOOGL CALL xk0nrxpvlxtv0rk	−43.01, −27.9%, 2026-03-27, GOOGL PUT i83o4578nx3v13m	0.27	74	142	total ≤ 0 (best 5 days −1; without them −74)
HOOD	ledger E2	−9.95, −27.9%, 2026-10-09, HOOD CALL 5dewoq9m3lm21bz	−9.95, −27.9%, 2026-10-09, HOOD CALL 5dewoq9m3lm21bz	0.00	10	1	total ≤ 0 (best 5 days −10; without them +0)
IBIT	ledger E2	−3.68, −3.9%, 2026-10-05, IBIT PUT mz572iaej6prqxd	−5.78, −11.9%, 2026-10-12, IBIT PUT ens8uikr9j4xd0u	0.00	9	6	total ≤ 0 (best 5 days −9; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	ledger E2	+80.92, +27.0%, 2026-03-19, IWM CALL 1794163he39p5c3	−80.34, −27.9%, 2026-03-09, IWM CALL 9417g6bc21he872	0.56	245	67	total ≤ 0 (best 5 days +108; without them −271)
META	ledger E2	+18.87, +10.2%, 2026-09-04, META CALL kmr87mdhotjnbzb	−68.38, −27.9%, 2026-09-16, META CALL u8myue00egjv6ch	0.32	183	39	total ≤ 0 (best 5 days −14; without them −170)
MSFT	ledger E2	+37.38, +9.5%, 2026-03-27, MSFT PUT 8o6nv14p53b063h	+37.38, +9.5%, 2026-03-27, MSFT PUT 8o6nv14p53b063h	∞	0	0	100%
MU	ledger E2	+30.51, +10.4%, 2026-10-07, MU CALL z6ss0x4wo8gjj9g	−39.30, −27.9%, 2026-10-09, MU PUT cmzm0ywaiod9jxm	1.25	39	1	100%
NDXP	ledger E2	+1,753.39, +122.4%, 2026-07-06, NDXP PUT fugmhxmitbyc9jl	−589.81, −27.2%, 2026-06-12, NDXP PUT 744n57ni3vvbc5v	2.49	780	23	142%
QQQ	ledger E2	+150.16, +104.2%, 2026-05-11, QQQ PUT n65jjs8fech846i	−100.91, −26.5%, 2026-07-10, QQQ CALL g8rhxc7ftrxf8ox	0.71	750	154	total ≤ 0 (best 5 days +541; without them −985)
RUTW	ledger E2	+355.75, +39.1%, 2026-07-13, RUTW PUT hh5gls5egcdlfbb	−301.13, −27.2%, 2026-07-28, RUTW PUT kekqioeqylgb4kl	2.23	301	14	115%
SLV	ledger E2	+7.07, +19.3%, 2026-09-11, SLV CALL m2eeuqdcs6jl3xr	−2.52, −3.9%, 2026-10-07, SLV CALL 2ti9rbrixeb0oze	4.29	3	1	100%
SMCI	ledger E2	−18.78, −27.9%, 2026-10-09, SMCI PUT u0lsesrgx6ikwsj	−18.78, −27.9%, 2026-10-09, SMCI PUT u0lsesrgx6ikwsj	0.00	19	1	total ≤ 0 (best 5 days −19; without them +0)
SMH	ledger E2	−5.80, −3.9%, 2026-10-12, SMH PUT c4i07emq4fnytre	−5.80, −3.9%, 2026-10-12, SMH PUT c4i07emq4fnytre	0.00	6	1	total ≤ 0 (best 5 days −6; without them +0)
SPXW	ledger E2	+824.13, +56.4%, 2026-09-29, SPXW CALL 4cwxg2xbhvss79h	−777.11, −27.2%, 2026-03-16, SPXW CALL r98xgo27bdx4g5f	0.93	3,099	159	total ≤ 0 (best 5 days +3,304; without them −4,070)
SPY	ledger E2	+39.10, +20.6%, 2026-08-18, SPY PUT kgrcrt2va87cvcl	−158.15, −26.5%, 2026-03-27, SPY PUT 5zy0jk44ra6yh18	0.34	752	158	total ≤ 0 (best 5 days +112; without them −864)
TSLA	ledger E2	+16.54, +6.2%, 2026-09-04, TSLA PUT mpxpmm8qv0ehcv0	−16.12, −3.9%, 2026-08-21, TSLA CALL v71km1xj03uovgs	1.00	33	123	322500%
UBER	ledger E2	−7.67, −27.9%, 2026-10-09, UBER CALL lk1w0emiuxfqtk7	−7.67, −27.9%, 2026-10-09, UBER CALL lk1w0emiuxfqtk7	0.00	8	1	total ≤ 0 (best 5 days −8; without them +0)
UCO	ledger E2	−20.49, −27.9%, 2026-03-20, UCO CALL 0c63wm0dox7wy65	−20.49, −27.9%, 2026-03-20, UCO CALL 0c63wm0dox7wy65	0.00	20	1	total ≤ 0 (best 5 days −20; without them +0)
USO	ledger E2	+39.30, +27.5%, 2026-03-20, USO CALL 4ug6m04x120lnp8	+39.30, +27.5%, 2026-03-20, USO CALL 4ug6m04x120lnp8	∞	0	0	100%
all names	ledger E2	+1,753.39, +122.4%, 2026-07-06, NDXP PUT fugmhxmitbyc9jl	−777.11, −27.2%, 2026-03-16, SPXW CALL r98xgo27bdx4g5f	1.07	3,800	88	424%

Random baseline: time-matched random minutes, trail engine: MFE percentile 53 [47.5, 59.2] (Epoch 1) and 48 (Epoch 2).

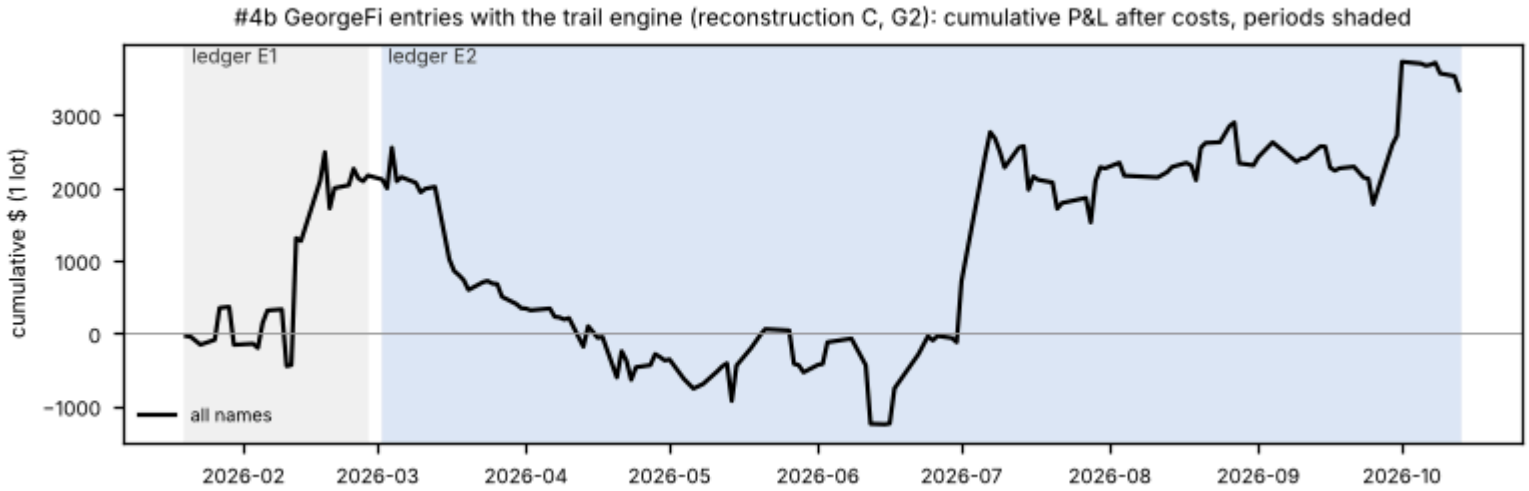
Exit reasons (share of trades):

- ledger E1: TRAIL 48.7%; BREAKEVEN 32.3%; STOP 16.4%; TIME_STOP 1.6%; CLOSE 1.1%
- ledger E2: TRAIL 39.0%; BREAKEVEN 37.0%; STOP 23.4%; TIME_STOP 0.6%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	AAPL	3	66.7	+2.77	+0.20	+1

year	name	trades	win %	mean %	mean $	total $
2026	AMD	1	0.0	−27.94	−37.02	−37
2026	AMZN	6	16.7	−4.65	−7.65	−46
2026	AVGO	5	40.0	−9.00	−3.44	−17
2026	BAC	1	0.0	−27.94	−4.54	−5
2026	COIN	6	50.0	−9.10	−17.86	−107
2026	CRCL	12	0.0	−27.94	−47.57	−571
2026	CRM	1	100.0	+62.43	+49.54	+50
2026	CRML	14	78.6	+5.65	−0.59	−8
2026	CVNA	1	0.0	−27.94	−205.75	−206
2026	GLD	17	47.1	+7.49	+5.49	+93
2026	GOOGL	13	15.4	−10.77	−10.71	−139
2026	HOOD	4	50.0	−8.24	−9.69	−39
2026	IBIT	2	0.0	−7.92	−4.73	−9
2026	IWM	84	58.3	+9.70	+21.84	+1,834
2026	META	17	35.3	−4.40	−10.79	−183
2026	MSFT	1	100.0	+9.51	+37.38	+37
2026	MU	4	50.0	+0.31	+3.02	+12
2026	NDXP	16	56.2	+17.18	+149.80	+2,397
2026	NVDA	14	57.1	+14.71	+11.90	+167
2026	OKLO	1	0.0	−27.94	−50.70	−51
2026	QQQ	166	34.3	−1.53	−4.27	−709
2026	RDDT	2	50.0	+13.84	+22.76	+46
2026	RKLB	28	64.3	+11.45	+9.16	+256
2026	RUTW	31	64.5	+7.46	+40.94	+1,269
2026	SLV	6	33.3	+2.39	−2.10	−13
2026	SMCI	1	0.0	−27.94	−18.78	−19
2026	SMH	1	0.0	−3.92	−5.80	−6
2026	SPXW	265	43.8	+0.79	−0.16	−44
2026	SPY	154	37.0	−0.96	−4.07	−626
2026	TSLA	8	62.5	+2.11	+0.00	+0
2026	UBER	1	0.0	−27.94	−7.67	−8
2026	UCO	1	0.0	−27.94	−20.49	−20
2026	USO	1	100.0	+27.52	+39.30	+39
2026	all names	888	43.2	+1.24	+3.76	+3,339



Amendments, deviations and integrity checks. —

What it showed. The entries are as good as random minutes once the exit is fixed.

#5 Deep research round 1: flow rule F1 on SPY (H2 family)

Folder: `research/flow_predictors/ (@7dfea57); report "Finding a 0DTE options edge"`. **Commits:** — / — / 96a2c09…fb4205f, 5208c8f. **Dates:** 2026-10-13.

Hypothesis. Options-flow imbalances (net premium, net delta, market tide, dark pool) lead the underlying.

Rule. F1: the flow predictor's signal minute on SPY → buy the 0DTE contract in its direction, trail engine (as #4b).

Fills and costs. Spread model of #2 (1% half-spread + $0.70 a side), entry at the VWAP of the first option bar after the signal.

Data. SPY, Unusual Whales flow and bars 2025-07-01 → 2026-10-09; F1 = 15-minute net premium against its trailing-20-session 20th/80th percentiles, every 5 minutes 09:45–15:00, at most 2 a day.

Periods and holdouts. One span (descriptive test 5).

Pass rule (as pre-registered). Multiple-testing correction across 62 cells (exploratory).

Verdict. Rejected: 0 of 62 cells survive correction; forward ICs ≤ 0.02 (registry H2). The other round-1 notes (GEX regime and exit grid of 576 configurations, short-vol session T1–T5) kept results by configuration only; their verdicts: H4 rejected, H6 rejected, H7 0 of 576 exit configurations positive (White p = 1), H8 every option expression negative.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	2025-07→2026-10	638	319 / 331	35.9	+13.9, +21.94	−13.2, −18.71	−2.9, −3.90	**−3.45**, −4.12	[−4.53, −2.12]; [−5.77, −2.02]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	2025-07→2026-10	+175.91, +84.6%, 2025-12-18, SPY251218C00676000	−83.79, −27.2%, 2026-07-29, SPY260729C00736000	0.66	2,631	331	total ≤ 0 (best 5 days +634; without them −3,261)

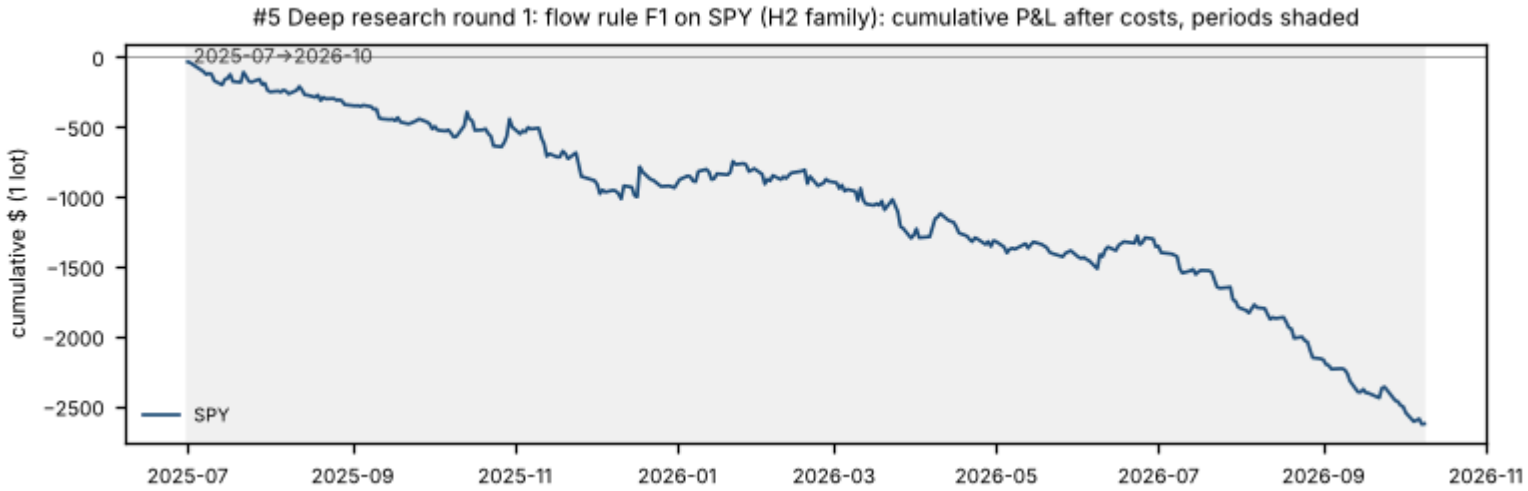
Random baseline: random minutes matched per day, random side, same exits and costs (574 trades): −$3.34 a trade, −2.26% (the rule −$4.12, −2.38%).

Exit reasons (share of trades):

- 2025-07→2026-10: BREAKEVEN 37.8%; TRAIL 35.9%; STOP 26.0%; TIME_STOP 0.3%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPY	250	36.4	−4.04	−3.75	−938
2026	SPY	388	35.6	−3.07	−4.35	−1,689



Amendments, deviations and integrity checks. —

What it showed. The one flow rule with a trade list loses after costs.

#6a T-A: the 13:00 short ATM 0DTE straddle, SPY/QQQ/IWM, bought back 15:55 (H9)

Folder: `research/thetadata/straddle_nbbo/ (@7dfea57)`. **Commits:** — / gate v2 924aee9 / ebde6ae. **Dates:** 2026-10-13.

Hypothesis. Selling the 13:00 at-the-money straddle harvests intraday variance premium after measured costs.

Rule. Sell the same-day ATM straddle at the 13:00 bids, buy it back at the 15:55 asks.

Fills and costs. Measured ThetaData NBBO both ways (the round trip $12.93 a SPY/QQQ straddle); return as % of the credit received.

Data. SPY, QQQ, IWM 0DTE, 339 sessions each, 2025-06-02 → 2026-10-09.

Periods and holdouts. One span; no holdout (“previously analysed”).

Pass rule (as pre-registered). Gate v2: a positive lower bound at measured NBBO (T-A).

Verdict. FAIL: +$8.60 a straddle [−6.07, +22.45] SPY/QQQ (n 678); IWM −$9.50 [−18.69, −0.44].

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2025-06→2026-10	339	339 / 352	57.2	+38.2, +41.81	−72.0, −78.15	+14.5, +12.20	**−8.94**, −9.50	[−16.30, −3.87]; [−19.66, −2.02]
QQQ	2025-06→2026-10	339	339 / 352	65.2	+38.6, +88.50	−54.6, −134.05	+19.6, +45.20	**+6.15**, +11.03	[+1.12, +10.94]; [−3.30, +23.67]
SPY	2025-06→2026-10	339	339 / 352	64.3	+39.0, +75.80	−57.3, −119.28	+19.4, +36.20	**+4.64**, +6.17	[−0.30, +8.72]; [−6.78, +16.87]
all names	2025-06→2026-10	1,017	339 / 352	62.2	+38.6, +69.82	−62.0, −108.29	+17.9, +25.20	**+0.62**, +2.57	[−4.33, +4.41]; [−8.76, +11.76]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
IWM	2025-06→2026-10	+328.20, +80.8%, 2026-07-29, IWM 13:00 short straddle 289	−405.80, −240.1%, 2026-06-11, IWM 13:00 short straddle 285	0.72	4,041	346	total ≤ 0 (best 5 days +1,025; without them −4,246)
QQQ	2025-06→2026-10	+343.20, +76.6%, 2026-06-12, QQQ 13:00 short straddle 722	−1,186.80, −225.2%, 2026-06-11, QQQ 13:00 short straddle 700	1.24	2,612	140	42%
SPY	2025-06→2026-10	+319.20, +66.8%, 2025-09-17, SPY 13:00 short straddle 658	−766.80, −197.1%, 2026-06-11, SPY 13:00 short straddle 727	1.14	2,399	148	59%
all names	2025-06→2026-10	+343.20, +76.6%, 2026-06-12, QQQ 13:00 short straddle 722	−1,186.80, −225.2%, 2026-06-11, QQQ 13:00 short straddle 700	1.06	6,712	212	135%

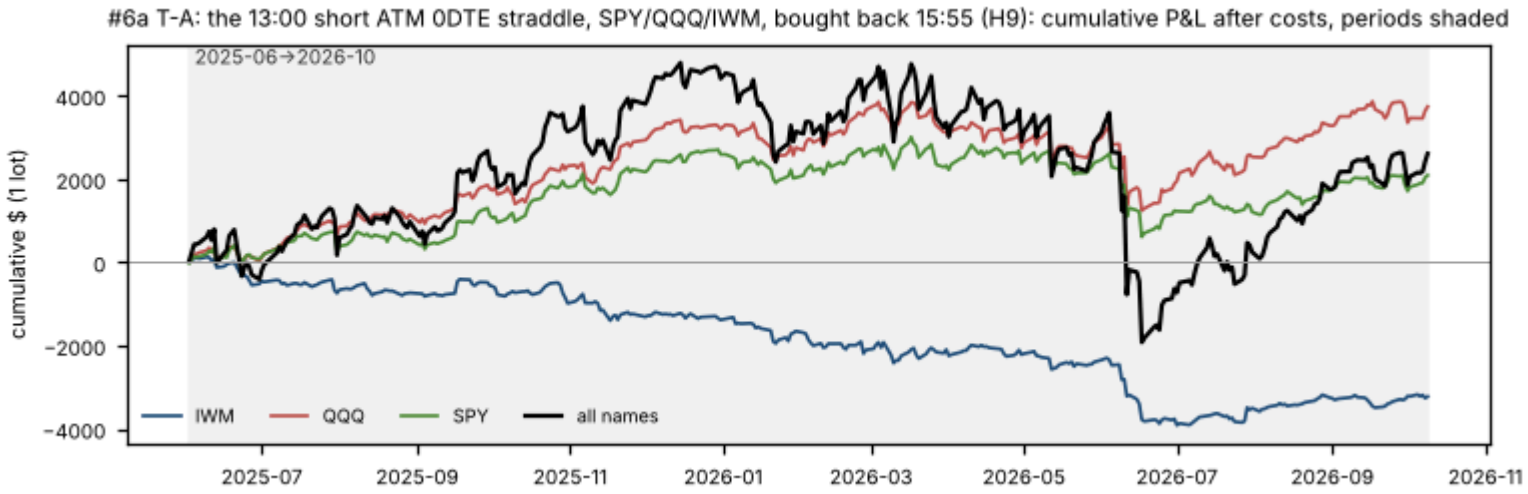
Random baseline: random entry times: +$13.10 [−2.42, +29.14] against a $12.93 round trip (registry H9).

Exit reasons (share of trades):

- 2025-06→2026-10: bought back 15:55 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	IWM	145	55.9	−10.23	−9.53	−1,382
2026	IWM	194	58.2	−7.98	−9.48	−1,839
2025	QQQ	145	69.7	+11.02	+22.07	+3,200
2026	QQQ	194	61.9	+2.51	+2.78	+540
2025	SPY	145	68.3	+9.48	+17.90	+2,595
2026	SPY	194	61.3	+1.02	−2.59	−503
2025	all names	435	64.6	+3.43	+10.14	+4,413
2026	all names	582	60.5	−1.48	−3.10	−1,803



Amendments, deviations and integrity checks. The modelled-spread result (+$8.36) was optimistic by about $5 a straddle.

What it showed. The 13:00 short straddle is positive on SPY/QQQ with an interval through zero and negative on IWM.

#6b T-B: gamma-conditioned last-30-minute momentum in SPY shares (H10)

Folder: `research/thetadata/index_tests/tb_momentum/ (@7dfea57)`. **Commits:** — / — / b209f7b. **Dates:** 2026-10-13.

Hypothesis. On negative-gamma days the last 30 minutes continue the session's move.

Rule. On days with dealer gamma below the flip (prior-day UW levels): 100 SPY in the direction of the open→15:30 move, 15:30 → official close.

Fills and costs. Net of the study's cost per day (0.4 bp).

Data. SPY, 299 negative-gamma days 2024-05-14 → 2026-10-07.

Periods and holdouts. One span.

Pass rule (as pre-registered). T-B: positive mean with a positive lower bound.

Verdict. FAIL (wrong sign): −3.07 bp a day [−5.73, −0.52], Sharpe −2.15.

Metrics (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	negative-gamma days 2024-05→2026-10	299	299 / 620	46.2	+0.127, +76.97	−0.166, −102.36	−0.014, −8.33	**−0.031**, −19.59	[−0.056, −0.003]; [−34.03, −4.03]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	negative-gamma days 2024-05→2026-10	+522.48, +1.0%, 2025-04-04, 100 SPY short 15:30→close	−699.16, −1.2%, 2025-03-04, 100 SPY long 15:30→close	0.64	6,320	620	total ≤ 0 (best 5 days +2,015; without them −7,874)

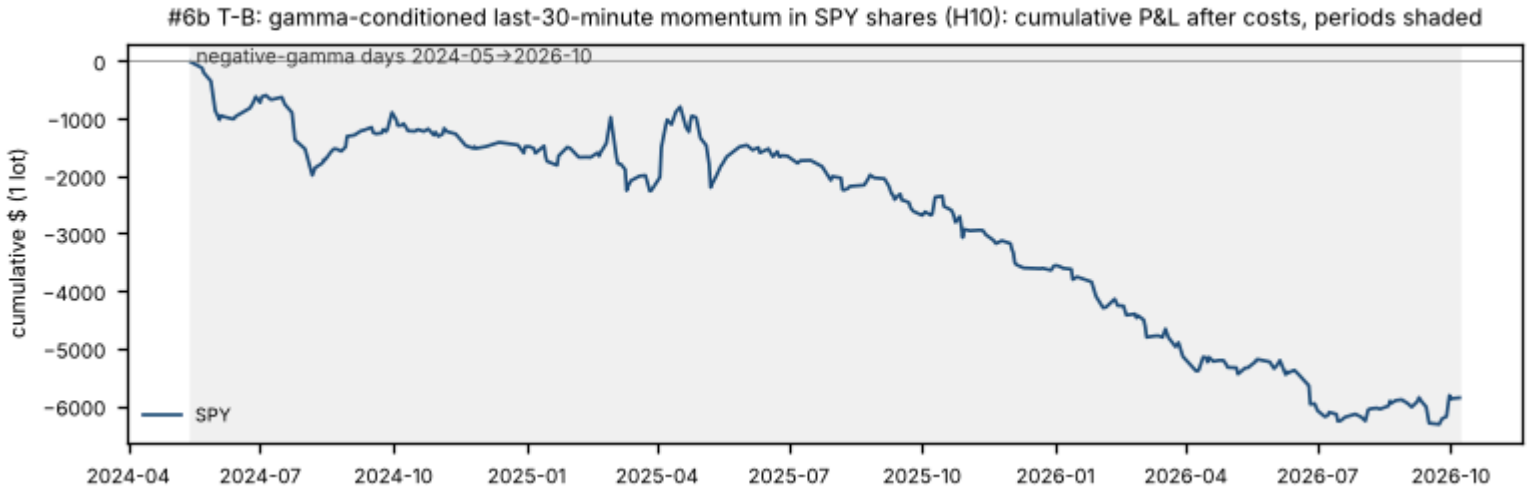
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- negative-gamma days 2024-05→2026-10: official close 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	SPY	77	50.6	−0.04	−19.36	−1,491
2025	SPY	125	43.2	−0.02	−16.58	−2,073
2026	SPY	97	46.4	−0.03	−23.66	−2,295



Amendments, deviations and integrity checks. The two gamma proxies agree on the regime 52% of days.

What it showed. The effect has the wrong sign.

#6c T-C: the 13:00 SPXW short straddle bought back 15:55 (H11)

Folder: `research/thetadata/index_tests/tc_spxw_straddle/ (@7dfea57)`. **Commits:** — / — / b209f7b. **Dates:** 2026-10-13.

Hypothesis. As T-A on SPXW (cash-settled).

Rule. Sell the SPXW ATM straddle at 13:00 (bids), buy back at 15:55 (asks).

Fills and costs. Measured NBBO; return as % of the credit.

Data. SPXW 0DTE, 339 sessions 2025-06-02 → 2026-10-09.

Periods and holdouts. One span.

Pass rule (as pre-registered). T-C: positive lower bound.

Verdict. FAIL: +$72.4 a day [−75.8, +214.2], max drawdown −$21,180, worst day −$8,499.

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2025-06→2026-10	339	339 / 352	63.4	+46.5, +832.56	−66.7, −1,245.53	+26.2, +396.00	**+5.11**, +72.43	[−0.06, +9.01]; [−55.98, +173.90]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2025-06→2026-10	+3,371.00, +71.7%, 2025-09-17, SPXW 13:00 short straddle 6590.0	−8,499.00, −230.3%, 2026-06-11, SPXW 13:00 short straddle 7275.0	1.16	21,181	91	54%

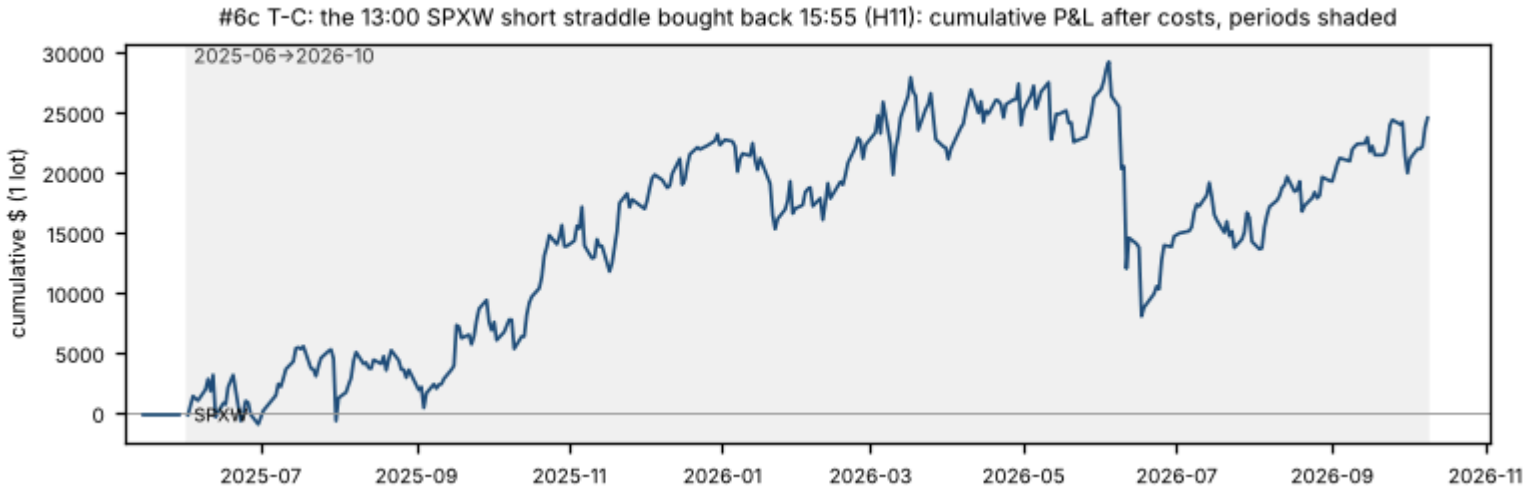
Random baseline: random entry: +$40.6 [−73.4, +164.2] against a $68.9 cost.

Exit reasons (share of trades):

- 2025-06→2026-10: bought back 15:55 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPXW	145	63.4	+7.63	+153.76	+22,295
2026	SPXW	194	63.4	+3.22	+11.64	+2,259



Amendments, deviations and integrity checks. —

What it showed. Positive mean, interval through zero, large tail days.

#6d T-E: the 13:00 SPXW ATM straddle held to PM settlement, ex FOMC/CPI/NFP (H13)

Folder: `research/thetadata/hwr_grid/spxw_v2/ (@7dfea57)`. **Commits:** protocol sha256 in protocol_sha256.txt / — / 01ee3f3, 5f253c6 (full chains). **Dates:** 2026-10-14.

Hypothesis. Holding the short straddle to cash settlement adds the last hour's decay.

Rule. Sell the SPXW ATM straddle at 13:00, hold to the PM settlement (Cboe close), skip FOMC/CPI/NFP days.

Fills and costs. Measured NBBO in; settlement value out.

Data. SPXW 0DTE 2022-05-13 → 2026-10-09 on complete chains.

Periods and holdouts. TRAIN 2022-05→2024 / TEST 2025→2026-10 (TEST read).

Pass rule (as pre-registered). T-E gate on TEST.

Verdict. FAIL: +$26.94 a day [−136.73, +170.75]; 2025-04-09 alone −$33,667 a lot; trails random entry in both windows.

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	TRAIN	575	575 / 683	60.2	+54.3, +881.35	−81.0, −1,228.82	+22.1, +354.00	**+0.42**, +40.95	[−6.12, +6.49]; [−56.54, +135.92]
SPXW	TEST	388	388 / 457	67.0	+52.8, +997.84	−80.7, −1,767.52	+28.1, +487.00	**+8.75**, +85.56	[+1.86, +14.77]; [−115.71, +252.73]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	TRAIN	+2,927.00, +96.3%, 2022-05-13, SPXW 13:00 4025P/4025C	−9,211.00, −860.8%, 2024-04-04, SPXW 13:00 5250P/5250C	1.08	22,411	526	58%
SPXW	TEST	+4,100.00, +96.5%, 2025-03-06, SPXW 13:00 5740P/5740C	−34,022.00, −392.4%, 2025-04-09, SPXW 13:00 5030P/5030C	1.15	38,494	175	53%

Random baseline: not available — no random baseline was computed by this study

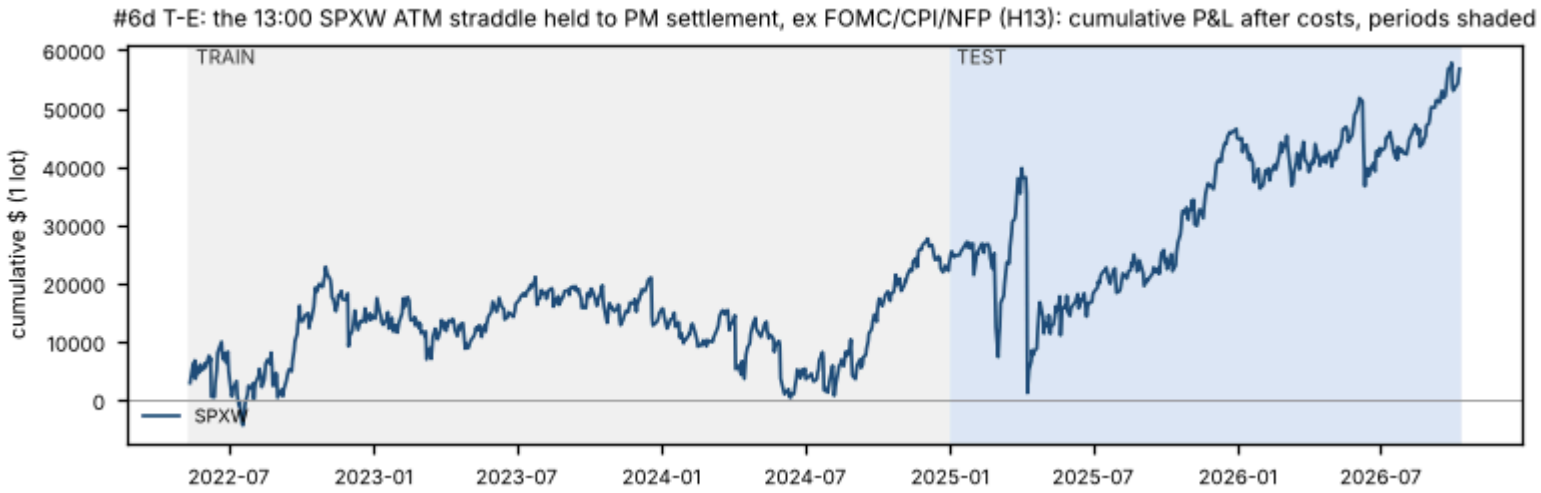
Exit reasons (share of trades):

- TRAIN: settle 100.0%
- TEST: settle 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	140	61.4	+4.60	+103.02	+14,423
2023	SPXW	217	58.5	−1.47	+1.88	+409
2024	SPXW	218	61.0	−0.38	+39.98	+8,716

year	name	trades	win %	mean %	mean $	total $
2025	SPXW	218	69.3	+10.33	+97.98	+21,359
2026	SPXW	170	64.1	+6.73	+69.63	+11,837



Amendments, deviations and integrity checks. Re-run on complete chains (5f253c6, 8fb5bad): no TEST figure changed.

What it showed. One day can erase years; no edge.

#6e H15 carry lane: the 14:05 SPXW short strangle at ±0.5 expected move, held to settlement

Folder: `research/thetadata/hwr_grid/spxw_v2/ (@7dfea57); report "High win rate 0DTE lanes"`. **Commits:** protocol_sha256.txt / — / 01ee3f3, 97e685a, 6691368. **Dates:** 2026-10-14.

Hypothesis. A 70%+ win-rate short strangle with a positive cost-realistic lower bound exists.

Rule. 14:05 sell the SPXW strangle at ±0.5 × the 14:05 expected move, held to cash settlement, skip FOMC and month-end (selection S2 on TRAIN).

Fills and costs. Measured NBBO in; settlement out.

Data. SPXW 0DTE 2022-05-13 → 2026-10-09.

Periods and holdouts. TRAIN (selection) / TEST (read). The deciding test T-G (forward paper ≥ 150 sessions) was never started.

Pass rule (as pre-registered). Positive TEST lower bound after the two selections; then T-G forward.

Verdict. Open, NOT promotable: TEST +$151.17 a lot [+42.15, +251.73], win 70.8%; the pre-registered reselection on complete TRAIN picks a different S2 that fails (+$52.54 [−22.24, +122.49]); ±10% shock ≈ −$65k a lot.

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	TRAIN	606	606 / 682	72.1	+83.9, +528.42	−166.6, −921.84	+97.4, +348.00	**+14.03**, +123.98	[+1.62, +24.85]; [+60.26, +188.45]
SPXW	TEST	407	407 / 457	70.8	+86.5, +646.35	−147.6, −1,047.27	+95.9, +353.00	**+18.02**, +151.17	[+3.54, +29.89]; [+34.16, +263.93]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	TRAIN	+1,828.00, +99.9%, 2024-08-05, SPXW 14:05 5175P/5215C	−7,146.00, −1,125.4%, 2024-04-04, SPXW 14:05 5225P/5240C	1.48	10,092	91	10%
SPXW	TEST	+4,578.00, +100.0%, 2025-04-10, SPXW 14:05 5245P/5335C	−7,131.00, −735.2%, 2026-03-09, SPXW 14:05 6690P/6715C	1.49	13,022	157	28%

Random baseline: matched random: +$22.60 [−147.61, +261.12].

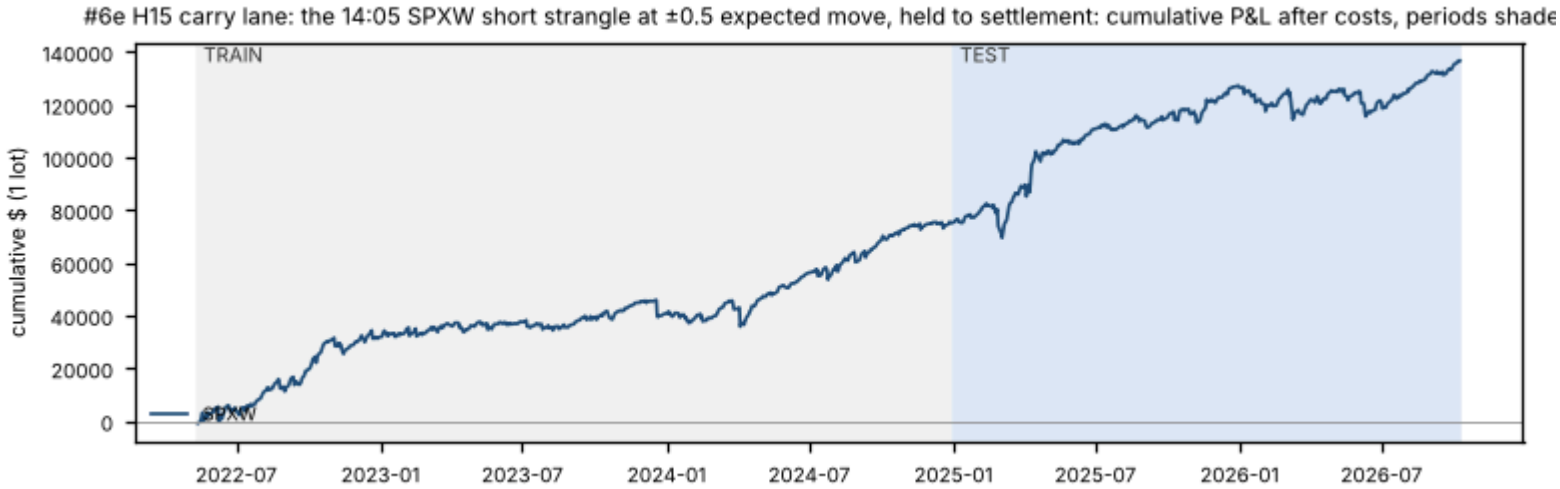
Exit reasons (share of trades):

- TRAIN: settle 100.0%
- TEST: settle 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	146	73.3	+20.73	+217.40	+31,741

year	name	trades	win %	mean %	mean $	total $
2023	SPXW	228	67.1	+5.43	+38.84	+8,856
2024	SPXW	232	76.3	+18.25	+148.84	+34,532
2025	SPXW	228	71.9	+22.53	+228.60	+52,121
2026	SPXW	179	69.3	+12.29	+52.54	+9,404



Amendments, deviations and integrity checks. All historical data has been used for selection; no further historical re-selection allowed (registry).

What it showed. Positive on TEST, but the reselection fails and the tail risk is unbounded.

#7a S01 Quick Scalps (owner baseline) on index 0DTE

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** protocol 113cad7 / — / 5e90201. **Dates:** 2026-10-14.

Hypothesis. The owner's Quick Scalps signal (RSI14 cross 55/45 + EMA9 lead, 5m EMA20 + VWAP confirm, 15m EMA alignment) predicts 0DTE moves.

Rule. Signal → buy the 0DTE contract in a delta band at the ask; exits: stop, two targets, VWAP loss, invalidation, time.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE 2022-11-11 → 2026-10-08.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Positive net expectancy with a positive lower bound (protocol).

Verdict. REJECTED: −$4.19 a contract [−5.68, −2.81].

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	685	274 / 638	33.3	+36.5, +20.47	−28.3, −15.40	−27.5, −10.40	**−6.74**, −3.46	[−9.19, −4.39]; [−4.91, −2.13]
QQQ	2022-11→2026-10	1,075	421 / 1,009	35.0	+36.5, +39.99	−26.4, −27.36	−23.4, −15.40	**−4.38**, −3.80	[−6.34, −2.71]; [−5.82, −1.96]
SPY	2022-11→2026-10	1,126	444 / 1,009	34.3	+35.7, +32.62	−25.2, −24.61	−19.5, −13.40	**−4.32**, −4.99	[−6.19, −2.60]; [−7.33, −3.05]
all names	2022-11→2026-10	2,886	614 / 1,009	34.3	+36.2, +32.62	−26.4, −23.41	−22.7, −12.40	**−4.92**, −4.19	[−6.31, −3.69]; [−5.79, −2.88]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+71.10, +50.1%, 2026-06-12, IWM 2026-06-12 292.0CALL	−60.40, −30.2%, 2025-04-07, IWM 2025-04-07 178.0PUT	0.66	2,450	571	total ≤ 0 (best 5 days +373; without them −2,745)
QQQ	2022-11→2026-10	+345.60, +111.8%, 2026-06-09, QQQ 2026-06-09 692.0PUT	−157.40, −34.5%, 2025-04-07, QQQ 2025-04-07 417.0PUT	0.79	4,610	1,007	total ≤ 0 (best 5 days +1,370; without them −5,457)
SPY	2022-11→2026-10	+144.60, +51.3%, 2025-03-11, SPY 2025-03-11 557.0PUT	−115.40, −25.1%, 2025-04-09, SPY 2025-04-09 495.0PUT	0.69	5,838	972	total ≤ 0 (best 5 days +751; without them −6,372)
all names	2022-11→2026-10	+345.60, +111.8%, 2026-06-09, QQQ 2026-06-09 692.0PUT	−157.40, −34.5%, 2025-04-07, QQQ 2025-04-07 417.0PUT	0.73	12,320	991	total ≤ 0 (best 5 days +1,506; without them −13,587)

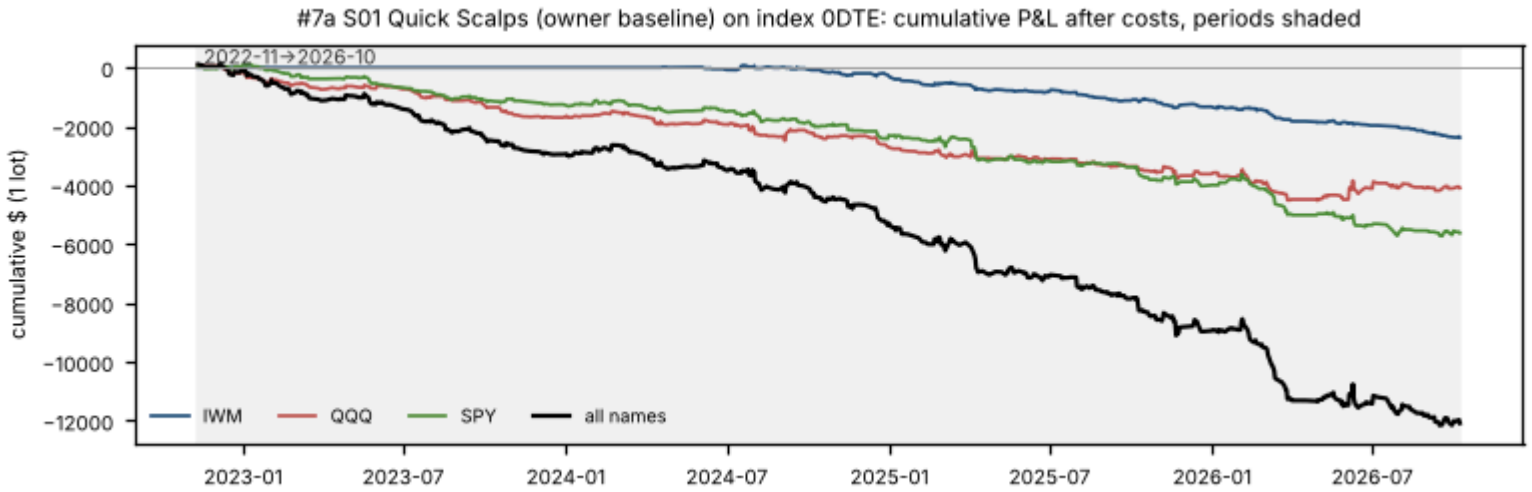
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-11→2026-10: stop 51.9%; target_2 26.3%; vwap_loss 15.3%; time 5.4%; invalidation 1.1%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	157	35.7	−5.66	−1.77	−277
2025	IWM	311	33.8	−5.80	−3.38	−1,051
2026	IWM	217	30.9	−8.85	−4.81	−1,044
2022	QQQ	34	32.4	−5.95	−4.96	−169
2023	QQQ	293	31.1	−7.48	−5.21	−1,527
2024	QQQ	287	35.5	−4.04	−3.50	−1,004
2025	QQQ	241	37.3	−2.60	−3.80	−915
2026	QQQ	220	37.3	−2.41	−2.15	−473
2022	SPY	40	42.5	+1.63	+0.99	+40
2023	SPY	285	31.9	−6.51	−4.52	−1,289
2024	SPY	281	34.5	−3.73	−3.74	−1,050
2025	SPY	282	35.8	−3.94	−6.02	−1,698
2026	SPY	238	33.6	−3.83	−6.82	−1,624
2022	all names	74	37.8	−1.85	−1.74	−129
2023	all names	578	31.5	−7.00	−4.87	−2,816
2024	all names	725	35.2	−4.27	−3.22	−2,331
2025	all names	834	35.5	−4.25	−4.39	−3,664
2026	all names	675	33.9	−4.98	−4.65	−3,141



Amendments, deviations and integrity checks. Status ≠ filled rows (no chain, in position, no expiry) are not trades.

What it showed. The baseline loses $4 a contract after costs.

#7b S02 Quick Scalps: 41-variant grid, walk-forward OOS (lower-bound selection)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 6bf5934. **Dates:** 2026-10-14.

Hypothesis. A single-factor variant of the baseline holds out of sample.

Rule. Each year, the variant with the best lower bound on the years before is traded the next year (walk-forward).

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE, OOS 2024-01 → 2026-10.

Periods and holdouts. Walk-forward OOS years 2024–2026.

Pass rule (as pre-registered). Positive OOS lower bound, better than matched random.

Verdict. Nothing promoted: OOS −$3.81 [−5.23, −2.35]; no variant positive or better than random.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	walk-forward OOS 2024-01→2026-10	935	314 / 638	32.2	+35.3, +18.96	−28.3, −14.23	−27.5, −8.40	**−7.85**, −3.55	[−10.10, −5.79]; [−4.73, −2.46]
QQQ	walk-forward OOS 2024-01→2026-10	1,117	352 / 715	36.6	+35.3, +39.77	−26.7, −27.00	−22.5, −11.40	**−3.97**, −2.55	[−5.94, −2.21]; [−4.85, −0.27]
SPY	walk-forward OOS 2024-01→2026-10	1,148	373 / 715	35.1	+35.0, +31.48	−25.9, −25.09	−19.7, −11.40	**−4.55**, −5.23	[−6.76, −2.77]; [−8.05, −2.99]
all names	walk-forward OOS 2024-01→2026-10	3,200	488 / 715	34.8	+35.2, +31.14	−26.9, −22.44	−23.6, −10.40	**−5.31**, −3.81	[−6.87, −3.98]; [−5.45, −2.46]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	walk-forward OOS 2024-01→2026-10	+108.60, +197.5%, 2025-02-07, IWM 227.5PUT no_15m	−120.40, −71.2%, 2025-09-17, IWM 239.0PUT no_15m	0.63	3,339	634	total ≤ 0 (best 5 days +531; without them −3,846)
QQQ	walk-forward OOS 2024-01→2026-10	+345.60, +111.8%, 2026-06-09, QQQ 692.0PUT no_15m	−157.40, −34.5%, 2025-04-07, QQQ 417.0PUT no_15m	0.85	2,992	668	total ≤ 0 (best 5 days +1,679; without them −4,532)
SPY	walk-forward OOS 2024-01→2026-10	+172.10, +40.0%, 2025-04-10, SPY 516.0PUT no_15m	−163.40, −23.6%, 2025-04-07, SPY 496.0PUT no_15m	0.68	6,234	692	total ≤ 0 (best 5 days +914; without them −6,922)
all names	walk-forward OOS 2024-01→2026-10	+345.60, +111.8%, 2026-06-09, QQQ 692.0PUT no_15m	−163.40, −23.6%, 2025-04-07, SPY 496.0PUT no_15m	0.74	12,295	673	total ≤ 0 (best 5 days +1,987; without them −14,163)

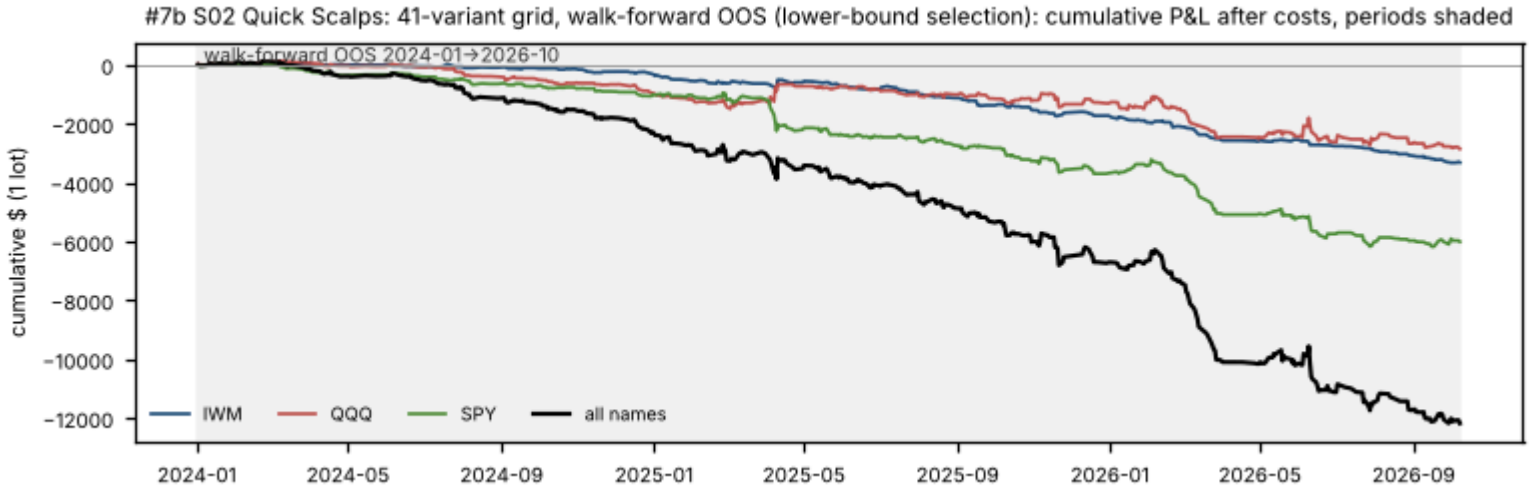
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- walk-forward OOS 2024-01→2026-10: stop 52.8%; target_2 24.4%; vwap_loss 14.3%; ema_target 4.2%; time 3.6%; invalidation 0.8%; settlement 0.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	150	31.3	−10.02	−2.09	−313
2025	IWM	475	33.7	−6.10	−2.97	−1,413
2026	IWM	310	30.3	−9.46	−5.13	−1,589
2024	QQQ	317	32.2	−6.64	−3.03	−960
2025	QQQ	429	40.8	−1.82	−0.81	−347
2026	QQQ	371	35.6	−4.17	−4.17	−1,546
2024	SPY	300	31.3	−6.87	−3.36	−1,008
2025	SPY	475	38.3	−3.68	−5.61	−2,667
2026	SPY	373	34.0	−3.79	−6.26	−2,334
2024	all names	767	31.7	−7.39	−2.97	−2,281
2025	all names	1,379	37.5	−3.94	−3.21	−4,427
2026	all names	1,054	33.5	−5.59	−5.19	−5,469



Amendments, deviations and integrity checks. —

What it showed. Variant selection does not rescue the signal.

#7c S03 Quick Scalps with weekly expirations (1–7 days)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. The signal works on weekly contracts.

Rule. Quick Scalps signal → nearest Friday contract, OI > 250.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM weekly chains, sampled sessions 2017-06 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S01.

Verdict. Nothing promoted: +$1.27 a contract [−3.69, +6.25] (= random).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2017-06→2026-10	239	141 / 2,394	40.6	+27.2, +39.75	−15.0, −23.63	−4.8, −7.40	**+2.14**, +2.09	[−1.42, +5.76]; [−2.85, +8.14]
QQQ	2017-06→2026-10	217	135 / 2,386	36.9	+23.0, +64.11	−16.3, −40.82	−7.7, −14.40	**−1.82**, −2.13	[−4.84, +1.90]; [−11.95, +9.23]
SPY	2017-06→2026-10	362	221 / 2,393	36.2	+28.4, +70.90	−16.9, −35.87	−7.2, −9.40	**−0.50**, +2.76	[−2.94, +2.05]; [−3.17, +9.82]
all names	2017-06→2026-10	818	376 / 2,399	37.7	+26.6, +59.33	−16.2, −33.79	−6.5, −9.40	**−0.08**, +1.27	[−1.87, +1.70]; [−2.78, +5.63]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2017-06→2026-10	+167.60, +39.2%, 2025-04-08, IWM 2025-04-11 176.0PUT	−100.40, −22.1%, 2020-06-12, IWM 2020-06-19 134.5PUT	1.15	631	1,446	146%
QQQ	2017-06→2026-10	+415.60, +39.7%, 2025-04-08, QQQ 2025-04-11 423.0PUT	−168.40, −33.8%, 2026-02-05, QQQ 2026-02-06 594.0PUT	0.92	1,953	1,975	total ≤ 0 (best 5 days +1,651; without them −2,114)
SPY	2017-06→2026-10	+463.10, +120.0%, 2020-05-15, SPY 2020-05-22 286.0CALL	−227.40, −31.9%, 2020-03-10, SPY 2020-03-13 273.0PUT	1.12	1,057	919	177%
all names	2017-06→2026-10	+463.10, +120.0%, 2020-05-15, SPY 2020-05-22 286.0CALL	−227.40, −31.9%, 2020-03-10, SPY 2020-03-13 273.0PUT	1.06	1,214	1,595	235%

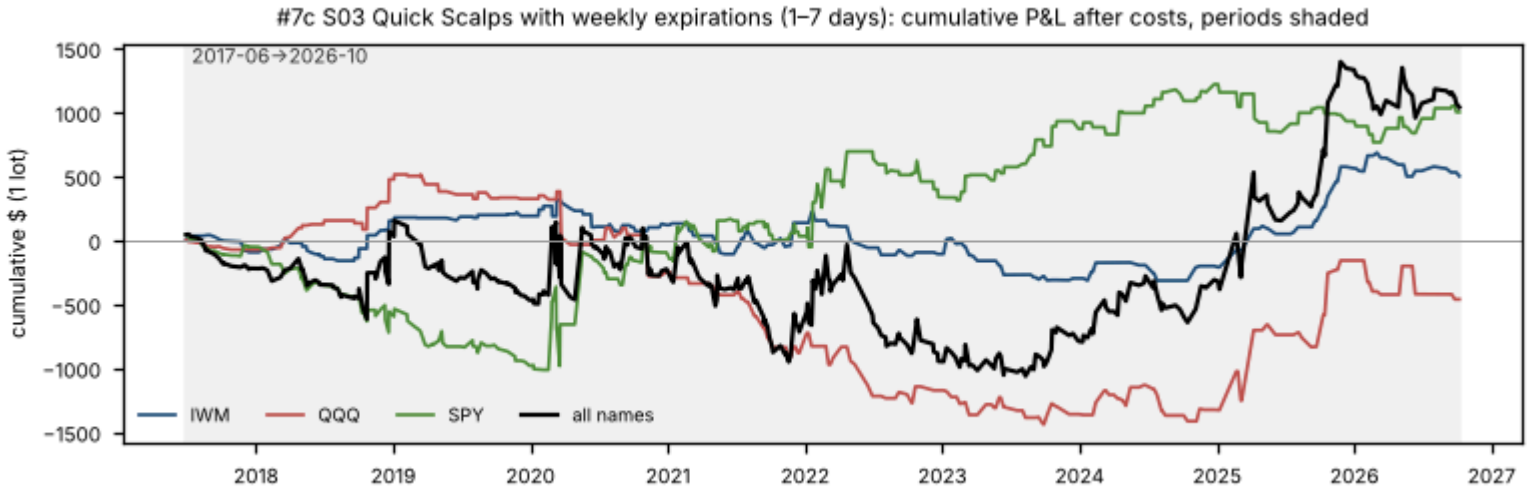
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2017-06→2026-10: vwap_loss 50.6%; data_end 17.6%; target_2 17.5%; stop 8.6%; invalidation 5.4%; time 0.4%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	IWM	20	35.0	−4.52	−4.73	−95
2018	IWM	30	50.0	+7.35	+8.18	+245
2019	IWM	15	40.0	+1.58	+2.57	+39
2020	IWM	34	35.3	+1.82	−2.09	−71
2021	IWM	29	37.9	+1.78	+0.58	+17
2022	IWM	20	20.0	−3.67	−12.17	−243
2023	IWM	31	35.5	−2.78	−6.16	−191
2024	IWM	17	41.2	+2.55	+5.69	+97
2025	IWM	24	66.7	+15.82	+31.81	+763
2026	IWM	19	42.1	−0.99	−3.19	−61
2017	QQQ	14	14.3	−9.39	−5.29	−74
2018	QQQ	21	76.2	+14.05	+26.15	+549
2019	QQQ	16	37.5	−9.44	−9.49	−152
2020	QQQ	32	37.5	−4.50	−17.85	−571
2021	QQQ	27	18.5	−12.66	−23.49	−634
2022	QQQ	26	30.8	−1.24	−11.23	−292
2023	QQQ	26	34.6	−1.36	−7.17	−186
2024	QQQ	23	30.4	+1.96	+1.51	+35
2025	QQQ	22	54.5	+7.67	+53.03	+1,167
2026	QQQ	10	30.0	−6.81	−30.35	−304
2017	SPY	32	25.0	−5.54	−1.62	−52
2018	SPY	75	29.3	−3.79	−6.74	−506
2019	SPY	46	21.7	−8.48	−9.25	−425
2020	SPY	39	41.0	+5.28	+22.73	+886
2021	SPY	51	41.2	−0.74	+2.49	+127
2022	SPY	36	38.9	+2.84	+8.48	+305
2023	SPY	30	53.3	+7.92	+17.80	+534
2024	SPY	21	57.1	+7.82	+16.60	+349
2025	SPY	17	35.3	−3.13	−16.99	−289
2026	SPY	15	40.0	+3.50	+4.73	+71
2017	all names	66	25.8	−6.05	−3.34	−220
2018	all names	126	42.1	+1.83	+2.29	+289
2019	all names	77	28.6	−6.72	−7.00	−539
2020	all names	105	38.1	+1.18	+2.32	+244
2021	all names	107	34.6	−3.06	−4.58	−490
2022	all names	82	31.7	−0.04	−2.81	−230
2023	all names	87	41.4	+1.33	+1.80	+157
2024	all names	61	42.6	+4.14	+7.87	+480
2025	all names	63	54.0	+7.86	+26.05	+1,641
2026	all names	44	38.6	−0.78	−6.66	−293



Amendments, deviations and integrity checks. A cut-short pull is a uniform sample by strata (d706328).

What it showed. Indistinguishable from zero.

#7d S09 Quick Scalps on SPXW 0DTE

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** 113cad7 / — / 7167541. **Dates:** 2026-10-14.

Hypothesis. The baseline on SPXW.

Rule. As S01, signals on SPX values with SPY volume.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO; SPXW per its fees

Data. SPXW 0DTE 2022-05-19 → 2026-10-08, 1,229 fills on 481 days.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S01.

Verdict. REJECTED: −$46.18 a contract [−63.19, −30.02], −$24 against matched random.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2022-05→2026-10	1,229	481 / 1,135	33.8	+37.8, +317.77	−25.0, −231.74	−22.1, −141.40	**−3.81**, −46.18	[−5.48, −2.15]; [−62.89, −30.83]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2022-05→2026-10	+1,483.60, +41.0%, 2025-04-09, SPXW 2025-04-09 5380.0CALL	−1,001.40, −40.5%, 2026-03-09, SPXW 2026-03-09 6640.0PUT	0.70	57,686	1,135	total ≤ 0 (best 5 days +8,081; without them −64,842)

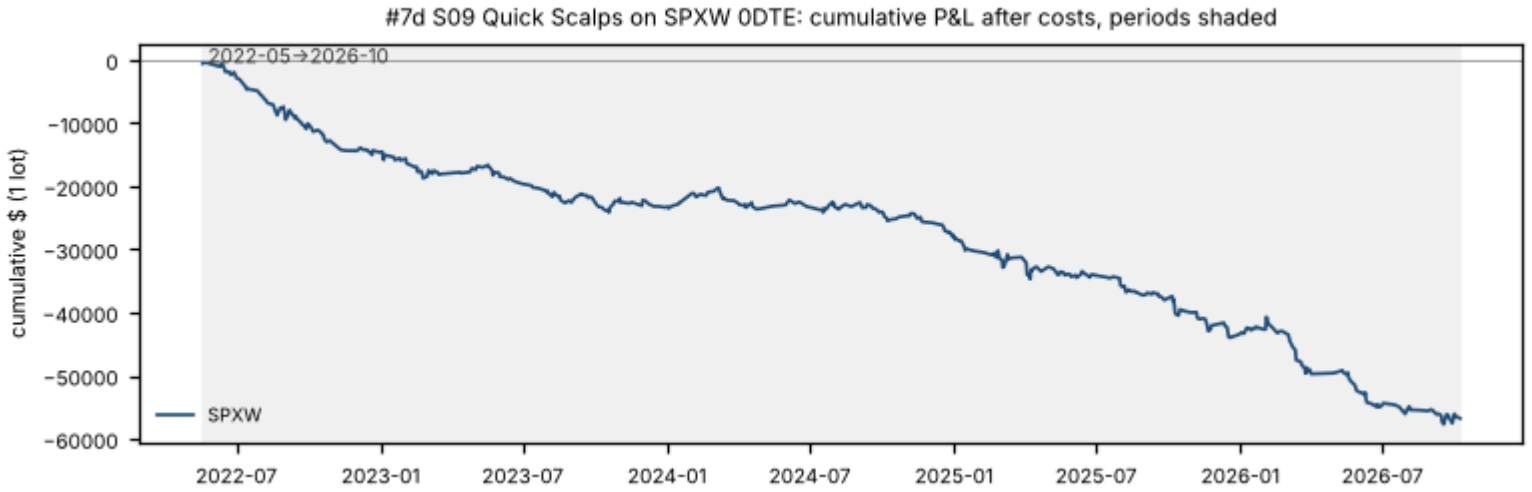
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-05→2026-10: stop 52.7%; target_2 26.2%; vwap_loss 15.3%; time 3.3%; invalidation 2.4%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	159	27.7	−7.46	−91.62	−14,568
2023	SPXW	300	34.3	−3.47	−29.22	−8,765
2024	SPXW	279	37.6	−1.20	−15.80	−4,408
2025	SPXW	266	33.8	−5.33	−58.93	−15,675
2026	SPXW	225	32.4	−3.14	−59.31	−13,345



Amendments, deviations and integrity checks. —

What it showed. Loses on SPXW too.

#7e S10 Short premium: fade the Quick Scalps signal (single contract, 2× stop, to 15:50)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A4 912b415 / rerun 1ecdfd5 / cf124a0, 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. Selling the contract the signal would buy earns its premium.

Rule. Sell the signal's contract at the bid, buy back at 15:50 or at a 2× credit stop (the best cell).

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 0DTE 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Positive lower bound and better than random.

Verdict. Nothing promoted: breakeven at best (+$0.52 a contract).

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	685	274 / 638	51.4	+80.4, +48.29	−104.8, −53.66	+14.1, +7.60	**−9.63**, −1.27	[−20.09, +0.58]; [−6.87, +4.01]
QQQ	2022-11→2026-10	1,075	421 / 1,009	56.3	+72.3, +79.79	−98.6, −104.04	+24.4, +19.60	**−2.40**, −0.58	[−10.34, +7.08]; [−11.23, +10.76]
SPY	2022-11→2026-10	1,126	444 / 1,009	57.1	+73.0, +74.48	−99.9, −92.94	+32.3, +21.60	**−1.17**, +2.67	[−8.30, +5.93]; [−4.58, +10.41]
all names	2022-11→2026-10	2,886	614 / 1,009	55.4	+74.4, +70.73	−100.7, −86.83	+28.1, +17.60	**−3.63**, +0.52	[−9.79, +2.83]; [−5.54, +7.20]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+185.60, +98.7%, 2025-04-09, IWM short 173.0PUT	−195.40, −107.4%, 2025-09-17, IWM short 241.0CALL	0.95	3,163	362	total ≤ 0 (best 5 days +1,283; without them −2,156)
QQQ	2022-11→2026-10	+418.60, +92.6%, 2025-04-07, QQQ short 417.0PUT	−426.40, −138.9%, 2026-06-09, QQQ short 692.0PUT	0.99	7,281	399	total ≤ 0 (best 5 days +3,940; without them −4,567)
SPY	2022-11→2026-10	+455.60, +99.3%, 2025-04-09, SPY short 495.0PUT	−414.40, −101.3%, 2025-04-11, SPY short 527.0CALL	1.07	3,255	390	116%
all names	2022-11→2026-10	+455.60, +99.3%, 2025-04-09, SPY short 495.0PUT	−426.40, −138.9%, 2026-06-09, QQQ short 692.0PUT	1.01	9,276	364	470%

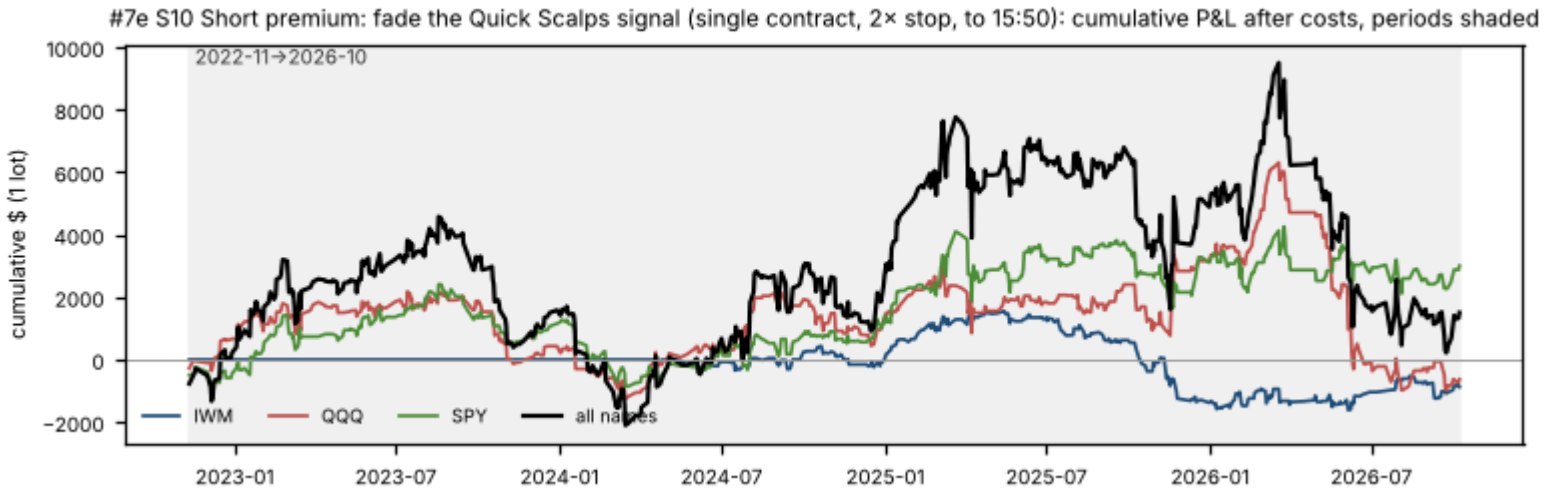
Random baseline: the study's matched random controls (5 seeds, same symbols, sessions and exits, 14,218 trades): mean −5.71 dollars and −10.27% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: session_cutoff 61.9%; stop 38.1%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	157	55.4	−3.84	+0.08	+13
2025	IWM	311	46.9	−17.46	−4.48	−1,392
2026	IWM	217	54.8	−2.58	+2.33	+506
2022	QQQ	34	67.6	+21.67	+19.69	+669
2023	QQQ	293	53.9	−2.85	−0.81	−237
2024	QQQ	287	59.6	−1.80	+2.90	+832
2025	QQQ	241	59.3	+2.56	+7.87	+1,896
2026	QQQ	220	50.0	−11.75	−17.21	−3,787
2022	SPY	40	50.0	−9.73	−11.18	−447
2023	SPY	285	58.9	+4.07	+5.61	+1,598
2024	SPY	281	57.7	−6.70	+0.88	+248
2025	SPY	282	58.2	+1.43	+6.05	+1,706
2026	SPY	238	54.2	−2.54	−0.43	−101
2022	all names	74	58.1	+4.69	+3.01	+222
2023	all names	578	56.4	+0.56	+2.35	+1,361
2024	all names	725	57.9	−4.14	+1.51	+1,093
2025	all names	834	54.3	−5.29	+2.65	+2,209
2026	all names	675	53.0	−5.55	−5.01	−3,382



Amendments, deviations and integrity checks. The vertical rows were superseded after the protective-leg fix; the single-contract rows (shown) are unchanged.

What it showed. The fade is about breakeven.

#7f S12 Intraday momentum: first-half-hour direction → 10:00–15:00 long option

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 fe93f87 / — / 2430c9b. **Dates:** 2026-10-14.

Hypothesis. The first half hour's direction continues (Gao et al.).

Rule. Buy the 0DTE contract in the first-half-hour direction at 10:00, exit 15:00.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 2022-11 → 2026-10 (options).

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). Two-part promotion rule (A1).

Verdict. Nothing promoted (QQQ-only underlying effect, not batch-robust; option cell negative).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	388	388 / 638	28.1	+115.2, +42.25	−73.7, −23.32	−64.3, −14.90	**−20.62**, −4.90	[−31.88, −9.17]; [−9.04, −0.40]
QQQ	2022-11→2026-10	920	920 / 1,010	30.9	+95.4, +66.78	−64.1, −36.51	−44.4, −20.40	**−14.88**, −4.62	[−21.50, −8.59]; [−9.62, +0.68]
SPY	2022-11→2026-10	906	906 / 1,010	30.8	+99.1, +64.33	−62.7, −34.22	−43.3, −18.40	**−12.89**, −3.87	[−18.58, −5.60]; [−7.90, +1.30]
all names	2022-11→2026-10	2,214	961 / 1,010	30.4	+100.1, +61.78	−65.3, −33.19	−47.1, −18.40	**−15.07**, −4.36	[−20.45, −9.22]; [−8.13, +0.06]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+370.60, +741.2%, 2026-07-29, IWM 293.0PUT	−109.40, −100.4%, 2024-09-18, IWM 222.0CALL	0.71	2,237	636	total ≤ 0 (best 5 days +1,184; without them −3,084)
QQQ	2022-11→2026-10	+1,187.60, +860.6%, 2026-07-29, QQQ 678.0PUT	−177.40, −78.5%, 2025-04-07, QQQ 424.0CALL	0.82	6,078	1,008	total ≤ 0 (best 5 days +3,412; without them −7,666)
SPY	2022-11→2026-10	+971.60, +299.0%, 2025-04-09, SPY 532.0CALL	−222.40, −98.0%, 2025-04-08, SPY 502.0CALL	0.84	4,959	1,010	total ≤ 0 (best 5 days +3,040; without them −6,547)
all names	2022-11→2026-10	+1,187.60, +860.6%, 2026-07-29, QQQ 678.0PUT	−222.40, −98.0%, 2025-04-08, SPY 502.0CALL	0.81	12,425	1,010	total ≤ 0 (best 5 days +7,095; without them −16,757)

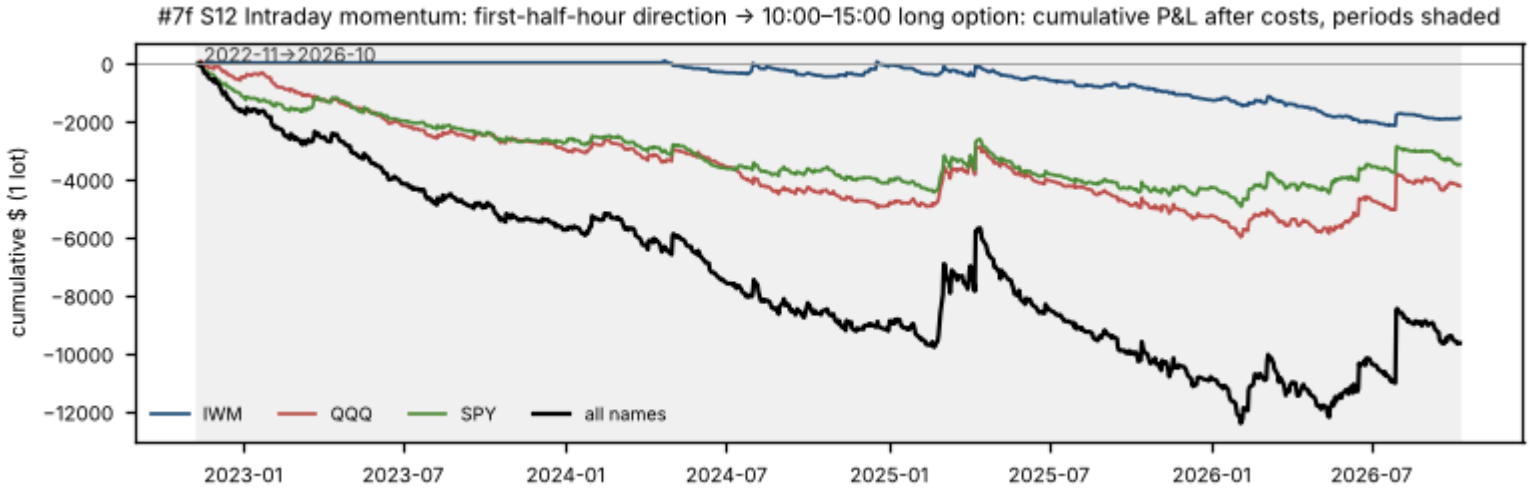
Random baseline: the study's matched random controls (5 seeds, same symbols, sessions and exits, 10,984 trades): mean −5.53 dollars and −14.32% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: session_cutoff 98.5%; settlement 1.5%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	98	30.6	−14.28	−1.41	−138
2025	IWM	166	27.1	−26.24	−6.90	−1,145
2026	IWM	124	27.4	−18.10	−4.97	−617
2022	QQQ	31	16.1	−26.55	−15.59	−483
2023	QQQ	225	29.8	−25.79	−11.09	−2,496
2024	QQQ	236	28.0	−18.94	−8.35	−1,970
2025	QQQ	239	33.1	−10.22	−1.30	−312
2026	QQQ	189	35.4	−0.80	+5.33	+1,007
2022	SPY	33	9.1	−60.86	−34.79	−1,148
2023	SPY	217	27.6	−16.78	−7.52	−1,633
2024	SPY	233	30.0	−13.07	−5.92	−1,379
2025	SPY	236	31.8	−11.06	−0.51	−120
2026	SPY	187	38.0	−2.00	+4.13	+773
2022	all names	64	12.5	−44.24	−25.49	−1,632
2023	all names	442	28.7	−21.36	−9.34	−4,129
2024	all names	567	29.3	−15.72	−6.15	−3,488
2025	all names	641	31.0	−14.68	−2.46	−1,577
2026	all names	500	34.4	−5.54	+2.33	+1,164



Amendments, deviations and integrity checks. seed 0 = the signal; 11–15 random.

What it showed. The option expression loses.

#7g S12b 10:00 long option in the first-half-hour direction

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5.1 b9b9206 / — / be991de. **Dates:** 2026-10-14.

Hypothesis. As S12 with the baseline rule's contract.

Rule. Buy the baseline-rule 0DTE contract at 10:00 in the first-half-hour direction; exit 15:00.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S12.

Verdict. Nothing promoted: −$4.87 a contract [−14.18, +4.87].

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2022-11→2026-10	564	564 / 639	32.8	+143.1, +109.85	−83.1, −60.77	−75.5, −41.40	**−8.94**, −4.81	[−19.12, +4.50]; [−13.87, +6.44]
QQQ	2022-11→2026-10	964	964 / 1,010	32.7	+149.0, +207.97	−76.8, −109.79	−62.1, −71.90	**−3.05**, −5.96	[−9.98, +4.00]; [−16.14, +4.07]
SPY	2022-11→2026-10	971	971 / 1,010	34.4	+143.7, +168.87	−77.0, −94.40	−61.0, −58.40	**−1.04**, −3.84	[−7.66, +6.24]; [−12.79, +7.75]
all names	2022-11→2026-10	2,499	971 / 1,010	33.4	+145.6, +170.54	−78.3, −92.74	−64.6, −55.40	**−3.60**, −4.87	[−9.45, +2.93]; [−12.95, +4.17]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	2022-11→2026-10	+858.60, +397.5%, 2025-04-09, IWM 176.0CALL	−181.40, −100.8%, 2025-04-23, IWM 194.0CALL	0.88	3,344	425	total ≤ 0 (best 5 days +3,043; without them −5,754)
QQQ	2022-11→2026-10	+2,477.60, +482.0%, 2025-04-09, QQQ 423.0CALL	−413.40, −99.9%, 2026-06-25, QQQ 704.0PUT	0.92	7,404	987	total ≤ 0 (best 5 days +6,398; without them −12,140)
SPY	2022-11→2026-10	+2,330.60, +411.0%, 2025-04-09, SPY 502.0CALL	−377.40, −100.4%, 2025-04-08, SPY 524.0CALL	0.94	4,483	835	total ≤ 0 (best 5 days +5,593; without them −9,323)
all names	2022-11→2026-10	+2,477.60, +482.0%, 2025-04-09, QQQ 423.0CALL	−413.40, −99.9%, 2026-06-25, QQQ 704.0PUT	0.92	13,595	987	total ≤ 0 (best 5 days +13,068; without them −25,251)

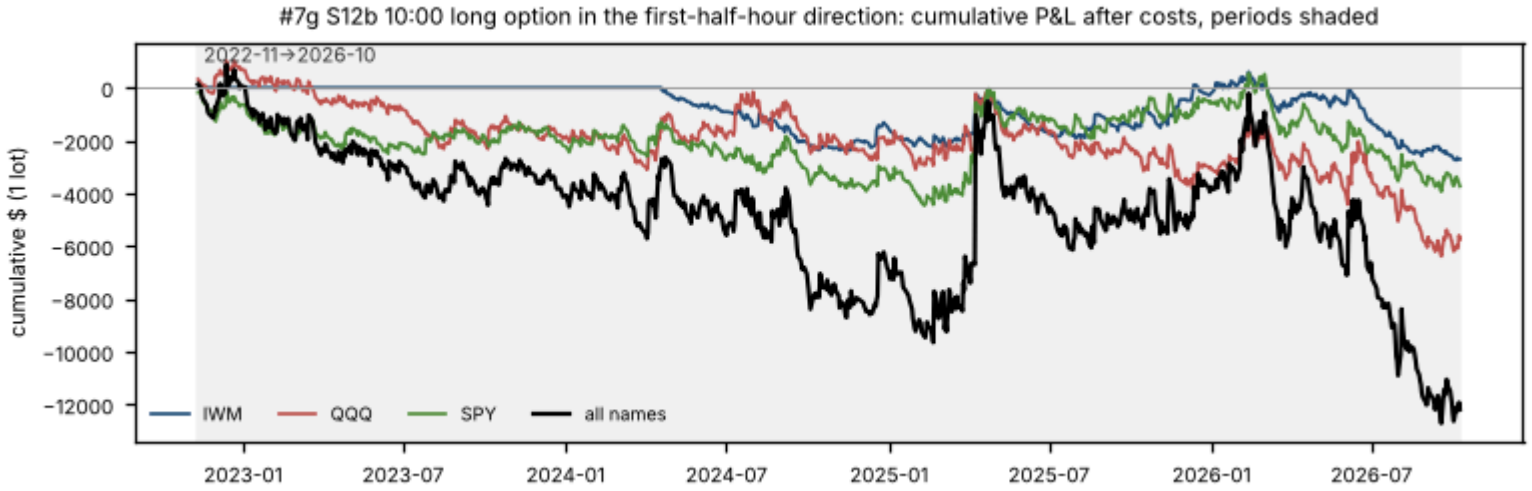
Random baseline: the study's matched random controls (5 seeds, same symbols, sessions and exits, 12,446 trades): mean −11.99 dollars and −9.75% a trade (no percentile was computed).

Exit reasons (share of trades):

- 2022-11→2026-10: time 95.5%; settlement 4.5%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	144	29.9	−17.42	−10.60	−1,527
2025	IWM	227	35.2	+0.80	+5.78	+1,312
2026	IWM	193	32.1	−14.05	−12.93	−2,496
2022	QQQ	31	45.2	+15.39	+21.73	+674
2023	QQQ	247	30.0	−10.02	−10.61	−2,620
2024	QQQ	248	32.7	+2.78	−0.13	−32
2025	QQQ	246	33.7	−3.65	−4.25	−1,046
2026	QQQ	192	32.8	−3.80	−14.15	−2,717
2022	SPY	33	27.3	−5.41	−19.70	−650
2023	SPY	248	33.5	−7.24	−6.21	−1,540
2024	SPY	249	34.9	−0.27	−4.77	−1,187
2025	SPY	247	37.7	+5.91	+11.57	+2,857
2026	SPY	194	32.0	−2.23	−16.55	−3,211
2022	all names	64	35.9	+4.67	+0.37	+23
2023	all names	495	31.7	−8.63	−8.40	−4,160
2024	all names	641	32.9	−2.94	−4.28	−2,745
2025	all names	720	35.6	+1.03	+4.34	+3,123
2026	all names	579	32.3	−6.69	−14.55	−8,424



Amendments, deviations and integrity checks. —

What it showed. Negative.

#7h S15 VRP-conditioned 10:00 short ATM 0DTE straddle (unconditional row)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 6dfba60 / — / 6ed12b6. **Dates:** 2026-10-14.

Hypothesis. Rich implied vs trailing realised volatility marks sellable straddles.

Rule. Sell the 10:00 ATM straddle at the bids, buy back at 15:50 at the asks.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ 0DTE, 970/971 sessions 2022-11 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). A conditioned cell with a positive lower bound.

Verdict. Nothing promoted: unconditional −$15.46 a straddle [−32.68, +0.29]; rich implied forecasts large moves.

Metrics (return = % of the credit received; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2022-11→2026-10	971	971 / 1,010	58.1	+44.1, +144.20	−74.2, −236.12	+13.8, +41.20	**−5.49**, −15.22	[−10.37, −0.97]; [−34.86, +1.25]
SPY	2022-11→2026-10	970	970 / 1,010	57.9	+43.1, +121.18	−72.3, −204.23	+13.9, +36.20	**−5.44**, −15.70	[−9.91, −1.65]; [−34.02, −1.83]
all names	2022-11→2026-10	1,941	971 / 1,010	58.0	+43.6, +132.71	−73.2, −220.16	+13.8, +39.20	**−5.47**, −15.46	[−9.93, −1.42]; [−33.96, −0.83]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2022-11→2026-10	+660.20, +75.7%, 2024-08-05, QQQ 10:00 short straddle 433.0	−3,323.80, −279.1%, 2025-04-09, QQQ 10:00 short straddle 421.0	0.85	19,527	1,003	total ≤ 0 (best 5 days +2,652; without them −17,426)
SPY	2022-11→2026-10	+658.20, +76.2%, 2024-08-05, SPY 10:00 short straddle 515.0	−3,099.80, −230.8%, 2025-04-09, SPY 10:00 short straddle 500.0	0.82	17,307	999	total ≤ 0 (best 5 days +2,515; without them −17,741)
all names	2022-11→2026-10	+660.20, +75.7%, 2024-08-05, QQQ 10:00 short straddle 433.0	−3,323.80, −279.1%, 2025-04-09, QQQ 10:00 short straddle 421.0	0.83	35,674	1,003	total ≤ 0 (best 5 days +5,055; without them −35,055)

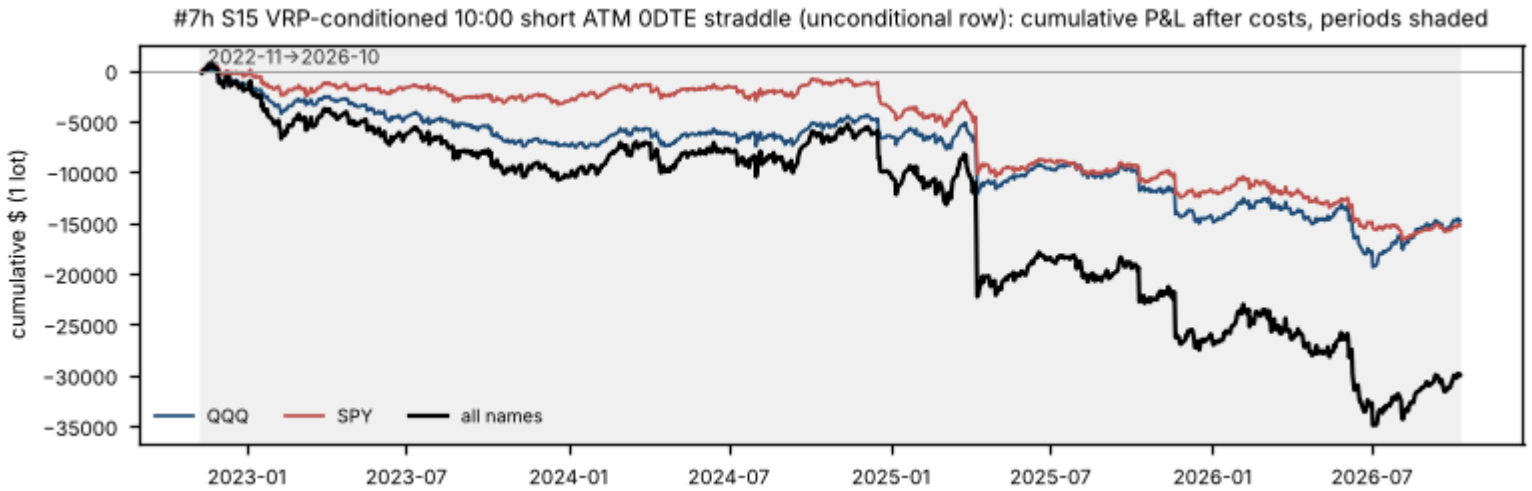
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2022-11→2026-10: bought back 15:50 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	45.5	−15.07	−40.22	−1,327
2023	QQQ	248	51.6	−12.00	−24.15	−5,990
2024	QQQ	249	63.5	+0.66	+3.55	+884
2025	QQQ	247	58.3	−7.12	−32.12	−7,935
2026	QQQ	194	61.3	−1.38	−2.09	−405
2022	SPY	33	51.5	−6.35	−15.86	−523
2023	SPY	248	57.3	−6.51	−10.36	−2,568
2024	SPY	248	63.7	−1.15	−2.22	−550
2025	SPY	247	54.7	−7.84	−33.25	−8,212
2026	SPY	194	56.7	−6.35	−17.38	−3,372
2022	all names	66	48.5	−10.71	−28.04	−1,851
2023	all names	496	54.4	−9.25	−17.26	−8,559
2024	all names	497	63.6	−0.24	+0.67	+333
2025	all names	494	56.5	−7.48	−32.68	−16,146
2026	all names	388	59.0	−3.87	−9.74	−3,777



Amendments, deviations and integrity checks. —

What it showed. The 10:00 short straddle loses.

#7i S16b Prior-day fade with a weekly contract (time exit)

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A5 fe93f87 / — / 5cabc13, 3c8d6cb. **Dates:** 2026-10-14.

Hypothesis. The prior session’s close-to-close move reverses.

Rule. Buy the weekly contract against the prior move at the 09:31 close; time exit.

Fills and costs. $0.70 a contract each side (protocol 113cad7), entry at the ask, exit at the bid, 1-minute NBBO

Data. SPY/QQQ/IWM weekly chains 2017-06 → 2026-10.

Periods and holdouts. Walk-forward by calendar year (protocol 113cad7); reads counted in holdout B (ledger).

Pass rule (as pre-registered). As S12.

Verdict. Nothing promoted: −$9.91 a contract [−19.31, −0.26].

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	2017-06→2026-10	287	287 / 2,411	36.6	+47.6, +76.59	−42.0, −65.66	−17.5, −23.40	**−9.21**, −13.62	[−14.76, −3.11]; [−22.82, −3.73]
QQQ	2017-06→2026-10	290	290 / 2,411	38.3	+46.3, +134.29	−41.5, −104.91	−14.3, −28.90	**−7.85**, −13.36	[−13.53, −1.97]; [−30.61, +4.83]
SPY	2017-06→2026-10	464	464 / 2,419	38.1	+57.5, +131.77	−42.9, −90.10	−15.9, −24.40	**−4.60**, −5.46	[−11.77, +1.69]; [−21.15, +9.49]
all names	2017-06→2026-10	1,041	699 / 2,419	37.8	+51.7, +117.74	−42.3, −87.33	−16.4, −25.40	**−6.78**, −9.91	[−11.05, −3.10]; [−18.95, −0.47]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
IWM	2017-06→2026-10	+375.60, +220.9%, 2026-05-20, IWM 275.0CALL	−278.40, −72.9%, 2025-04-08, IWM 186.0CALL	0.67	4,113	1,453	total ≤ 0 (best 5 days +1,234; without them −5,142)
QQQ	2017-06→2026-10	+1,174.60, +123.6%, 2025-04-08, QQQ 435.0PUT	−445.40, −66.1%, 2026-06-12, QQQ 713.0PUT	0.79	3,887	2,362	total ≤ 0 (best 5 days +2,966; without them −6,839)
SPY	2017-06→2026-10	+1,274.60, +404.6%, 2022-10-13, SPY 350.0CALL	−416.40, −89.9%, 2022-05-04, SPY 415.0PUT	0.90	4,600	2,387	total ≤ 0 (best 5 days +4,189; without them −6,725)
all names	2017-06→2026-10	+1,274.60, +404.6%, 2022-10-13, SPY 350.0CALL	−445.40, −66.1%, 2026-06-12, QQQ 713.0PUT	0.82	11,188	2,388	total ≤ 0 (best 5 days +4,588; without them −14,905)

Random baseline: the study’s matched random controls (5 seeds, same symbols, sessions and exits, 5,218 trades): mean −9.80 dollars and −4.48% a trade (no percentile was computed).

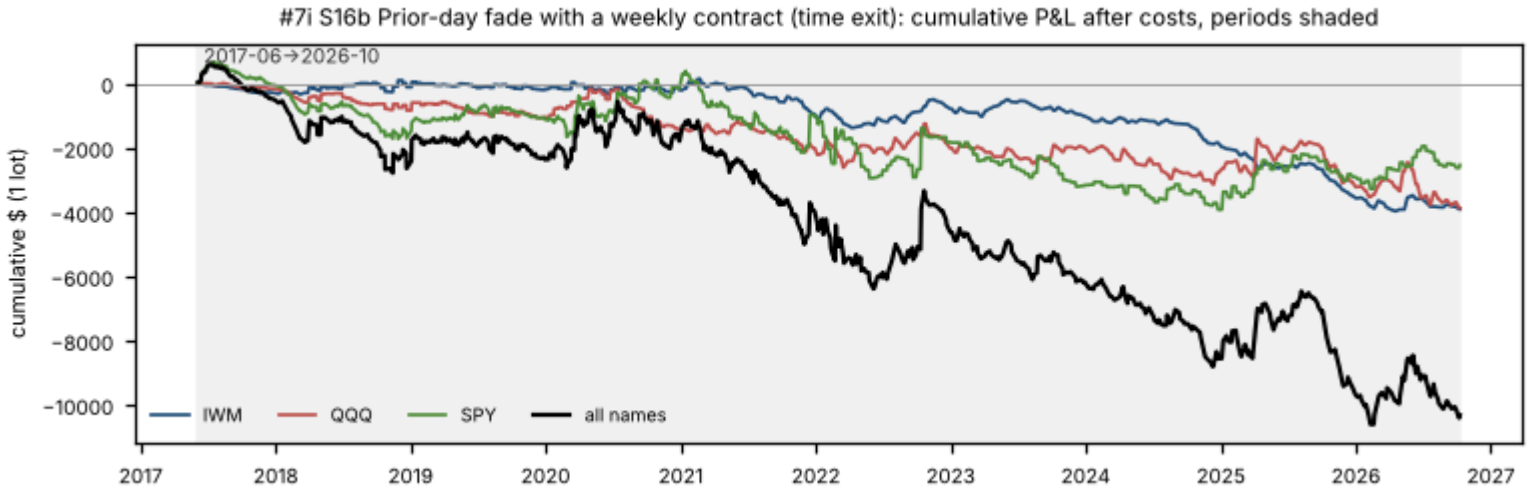
Exit reasons (share of trades):

- 2017-06→2026-10: data_end 88.1%; time 11.9%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	IWM	16	18.8	−29.60	−17.96	−287
2018	IWM	30	50.0	+7.71	+6.00	+180
2019	IWM	32	34.4	−4.32	−1.84	−59
2020	IWM	28	42.9	−1.86	−1.26	−35
2021	IWM	31	38.7	−15.39	−24.24	−751
2022	IWM	32	50.0	−1.03	+5.26	+168
2023	IWM	31	38.7	−2.67	−7.37	−228
2024	IWM	31	29.0	−15.87	−26.63	−825
2025	IWM	32	21.9	−31.25	−53.93	−1,726
2026	IWM	24	33.3	−5.23	−14.32	−344
2017	QQQ	17	23.5	−21.99	−9.34	−159
2018	QQQ	31	32.3	−18.24	−21.72	−673
2019	QQQ	31	29.0	−8.21	−6.88	−213
2020	QQQ	30	46.7	+1.27	−13.13	−394
2021	QQQ	31	38.7	−9.14	−18.17	−563
2022	QQQ	32	43.8	−1.60	−0.15	−5
2023	QQQ	31	38.7	−2.45	+0.21	+7
2024	QQQ	31	38.7	−7.82	−25.50	−790
2025	QQQ	32	40.6	−10.49	−13.31	−426
2026	QQQ	24	45.8	−5.52	−27.32	−656
2017	SPY	58	43.1	−0.06	−0.71	−41
2018	SPY	62	33.9	−21.05	−26.74	−1,658
2019	SPY	64	34.4	+4.95	+9.12	+583
2020	SPY	60	45.0	+1.16	+16.17	+970
2021	SPY	64	29.7	−10.72	−13.95	−893
2022	SPY	37	40.5	−3.63	−21.16	−783
2023	SPY	32	34.4	−15.11	−41.74	−1,336
2024	SPY	31	35.5	−8.72	−24.66	−764
2025	SPY	31	48.4	+8.62	+32.83	+1,018
2026	SPY	25	44.0	+3.74	+14.72	+368
2017	all names	91	35.2	−9.35	−5.36	−487
2018	all names	123	37.4	−13.33	−17.49	−2,151
2019	all names	127	33.1	−0.60	+2.45	+311
2020	all names	118	44.9	+0.47	+4.58	+541
2021	all names	126	34.1	−11.48	−17.52	−2,207
2022	all names	101	44.6	−2.16	−6.13	−619
2023	all names	94	37.2	−6.83	−16.57	−1,558

year	name	trades	win %	mean %	mean $	total $
2024	all names	93	34.4	−10.80	−25.59	−2,380
2025	all names	95	36.8	−11.25	−11.94	−1,134
2026	all names	73	41.1	−2.25	−8.65	−631



Amendments, deviations and integrity checks. —

What it showed. Negative.

#7j-atm_put S17 Index put-writing: monthly ATM cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly ATM put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs. Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	80.2	+1.9, +495.15	−4.1, −1,004.52	+1.6, +215.60	**+0.72**, +198.81	[+0.20, +1.16]; [+56.08, +321.56]
SPY	2012-06→2026-10	167	166 / 3,633	82.6	+1.5, +526.40	−3.9, −1,354.73	+1.2, +329.60	**+0.59**, +199.73	[+0.07, +1.01]; [+19.36, +346.51]
all names	2012-06→2026-10	334	167 / 3,633	81.4	+1.7, +511.00	−4.0, −1,168.33	+1.4, +266.60	**+0.65**, +199.27	[+0.16, +1.07]; [+41.16, +332.55]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+2,116.60, +3.0%, 2026-07-17, QQQ 2026-08-21 695.0	−5,457.40, −23.7%, 2020-02-21, QQQ 2020-03-20 230.0	2.00	7,626	581	24%
SPY	2012-06→2026-10	+1,692.60, +2.3%, 2026-05-15, SPY 2026-07-17 735.0	−9,871.40, −29.6%, 2020-02-21, SPY 2020-03-20 333.0	1.85	9,871	586	22%
all names	2012-06→2026-10	+2,116.60, +3.0%, 2026-07-17, QQQ 2026-08-21 695.0	−9,871.40, −29.6%, 2020-02-21, SPY 2020-03-20 333.0	1.92	15,329	581	23%

Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

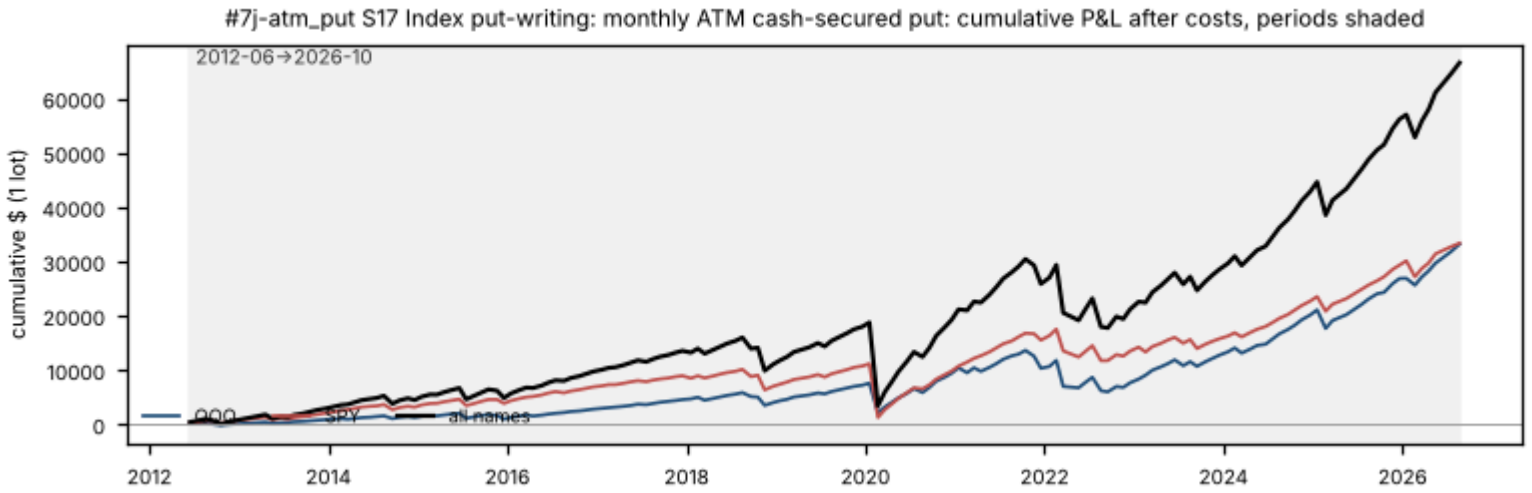
- 2012-06→2026-10: held to expiration 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+0.31	+15.10	+106
2013	QQQ	12	91.7	+0.87	+64.31	+772
2014	QQQ	12	75.0	+0.43	+35.72	+429

year	name	trades	win %	mean %	mean $	total $
2015	QQQ	12	66.7	−0.15	−25.02	−300
2016	QQQ	12	91.7	+1.40	+152.14	+1,826
2017	QQQ	12	91.7	+1.05	+146.27	+1,755
2018	QQQ	12	66.7	−0.21	−44.90	−539
2019	QQQ	11	90.9	+1.58	+290.87	+3,200
2020	QQQ	12	83.3	+0.71	+182.93	+2,195
2021	QQQ	12	66.7	+0.33	+75.27	+903
2022	QQQ	11	54.5	−0.52	−241.67	−2,658
2023	QQQ	12	83.3	+1.31	+422.85	+5,074
2024	QQQ	12	91.7	+1.31	+616.02	+7,392
2025	QQQ	11	90.9	+1.08	+609.96	+6,710
2026	QQQ	7	85.7	+1.33	+905.31	+6,337
2012	SPY	7	71.4	+0.74	+99.03	+693
2013	SPY	12	83.3	+0.71	+114.52	+1,374
2014	SPY	12	83.3	+0.53	+95.02	+1,140
2015	SPY	12	83.3	+0.29	+56.14	+674
2016	SPY	12	91.7	+1.24	+253.02	+3,036
2017	SPY	12	91.7	+0.70	+171.14	+2,054
2018	SPY	12	66.7	−0.53	−160.32	−1,924
2019	SPY	11	90.9	+1.17	+338.69	+3,726
2020	SPY	12	83.3	+0.01	−77.65	−932
2021	SPY	12	83.3	+1.18	+474.35	+5,692
2022	SPY	11	63.6	−0.32	−183.45	−2,018
2023	SPY	12	75.0	+0.48	+188.85	+2,266
2024	SPY	12	91.7	+1.05	+577.77	+6,933
2025	SPY	11	90.9	+0.95	+599.87	+6,599
2026	SPY	7	85.7	+0.80	+577.46	+4,042
2012	all names	14	71.4	+0.52	+57.06	+799
2013	all names	24	87.5	+0.79	+89.41	+2,146
2014	all names	24	79.2	+0.48	+65.37	+1,569
2015	all names	24	75.0	+0.07	+15.56	+373
2016	all names	24	91.7	+1.32	+202.58	+4,862
2017	all names	24	91.7	+0.88	+158.70	+3,809
2018	all names	24	66.7	−0.37	−102.61	−2,463
2019	all names	22	90.9	+1.37	+314.78	+6,925
2020	all names	24	83.3	+0.36	+52.64	+1,263
2021	all names	24	75.0	+0.75	+274.81	+6,595
2022	all names	22	59.1	−0.42	−212.56	−4,676
2023	all names	24	79.2	+0.89	+305.85	+7,340
2024	all names	24	91.7	+1.18	+596.89	+14,325

year	name	trades	win %	mean %	mean $	total $
2025	all names	22	90.9	+1.02	+604.92	+13,308
2026	all names	14	85.7	+1.06	+741.39	+10,379



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#7j-put_30d S17 Index put-writing: monthly 0.30-delta cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly 0.30-delta put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs. Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	85.6	+1.2, +317.04	−3.7, −835.30	+1.0, +155.60	**+0.51**, +151.44	[+0.11, +0.86]; [+43.50, +238.72]
SPY	2012-06→2026-10	167	166 / 3,633	85.6	+1.0, +337.16	−3.3, −1,091.40	+0.8, +222.60	**+0.40**, +131.86	[−0.08, +0.72]; [−22.88, +240.91]
all names	2012-06→2026-10	334	167 / 3,633	85.6	+1.1, +327.10	−3.5, −963.35	+0.9, +185.60	**+0.46**, +141.65	[+0.02, +0.79]; [+15.02, +238.25]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+1,426.60, +2.1%, 2026-05-15, QQQ 2026-07-17 675.0	−5,031.40, −22.5%, 2020-02-21, QQQ 2020-03-20 224.0	2.26	5,031	368	22%
SPY	2012-06→2026-10	+1,115.60, +1.6%, 2026-05-15, SPY 2026-07-17 715.0	−9,371.40, −28.7%, 2020-02-21, SPY 2020-03-20 326.0	1.84	9,371	408	20%
all names	2012-06→2026-10	+1,426.60, +2.1%, 2026-05-15, QQQ 2026-07-17 675.0	−9,371.40, −28.7%, 2020-02-21, SPY 2020-03-20 326.0	2.02	14,403	343	21%

Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

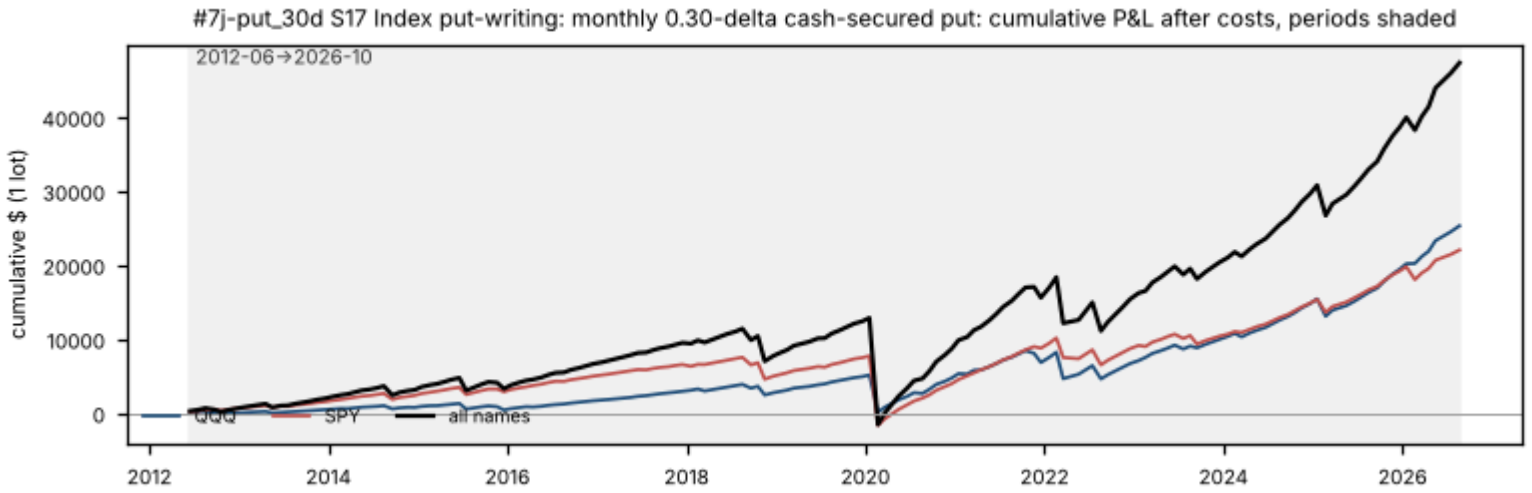
- 2012-06→2026-10: held to expiration 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+0.09	+1.53	+11
2013	QQQ	12	91.7	+0.50	+36.47	+438
2014	QQQ	12	75.0	+0.32	+25.47	+306

year	name	trades	win %	mean %	mean $	total $
2015	QQQ	12	75.0	−0.22	−29.03	−348
2016	QQQ	12	91.7	+0.98	+104.35	+1,252
2017	QQQ	12	100.0	+0.77	+106.52	+1,278
2018	QQQ	12	75.0	−0.07	−16.65	−200
2019	QQQ	11	100.0	+1.10	+198.60	+2,185
2020	QQQ	12	83.3	−0.10	−15.90	−191
2021	QQQ	12	75.0	+0.58	+174.77	+2,097
2022	QQQ	11	81.8	+0.17	−13.04	−143
2023	QQQ	12	83.3	+0.87	+274.35	+3,292
2024	QQQ	12	91.7	+0.87	+400.85	+4,810
2025	QQQ	11	90.9	+0.75	+419.60	+4,616
2026	QQQ	7	100.0	+1.32	+841.17	+5,888
2012	SPY	7	85.7	+0.65	+87.46	+612
2013	SPY	12	83.3	+0.48	+77.35	+928
2014	SPY	12	91.7	+0.39	+69.35	+832
2015	SPY	12	83.3	+0.22	+41.60	+499
2016	SPY	12	91.7	+0.87	+175.39	+2,105
2017	SPY	12	91.7	+0.55	+130.68	+1,568
2018	SPY	12	66.7	−0.45	−130.90	−1,571
2019	SPY	11	90.9	+0.80	+227.78	+2,506
2020	SPY	12	91.7	−0.71	−293.65	−3,524
2021	SPY	12	91.7	+0.99	+400.10	+4,801
2022	SPY	11	72.7	+0.09	−5.31	−58
2023	SPY	12	75.0	+0.36	+139.35	+1,672
2024	SPY	12	91.7	+0.69	+373.18	+4,478
2025	SPY	11	90.9	+0.63	+392.24	+4,315
2026	SPY	7	85.7	+0.58	+408.17	+2,857
2012	all names	14	78.6	+0.37	+44.49	+623
2013	all names	24	87.5	+0.49	+56.91	+1,366
2014	all names	24	83.3	+0.35	+47.41	+1,138
2015	all names	24	79.2	+0.00	+6.29	+151
2016	all names	24	91.7	+0.92	+139.87	+3,357
2017	all names	24	95.8	+0.66	+118.60	+2,846
2018	all names	24	70.8	−0.26	−73.77	−1,771
2019	all names	22	95.5	+0.95	+213.19	+4,690
2020	all names	24	87.5	−0.40	−154.77	−3,715
2021	all names	24	83.3	+0.78	+287.43	+6,898
2022	all names	22	77.3	+0.13	−9.17	−202
2023	all names	24	79.2	+0.61	+206.85	+4,964
2024	all names	24	91.7	+0.78	+387.02	+9,288

year	name	trades	win %	mean %	mean $	total $
2025	all names	22	90.9	+0.69	+405.92	+8,930
2026	all names	14	92.9	+0.95	+624.67	+8,745



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#7j-put_15d S17 Index put-writing: monthly 0.15-delta cash-secured put

Folder: `research/compass/studies/, results/v1/ (@7dfea57)`. **Commits:** A7 c81522a / wording 9ea4adf / 5e7e1bd. **Dates:** 2026-10-15.

Hypothesis. Index put-writing earns the variance risk premium (Cboe PUT).

Rule. Sell the standard monthly 0.15-delta put at the bid at expiry roll, hold to expiration, cash-secured.

Fills and costs. Bid in; settlement out; $0.70 a contract.

Data. SPY and QQQ standard monthlies 2012-06 → 2026-10.

Periods and holdouts. One span (2012–2026); no holdout.

Pass rule (as pre-registered). A7 carry-lane rule (positive mean with a positive lower bound per cell).

Verdict. PASSES A7 for QQQ ATM, 0.30Δ and 0.15Δ and SPY ATM puts; alpha 0 against the index, crash risk stated (a carry lane, not an edge).

Metrics (return = % of the cash set aside; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	2012-06→2026-10	167	167 / 3,633	93.4	+0.6, +150.58	−3.9, −781.85	+0.5, +85.60	**+0.30**, +89.16	[−0.01, +0.51]; [+18.70, +137.29]
SPY	2012-06→2026-10	167	166 / 3,633	94.6	+0.5, +159.43	−4.6, −1,391.07	+0.4, +113.60	**+0.22**, +75.87	[−0.19, +0.45]; [−48.89, +147.70]
all names	2012-06→2026-10	334	167 / 3,633	94.0	+0.5, +155.03	−4.2, −1,056.00	+0.5, +106.10	**+0.26**, +82.52	[−0.10, +0.47]; [−15.66, +141.14]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	2012-06→2026-10	+718.60, +1.1%, 2026-05-15, QQQ 2026-07-17 635.0	−4,191.40, −19.6%, 2020-02-21, QQQ 2020-03-20 214.0	2.73	4,191	408	18%
SPY	2012-06→2026-10	+527.60, +0.8%, 2026-05-15, SPY 2026-07-17 675.0	−8,446.40, −26.8%, 2020-02-21, SPY 2020-03-20 315.0	2.01	8,543	1,336	17%
all names	2012-06→2026-10	+718.60, +1.1%, 2026-05-15, QQQ 2026-07-17 635.0	−8,446.40, −26.8%, 2020-02-21, SPY 2020-03-20 315.0	2.30	12,638	776	17%

Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

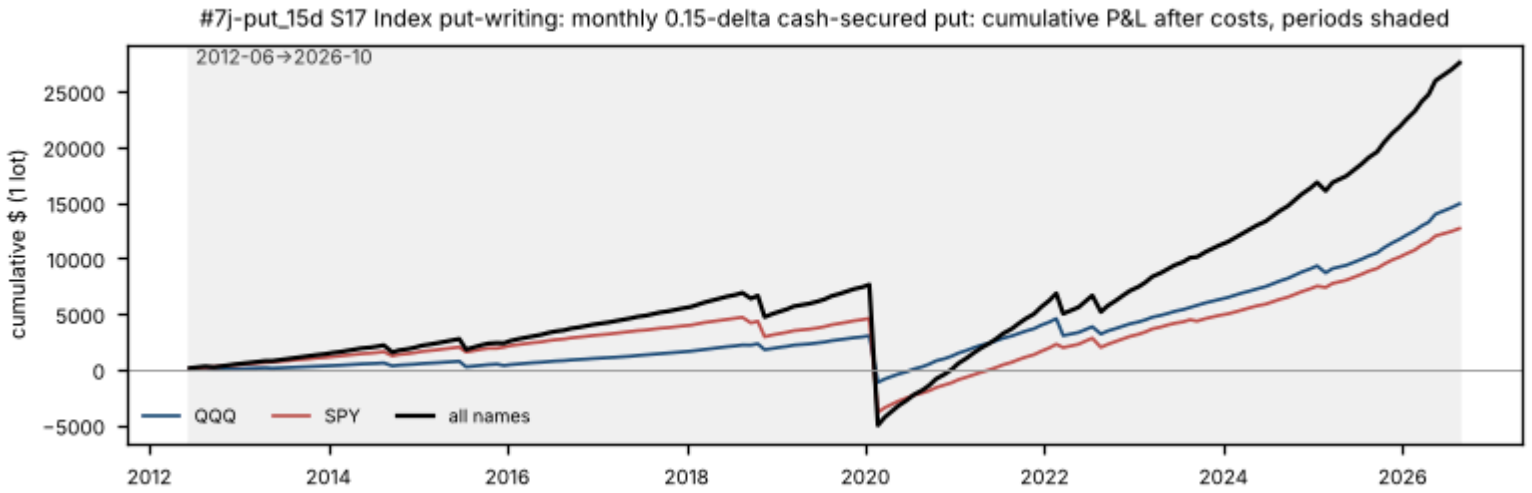
- 2012-06→2026-10: held to expiration 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	85.7	+0.16	+7.89	+55
2013	QQQ	12	91.7	+0.30	+21.31	+256
2014	QQQ	12	91.7	+0.17	+12.93	+155

year	name	trades	win %	mean %	mean $	total $
2015	QQQ	12	83.3	−0.06	−9.52	−114
2016	QQQ	12	100.0	+0.51	+52.43	+629
2017	QQQ	12	100.0	+0.37	+50.10	+601
2018	QQQ	12	83.3	+0.17	+26.43	+317
2019	QQQ	11	100.0	+0.54	+93.33	+1,027
2020	QQQ	12	91.7	−0.68	−144.48	−1,734
2021	QQQ	12	100.0	+0.72	+231.18	+2,774
2022	QQQ	11	81.8	+0.14	+12.42	+137
2023	QQQ	12	100.0	+0.57	+182.77	+2,193
2024	QQQ	12	100.0	+0.52	+228.43	+2,741
2025	QQQ	11	90.9	+0.46	+241.69	+2,659
2026	QQQ	7	100.0	+0.76	+456.31	+3,194
2012	SPY	7	100.0	+0.45	+59.17	+414
2013	SPY	12	100.0	+0.35	+55.18	+662
2014	SPY	12	91.7	+0.20	+35.27	+423
2015	SPY	12	91.7	+0.23	+41.93	+503
2016	SPY	12	100.0	+0.45	+87.18	+1,046
2017	SPY	12	100.0	+0.30	+71.35	+856
2018	SPY	12	83.3	−0.24	−65.57	−787
2019	SPY	11	100.0	+0.45	+123.33	+1,357
2020	SPY	12	91.7	−1.38	−472.40	−5,669
2021	SPY	12	100.0	+0.59	+234.43	+2,813
2022	SPY	11	81.8	+0.36	+123.05	+1,354
2023	SPY	12	91.7	+0.40	+157.02	+1,884
2024	SPY	12	100.0	+0.38	+199.10	+2,389
2025	SPY	11	90.9	+0.44	+258.96	+2,849
2026	SPY	7	100.0	+0.55	+367.89	+2,575
2012	all names	14	92.9	+0.31	+33.53	+469
2013	all names	24	95.8	+0.32	+38.25	+918
2014	all names	24	91.7	+0.18	+24.10	+578
2015	all names	24	87.5	+0.09	+16.20	+389
2016	all names	24	100.0	+0.48	+69.81	+1,675
2017	all names	24	100.0	+0.34	+60.73	+1,457
2018	all names	24	83.3	−0.04	−19.57	−470
2019	all names	22	100.0	+0.49	+108.33	+2,383
2020	all names	24	91.7	−1.03	−308.44	−7,403
2021	all names	24	100.0	+0.66	+232.81	+5,587
2022	all names	22	81.8	+0.25	+67.74	+1,490
2023	all names	24	95.8	+0.49	+169.89	+4,077
2024	all names	24	100.0	+0.45	+213.77	+5,130

year	name	trades	win %	mean %	mean $	total $
2025	all names	22	90.9	+0.45	+250.33	+5,507
2026	all names	14	100.0	+0.66	+412.10	+5,769



Amendments, deviations and integrity checks. Covered calls not significantly negative (9ea4adf).

What it showed. Put-writing earned its premium like the index it is short; a carry, not an edge.

#8a V1 T0: short every monthly ATM straddle across the optionable universe

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A8 29f20e3 (A8.1–A8.4) / — / aff15d9, df4fe32. **Dates:** 2026-10-15.

Hypothesis. Single-name monthly straddles are overpriced (volatility risk premium).

Rule. Sell the ATM straddle of every optionable name at the cycle's entry, hold to expiration (adverse fill: one tick worse).

Fills and costs. $0.70 a contract each side; mid ± one tick ("adv").

Data. The optionable universe 2023-11 → 2026-08, 44,447 name-cycles.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A8: negative T0 with sorts (T1 IV/RV, T2 term slope, T3 skew) adding a positive long-short.

Verdict. Rejected: T0 −13.2% of premium dev+val [−22.4, −5.0], −23.6% hold-out; T1–T3 add nothing.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
ADR	development	731	14 / 283	50.8	+46.8, +306.71	−77.2, −433.68	+0.6, +4.60	**−14.31**, −57.92	[−29.64, −4.41]; [−175.12, +24.87]
Common Stock	development	12,281	14 / 283	54.6	+47.3, +505.35	−73.7, −746.99	+8.1, +35.60	**−7.69**, −63.56	[−9.11, +0.46]; [−110.62, +9.05]
ETF	development	2,642	14 / 283	50.5	+44.5, +257.26	−61.3, −253.01	+0.8, +2.60	**−7.86**, +4.83	[−18.67, +8.12]; [−60.93, +83.28]
Structured Product	development	12	12 / 283	75.0	+50.5, +154.93	−43.3, −109.73	+34.5, +94.60	**+27.04**, +88.77	[−7.09, +49.70]; [−15.04, +128.61]
Unit	development	51	14 / 283	58.8	+40.6, +38.37	−95.8, −94.92	+13.2, +10.60	**−15.55**, −16.52	[−32.77, +21.79]; [−26.56, +23.01]
stock	development	14	14 / 283	57.1	+54.5, +104.10	−49.3, −78.57	+18.2, +25.60	**+10.02**, +25.81	[−18.20, +40.54]; [−27.96, +81.22]
all names	development	15,731	14 / 283	53.7	+46.8, +455.05	−71.7, −640.06	+6.7, +23.60	**−8.01**, −51.47	[−10.10, +0.41]; [−91.60, +11.71]
ADR	validation	761	12 / 239	50.2	+43.9, +357.50	−85.8, −581.61	+0.9, +3.60	**−20.69**, −110.21	[−32.48, −5.20]; [−190.84, +30.48]
Common Stock	validation	12,557	12 / 239	48.5	+45.6, +573.59	−82.5, −939.24	−3.0, −13.40	**−20.38**, −205.54	[−35.64, −5.51]; [−395.03, −27.17]
ETF	validation	2,832	12 / 239	45.4	+43.2, +288.52	−78.8, −497.04	−8.1, −21.40	**−23.43**, −140.32	[−52.31, −1.38]; [−395.08, +27.44]
Other	validation	2	2 / 21	0.0	—, —	−101.1, −336.90	−101.1, −336.90	**−101.09**, −336.90	n/a (< 60 sessions); n/a (< 60 sessions)
Structured Product	validation	12	12 / 239	58.3	+43.8, +240.46	−121.6, −748.00	+21.0, +106.10	**−25.09**, −171.40	[−139.90, +22.16]; [−757.83, +126.42]
Unit	validation	45	12 / 239	57.8	+50.2, +73.02	−76.6, −91.56	+25.0, +26.60	**−3.33**, +3.53	[−43.01, +37.34]; [−49.35, +53.67]
stock	validation	23	12 / 239	34.8	+44.5, +150.10	−103.9, −377.07	−34.4, −121.40	**−52.24**, −193.70	[−104.78, −7.46]; [−283.60, −13.73]
all names	validation	16,232	12 / 239	48.0	+45.1, +513.59	−82.0, −838.93	−3.8, −15.40	**−20.93**, −189.08	[−38.05, −5.29]; [−374.52, −19.12]
ADR	hold-out	588	8 / 155	46.3	+40.7, +556.16	−82.1, −793.25	−10.2, −34.40	**−25.31**, −169.03	[−46.43, −12.36]; [−504.30, +6.57]
Common Stock	hold-out	9,500	8 / 155	46.0	+43.8, +772.61	−78.3, −1,105.96	−8.2, −45.40	**−22.12**, −241.42	[−26.76, −14.02]; [−378.73, −38.93]
ETF	hold-out	2,296	8 / 155	37.1	+40.2, +375.90	−77.9, −662.00	−24.7, −78.90	**−34.11**, −277.31	[−53.44, −22.80]; [−496.93, −184.25]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Other	hold-out	11	7 / 155	45.5	+40.5, +347.40	−44.3, −419.90	−10.0, −170.40	**−5.76**, −71.13	[−53.95, +36.37]; [−460.90, +388.60]
Structured Product	hold-out	8	8 / 155	62.5	+44.8, +166.60	−25.2, −110.07	+16.2, +41.60	**+18.57**, +62.85	[−18.81, +44.93]; [−90.70, +131.46]
Unit	hold-out	34	8 / 155	44.1	+38.1, +59.73	−81.1, −105.77	−11.0, −19.40	**−28.55**, −32.75	[−55.00, +7.31]; [−69.69, +10.54]
stock	hold-out	47	8 / 155	42.6	+58.3, +659.05	−63.1, −623.03	−4.1, −41.40	**−11.47**, −77.46	[−51.92, +4.60]; [−611.61, +116.11]
all names	hold-out	12,484	8 / 155	44.4	+43.1, +697.78	−78.3, −993.71	−11.1, −50.40	**−24.41**, −243.08	[−32.05, −16.96]; [−369.04, −70.18]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
ADR	development	+4,424.60, +66.0%, 2024-07-22, ASML 2024-08-16 940.0 straddle	−11,415.40, −224.7%, 2024-01-22, ASML 2024-02-16 765.0 straddle	0.73	57,903	283	total ≤ 0 (best 5 days +30,869; without them −73,206)
Common Stock	development	+19,754.60, +36.9%, 2024-03-18, MSTR 2024-04-19 1500.0 straddle	−95,327.40, −739.7%, 2024-02-20, MSTR 2024-03-15 705.0 straddle	0.81	849,895	283	total ≤ 0 (best 5 days +263,308; without them −1,043,946)
ETF	development	+2,698.60, +41.9%, 2024-11-18, MSTX 2024-12-20 115.0 straddle	−5,421.40, −209.3%, 2024-05-20, NVDX 2024-06-21 105.0 straddle	1.04	118,222	282	958%
Structured Product	development	+489.60, +90.8%, 2024-12-23, VXX 2025-01-17 44.0 straddle	−182.40, −97.0%, 2024-04-22, VXX 2024-05-17 14.5 straddle	4.24	182	109	105%
Unit	development	+116.60, +72.9%, 2024-08-19, WES 2024-09-20 39.0 straddle	−444.40, −191.1%, 2023-11-20, NEP 2023-12-15 23.5 straddle	0.58	892	217	total ≤ 0 (best 5 days +665; without them −1,507)
stock	development	+190.60, +88.7%, 2024-10-21, SPLG 2024-11-15 69.0 straddle	−155.40, −84.0%, 2024-04-22, SPLG 2024-05-17 59.0 straddle	1.77	407	237	196%
all names	development	+19,754.60, +36.9%, 2024-03-18, MSTR 2024-04-19 1500.0 straddle	−95,327.40, −739.7%, 2024-02-20, MSTR 2024-03-15 705.0 straddle	0.83	935,320	283	total ≤ 0 (best 5 days +337,989; without them −1,147,618)
ADR	validation	+4,875.60, +87.1%, 2025-05-19, ASML 2025-06-20 750.0 straddle	−23,248.40, −315.7%, 2025-12-22, ASML 2026-01-16 1055.0 straddle	0.62	99,037	195	total ≤ 0 (best 5 days +25,684; without them −109,551)
Common Stock	validation	+21,041.60, +68.0%, 2025-05-19, BKNG 2025-06-20 5380.0 straddle	−45,598.40, −100.8%, 2025-04-21, BKNG 2025-05-16 4440.0 straddle	0.58	2,580,922	239	total ≤ 0 (best 5 days +57,393; without them −2,638,314)
ETF	validation	+4,381.60, +87.3%, 2025-08-18, ETHU 2025-09-19 155.0 straddle	−8,802.40, −210.3%, 2025-04-21, VGT 2025-05-16 490.0 straddle	0.48	418,065	239	total ≤ 0 (best 5 days +121,592; without them −518,978)
Other	validation	−199.40, −67.6%, 2025-06-23, RDW 2025-07-18 15.0 straddle	−474.40, −134.6%, 2025-07-21, RDW 2025-08-15 17.0 straddle	0.00	674	21	total ≤ 0 (best 5 days −674; without them +0)
Structured Product	validation	+532.60, +70.8%, 2025-05-19, VXX 2025-06-20 52.5 straddle	−2,174.40, −453.9%, 2025-03-24, VXX 2025-04-17 45.5 straddle	0.45	3,233	195	total ≤ 0 (best 5 days +1,471; without them −3,528)
Unit	validation	+201.60, +92.7%, 2025-07-21, MPLX 2025-08-15 50.0 straddle	−261.40, −126.0%, 2025-03-24, MPLX 2025-04-17 55.0 straddle	1.09	944	105	816%
stock	validation	+410.60, +56.6%, 2025-07-21, HIMZ 2025-08-15 19.0 straddle	−1,176.40, −162.3%, 2025-12-22, Q 2026-01-16 80.0 straddle	0.21	4,606	215	total ≤ 0 (best 5 days +705; without them −5,161)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	validation	+21,041.60, +68.0%, 2025-05-19, BKNG 2025-06-20 5380.0 straddle	−45,598.40, −100.8%, 2025-04-21, BKNG 2025-05-16 4440.0 straddle	0.57	3,069,201	239	total ≤ 0 (best 5 days +112,574; without them −3,181,775)
ADR	hold-out	+21,362.60, +87.6%, 2026-07-20, ASML 2026-08-21 1740.0 straddle	−29,899.40, −178.8%, 2026-05-18, ASML 2026-06-18 1470.0 straddle	0.60	125,168	155	total ≤ 0 (best 5 days +21,649; without them −121,041)
Common Stock	hold-out	+30,840.60, +58.9%, 2026-07-20, SNDK 2026-08-21 1390.0 straddle	−65,572.40, −189.2%, 2026-01-20, BKNG 2026-02-20 5050.0 straddle	0.60	2,538,940	155	total ≤ 0 (best 5 days −797,645; without them −1,495,846)
ETF	hold-out	+7,361.60, +95.7%, 2026-07-20, SMH 2026-08-21 560.0 straddle	−47,329.40, −266.6%, 2026-05-18, MUU 2026-06-18 445.0 straddle	0.33	646,925	131	total ≤ 0 (best 5 days −168,315; without them −468,381)
Other	hold-out	+743.60, +43.0%, 2026-05-18, FCFS 2026-06-18 220.0 straddle	−1,185.40, −67.5%, 2026-04-20, FCFS 2026-05-15 200.0 straddle	0.69	1,583	111	total ≤ 0 (best 5 days +1,079; without them −1,862)
Structured Product	hold-out	+351.60, +60.6%, 2026-01-20, VXX 2026-02-20 29.5 straddle	−216.40, −43.6%, 2026-02-23, VXX 2026-03-20 29.5 straddle	2.52	216	85	166%
Unit	hold-out	+193.60, +73.8%, 2026-04-20, MPLX 2026-05-15 55.0 straddle	−297.40, −241.8%, 2026-01-20, EPD 2026-02-20 32.5 straddle	0.45	1,458	155	total ≤ 0 (best 5 days +303; without them −1,416)
stock	hold-out	+2,136.60, +74.8%, 2026-07-20, Q 2026-08-21 135.0 straddle	−1,753.40, −174.5%, 2026-06-22, SOLS 2026-07-17 85.0 straddle	0.78	9,338	155	total ≤ 0 (best 5 days +6,697; without them −10,338)
all names	hold-out	+30,840.60, +58.9%, 2026-07-20, SNDK 2026-08-21 1390.0 straddle	−65,572.40, −189.2%, 2026-01-20, BKNG 2026-02-20 5050.0 straddle	0.56	3,291,177	155	total ≤ 0 (best 5 days −1,258,337; without them −1,776,277)

Random baseline: not available — no random baseline was computed by this study

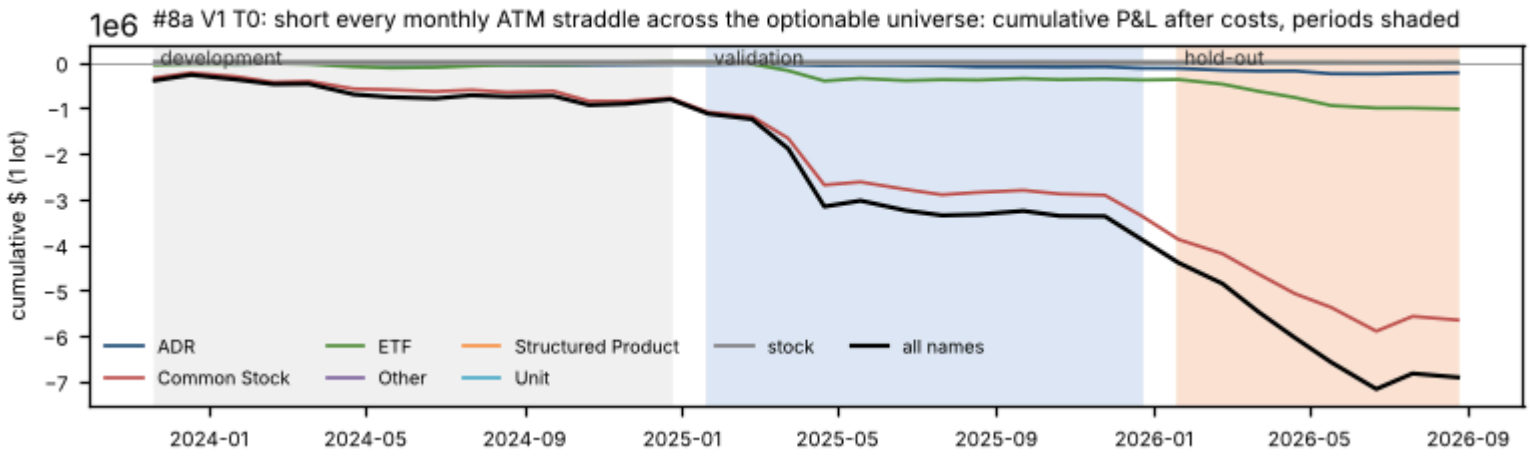
Exit reasons (share of trades):

- development: held to expiration 100.0%
- validation: held to expiration 100.0%
- hold-out: held to expiration 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	ADR	106	47.2	−14.16	−69.15	−7,330
2024	ADR	625	51.4	−14.34	−56.01	−35,007
2025	ADR	761	50.2	−20.69	−110.21	−83,866
2026	ADR	588	46.3	−25.31	−169.03	−99,392
2023	Common Stock	1,681	51.9	−23.11	−135.18	−227,229
2024	Common Stock	10,600	55.0	−5.24	−52.21	−553,408
2025	Common Stock	12,557	48.5	−20.38	−205.54	−2,580,922
2026	Common Stock	9,500	46.0	−22.12	−241.42	−2,293,491
2023	ETF	404	39.6	−33.22	−92.67	−37,440

year	name	trades	win %	mean %	mean $	total $
2024	ETF	2,238	52.5	−3.28	+22.43	+50,202
2025	ETF	2,832	45.4	−23.43	−140.32	−397,386
2026	ETF	2,296	37.1	−34.11	−277.31	−636,696
2025	Other	2	0.0	−101.09	−336.90	−674
2026	Other	11	45.5	−5.76	−71.13	−782
2023	Structured Product	2	50.0	+16.33	+34.10	+68
2024	Structured Product	10	80.0	+29.18	+99.70	+997
2025	Structured Product	12	58.3	−25.09	−171.40	−2,057
2026	Structured Product	8	62.5	+18.57	+62.85	+503
2023	Unit	7	71.4	+2.15	−44.97	−315
2024	Unit	44	56.8	−18.36	−11.99	−528
2025	Unit	45	57.8	−3.33	+3.53	+159
2026	Unit	34	44.1	−28.55	−32.75	−1,114
2023	stock	2	50.0	−24.62	−33.40	−67
2024	stock	12	58.3	+15.79	+35.68	+428
2025	stock	23	34.8	−52.24	−193.70	−4,455
2026	stock	47	42.6	−11.47	−77.46	−3,641
2023	all names	2,202	49.5	−24.42	−123.67	−272,313
2024	all names	13,529	54.4	−5.34	−39.72	−537,315
2025	all names	16,232	48.0	−20.93	−189.08	−3,069,201
2026	all names	12,484	44.4	−24.41	−243.08	−3,034,614



Amendments, deviations and integrity checks. Corporate actions detected from the options market (A8.1–A8.3c).

What it showed. Selling every straddle lost; the spread takes the premium.

#8b V2a: short ATM straddle across the earnings report

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A9 47c2c2f / — / 4ed84d4, 152f786. **Dates:** 2026-10-15.

Hypothesis. Earnings straddles are overpriced.

Rule. Sell the ATM straddle at the close before the report, buy back after it (adverse fill).

Fills and costs. $0.70 a contract each side; adverse tick.

Data. 1,578 names, reports 2023-11 → 2026-09.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A9.

Verdict. Rejected: −26.1% dev+val [−28.5, −23.3], −34.2% hold-out.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	development	3,558	234 / 276	39.2	+27.6, +306.54	−60.1, −615.66	−13.1, −54.80	**−25.67**, −253.83	[−28.33, −22.08]; [−290.45, −216.70]
single names	validation	4,244	218 / 246	32.5	+26.2, +359.56	−56.5, −550.99	−18.1, −88.80	**−29.62**, −255.34	[−33.80, −24.38]; [−317.68, −195.13]
single names	hold-out	3,682	165 / 191	25.6	+23.6, +344.83	−56.5, −630.33	−24.4, −149.30	**−35.95**, −380.58	[−38.94, −30.24]; [−413.48, −334.88]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	development	+5,620.60, +31.6%, 2024-05-02, BKNG 2024-05-03 3475.0	−23,115.40, −129.2%, 2024-02-22, BKNG 2024-02-23 3900.0	0.32	907,257	266	total ≤ 0 (best 5 days +22,304; without them −925,432)
single names	validation	+26,832.60, +84.4%, 2025-02-20, BKNG 2025-02-21 5020.0	−18,093.80, −406.1%, 2025-09-09, SNPS 2025-09-12 605.0	0.31	1,085,137	244	total ≤ 0 (best 5 days +43,589; without them −1,127,248)
single names	hold-out	+5,017.20, +30.3%, 2026-08-05, SNDK 2026-08-07 1350.0	−22,302.80, −330.7%, 2026-05-04, STRL 2026-05-15 530.0	0.19	1,408,044	191	total ≤ 0 (best 5 days +19,461; without them −1,420,756)

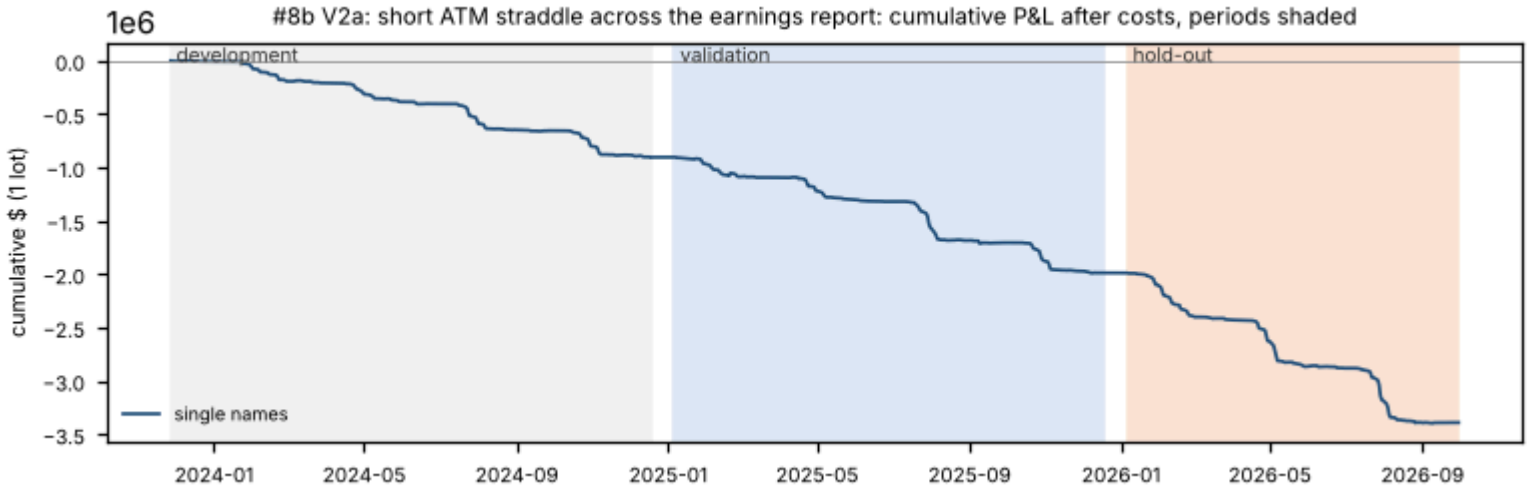
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: closed after the report 100.0%
- validation: closed after the report 100.0%
- hold-out: closed after the report 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	single names	71	47.9	−20.34	−101.65	−7,217
2024	single names	3,487	39.1	−25.77	−256.93	−895,910
2025	single names	4,244	32.5	−29.62	−255.34	−1,083,659
2026	single names	3,682	25.6	−35.95	−380.58	−1,401,295



Amendments, deviations and integrity checks. Friday-only expiry probe checked on all names (95c90f9: no change).

What it showed. Selling earnings straddles lost a quarter of the premium.

#8c V2b: long ATM straddle into the earnings report

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A9 47c2c2f / — / 152f786. **Dates:** 2026-10-15.

Hypothesis. Buying into the report captures the run-up in implied volatility.

Rule. Buy the ATM straddle five sessions before the report, sell at the report.

Fills and costs. As V2a.

Data. As V2a.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A9.

Verdict. Rejected: −20.9% / −29.3%.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	development	3,485	234 / 277	20.3	+18.2, +211.15	−22.8, −231.84	−13.8, −76.80	**−14.51**, −141.84	[−16.48, −11.70]; [−167.85, −104.26]
single names	validation	4,100	217 / 246	9.0	+20.2, +277.25	−34.6, −370.66	−27.0, −196.80	**−29.71**, −312.50	[−33.51, −25.43]; [−359.55, −266.38]
single names	hold-out	3,543	164 / 191	10.1	+23.8, +486.49	−37.2, −435.80	−29.1, −234.80	**−31.05**, −342.34	[−34.85, −25.85]; [−396.10, −270.55]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	development	+8,232.20, +93.4%, 2024-10-30, TPL 2024-11-15 1150.0	−11,697.80, −50.7%, 2024-04-22, MSTR 2024-05-03 1320.0	0.23	494,784	277	total ≤ 0 (best 5 days +22,371; without them −516,696)
single names	validation	+10,492.20, +99.0%, 2025-02-18, AXON 2025-03-21 710.0	−10,932.80, −32.1%, 2025-07-22, BKNG 2025-08-01 5775.0	0.07	1,281,270	246	total ≤ 0 (best 5 days +3,772; without them −1,285,042)
single names	hold-out	+19,712.20, +71.8%, 2026-05-21, AZO 2026-06-18 3450.0	−11,372.80, −36.5%, 2026-02-10, BKNG 2026-02-20 4300.0	0.13	1,212,924	191	total ≤ 0 (best 5 days +27,492; without them −1,240,416)

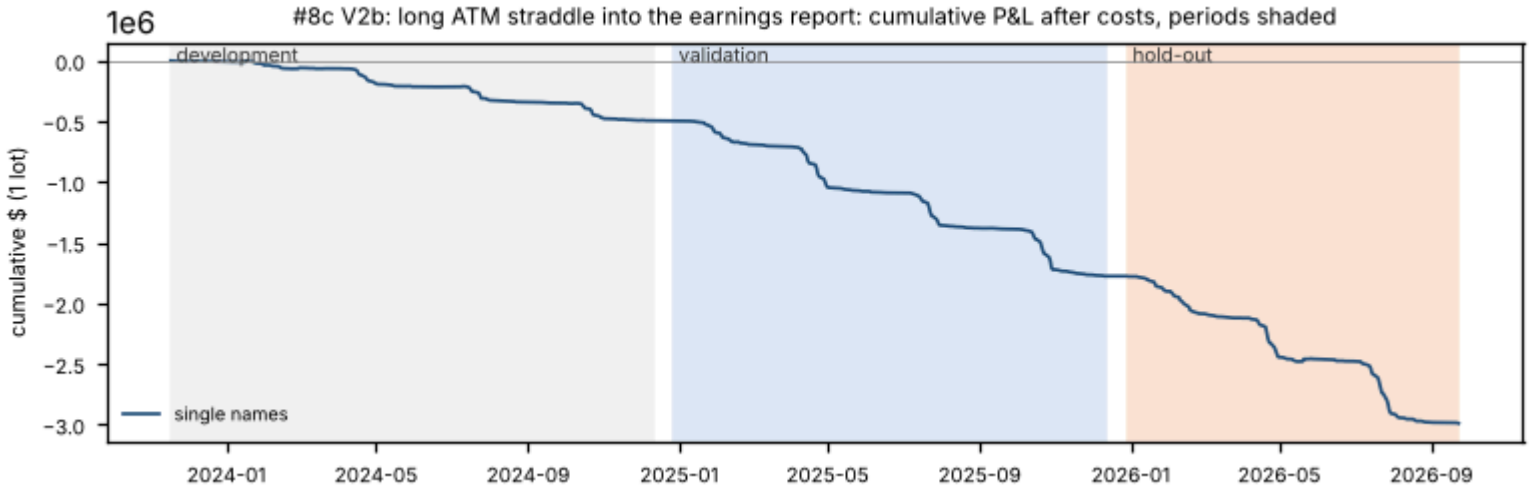
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: closed at the report 100.0%
- validation: closed at the report 100.0%
- hold-out: closed at the report 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	single names	76	15.8	−9.58	−98.33	−7,473
2024	single names	3,417	20.4	−14.66	−143.04	−488,758
2025	single names	4,101	9.0	−29.72	−312.65	−1,282,158
2026	single names	3,534	10.1	−31.03	−342.43	−1,210,131



Amendments, deviations and integrity checks. —

What it showed. Both sides of the earnings straddle lose.

#8d V3a: single-name 0DTE short ATM straddle 10:00 → 15:50

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A10–A10.4 9c476c5, 048d33a / — / e4e4cea. **Dates:** 2026-10-16.

Hypothesis. Single-name same-day premium is overpriced.

Rule. Sell the ATM straddle at 10:00, buy back 15:50 (adverse).

Fills and costs. $0.70 a contract each side; adverse tick.

Data. 11 Mon/Wed/Fri names 2017-06 → 2026-10.

Periods and holdouts. development / validation / hold-out (each hold-out opened once).

Pass rule (as pre-registered). A10.

Verdict. Rejected: −19.8% dev+val [−25.5, −14.7], −7.4% hold-out.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
AAPL	development	283	283 / 1,446	49.1	+44.7, +82.80	−81.5, −153.15	−1.9, −2.80	**−19.52**, −37.26	[−28.35, −5.92]; [−58.95, −13.76]
AMD	development	284	284 / 1,446	48.2	+35.7, +44.08	−79.9, −74.50	−3.9, −1.80	**−24.12**, −17.30	[−32.87, −14.20]; [−29.55, −6.15]
AMZN	development	283	283 / 1,446	51.6	+40.2, +799.45	−81.3, −1,487.27	+3.1, +21.20	**−18.63**, −307.55	[−26.79, −7.45]; [−523.37, −98.21]
AVGO	development	284	284 / 1,446	31.0	+33.2, +130.00	−96.9, −369.60	−38.4, −128.80	**−56.57**, −214.80	[−67.43, −43.02]; [−265.19, −161.55]
GOOGL	development	283	283 / 1,446	41.7	+36.5, +470.72	−89.5, −1,072.12	−14.9, −106.80	**−36.98**, −428.81	[−46.44, −24.19]; [−586.02, −274.06]
INTC	development	283	283 / 1,446	39.9	+31.3, +17.65	−91.2, −44.69	−17.4, −8.80	**−42.26**, −19.80	[−54.48, −29.44]; [−26.32, −13.52]
META	development	284	284 / 1,446	52.1	+42.2, +106.42	−71.2, −168.63	+3.8, +10.20	**−12.07**, −25.29	[−18.45, −2.33]; [−44.89, −3.05]
MSFT	development	284	284 / 1,446	43.7	+37.0, +66.68	−92.9, −140.57	−12.0, −13.30	**−36.16**, −50.08	[−46.26, −24.48]; [−69.69, −31.86]
MU	development	284	284 / 1,446	50.7	+35.5, +30.07	−84.4, −69.10	+0.5, +0.20	**−23.59**, −18.82	[−32.65, −12.50]; [−27.11, −10.11]
NVDA	development	283	283 / 1,446	55.8	+40.2, +165.80	−72.5, −331.71	+9.6, +34.20	**−9.58**, −53.95	[−17.63, −1.10]; [−96.45, −10.54]
TSLA	development	282	282 / 1,446	57.1	+46.6, +479.89	−85.1, −861.43	+17.0, +97.20	**−9.89**, −95.64	[−18.72, −1.20]; [−217.76, +20.18]
all names	development	3,117	284 / 1,446	47.4	+39.0, +227.00	−85.1, −423.22	−5.8, −7.80	**−26.32**, −115.32	[−30.82, −19.20]; [−151.58, −75.22]
AAPL	validation	102	102 / 512	60.8	+47.6, +74.26	−65.0, −106.85	+21.7, +36.70	**+3.45**, +3.24	[−2.56, +19.29]; [−6.38, +29.09]
AMD	validation	102	102 / 512	57.8	+46.2, +98.81	−79.8, −165.47	+19.3, +30.20	**−6.93**, −12.60	[−16.69, +10.03]; [−41.08, +20.78]
AMZN	validation	102	102 / 512	52.9	+43.6, +74.27	−70.4, −109.76	+4.7, +7.70	**−10.03**, −12.33	[−18.30, +6.15]; [−25.37, +13.60]
AVGO	validation	101	101 / 512	47.5	+36.7, +323.08	−88.1, −802.71	−1.4, −13.80	**−28.79**, −267.68	[−46.02, −10.45]; [−449.15, −98.76]
GOOGL	validation	102	102 / 512	52.9	+40.7, +53.50	−88.5, −110.76	+7.0, +8.70	**−20.12**, −23.80	[−33.58, −3.41]; [−37.76, −2.12]
INTC	validation	102	102 / 512	59.8	+31.7, +15.30	−90.9, −39.65	+10.4, +3.70	**−17.56**, −6.79	[−27.04, −2.17]; [−12.00, +0.00]
META	validation	102	102 / 512	56.9	+48.5, +222.82	−98.4, −389.30	+17.1, +60.20	**−14.87**, −41.23	[−33.14, +1.23]; [−122.21, +30.79]
MSFT	validation	102	102 / 512	50.0	+38.5, +116.28	−72.7, −200.02	+0.1, +0.20	**−17.09**, −41.87	[−32.02, −2.96]; [−79.24, −3.42]
MU	validation	102	102 / 512	54.9	+35.8, +43.56	−70.6, −84.06	+6.1, +7.70	**−12.21**, −14.00	[−19.70, +1.67]; [−24.70, +3.38]
NVDA	validation	101	101 / 512	54.5	+46.7, +237.09	−90.9, −650.23	+3.6, +11.20	**−16.01**, −167.04	[−30.76, +6.52]; [−367.08, −8.52]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
TSLA	validation	102	102 / 512	54.9	+51.0, +216.65	−72.0, −265.23	+13.4, +39.20	**−4.47**, −0.67	[−15.41, +12.65]; [−46.85, +65.81]
all names	validation	1,120	102 / 512	54.8	+42.6, +130.68	−80.7, −275.60	+7.9, +9.70	**−13.13**, −52.87	[−19.02, −2.17]; [−90.96, −11.87]
AAPL	hold-out	161	161 / 456	53.4	+44.2, +108.53	−80.5, −191.27	+5.3, +12.20	**−13.89**, −31.13	[−30.49, −1.41]; [−69.21, −0.05]
AMD	hold-out	116	116 / 456	60.3	+44.2, +253.19	−71.0, −325.63	+20.1, +76.20	**−1.50**, +23.66	[−15.86, +8.01]; [−42.34, +72.50]
AMZN	hold-out	161	161 / 456	60.9	+39.6, +99.20	−75.7, −184.61	+16.4, +34.20	**−5.53**, −11.86	[−18.28, +5.86]; [−40.24, +14.71]
AVGO	hold-out	159	159 / 456	53.5	+37.0, +183.58	−57.5, −321.06	+7.1, +32.20	**−7.00**, −51.28	[−15.63, +1.50]; [−106.34, −1.74]
GOOGL	hold-out	159	159 / 456	51.6	+45.3, +143.43	−73.7, −218.79	+4.3, +11.20	**−12.32**, −31.98	[−26.47, −2.39]; [−75.35, +0.07]
INTC	hold-out	116	116 / 456	51.7	+36.6, +57.92	−75.0, −94.35	+1.9, +2.20	**−17.28**, −15.59	[−34.85, −4.09]; [−38.02, +1.97]
META	hold-out	159	159 / 456	55.3	+37.0, +290.43	−71.8, −502.31	+3.5, +26.20	**−11.56**, −63.56	[−26.18, −3.73]; [−162.39, −11.26]
MSFT	hold-out	159	159 / 456	50.9	+44.7, +176.63	−78.0, −283.17	+6.7, +25.20	**−15.47**, −48.93	[−35.07, −2.94]; [−124.55, +1.12]
MU	hold-out	115	115 / 456	60.0	+41.2, +535.04	−89.2, −742.39	+11.2, +55.20	**−10.96**, +24.07	[−28.77, +5.92]; [−153.77, +232.75]
NVDA	hold-out	158	158 / 456	63.3	+44.9, +108.63	−82.0, −185.61	+21.4, +52.20	**−1.71**, +0.62	[−16.12, +10.93]; [−32.21, +30.32]
TSLA	hold-out	159	159 / 456	59.7	+47.1, +290.07	−87.8, −524.97	+21.5, +115.20	**−7.21**, −37.99	[−20.89, +5.22]; [−117.97, +40.83]
all names	hold-out	1,622	161 / 456	56.4	+42.1, +200.02	−76.1, −315.12	+10.1, +30.20	**−9.47**, −24.84	[−19.41, −2.80]; [−64.98, +5.67]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AAPL	development	+436.20, +59.5%, 2020-08-28, AAPL 10:00 short straddle 505.0	−1,149.80, −146.5%, 2020-03-20, AAPL 10:00 short straddle 250.0	0.52	10,672	1,446	total ≤ 0 (best 5 days +1,609; without them −12,153)
AMD	development	+221.20, +73.2%, 2022-05-06, AMD 10:00 short straddle 95.0	−840.80, −263.2%, 2022-02-11, AMD 10:00 short straddle 123.0	0.55	5,479	1,446	total ≤ 0 (best 5 days +1,017; without them −5,930)
AMZN	development	+3,217.20, +46.9%, 2020-09-04, AMZN 10:00 short straddle 3275.0	−7,895.80, −413.4%, 2021-10-15, AMZN 10:00 short straddle 3310.0	0.57	91,362	1,446	total ≤ 0 (best 5 days +13,705; without them −100,742)
AVGO	development	+410.20, +55.1%, 2022-06-10, AVGO 10:00 short straddle 545.0	−1,893.80, −244.4%, 2022-08-26, AVGO 10:00 short straddle 547.5	0.16	61,600	1,446	total ≤ 0 (best 5 days +1,842; without them −62,844)
GOOGL	development	+2,031.20, +67.9%, 2022-07-15, GOOGL 10:00 short straddle 2225.0	−6,483.80, −498.8%, 2021-09-10, GOOGL 10:00 short straddle 2890.0	0.31	122,845	1,446	total ≤ 0 (best 5 days +8,907; without them −130,261)
INTC	development	+92.20, +60.5%, 2020-07-24, INTC 10:00 short straddle 50.0	−384.80, −538.2%, 2018-03-29, INTC 10:00 short straddle 50.0	0.26	5,603	1,446	total ≤ 0 (best 5 days +325; without them −5,928)
META	development	+322.20, +64.0%, 2021-12-03, META 10:00 short straddle 305.0	−893.80, −222.1%, 2020-06-26, META 10:00 short straddle 227.5	0.69	7,534	1,446	total ≤ 0 (best 5 days +1,520; without them −8,703)
MSFT	development	+265.20, +49.8%, 2022-01-21, MSFT 10:00 short straddle 300.0	−846.80, −227.3%, 2022-08-26, MSFT 10:00 short straddle 280.0	0.37	14,274	1,446	total ≤ 0 (best 5 days +1,187; without them −15,410)
MU	development	+118.20, +64.2%, 2022-01-21, MU 10:00 short straddle 82.5	−343.80, −302.9%, 2020-06-12, MU 10:00 short straddle 50.0	0.45	5,424	1,446	total ≤ 0 (best 5 days +432; without them −5,776)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
NVDA	development	+660.20, +75.9%, 2020-05-22, NVDA 10:00 short straddle 360.0	−2,159.80, −244.0%, 2020-11-06, NVDA 10:00 short straddle 557.5	0.63	15,583	1,441	total ≤ 0 (best 5 days +2,817; without them −18,084)
TSLA	development	+2,872.20, +50.6%, 2020-08-21, TSLA 10:00 short straddle 2060.0	−12,864.80, −442.9%, 2020-07-10, TSLA 10:00 short straddle 1385.0	0.74	28,054	1,441	total ≤ 0 (best 5 days +10,845; without them −37,816)
all names	development	+3,217.20, +46.9%, 2020-09-04, AMZN 10:00 short straddle 3275.0	−12,864.80, −442.9%, 2020-07-10, TSLA 10:00 short straddle 1385.0	0.48	361,761	1,446	total ≤ 0 (best 5 days +27,174; without them −386,617)
AAPL	validation	+196.20, +54.8%, 2024-08-02, AAPL 10:00 short straddle 222.5	−453.80, −214.6%, 2024-12-20, AAPL 10:00 short straddle 247.5	1.08	601	120	233%
AMD	validation	+258.20, +79.9%, 2024-01-26, AMD 10:00 short straddle 177.5	−875.80, −329.9%, 2024-01-19, AMD 10:00 short straddle 162.5	0.82	1,812	453	total ≤ 0 (best 5 days +1,173; without them −2,459)
AMZN	validation	+199.20, +74.3%, 2023-10-27, AMZN 10:00 short straddle 128.0	−339.80, −244.5%, 2024-02-09, AMZN 10:00 short straddle 170.0	0.76	1,441	512	total ≤ 0 (best 5 days +802; without them −2,060)
AVGO	validation	+1,856.20, +67.0%, 2024-06-21, AVGO 10:00 short straddle 1680.0	−4,833.80, −406.2%, 2023-05-26, AVGO 10:00 short straddle 755.0	0.36	27,435	512	total ≤ 0 (best 5 days +5,475; without them −32,511)
GOOGL	validation	+127.20, +76.9%, 2024-12-27, GOOGL 10:00 short straddle 192.5	−507.80, −206.0%, 2024-12-20, GOOGL 10:00 short straddle 185.0	0.54	2,555	512	total ≤ 0 (best 5 days +591; without them −3,019)
INTC	validation	+52.20, +50.0%, 2024-08-02, INTC 10:00 short straddle 21.0	−177.80, −219.5%, 2024-03-08, INTC 10:00 short straddle 46.0	0.57	754	512	total ≤ 0 (best 5 days +193; without them −886)
META	validation	+964.20, +84.8%, 2024-02-02, META 10:00 short straddle 475.0	−1,495.80, −323.8%, 2024-07-05, META 10:00 short straddle 522.5	0.75	5,116	512	total ≤ 0 (best 5 days +3,123; without them −7,329)
MSFT	validation	+334.20, +66.6%, 2024-08-02, MSFT 10:00 short straddle 407.5	−773.80, −358.2%, 2023-04-06, MSFT 10:00 short straddle 282.5	0.58	4,964	512	total ≤ 0 (best 5 days +1,265; without them −5,536)
MU	validation	+133.20, +67.4%, 2024-04-05, MU 10:00 short straddle 123.0	−358.80, −138.3%, 2024-08-02, MU 10:00 short straddle 96.0	0.63	1,484	512	total ≤ 0 (best 5 days +550; without them −1,978)
NVDA	validation	+1,066.20, +77.0%, 2024-05-03, NVDA 10:00 short straddle 887.5	−6,704.80, −541.8%, 2024-04-19, NVDA 10:00 short straddle 840.0	0.44	17,933	378	total ≤ 0 (best 5 days +4,315; without them −21,186)
TSLA	validation	+571.20, +68.8%, 2024-11-15, TSLA 10:00 short straddle 317.5	−1,071.80, −291.3%, 2023-01-27, TSLA 10:00 short straddle 165.0	0.99	2,569	343	total ≤ 0 (best 5 days +2,332; without them −2,401)
all names	validation	+1,856.20, +67.0%, 2024-06-21, AVGO 10:00 short straddle 1680.0	−6,704.80, −541.8%, 2024-04-19, NVDA 10:00 short straddle 840.0	0.58	61,185	512	total ≤ 0 (best 5 days +9,262; without them −68,474)
AAPL	hold-out	+334.20, +60.7%, 2026-09-09, AAPL 10:00 short straddle 317.5	−735.80, −419.3%, 2025-10-10, AAPL 10:00 short straddle 255.0	0.65	6,733	443	total ≤ 0 (best 5 days +1,168; without them −6,180)
AMD	hold-out	+843.20, +58.6%, 2026-06-12, AMD 10:00 short straddle 510.0	−2,434.80, −199.6%, 2026-07-02, AMD 10:00 short straddle 545.0	1.18	3,533	252	131%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
AMZN	hold-out	+301.20, +72.4%, 2026-08-03, AMZN 10:00 short straddle 285.0	−915.80, −495.0%, 2025-10-10, AMZN 10:00 short straddle 227.5	0.84	3,973	293	total ≤ 0 (best 5 days +1,373; without them −3,282)
AVGO	hold-out	+436.20, +57.8%, 2026-05-15, AVGO 10:00 short straddle 425.0	−3,668.80, −137.2%, 2026-09-02, AVGO 10:00 short straddle 375.0	0.66	8,212	447	total ≤ 0 (best 5 days +2,044; without them −10,198)
GOOGL	hold-out	+395.20, +80.7%, 2026-02-06, GOOGL 10:00 short straddle 322.5	−1,530.80, −385.6%, 2026-08-05, GOOGL 10:00 short straddle 380.0	0.70	6,158	277	total ≤ 0 (best 5 days +1,609; without them −6,694)
INTC	hold-out	+236.20, +71.6%, 2026-04-24, INTC 10:00 short straddle 82.0	−897.80, −259.9%, 2026-07-02, INTC 10:00 short straddle 130.0	0.66	2,361	456	total ≤ 0 (best 5 days +967; without them −2,776)
META	hold-out	+1,093.20, +76.7%, 2025-04-11, META 10:00 short straddle 542.5	−2,419.80, −223.0%, 2026-09-21, META 10:00 short straddle 712.5	0.72	13,784	438	total ≤ 0 (best 5 days +3,812; without them −13,918)
MSFT	hold-out	+410.20, +63.4%, 2026-07-31, MSFT 10:00 short straddle 462.5	−954.80, −270.1%, 2026-06-17, MSFT 10:00 short straddle 390.0	0.65	11,145	438	total ≤ 0 (best 5 days +1,775; without them −9,555)
MU	hold-out	+2,452.20, +71.7%, 2026-06-12, MU 10:00 short straddle 985.0	−7,239.80, −237.0%, 2026-07-02, MU 10:00 short straddle 1060.0	1.08	9,849	361	319%
NVDA	hold-out	+269.20, +83.1%, 2026-06-08, NVDA 10:00 short straddle 207.5	−805.80, −380.1%, 2025-10-10, NVDA 10:00 short straddle 195.0	1.01	2,489	155	1291%
TSLA	hold-out	+796.20, +77.1%, 2025-11-21, TSLA 10:00 short straddle 395.0	−1,994.80, −427.6%, 2026-04-15, TSLA 10:00 short straddle 370.0	0.82	10,250	456	total ≤ 0 (best 5 days +3,457; without them −9,498)
all names	hold-out	+2,452.20, +71.7%, 2026-06-12, MU 10:00 short straddle 985.0	−7,239.80, −237.0%, 2026-07-02, MU 10:00 short straddle 1060.0	0.82	60,074	443	total ≤ 0 (best 5 days +18,099; without them −58,386)

Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: bought back 15:50 100.0%
- validation: bought back 15:50 100.0%
- hold-out: bought back 15:50 100.0%

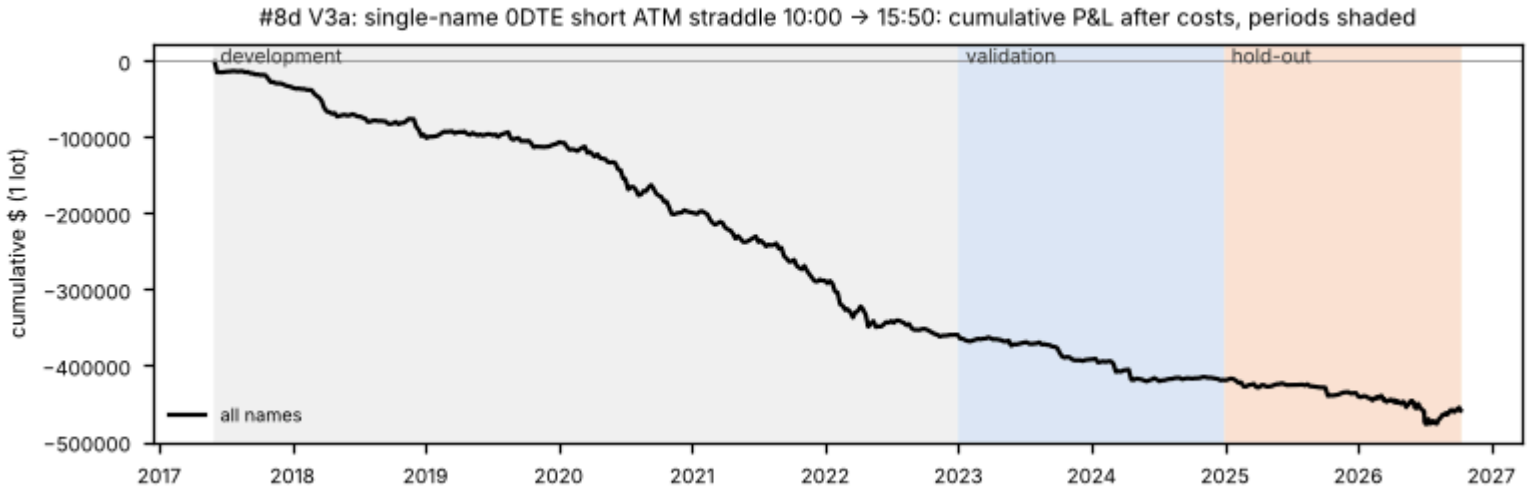
By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	AAPL	30	53.3	−41.28	−30.00	−900
2018	AAPL	51	45.1	−27.59	−40.58	−2,070
2019	AAPL	51	58.8	−7.30	−15.82	−807
2020	AAPL	49	42.9	−19.01	−87.27	−4,276
2021	AAPL	51	49.0	−15.45	−14.17	−723
2022	AAPL	51	47.1	−15.43	−34.68	−1,769
2023	AAPL	51	58.8	+4.32	+7.36	+375

year	name	trades	win %	mean %	mean $	total $
2024	AAPL	51	62.7	+2.58	−0.88	−45
2025	AAPL	50	50.0	−25.15	−55.36	−2,768
2026	AAPL	111	55.0	−8.82	−20.21	−2,244
2017	AMD	30	43.3	−41.95	−10.37	−311
2018	AMD	51	37.3	−48.34	−20.47	−1,044
2019	AMD	51	45.1	−20.77	−9.49	−484
2020	AMD	50	48.0	−23.67	−28.68	−1,434
2021	AMD	51	51.0	−17.02	−21.35	−1,089
2022	AMD	51	62.7	−0.30	−10.82	−552
2023	AMD	51	54.9	−11.14	−10.35	−528
2024	AMD	51	60.8	−2.73	−14.86	−758
2025	AMD	50	54.0	−14.60	−22.26	−1,113
2026	AMD	66	65.2	+8.42	+58.44	+3,857
2017	AMZN	30	46.7	−50.16	−325.10	−9,753
2018	AMZN	51	52.9	−21.56	−382.68	−19,517
2019	AMZN	51	64.7	−2.16	−32.78	−1,672
2020	AMZN	50	44.0	−19.62	−446.40	−22,320
2021	AMZN	51	51.0	−22.50	−441.33	−22,508
2022	AMZN	50	48.0	−8.57	−225.36	−11,268
2023	AMZN	51	52.9	−10.08	−9.27	−473
2024	AMZN	51	52.9	−9.98	−15.39	−785
2025	AMZN	50	70.0	−4.94	−3.30	−165
2026	AMZN	111	56.8	−5.80	−15.71	−1,744
2017	AVGO	30	33.3	−92.88	−206.43	−6,193
2018	AVGO	51	31.4	−33.57	−90.90	−4,636
2019	AVGO	51	49.0	−18.78	−40.39	−2,060
2020	AVGO	50	20.0	−69.24	−269.16	−13,458
2021	AVGO	51	15.7	−100.21	−403.25	−20,566
2022	AVGO	51	37.3	−39.94	−276.27	−14,090
2023	AVGO	51	43.1	−41.25	−351.58	−17,931
2024	AVGO	50	52.0	−16.07	−182.10	−9,105
2025	AVGO	50	56.0	−7.93	−38.34	−1,917
2026	AVGO	109	52.3	−6.58	−57.22	−6,237
2017	GOOGL	30	33.3	−61.82	−291.67	−8,750
2018	GOOGL	51	33.3	−47.24	−464.78	−23,704
2019	GOOGL	51	51.0	−15.37	−85.27	−4,349
2020	GOOGL	50	42.0	−34.86	−428.38	−21,419
2021	GOOGL	51	43.1	−45.69	−742.35	−37,860
2022	GOOGL	50	44.0	−26.92	−505.46	−25,273
2023	GOOGL	51	56.9	−19.95	−15.88	−810

year	name	trades	win %	mean %	mean $	total $
2024	GOOGL	51	49.0	−20.29	−31.72	−1,618
2025	GOOGL	50	58.0	−13.96	−13.96	−698
2026	GOOGL	109	48.6	−11.57	−40.25	−4,387
2017	INTC	29	20.7	−72.89	−18.56	−538
2018	INTC	51	29.4	−86.87	−42.88	−2,187
2019	INTC	51	52.9	−15.12	−6.33	−323
2020	INTC	50	42.0	−38.99	−22.40	−1,120
2021	INTC	51	45.1	−36.63	−21.84	−1,114
2022	INTC	51	41.2	−16.20	−6.31	−322
2023	INTC	51	52.9	−21.22	−8.58	−438
2024	INTC	51	66.7	−13.91	−5.00	−255
2025	INTC	50	42.0	−36.62	−18.96	−948
2026	INTC	66	59.1	−2.63	−13.04	−861
2017	META	30	43.3	−34.02	−32.30	−969
2018	META	51	41.2	−21.68	−41.13	−2,098
2019	META	51	66.7	+5.30	+10.49	+535
2020	META	50	48.0	−22.99	−63.46	−3,173
2021	META	51	56.9	−2.22	−7.39	−377
2022	META	51	52.9	−6.05	−21.60	−1,102
2023	META	51	49.0	−27.44	−84.80	−4,325
2024	META	51	64.7	−2.30	+2.34	+119
2025	META	50	38.0	−15.69	−81.48	−4,074
2026	META	109	63.3	−9.66	−55.34	−6,032
2017	MSFT	30	43.3	−87.18	−33.70	−1,011
2018	MSFT	51	37.3	−60.07	−49.23	−2,511
2019	MSFT	51	49.0	−13.47	−13.23	−675
2020	MSFT	50	40.0	−28.09	−51.70	−2,585
2021	MSFT	51	45.1	−20.39	−52.66	−2,686
2022	MSFT	51	47.1	−28.60	−93.25	−4,756
2023	MSFT	51	49.0	−25.22	−61.53	−3,138
2024	MSFT	51	51.0	−8.95	−22.21	−1,133
2025	MSFT	50	42.0	−18.65	−58.92	−2,946
2026	MSFT	109	55.0	−14.01	−44.35	−4,834
2017	MU	30	36.7	−61.78	−26.37	−791
2018	MU	51	47.1	−19.64	−14.43	−736
2019	MU	51	51.0	−10.75	−5.90	−301
2020	MU	50	54.0	−43.78	−37.18	−1,859
2021	MU	51	56.9	−16.62	−24.21	−1,235
2022	MU	51	52.9	−5.07	−8.29	−423
2023	MU	51	54.9	−11.48	−9.66	−493

year	name	trades	win %	mean %	mean $	total $
2024	MU	51	54.9	−12.93	−18.33	−935
2025	MU	50	46.0	−31.48	−51.64	−2,582
2026	MU	65	70.8	+4.83	+82.31	+5,350
2017	NVDA	30	50.0	−25.80	−73.50	−2,205
2018	NVDA	51	56.9	−8.07	−19.43	−991
2019	NVDA	51	68.6	+6.07	+15.16	+773
2020	NVDA	50	36.0	−36.95	−250.40	−12,520
2021	NVDA	50	68.0	+7.50	+27.90	+1,395
2022	NVDA	51	52.9	−7.10	−33.72	−1,720
2023	NVDA	51	54.9	−17.64	−82.23	−4,194
2024	NVDA	50	54.0	−14.34	−253.54	−12,677
2025	NVDA	50	66.0	−0.98	+6.44	+322
2026	NVDA	108	62.0	−2.05	−2.08	−224
2017	TSLA	30	63.3	−27.24	−89.00	−2,670
2018	TSLA	51	54.9	−8.86	−61.64	−3,144
2019	TSLA	51	58.8	−8.15	−47.72	−2,434
2020	TSLA	49	61.2	−1.24	−129.90	−6,365
2021	TSLA	51	54.9	−5.32	−47.00	−2,397
2022	TSLA	50	52.0	−15.42	−199.22	−9,961
2023	TSLA	51	54.9	−7.48	−6.74	−344
2024	TSLA	51	54.9	−1.47	+5.40	+275
2025	TSLA	50	66.0	−4.35	−5.18	−259
2026	TSLA	109	56.9	−8.52	−53.05	−5,782
2017	all names	329	42.6	−54.22	−103.62	−34,091
2018	all names	561	42.4	−34.86	−111.65	−62,636
2019	all names	561	56.0	−9.14	−21.02	−11,795
2020	all names	548	43.4	−30.84	−165.20	−90,529
2021	all names	560	48.8	−25.02	−159.21	−89,158
2022	all names	558	48.9	−15.41	−127.66	−71,234
2023	all names	561	52.9	−17.14	−57.57	−32,297
2024	all names	559	56.7	−9.10	−48.15	−26,915
2025	all names	550	53.5	−15.85	−31.18	−17,148
2026	all names	1,072	57.8	−6.19	−21.58	−23,139



Amendments, deviations and integrity checks. —

What it showed. Negative.

#8e V8: month-end 13:00 long straddle on unread chains

Folder: `research/compass/intel/, studies/v*.py, results/v1/ (@7dfea57)`. **Commits:** A15 8bc3baf / — / e4e4cea. **Dates:** 2026-10-16.

Hypothesis. Month-end afternoons move more than priced (from V7).

Rule. Buy the ATM straddle at 13:00 on the month's last session, hold to settlement.

Fills and costs. Ask in (adverse); settlement out.

Data. SPXW 2017-06 → 2022-04, SPY 2017-06 → 2022-10 (chains no study had read).

Periods and holdouts. One unread span.

Pass rule (as pre-registered). A15.

Verdict. Rejected: SPXW −6.6% [−24.5, +12.6], SPY −8.4% [−27.2, +11.3].

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	2017-06→2022 (unread chains)	56	56 / 1,251	41.1	+62.0, +811.30	−57.5, −807.70	−24.6, −288.40	**−8.42**, −142.76	[−26.64, +11.50]; [−474.45, +119.46]
SPY	2017-06→2022 (unread chains)	49	49 / 1,382	36.7	+64.6, +108.32	−53.2, −96.51	−30.4, −40.40	**−9.96**, −21.27	[−24.94, +12.12]; [−53.44, +19.42]
all names	2017-06→2022 (unread chains)	105	62 / 1,382	39.0	+63.1, +502.67	−55.4, −463.22	−29.1, −55.90	**−9.14**, −86.06	[−23.88, +10.99]; [−259.18, +99.44]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	2017-06→2022 (unread chains)	+3,221.60, +154.1%, 2022-03-31, SPXW 13:00 long straddle 4585.0	−3,552.40, −78.9%, 2020-02-28, SPXW 13:00 long straddle 2945.0	0.70	15,908	971	total ≤ 0 (best 5 days +11,965; without them −19,959)
SPY	2017-06→2022 (unread chains)	+302.60, +133.3%, 2022-03-31, SPY 13:00 long straddle 457.0	−378.40, −78.5%, 2020-02-28, SPY 13:00 long straddle 295.0	0.65	1,872	1,190	total ≤ 0 (best 5 days +1,330; without them −2,372)
all names	2017-06→2022 (unread chains)	+3,221.60, +154.1%, 2022-03-31, SPXW 13:00 long straddle 4585.0	−3,552.40, −78.9%, 2020-02-28, SPXW 13:00 long straddle 2945.0	0.70	17,762	1,102	total ≤ 0 (best 5 days +13,096; without them −22,132)

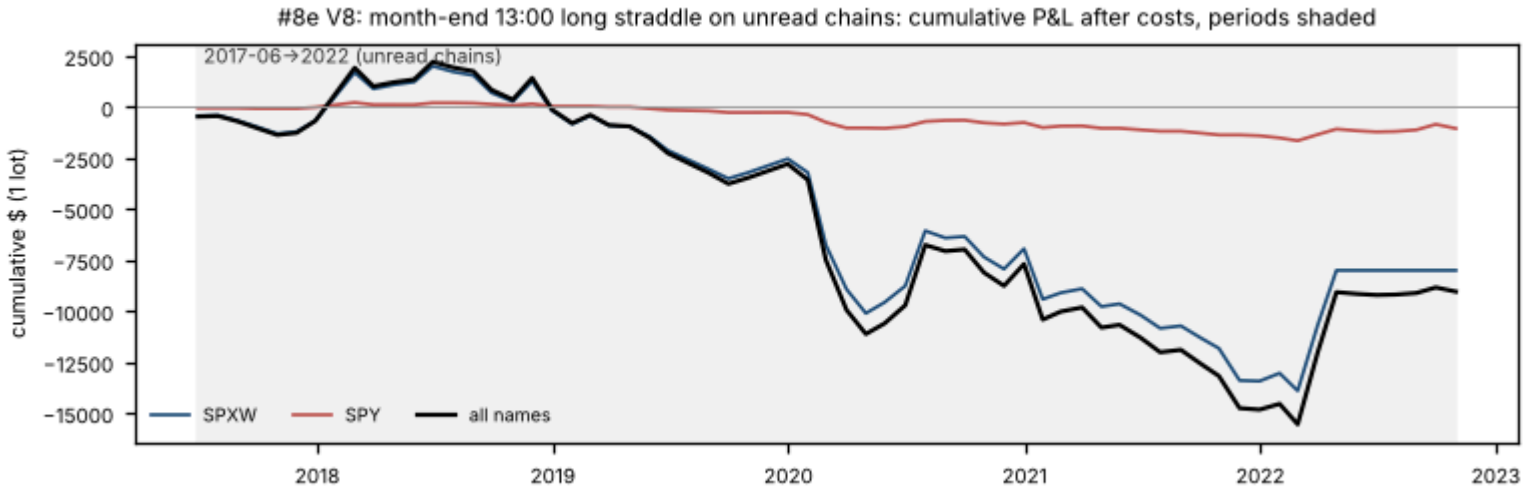
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- 2017-06→2022 (unread chains): held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	SPXW	7	42.9	−25.04	−94.97	−665
2018	SPXW	11	45.5	+10.59	+45.05	+496
2019	SPXW	10	30.0	−23.51	−235.90	−2,359
2020	SPXW	12	41.7	−2.99	−367.15	−4,406
2021	SPXW	12	33.3	−31.25	−540.48	−6,486
2022	SPXW	4	75.0	+58.35	+1,356.35	+5,425
2017	SPY	3	33.3	−5.21	−1.40	−4
2018	SPY	9	33.3	+14.41	+5.38	+48
2019	SPY	6	16.7	−53.15	−49.90	−299
2020	SPY	11	45.5	−0.24	−44.95	−494
2021	SPY	10	30.0	−42.53	−64.80	−648
2022	SPY	10	50.0	+14.48	+35.55	+356
2017	all names	10	40.0	−19.09	−66.90	−669
2018	all names	20	40.0	+12.31	+27.20	+544
2019	all names	16	25.0	−34.63	−166.15	−2,658
2020	all names	23	43.5	−1.67	−213.05	−4,900
2021	all names	22	31.8	−36.38	−324.26	−7,134
2022	all names	14	57.1	+27.01	+412.92	+5,781



Amendments, deviations and integrity checks. Read once on unread chains.

What it showed. The month-end effect V7 suggested did not appear on unread data.

#9a Lifecycle P1: Quick Scalps → cash-secured put sale (portfolio)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** protocol 1f99d75 / L-A4 8bd322a / 425c7f4, b9d04b5. **Dates:** 2026-10-15.

Hypothesis. The Tier I finalists, run as a portfolio, earn after costs.

Rule. QS signal → sell a cash-secured put (s20 policy), one unit, the lifecycle position engine.

Fills and costs. $0.70 a contract each side, NBBO.

Data. SPY/QQQ/IWM 0DTE 2022-11 → 2026-10.

Periods and holdouts. development + validation / hold-out 2025-10-01 → 2026-10-09 (opened once).

Pass rule (as pre-registered). R1–R6 promotion rules (Holm; R2 random-day control).

Verdict. Rejected: hold-out +$987 (about 1% a year, collateral-bound).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	development + validation	88	78 / 318	56.8	+0.1, +25.15	−0.2, −38.64	+0.0, +9.10	**−0.01**, −2.39	[−0.05, +0.02]; [−10.81, +5.62]
QQQ	development + validation	165	157 / 738	61.8	+0.1, +44.54	−0.2, −60.21	+0.1, +26.60	**+0.01**, +4.54	[−0.01, +0.03]; [−3.74, +12.04]
SPY	development + validation	155	148 / 742	62.6	+0.1, +45.51	−0.1, −60.64	+0.1, +26.60	**+0.01**, +5.79	[−0.00, +0.03]; [−2.86, +13.35]
all names	development + validation	408	269 / 742	61.0	+0.1, +41.02	−0.1, −55.21	+0.1, +21.10	**+0.01**, +3.52	[−0.01, +0.02]; [−1.36, +7.79]
IWM	hold-out 2025-10→2026-10	51	48 / 259	49.0	+0.1, +31.76	−0.2, −43.17	−0.1, −14.40	**−0.02**, −6.44	[−0.06, +0.02]; [−16.78, +4.00]
QQQ	hold-out 2025-10→2026-10	32	32 / 252	81.2	+0.1, +70.56	−0.2, −99.40	+0.1, +58.60	**+0.06**, +38.69	[+0.02, +0.10]; [+16.00, +68.00]
SPY	hold-out 2025-10→2026-10	39	35 / 240	66.7	+0.1, +50.48	−0.1, −95.02	+0.0, +33.60	**+0.00**, +1.98	[−0.04, +0.04]; [−25.34, +26.50]
all names	hold-out 2025-10→2026-10	122	89 / 262	63.1	+0.1, +51.18	−0.2, −65.64	+0.1, +29.10	**+0.01**, +8.09	[−0.02, +0.03]; [−6.89, +21.87]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	development + validation	+97.60, +0.4%, 2025-09-17, IWM CSP m85	−85.40, −0.4%, 2024-07-17, IWM CSP m16	0.86	526	318	total ≤ 0 (best 5 days +280; without them −491)
QQQ	development + validation	+170.60, +0.4%, 2025-04-11, QQQ CSP m181	−168.40, −0.4%, 2025-04-23, QQQ CSP m54	1.20	523	217	77%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	development + validation	+242.60, +0.5%, 2025-04-11, SPY CSP m75	−198.40, −0.4%, 2025-04-10, SPY CSP m309	1.26	459	218	76%
all names	development + validation	+242.60, +0.5%, 2025-04-11, SPY CSP m75	−198.40, −0.4%, 2025-04-10, SPY CSP m309	1.16	970	200	70%
IWM	hold-out 2025-10→2026-10	+90.60, +0.3%, 2026-06-12, IWM CSP m21	−115.40, −0.5%, 2025-10-29, IWM CSP m139	0.71	334	248	total ≤ 0 (best 5 days +267; without them −595)
QQQ	hold-out 2025-10→2026-10	+135.60, +0.2%, 2026-08-03, QQQ CSP m15	−167.40, −0.3%, 2025-10-29, QQQ CSP m44	3.08	216	129	48%
SPY	hold-out 2025-10→2026-10	+90.60, +0.1%, 2026-04-01, SPY CSP m38	−198.40, −0.3%, 2025-10-14, SPY CSP m283	1.06	514	147	532%
all names	hold-out 2025-10→2026-10	+135.60, +0.2%, 2026-08-03, QQQ CSP m15	−198.40, −0.3%, 2025-10-14, SPY CSP m283	1.33	834	171	71%

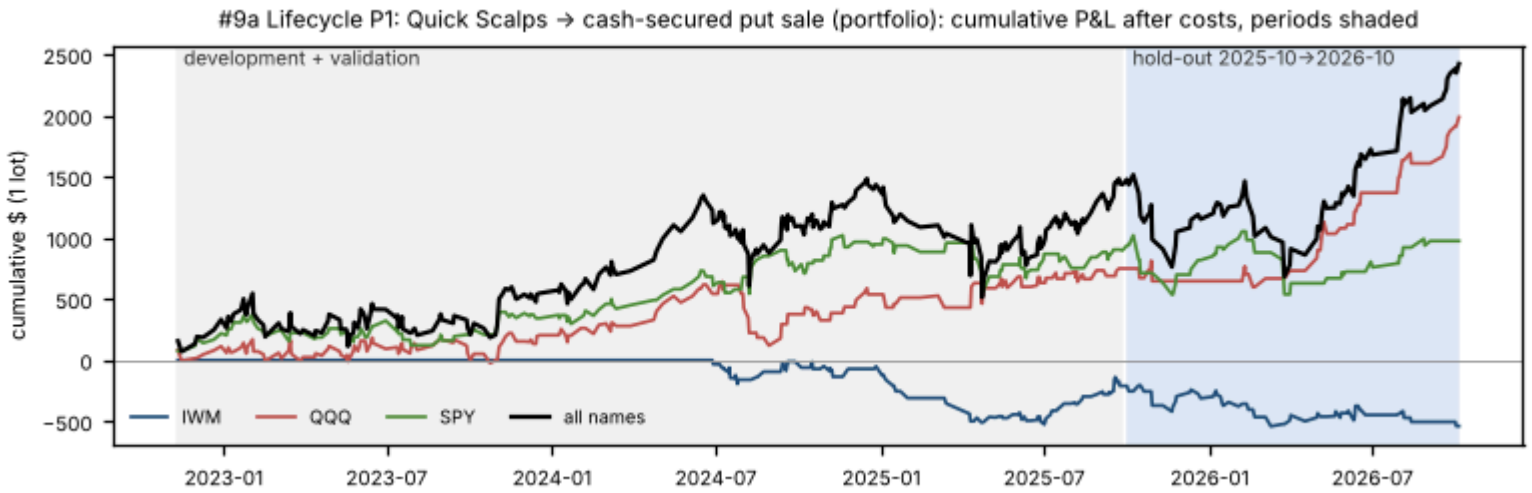
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development + validation: policy 100.0%
- hold-out 2025-10→2026-10: policy 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	36	58.3	−0.01	−1.40	−50
2025	IWM	66	54.5	−0.02	−3.97	−262
2026	IWM	37	48.6	−0.02	−6.13	−227
2022	QQQ	6	83.3	+0.07	+18.93	+114
2023	QQQ	76	57.9	+0.00	+1.19	+91
2024	QQQ	48	62.5	+0.01	+6.95	+334
2025	QQQ	40	65.0	+0.01	+2.72	+109
2026	QQQ	27	85.2	+0.07	+49.67	+1,341
2022	SPY	7	85.7	+0.08	+31.31	+219
2023	SPY	59	55.9	+0.01	+2.09	+123
2024	SPY	54	68.5	+0.02	+11.20	+605
2025	SPY	47	61.7	−0.01	−2.74	−129
2026	SPY	27	66.7	+0.01	+5.79	+156
2022	all names	13	84.6	+0.08	+25.60	+333
2023	all names	135	57.0	+0.00	+1.59	+214
2024	all names	138	63.8	+0.01	+6.44	+888
2025	all names	153	59.5	−0.01	−1.84	−282
2026	all names	91	64.8	+0.01	+13.96	+1,271



Amendments, deviations and integrity checks. A superseded hold-out file is kept separately.

What it showed. The put sale earns cents on the cash it ties up.

#9b Lifecycle P2: Quick Scalps → credit put spread to expiry (portfolio)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** 1f99d75 / L-A4 / d8a7ebb. **Dates:** 2026-10-15.

Hypothesis. As P1 with a defined-risk spread.

Rule. QS signal → credit put spread held to expiry.

Fills and costs. As P1.

Data. SPY/QQQ/IWM/SPXW 0DTE.

Periods and holdouts. As P1.

Pass rule (as pre-registered). As P1.

Verdict. Rejected: hold-out −$6,805.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	development + validation	80	80 / 335	93.8	+12.1, +105.99	−68.3, −621.20	+7.4, +68.00	**+7.11**, +60.54	[+4.68, +12.45]; [+38.69, +107.46]
QQQ	development + validation	173	173 / 738	89.6	+13.2, +117.62	−108.8, −965.87	+9.1, +83.00	**+0.52**, +4.89	[−6.78, +7.22]; [−61.15, +64.64]
SPXW	development + validation	182	182 / 844	85.2	+12.1, +106.90	−79.7, −702.76	+11.6, +103.60	**−1.56**, −13.21	[−7.24, +3.13]; [−63.22, +28.20]
SPY	development + validation	169	169 / 742	92.3	+12.8, +113.84	−98.9, −836.83	+8.5, +78.00	**+4.17**, +40.71	[−2.65, +10.65]; [−18.78, +97.33]
all names	development + validation	604	299 / 844	89.6	+12.6, +111.85	−91.1, −799.13	+9.7, +85.80	**+1.79**, +16.83	[−2.76, +5.49]; [−22.82, +49.32]
IWM	hold-out 2025-10→2026-10	53	53 / 259	79.2	+7.0, +60.59	−87.9, −802.14	+5.6, +48.00	**−12.68**, −118.47	[−24.08, −3.92]; [−222.66, −37.98]
QQQ	hold-out 2025-10→2026-10	48	48 / 252	95.8	+16.1, +140.89	−101.6, −896.50	+9.7, +83.00	**+11.17**, +97.67	[−2.07, +20.92]; [−22.54, +185.73]
SPXW	hold-out 2025-10→2026-10	47	47 / 261	89.4	+11.2, +99.79	−99.7, −882.06	+11.0, +98.60	**−0.64**, −4.66	[−11.14, +8.75]; [−97.84, +78.53]
SPY	hold-out 2025-10→2026-10	48	48 / 261	87.5	+15.4, +142.83	−199.7, −1,832.25	+9.1, +83.00	**−11.45**, −104.05	[−45.74, +7.48]; [−422.46, +71.95]
all names	hold-out 2025-10→2026-10	196	91 / 262	87.8	+12.5, +111.72	−119.5, −1,084.18	+9.1, +78.30	**−3.65**, −34.72	[−16.19, +4.15]; [−150.83, +35.98]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	development + validation	+1,564.50, +178.8%, 2024-07-25, IWM CPS m163	−894.00, −101.6%, 2024-07-24, IWM CPS m140	2.56	894	45	68%
QQQ	development + validation	+1,759.50, +187.2%, 2024-03-06, QQQ CPS m226	−2,160.50, −238.7%, 2023-02-03, QQQ CPS m105	1.05	6,229	695	654%
SPXW	development + validation	+158.60, +18.9%, 2024-07-31, SPXW CPS m76	−942.80, −100.3%, 2024-07-12, SPXW CPS m311	0.87	5,324	416	total ≤ 0 (best 5 days +758; without them −3,163)
SPY	development + validation	+2,454.50, +283.8%, 2025-06-05, SPY CPS m85	−1,510.50, −168.8%, 2023-02-03, SPY CPS m92	1.63	4,848	304	75%
all names	development + validation	+2,454.50, +283.8%, 2025-06-05, SPY CPS m85	−2,160.50, −238.7%, 2023-02-03, QQQ CPS m105	1.20	10,333	505	85%
IWM	hold-out 2025-10→2026-10	+108.00, +12.2%, 2026-03-31, IWM CPS m225	−2,360.50, −256.6%, 2026-01-16, IWM CPS m187	0.29	6,753	241	total ≤ 0 (best 5 days +515; without them −6,794)
QQQ	hold-out 2025-10→2026-10	+1,774.50, +196.1%, 2026-02-10, QQQ CPS m61	−909.00, −101.6%, 2026-08-28, QQQ CPS m123	3.61	909	56	70%
SPXW	hold-out 2025-10→2026-10	+138.60, +16.1%, 2026-08-27, SPXW CPS m21	−897.80, −100.3%, 2025-10-14, SPXW CPS m283	0.95	1,826	203	total ≤ 0 (best 5 days +638; without them −857)
SPY	hold-out 2025-10→2026-10	+1,944.50, +203.6%, 2025-10-14, SPY CPS m283	−5,195.50, −558.7%, 2026-01-16, SPY CPS m316	0.55	8,720	250	total ≤ 0 (best 5 days +3,048; without them −8,043)
all names	hold-out 2025-10→2026-10	+1,944.50, +203.6%, 2025-10-14, SPY CPS m283	−5,195.50, −558.7%, 2026-01-16, SPY CPS m316	0.74	12,260	251	total ≤ 0 (best 5 days +5,166; without them −11,970)

Random baseline: not available — no random baseline was computed by this study

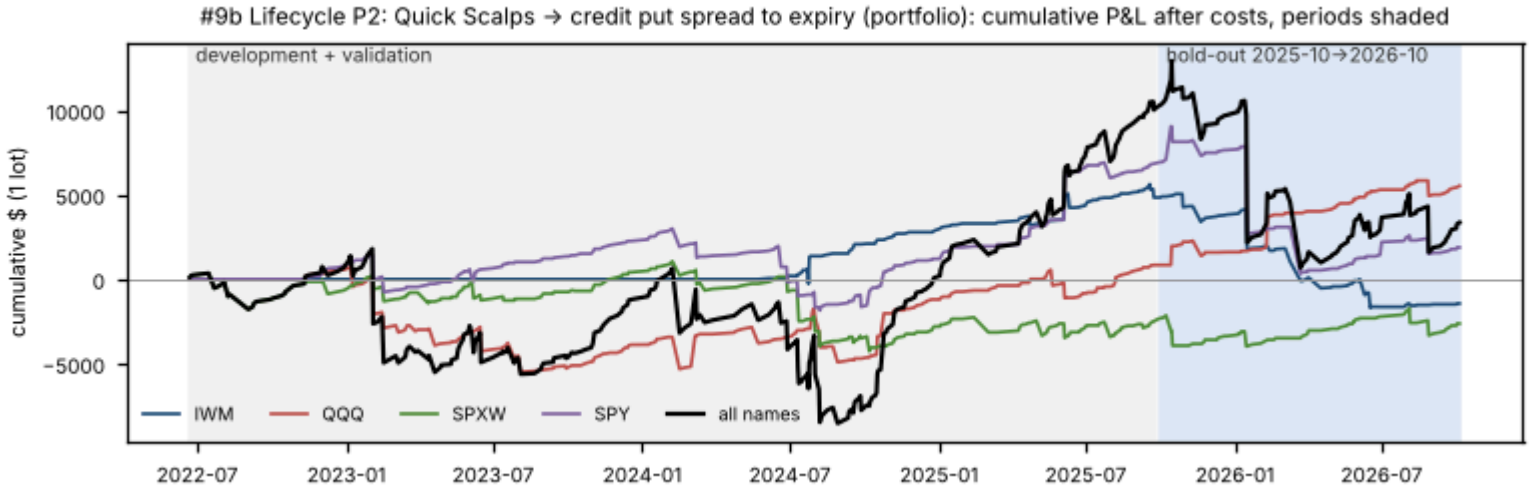
Exit reasons (share of trades):

- development + validation: policy 100.0%
- hold-out 2025-10→2026-10: policy 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	31	96.8	+11.37	+92.82	+2,877
2025	IWM	62	90.3	+1.82	+15.33	+950
2026	IWM	40	77.5	−14.20	−131.59	−5,263
2022	QQQ	6	100.0	+10.56	+94.70	+568
2023	QQQ	68	85.3	−7.00	−65.38	−4,446
2024	QQQ	53	88.7	+5.11	+50.58	+2,681
2025	QQQ	56	94.6	+5.65	+50.10	+2,806
2026	QQQ	38	97.4	+11.86	+103.29	+3,925
2022	SPXW	28	85.7	−2.76	−23.39	−655
2023	SPXW	51	86.3	+2.39	+20.97	+1,070
2024	SPXW	56	80.4	−6.50	−57.36	−3,212

year	name	trades	win %	mean %	mean $	total $
2025	SPXW	57	87.7	−1.27	−9.99	−569
2026	SPXW	37	91.9	+2.11	+20.07	+743
2022	SPY	7	100.0	+13.78	+112.79	+790
2023	SPY	54	92.6	+2.94	+27.27	+1,473
2024	SPY	61	86.9	−3.13	−17.25	−1,052
2025	SPY	58	94.8	+12.26	+111.06	+6,442
2026	SPY	37	89.2	−16.78	−155.81	−5,765
2022	all names	41	90.2	+2.01	+17.14	+703
2023	all names	173	87.9	−1.13	−11.00	−1,904
2024	all names	201	87.1	+0.34	+6.44	+1,294
2025	all names	233	91.8	+4.58	+41.32	+9,628
2026	all names	152	88.8	−4.34	−41.85	−6,361



Amendments, deviations and integrity checks. —

What it showed. Loses on the hold-out.

#9c Lifecycle Tier I, daily tier, L-A3, H-FLO (by configuration)

Folder: `research/compass/lifecycle/ (@7dfea57)`. **Commits:** 1f99d75 / L-A2, L-A3 889691a, L-A4 8bd322a, L-A4.1 f815d27 / 425c7f4, f6d5a98, d8a7ebb, 66bc5c9, 50de62a. **Dates:** 2026-10-15.

Hypothesis. Some of 12 setups × 8 structures × 116 management policies (Tier I) or daily setups (Tier D) survive a hold-out.

Rule. Grids; finalists chosen on development + validation.

Fills and costs. $0.70 a contract each side, NBBO.

Data. 0DTE SPY/QQQ/IWM/SPXW; index-ETF option EOD 2012-06 → 2026-10.

Periods and holdouts. hold-out 2025-10-01 → 2026-10-09, opened once per group.

Pass rule (as pre-registered). R1–R6.

Verdict. 0 promoted: Tier I no long-premium policy with a positive lower bound, finalists all put sales; L-A3 debit call spread +$64.37 [−18.16, +138.67] p 0.064 (fail); daily tier 19 finalists, all fail R2 (random day); H-FLO a regime descriptor, not an edge.

Metrics. not available — committed as configuration-level cells (grid_by_setup.csv, cells.csv); the trade rows are in the scratchpad (lifecycle/archive, 281 MB) and were not re-run for this report.

Amendments, deviations and integrity checks. Random-minute look-ahead corrected (L-A2) before the hold-out.

What it showed. No configuration survived the hold-out.

#10 ATM straddle samples and the independent short-premium prereg (branch compass-research-2026-10-14)

Folder: `research/compass_2026_10_14/ on origin/compass-research-2026-10-14`. **Commits:** short_premium PREREG 2f97a0c / — / BLOCKED. **Dates:** 2026-10-14 → 15.

Hypothesis. (Measurement) what short-dated ATM straddles cost on 10 names.

Rule. —

Fills and costs. —

Data. 8–19 sessions a name.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. No result: the short-premium study was blocked (ThetaData session invalidated off-box).

Metrics. not available — price samples only (no trade with an exit); the pre-registered study never ran.

Amendments, deviations and integrity checks. —

What it showed. Medians of ATM straddle prices; no strategy.

#11 Price-action catalogue and atlas (43 names)

Folder: `research/compass/intel/04, 06, 07 (@7dfea57)`. **Commits:** — / — / 048d33a, 0d706b6, 98fa5f0. **Dates:** 2026-10-15.

Hypothesis. (Descriptive) how price action behaves around levels.

Rule. —

Fills and costs. —

Data. 43 names 2017–2026 bars.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Descriptive; V6 tested its levels (rejected).

Metrics. not available — descriptive tables, no trades.

Amendments, deviations and integrity checks. —

What it showed. No strategy.

Part B. Studies after the fresh start

#12 Step 1: every 0DTE contract broken down

Folder: `step1/`. **Commits:** — / — / 2185eb9. **Dates:** 2026-10-16.

Hypothesis. (Descriptive) the anatomy of 1,176,935 same-day contracts over 3,687 sessions.

Rule. —

Fills and costs. —

Data. SPY, QQQ, IWM, SPX 0DTE.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Descriptive.

Metrics. not available — a contract inventory, no trades.

Amendments, deviations and integrity checks. Checked against the raw quotes (verify.py).

What it showed. No strategy.

#13 Step 2: buy near the money, sell at +100% or hold (within 0.5% of the price)

Folder: `step2/`. **Commits:** — / — / b15b56b. **Dates:** 2026-10-16.

Hypothesis. A same-day contract that can double pays for those that do not.

Rule. Buy every contract within 0.5% of the underlying at the ask at 9:31, 10:00, 10:30, 11:00, 12:00, 1:00, 2:00, 3:00, 3:30; sell the first minute the bid reaches 2× the price paid (at exactly 2×), else hold to the close (worth intrinsic).

Fills and costs. $0.65 a contract on the buy and on the sale; none when it expires worthless.

Data. SPY, QQQ, IWM, SPX 0DTE 2022-05-13 → 2026-10-09 (the near-the-money slice of 10,547,433 trades).

Periods and holdouts. pick 2022–24 / check 2025–26 (no holdout; holdout B months read).

Pass rule (as pre-registered). None pre-registered (a population study).

Verdict. Every group loses before and after fees; near the money 29% double, below the share needed to break even.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	pick 2022–24	6,960	177 / 182	37.5	+81.2, +54.07	−87.3, −49.62	−64.5, −14.30	**−24.05**, −10.71	[−28.15, −21.15]; [−12.89, −9.10]
QQQ	pick 2022–24	39,574	536 / 553	38.5	+81.3, +94.74	−86.6, −86.84	−60.6, −24.30	**−21.90**, −16.90	[−23.39, −20.69]; [−20.04, −14.50]
SPXW	pick 2022–24	109,721	662 / 683	41.3	+83.3, +914.78	−85.2, −810.27	−46.6, −120.65	**−15.61**, −97.77	[−16.94, −14.47]; [−116.73, −78.73]
SPY	pick 2022–24	46,425	536 / 553	39.0	+78.6, +87.63	−86.3, −78.41	−54.3, −16.65	**−21.94**, −13.58	[−23.48, −20.59]; [−15.77, −11.70]
all names	pick 2022–24	202,680	662 / 683	40.1	+81.8, +548.95	−85.8, −467.39	−51.8, −32.30	**−18.58**, −59.71	[−19.90, −17.44]; [−72.48, −48.05]
IWM	check 2025–26	20,177	444 / 457	40.0	+81.7, +61.46	−88.7, −55.58	−57.2, −12.65	**−20.51**, −8.74	[−22.95, −18.65]; [−10.37, −7.55]
QQQ	check 2025–26	48,342	444 / 457	39.2	+81.4, +140.39	−86.1, −125.05	−56.8, −27.65	**−20.52**, −21.07	[−22.70, −18.63]; [−23.99, −18.95]
SPXW	check 2025–26	106,236	444 / 457	41.1	+81.5, +1,168.53	−84.0, −982.07	−42.5, −105.65	**−15.96**, −97.67	[−17.79, −14.25]; [−123.19, −77.45]
SPY	check 2025–26	53,096	444 / 457	38.9	+78.5, +120.19	−85.5, −102.36	−51.6, −18.30	**−21.68**, −15.78	[−23.83, −19.76]; [−18.70, −13.65]
all names	check 2025–26	227,851	444 / 457	40.1	+80.8, +620.52	−85.2, −506.22	−48.9, −30.65	**−18.66**, −54.46	[−20.55, −16.99]; [−67.74, −44.13]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	pick 2022–24	+375.70, +99.7%, 2024-08-05, IWM 197.0C m1	−336.65, −100.2%, 2024-07-24, IWM 220.0C m30	0.65	74,836	180	total ≤ 0 (best 5 days +2,626; without them −77,159)
QQQ	pick 2022–24	+760.70, +99.8%, 2024-08-05, QQQ 423.0C m1	−700.65, −100.1%, 2024-08-05, QQQ 426.0P m1	0.68	669,065	551	total ≤ 0 (best 5 days +20,716; without them −689,459)
SPXW	pick 2022–24	+7,668.70, +100.0%, 2024-08-05, SPXW 5095.0C m1	−6,890.65, −100.0%, 2024-08-05, SPXW 5145.0P m1	0.79	10,727,818	683	total ≤ 0 (best 5 days +721,445; without them −11,449,263)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	pick 2022–24	+708.70, +99.8%, 2024-08-05, SPY 509.0C m1	−678.65, −100.1%, 2024-08-05, SPY 513.0P m1	0.72	630,388	553	total ≤ 0 (best 5 days +26,913; without them −657,302)
all names	pick 2022–24	+7,668.70, +100.0%, 2024-08-05, SPXW 5095.0C m1	−6,890.65, −100.0%, 2024-08-05, SPXW 5145.0P m1	0.79	12,101,482	683	total ≤ 0 (best 5 days +757,666; without them −12,859,148)
IWM	check 2025–26	+343.70, +99.6%, 2025-04-07, IWM 173.0C m1	−340.65, −100.2%, 2025-04-07, IWM 179.0P m60	0.74	176,478	457	total ≤ 0 (best 5 days +6,579; without them −183,016)
QQQ	check 2025–26	+786.70, +99.8%, 2025-04-09, QQQ 415.0C m1	−792.65, −100.1%, 2025-04-09, QQQ 418.0P m1	0.72	1,018,725	457	total ≤ 0 (best 5 days +53,297; without them −1,072,022)
SPXW	check 2025–26	+9,038.70, +100.0%, 2025-04-09, SPXW 4940.0C m1	−18,090.65, −100.0%, 2025-11-28, SPXW 6845.0P m1	0.83	10,392,716	457	total ≤ 0 (best 5 days +911,348; without them −11,286,896)
SPY	check 2025–26	+930.70, +99.9%, 2025-04-09, SPY 492.0C m1	−880.65, −100.1%, 2025-04-09, SPY 496.0P m1	0.75	837,710	457	total ≤ 0 (best 5 days +45,406; without them −883,041)
all names	check 2025–26	+9,038.70, +100.0%, 2025-04-09, SPXW 4940.0C m1	−18,090.65, −100.0%, 2025-11-28, SPXW 6845.0P m1	0.82	12,424,159	457	total ≤ 0 (best 5 days +1,009,839; without them −13,418,185)

Random baseline: not available — no random baseline was computed by this study

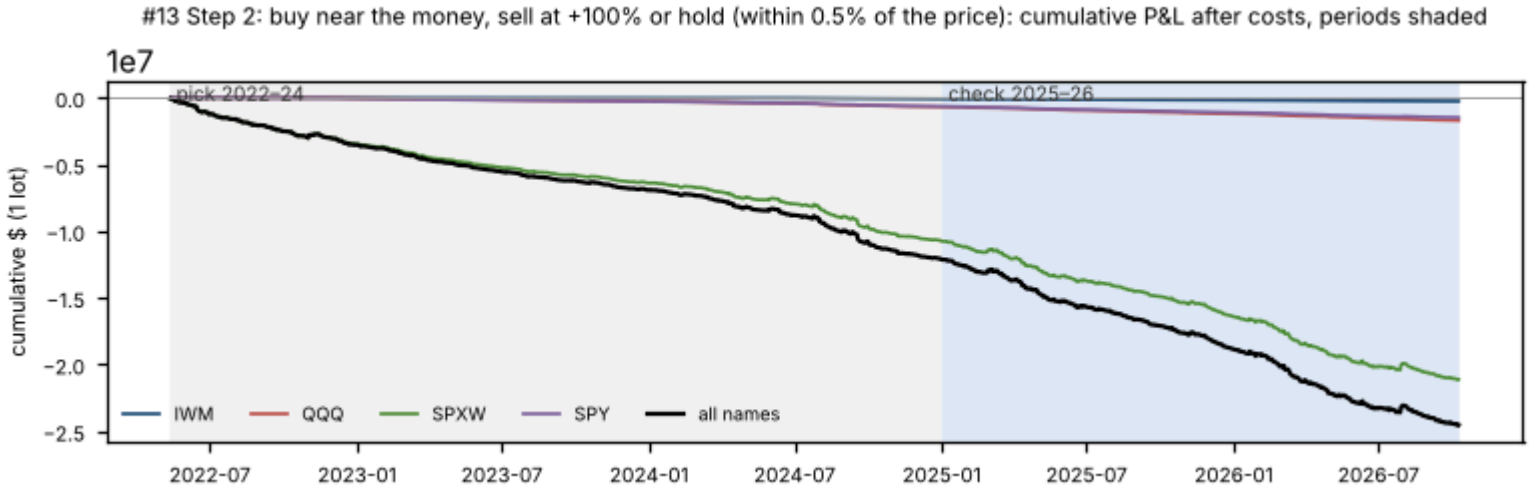
Exit reasons (share of trades):

- pick 2022–24: held to the close 70.8%; +100% target 29.2%
- check 2025–26: held to the close 71.1%; +100% target 28.9%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	6,960	37.5	−24.05	−10.71	−74,533
2025	IWM	10,251	40.0	−20.88	−8.88	−91,080
2026	IWM	9,926	40.0	−20.13	−8.60	−85,358
2022	QQQ	1,720	36.7	−23.50	−19.54	−33,612
2023	QQQ	15,750	39.3	−21.11	−14.02	−220,871
2024	QQQ	22,104	38.1	−22.33	−18.74	−414,261
2025	QQQ	24,779	38.3	−22.29	−21.20	−525,436
2026	QQQ	23,563	40.1	−18.66	−20.93	−493,289
2022	SPXW	22,444	40.9	−15.56	−153.19	−3,438,201
2023	SPXW	38,347	41.6	−15.36	−76.63	−2,938,671
2024	SPXW	48,930	41.2	−15.84	−88.92	−4,350,946
2025	SPXW	55,548	41.2	−15.93	−101.81	−5,655,180
2026	SPXW	50,688	41.1	−15.98	−93.13	−4,720,368

year	name	trades	win %	mean %	mean $	total $
2022	SPY	2,386	37.9	−21.77	−18.80	−44,867
2023	SPY	19,406	39.7	−21.31	−11.56	−224,382
2024	SPY	24,633	38.7	−22.45	−14.66	−361,140
2025	SPY	27,804	39.0	−21.77	−16.32	−453,673
2026	SPY	25,292	38.8	−21.58	−15.18	−383,961
2022	all names	26,550	40.3	−16.63	−132.45	−3,516,680
2023	all names	73,503	40.6	−18.16	−46.04	−3,383,923
2024	all names	102,627	39.7	−19.38	−50.68	−5,200,879
2025	all names	118,382	40.0	−19.06	−56.81	−6,725,369
2026	all names	109,469	40.2	−18.23	−51.91	−5,682,976



Amendments, deviations and integrity checks. Formulas: step2/summarize.py (tp100, fees), the committed loader step2.double.load.

What it showed. Near-the-money 0DTE buys lose 16–22% of premium a trade.

#14 Step 3: 108 exits × 9 conditions

Folder: `step3/`. **Commits:** — / — / 09885b6. **Dates:** 2026-10-16.

Hypothesis. A target, stop or time exit, or a condition known at the buy, makes step 2’s trades pay.

Rule. 108 exits × 9 entry conditions on step 2’s trades.

Fills and costs. $0.65 a contract each way.

Data. As step 2.

Periods and holdouts. discovery 2022–24 / check 2025–26.

Pass rule (as pre-registered). None pre-registered.

Verdict. Nothing made money in a way that held up; a −50% stop shrank losses; targets raised the win rate and lowered the average.

Metrics. not available — committed as per-rule summaries (exits.csv, exits_best.csv); the per-trade paths are committed but the exit grid was not re-run here.

Amendments, deviations and integrity checks. —

What it showed. No exit rule rescues the buys.

#15 Step 4: turn-up buy (1.5× the low) + 20% trailing stop, near the money

Folder: `step4/`. **Commits:** — / — / 6c0e60b. **Dates:** 2026-10-16.

Hypothesis. Buying the turn up off the low and selling near the high with a trailing stop finds an edge.

Rule. The first minute (9:36 → 15 minutes before the close) a contract's ask reaches 1.5× its low so far → buy at the ask; sell at the bid once it falls 20% from the highest bid since buying, else hold to the close. Strikes within 0.5% of the price.

Fills and costs. $0.65 a contract each way (none when worthless).

Data. SPY, QQQ, IWM, SPX 0DTE 2022–2026 (this slice of 12,149,834 trades).

Periods and holdouts. pick 2022–24 / check 2025–26.

Pass rule (as pre-registered). None pre-registered; 126 rules positive in both periods, the 11 biggest failed the stress test.

Verdict. A trailing stop beats holding but still loses.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	pick 2022–24	807	177 / 182	23.3	+37.0, +28.76	−22.0, −18.21	−20.1, −11.30	**−8.30**, −7.27	[−11.58, −5.37]; [−10.27, −4.59]
QQQ	pick 2022–24	4,603	536 / 553	27.4	+39.4, +52.74	−19.9, −26.58	−16.5, −15.30	**−3.66**, −4.89	[−4.88, −1.77]; [−6.69, −1.93]
SPXW	pick 2022–24	12,787	662 / 683	26.8	+36.8, +473.51	−19.3, −243.46	−16.6, −116.30	**−4.29**, −51.53	[−5.43, −2.91]; [−68.61, −30.35]
SPY	pick 2022–24	5,388	536 / 553	26.2	+36.7, +47.64	−20.6, −23.22	−17.2, −11.30	**−5.57**, −4.66	[−6.91, −3.68]; [−6.25, −2.04]
all names	pick 2022–24	23,585	662 / 683	26.6	+37.3, +280.19	−19.8, −142.87	−16.8, −26.30	**−4.59**, −30.20	[−5.59, −3.37]; [−41.20, −17.62]
IWM	check 2025–26	2,347	444 / 457	24.8	+41.3, +38.47	−21.4, −19.47	−18.4, −11.30	**−5.82**, −5.08	[−8.42, −2.59]; [−7.81, −2.07]
QQQ	check 2025–26	5,595	444 / 457	28.3	+41.5, +87.29	−19.7, −39.29	−15.7, −21.30	**−2.38**, −3.41	[−4.82, +0.08]; [−7.63, +1.56]
SPXW	check 2025–26	12,238	444 / 457	30.0	+41.0, +722.70	−18.9, −311.15	−14.6, −121.30	**−0.94**, −1.45	[−2.55, +1.19]; [−42.34, +52.40]
SPY	check 2025–26	6,078	444 / 457	29.3	+39.7, +72.29	−19.8, −32.35	−15.1, −12.30	**−2.41**, −1.72	[−4.16, −0.30]; [−5.36, +2.51]
all names	check 2025–26	26,258	444 / 457	29.0	+40.8, +385.98	−19.5, −160.81	−15.3, −24.30	**−2.02**, −2.26	[−3.63, −0.05]; [−22.69, +24.38]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	pick 2022–24	+195.70, +160.4%, 2024-07-11, IWM 207.0C m14	−241.30, −74.2%, 2024-09-03, IWM 217.0C m30	0.48	5,982	179	total ≤ 0 (best 5 days +1,030; without them −6,895)
QQQ	pick 2022–24	+603.70, +359.3%, 2023-08-29, QQQ 367.0C m13	−271.30, −52.8%, 2023-03-22, QQQ 313.0P m270	0.75	22,493	553	total ≤ 0 (best 5 days +7,132; without them −29,625)
SPXW	pick 2022–24	+7,302.70, +268.5%, 2024-05-31, SPXW 5180.0C m211	−4,491.30, −81.8%, 2023-09-21, SPXW 4350.0C m6	0.71	661,588	673	total ≤ 0 (best 5 days +145,957; without them −804,860)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	pick 2022–24	+795.70, +248.7%, 2024-08-01, SPY 555.0P m36	−331.30, −68.3%, 2024-09-18, SPY 565.0P m270	0.73	25,111	553	total ≤ 0 (best 5 days +6,716; without them −31,828)
all names	pick 2022–24	+7,302.70, +268.5%, 2024-05-31, SPXW 5180.0C m211	−4,491.30, −81.8%, 2023-09-21, SPXW 4350.0C m6	0.71	715,057	673	total ≤ 0 (best 5 days +154,181; without them −866,554)
IWM	check 2025–26	+639.70, +752.6%, 2025-02-21, IWM 225.0P m6	−246.30, −66.4%, 2025-08-01, IWM 212.5C m30	0.65	12,213	456	total ≤ 0 (best 5 days +3,614; without them −15,535)
QQQ	check 2025–26	+1,032.70, +983.5%, 2026-09-21, QQQ 730.0C m8	−275.30, −100.5%, 2025-07-30, QQQ 570.0P m104	0.88	30,063	323	total ≤ 0 (best 5 days +21,424; without them −40,512)
SPXW	check 2025–26	+21,608.70, +266.1%, 2025-04-07, SPXW 4875.0C m19	−39,941.30, −80.0%, 2025-04-09, SPXW 5340.0P m238	0.99	458,549	390	total ≤ 0 (best 5 days +421,847; without them −439,607)
SPY	check 2025–26	+1,875.70, +246.8%, 2025-04-07, SPY 490.0C m23	−2,363.30, −86.0%, 2025-04-09, SPY 521.0P m231	0.92	29,308	390	total ≤ 0 (best 5 days +20,771; without them −31,245)
all names	check 2025–26	+21,608.70, +266.1%, 2025-04-07, SPXW 4875.0C m19	−39,941.30, −80.0%, 2025-04-09, SPXW 5340.0P m238	0.98	525,009	390	total ≤ 0 (best 5 days +459,513; without them −518,756)

Random baseline: not available — no random baseline was computed by this study

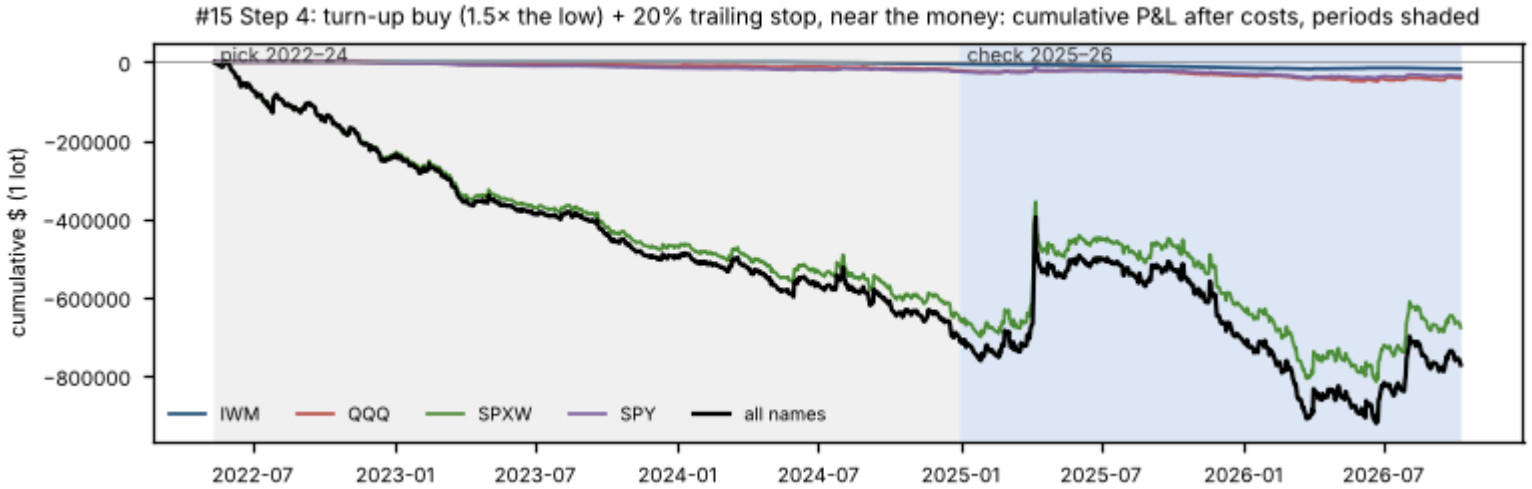
Exit reasons (share of trades):

- pick 2022–24: 20% trailing stop 99.5%; held to the close 0.5%
- check 2025–26: 20% trailing stop 99.4%; held to the close 0.6%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	807	23.3	−8.30	−7.27	−5,865
2025	IWM	1,182	23.0	−7.82	−8.19	−9,676
2026	IWM	1,165	26.7	−3.80	−1.93	−2,245
2022	QQQ	202	19.3	−7.67	−11.90	−2,405
2023	QQQ	1,842	27.7	−3.64	−4.45	−8,203
2024	QQQ	2,559	27.7	−3.36	−4.64	−11,886
2025	QQQ	2,874	27.1	−3.68	−4.29	−12,343
2026	QQQ	2,721	29.6	−1.00	−2.48	−6,745
2022	SPXW	2,622	25.2	−5.30	−90.87	−238,256
2023	SPXW	4,479	25.9	−4.97	−51.37	−230,081
2024	SPXW	5,686	28.2	−3.28	−33.52	−190,567
2025	SPXW	6,428	29.9	−0.93	+4.92	+31,604
2026	SPXW	5,810	30.0	−0.95	−8.50	−49,364

year	name	trades	win %	mean %	mean $	total $
2022	SPY	300	22.7	−7.11	−10.67	−3,200
2023	SPY	2,240	25.1	−6.60	−5.39	−12,063
2024	SPY	2,848	27.4	−4.59	−3.46	−9,848
2025	SPY	3,194	29.5	−2.72	−1.71	−5,458
2026	SPY	2,884	29.0	−2.07	−1.74	−5,016
2022	all names	3,124	24.6	−5.63	−78.06	−243,860
2023	all names	8,561	26.0	−5.11	−29.24	−250,346
2024	all names	11,900	27.6	−3.95	−18.33	−218,166
2025	all names	13,678	28.6	−2.52	+0.30	+4,127
2026	all names	12,580	29.4	−1.48	−5.04	−63,371



Amendments, deviations and integrity checks. Formulas copied from step4/summarize.py prep().

What it showed. The near-the-money turn-up buy with a 20% trail loses 1–8% of premium a trade.

#16 Step 5: the runner model (model A/B) and its top 1% picks

Folder: `step5/`. **Commits:** — / — / ae3c63d, 1227394, f9f2746, 491a0c7. **Dates:** 2026-10-16 → 17.

Hypothesis. What was visible at the turn tells runners from duds well enough to trade.

Rule. Gradient-boosted model A (P(bid ≥ 3× paid)) and B; trade the top 1% / 0.5% / 0.1%, three exits.

Fills and costs. $0.65 a contract each way.

Data. 1,602,401 turn-up and range-break buys; trained 2022–24, scored once on 2025–26.

Periods and holdouts. train 2022–24 / scored 2025–26.

Pass rule (as pre-registered). None pre-registered.

Verdict. Runners partly separable (top 1% reach 3× 26% vs 6%) but do not pay; every pick total made money almost all on 2025-11-20, and without its best 5 days it lost $620,114.

Metrics. not available — the picks are committed as daily sums (top1_daily.csv), not trades.

Amendments, deviations and integrity checks. Scored once.

What it showed. The model ranks runners but its picks do not pay without a handful of days.

#17 Step 6: the six pre-registered runner rules × three exits

Folder: `step6/`. **Commits:** 30babaa / — / 44b5547. **Dates:** 2026-10-23.

Hypothesis. A condition visible at the buy raises the odds of a +40% (family A) or +500% (family B) run enough to trade.

Rule. The top 3 rules of each family by Pick lift (one or two Part 2 buckets, ≥ 300 Pick trades on ≥ 60 dates), each buying every near-the-money population trade that meets it; exits (a) +40% / −30% / 60 minutes or 3:30, (b) 50% trail armed at +40%, out 3:30, (c) half at +40%, trail the rest 50%, out 3:30.

Fills and costs. $0.65 a contract each way (SPY/QQQ/IWM), SPXW per schedule; no sale fee at a $0 bid.

Data. 240,493 near-the-money 0DTE buys 2022-05 → 2026-10-09 (the 28 held-out months never read).

Periods and holdouts. pick 2022-05→2024 / check 2025→2026-10-09; the 28 holdout-B months UNOPENED.

Pass rule (as pre-registered). Check: average r > 0, its 20-session block lower end > 0, and above the 97.5th percentile of 500 cell-matched random draws.

Verdict. FAIL: 0 of 18 pass; all lose on Check (−3.3% to −21.9% a trade); 14 of 18 below the 20th random percentile.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A1a	pick 2022-05→2024	665	80 / 681	49.8	+35.7, +234.96	−33.3, −297.25	−0.5, −1.30	**+1.07**, −32.35	[−6.21, +10.10]; [−85.26, +64.39]
A1b	pick 2022-05→2024	665	80 / 681	31.9	+79.6, +611.09	−50.5, −418.65	−23.4, −41.30	**−9.02**, −90.37	[−25.54, +15.35]; [−198.72, +179.43]
A1c	pick 2022-05→2024	665	80 / 681	66.2	+65.4, +507.99	−153.8, −1,321.76	+14.1, +34.20	**−8.77**, −111.10	[−34.77, +29.18]; [−283.79, +319.36]
A2a	pick 2022-05→2024	1,081	186 / 678	43.1	+37.8, +323.47	−34.4, −270.96	−30.9, −20.30	**−3.29**, −14.71	[−10.74, +1.18]; [−93.67, +31.73]
A2b	pick 2022-05→2024	1,081	186 / 678	32.7	+141.1, +953.79	−50.4, −417.45	−22.7, −46.30	**+12.30**, +31.60	[−8.79, +40.13]; [−124.49, +238.45]
A2c	pick 2022-05→2024	1,081	186 / 678	64.0	+99.9, +712.93	−146.4, −1,191.32	+13.5, +38.89	**+11.28**, +27.68	[−15.39, +42.35]; [−188.37, +293.41]
A3a	pick 2022-05→2024	1,239	127 / 683	47.4	+37.6, +323.20	−33.7, −288.28	−14.7, −15.30	**+0.07**, +1.42	[−7.06, +6.68]; [−81.46, +73.56]
A3b	pick 2022-05→2024	1,239	127 / 683	28.6	+95.9, +898.21	−45.5, −356.22	−20.9, −51.30	**−5.11**, +2.19	[−26.65, +2.15]; [−270.58, +75.75]
A3c	pick 2022-05→2024	1,239	127 / 683	66.0	+68.8, +651.47	−145.3, −1,125.75	+15.1, +36.00	**−3.97**, +47.59	[−36.20, +11.73]; [−385.28, +200.43]
B1a	pick 2022-05→2024	2,426	121 / 443	42.3	+31.5, +630.72	−29.8, −598.56	−18.6, −382.55	**−3.88**, −78.17	[−5.32, −2.90]; [−104.92, −55.79]
B1b	pick 2022-05→2024	2,426	121 / 443	34.5	+106.5, +2,149.50	−52.9, −1,062.58	−21.6, −412.55	**+2.01**, +44.30	[−10.52, +14.04]; [−210.66, +297.90]
B1c	pick 2022-05→2024	2,426	121 / 443	58.4	+93.3, +1,885.77	−142.3, −2,856.32	+13.8, +252.89	**−4.69**, −86.51	[−16.90, +8.33]; [−328.31, +192.24]
B2a	pick 2022-05→2024	2,483	137 / 621	38.6	+30.0, +579.34	−27.6, −533.86	−15.5, −312.55	**−5.35**, −103.91	[−6.45, −4.60]; [−126.98, −88.86]
B2b	pick 2022-05→2024	2,483	137 / 621	34.8	+95.2, +1,817.29	−55.5, −1,074.00	−23.8, −432.55	**−2.99**, −66.76	[−11.44, +6.44]; [−247.19, +130.14]
B2c	pick 2022-05→2024	2,483	137 / 621	55.8	+90.1, +1,733.08	−142.6, −2,744.70	+10.8, +206.89	**−12.68**, −245.22	[−21.40, −3.60]; [−429.82, −54.34]
B3a	pick 2022-05→2024	752	60 / 85	43.8	+35.9, +320.37	−32.9, −203.52	−30.6, −23.30	**−2.80**, +25.68	[−10.87, −2.38]; [−75.73, +33.84]
B3b	pick 2022-05→2024	752	60 / 85	32.3	+104.5, +934.95	−50.7, −338.31	−22.0, −49.80	**−0.53**, +73.13	[−27.90, +5.39]; [−230.53, +152.66]
B3c	pick 2022-05→2024	752	60 / 85	55.9	+91.0, +801.74	−134.0, −857.15	+11.7, +24.10	**−8.30**, +69.36	[−49.01, −1.58]; [−402.35, +169.93]
A1a	check 2025→2026-10-09	1,035	101 / 451	36.6	+37.2, +430.01	−33.4, −287.39	−31.4, −33.30	**−7.54**, −24.69	[−15.50, −0.10]; [−159.16, +104.87]
A1b	check 2025→2026-10-09	1,035	101 / 451	27.1	+83.4, +332.19	−60.6, −660.70	−30.0, −73.30	**−21.53**, −391.13	[−42.34, +0.40]; [−621.95, −125.68]
A1c	check 2025→2026-10-09	1,035	101 / 451	53.4	+70.0, +418.67	−162.5, −1,617.02	+7.1, +6.40	**−38.24**, −529.36	[−80.04, −0.96]; [−823.99, −209.03]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A2a	check 2025→2026-10-09	1,502	155 / 456	32.2	+36.2, +378.08	−33.7, −318.23	−31.7, −40.80	**−11.17**, −93.85	[−15.01, −7.26]; [−188.26, +1.70]
A2b	check 2025→2026-10-09	1,502	155 / 456	35.8	+81.2, +882.58	−63.9, −596.57	−25.0, −46.30	**−11.89**, −66.75	[−30.37, +10.29]; [−318.50, +161.97]
A2c	check 2025→2026-10-09	1,502	155 / 456	55.3	+83.7, +935.37	−164.8, −1,565.82	+11.2, +19.30	**−27.31**, −182.01	[−59.32, +9.59]; [−620.91, +239.32]
A3a	check 2025→2026-10-09	2,435	146 / 451	37.1	+37.1, +364.92	−34.0, −305.79	−31.6, −35.30	**−7.62**, −57.06	[−11.88, −3.35]; [−116.97, +0.68]
A3b	check 2025→2026-10-09	2,435	146 / 451	28.7	+75.1, +514.83	−61.0, −622.63	−28.1, −60.30	**−21.90**, −296.11	[−32.27, −7.52]; [−508.70, −5.52]
A3c	check 2025→2026-10-09	2,435	146 / 451	54.0	+69.6, +546.69	−166.1, −1,674.08	+8.8, +11.00	**−38.83**, −474.77	[−56.61, −15.99]; [−825.13, −6.23]
B1a	check 2025→2026-10-09	6,350	197 / 457	41.6	+31.8, +685.75	−29.5, −642.56	−17.7, −432.55	**−3.97**, −89.48	[−4.79, −3.41]; [−107.57, −76.20]
B1b	check 2025→2026-10-09	6,350	197 / 457	35.0	+93.5, +2,028.38	−55.6, −1,238.88	−22.1, −442.55	**−3.34**, −94.06	[−8.46, +2.81]; [−198.26, +34.54]
B1c	check 2025→2026-10-09	6,350	197 / 457	59.0	+85.8, +1,874.18	−150.5, −3,356.98	+13.9, +292.89	**−11.03**, −269.36	[−16.70, −5.06]; [−398.31, −127.42]
B2a	check 2025→2026-10-09	5,406	136 / 446	39.7	+31.7, +703.69	−29.3, −655.79	−19.9, −452.55	**−5.07**, −116.37	[−6.44, −4.02]; [−152.63, −88.38]
B2b	check 2025→2026-10-09	5,406	136 / 446	35.4	+93.0, +2,087.62	−57.7, −1,324.39	−22.1, −462.55	**−4.29**, −115.10	[−9.49, +3.40]; [−237.67, +59.66]
B2c	check 2025→2026-10-09	5,406	136 / 446	57.3	+88.7, +2,007.17	−152.2, −3,498.43	+13.2, +276.89	**−14.07**, −341.32	[−21.04, −5.71]; [−528.55, −137.30]
B3a	check 2025→2026-10-09	5,454	321 / 457	37.4	+36.3, +376.73	−31.8, −325.65	−30.9, −41.30	**−6.35**, −63.19	[−7.74, −5.02]; [−120.75, −21.88]
B3b	check 2025→2026-10-09	5,454	321 / 457	31.5	+95.1, +1,032.89	−54.9, −556.86	−25.8, −65.30	**−7.66**, −56.38	[−12.26, −3.30]; [−130.94, +13.26]
B3c	check 2025→2026-10-09	5,454	321 / 457	53.6	+86.3, +955.64	−142.4, −1,426.40	+8.7, +17.10	**−19.87**, −150.21	[−27.20, −13.13]; [−287.45, −42.98]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
A1a	pick 2022-05→2024	+1,237.45, +39.9%, 2022-09-01, SPX 3955.0P m60	−1,412.55, −31.6%, 2022-05-19, SPX 3935.0P m90	0.78	36,426	681	total ≤ 0 (best 5 days +31,037; without them −52,548)
A1b	pick 2022-05→2024	+6,137.45, +204.6%, 2022-10-20, SPX 3755.0P m90	−3,402.55, −94.3%, 2022-05-17, SPX 4075.0P m90	0.68	82,630	681	total ≤ 0 (best 5 days +91,995; without them −152,094)
A1c	pick 2022-05→2024	+7,518.89, +200.0%, 2022-07-05, SPX 3725.0C m60	−6,805.11, −188.5%, 2022-05-17, SPX 4075.0P m90	0.75	115,765	681	total ≤ 0 (best 5 days +125,542; without them −199,422)
A2a	pick 2022-05→2024	+1,689.45, +39.9%, 2022-06-13, SPX 3750.0C m150	−1,322.55, −36.8%, 2022-05-19, SPX 3935.0P m210	0.90	65,459	660	total ≤ 0 (best 5 days +53,846; without them −69,751)
A2b	pick 2022-05→2024	+7,777.45, +436.9%, 2024-09-11, SPX 5445.0C m150	−3,622.55, −86.0%, 2022-05-24, SPX 3930.0P m60	1.11	86,722	596	477%
A2c	pick 2022-05→2024	+8,486.89, +476.8%, 2024-09-11, SPX 5445.0C m150	−7,245.11, −172.1%, 2022-05-24, SPX 3930.0P m60	1.06	113,190	543	651%
A3a	pick 2022-05→2024	+1,813.45, +39.9%, 2022-05-18, SPX 4050.0P m60	−1,562.55, −40.7%, 2022-10-21, SPX 3710.0P m90	1.01	59,987	587	2981%
A3b	pick 2022-05→2024	+8,687.45, +191.4%, 2022-05-18, SPX 4050.0P m60	−3,837.46, −99.9%, 2022-10-21, SPX 3710.0P m90	1.01	171,405	679	8340%
A3c	pick 2022-05→2024	+10,500.89, +231.3%, 2022-05-18, SPX 4050.0P m60	−7,674.93, −199.9%, 2022-10-21, SPX 3710.0P m90	1.12	187,483	633	497%
B1a	pick 2022-05→2024	+1,625.45, +39.9%, 2024-12-31, SPX 5955.0P m60	−2,557.46, −99.5%, 2023-09-21, SPX 4355.0C m1	0.77	189,790	442	total ≤ 0 (best 5 days +11,210; without them −200,843)
B1b	pick 2022-05→2024	+12,317.45, +443.1%, 2024-12-18, SPX 6090.0P m90	−3,431.28, −100.0%, 2024-04-15, SPX 5135.0C m1	1.06	233,599	324	525%
B1c	pick 2022-05→2024	+13,426.89, +483.0%, 2024-12-18, SPX 6090.0P m90	−6,862.55, −200.1%, 2024-04-15, SPX 5135.0C m1	0.93	392,636	443	total ≤ 0 (best 5 days +559,720; without them −769,604)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
B2a	pick 2022-05→2024	+1,533.45, +39.9%, 2024-10-04, SPX 5775.0P m1	−1,382.55, −37.8%, 2022-08-10, SPX 4165.0C m1	0.68	259,938	618	total ≤ 0 (best 5 days +8,995; without them −267,011)
B2b	pick 2022-05→2024	+12,317.45, +443.1%, 2024-12-18, SPX 6090.0P m90	−3,322.55, −94.1%, 2024-10-31, SPX 5740.0C m1	0.90	457,000	621	total ≤ 0 (best 5 days +395,720; without them −561,491)
B2c	pick 2022-05→2024	+13,426.89, +483.0%, 2024-12-18, SPX 6090.0P m90	−6,645.11, −188.2%, 2024-10-31, SPX 5740.0C m1	0.80	890,657	621	total ≤ 0 (best 5 days +390,586; without them −999,469)
B3a	pick 2022-05→2024	+1,281.45, +39.9%, 2024-09-06, SPX 5410.0C m90	−1,112.55, −35.2%, 2024-09-06, SPX 5435.0C m60	1.22	19,070	62	202%
B3b	pick 2022-05→2024	+11,797.45, +655.4%, 2024-12-18, SPX 6075.0P m150	−2,932.55, −92.8%, 2024-09-06, SPX 5435.0C m60	1.32	83,967	79	268%
B3c	pick 2022-05→2024	+12,514.89, +695.3%, 2024-12-18, SPX 6075.0P m150	−5,865.11, −185.6%, 2024-09-06, SPX 5435.0C m60	1.18	138,490	79	344%
A1a	check 2025→2026-10-09	+4,161.45, +40.0%, 2025-04-09, SPX 5085.0P m60	−3,532.55, −37.5%, 2025-04-04, SPX 5255.0P m90	0.86	75,185	368	total ≤ 0 (best 5 days +140,471; without them −166,024)
A1b	check 2025→2026-10-09	+3,477.45, +37.0%, 2025-04-04, SPX 5255.0P m90	−13,347.46, −100.0%, 2025-04-09, SPX 5125.0P m60	0.19	405,214	446	total ≤ 0 (best 5 days +40,691; without them −445,512)
A1c	check 2025→2026-10-09	+7,238.89, +76.9%, 2025-04-04, SPX 5255.0P m90	−26,694.93, −200.0%, 2025-04-09, SPX 5125.0P m60	0.30	591,518	384	total ≤ 0 (best 5 days +136,266; without them −684,148)
A2a	check 2025→2026-10-09	+2,647.45, +23.2%, 2025-04-11, SPX 5155.0C m90	−4,132.55, −31.1%, 2025-04-07, SPX 4900.0C m150	0.56	156,280	408	total ≤ 0 (best 5 days +96,136; without them −237,101)
A2b	check 2025→2026-10-09	+9,607.45, +84.2%, 2025-04-11, SPX 5155.0C m90	−6,502.55, −88.7%, 2025-04-08, SPX 5025.0C m270	0.83	388,541	409	total ≤ 0 (best 5 days +308,240; without them −408,502)
A2c	check 2025→2026-10-09	+14,168.89, +124.2%, 2025-04-11, SPX 5155.0C m90	−13,005.11, −177.4%, 2025-04-08, SPX 5025.0C m270	0.74	696,010	408	total ≤ 0 (best 5 days +430,849; without them −704,227)
A3a	check 2025→2026-10-09	+4,161.45, +40.0%, 2025-04-09, SPX 5085.0P m60	−3,532.55, −37.5%, 2025-04-04, SPX 5255.0P m90	0.70	218,542	383	total ≤ 0 (best 5 days +177,365; without them −316,304)
A3b	check 2025→2026-10-09	+5,297.45, +118.5%, 2025-10-10, SPX 6680.0P m150	−13,347.46, −100.0%, 2025-04-09, SPX 5125.0P m60	0.33	756,263	451	total ≤ 0 (best 5 days +152,628; without them −873,645)
A3c	check 2025→2026-10-09	+7,238.89, +76.9%, 2025-04-04, SPX 5255.0P m90	−26,694.93, −200.0%, 2025-04-09, SPX 5125.0P m60	0.38	1,202,103	451	total ≤ 0 (best 5 days +252,050; without them −1,408,120)
B1a	check 2025→2026-10-09	+1,957.45, +39.9%, 2026-06-17, SPX 7570.0P m60	−2,891.28, −100.0%, 2025-07-30, SPX 6360.0C m60	0.76	571,028	449	total ≤ 0 (best 5 days +22,656; without them −590,866)
B1b	check 2025→2026-10-09	+16,707.45, +843.8%, 2025-10-10, SPX 6770.0P m30	−5,517.55, −98.2%, 2026-09-16, SPX 7560.0C m1	0.88	699,558	401	total ≤ 0 (best 5 days +624,048; without them −1,221,308)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
B1c	check 2025→2026-10-09	+17,496.89, +883.7%, 2025-10-10, SPX 6770.0P m30	−11,035.11, −196.4%, 2026-09-16, SPX 7560.0C m1	0.80	1,792,998	434	total ≤ 0 (best 5 days +576,078; without them −2,286,495)
B2a	check 2025→2026-10-09	+2,301.45, +40.0%, 2026-02-20, SPX 6790.0C m1	−2,891.28, −100.0%, 2025-07-30, SPX 6360.0C m60	0.71	629,505	446	total ≤ 0 (best 5 days +22,604; without them −651,714)
B2b	check 2025→2026-10-09	+16,707.45, +843.8%, 2025-10-10, SPX 6770.0P m30	−5,762.55, −97.0%, 2026-02-20, SPX 6890.0P m1	0.87	916,065	251	total ≤ 0 (best 5 days +550,802; without them −1,173,037)
B2c	check 2025→2026-10-09	+17,496.89, +883.7%, 2025-10-10, SPX 6770.0P m30	−11,525.11, −194.0%, 2026-02-20, SPX 6890.0P m1	0.77	1,934,517	251	total ≤ 0 (best 5 days +521,335; without them −2,366,486)
B3a	check 2025→2026-10-09	+2,985.45, +40.0%, 2025-04-07, SPX 4925.0C m210	−4,022.55, −33.5%, 2025-04-07, SPX 4820.0C m1	0.69	348,301	456	total ≤ 0 (best 5 days +72,070; without them −416,712)
B3b	check 2025→2026-10-09	+23,857.45, +381.1%, 2025-04-08, SPX 5295.0P m60	−7,837.46, −99.2%, 2025-04-04, SPX 5215.0C m1	0.85	486,300	390	total ≤ 0 (best 5 days +678,695; without them −986,191)
B3c	check 2025→2026-10-09	+26,358.89, +421.1%, 2025-04-08, SPX 5295.0P m60	−15,674.93, −198.4%, 2025-04-04, SPX 5215.0C m1	0.77	976,739	412	total ≤ 0 (best 5 days +852,538; without them −1,671,786)

Random baseline: 500 cell-matched random draws per rule and exit (step6/out/results.csv): the rule's percentile is printed there; on Check 14 of 18 rule × exit cells sit below the 20th.

Exit reasons (share of trades):

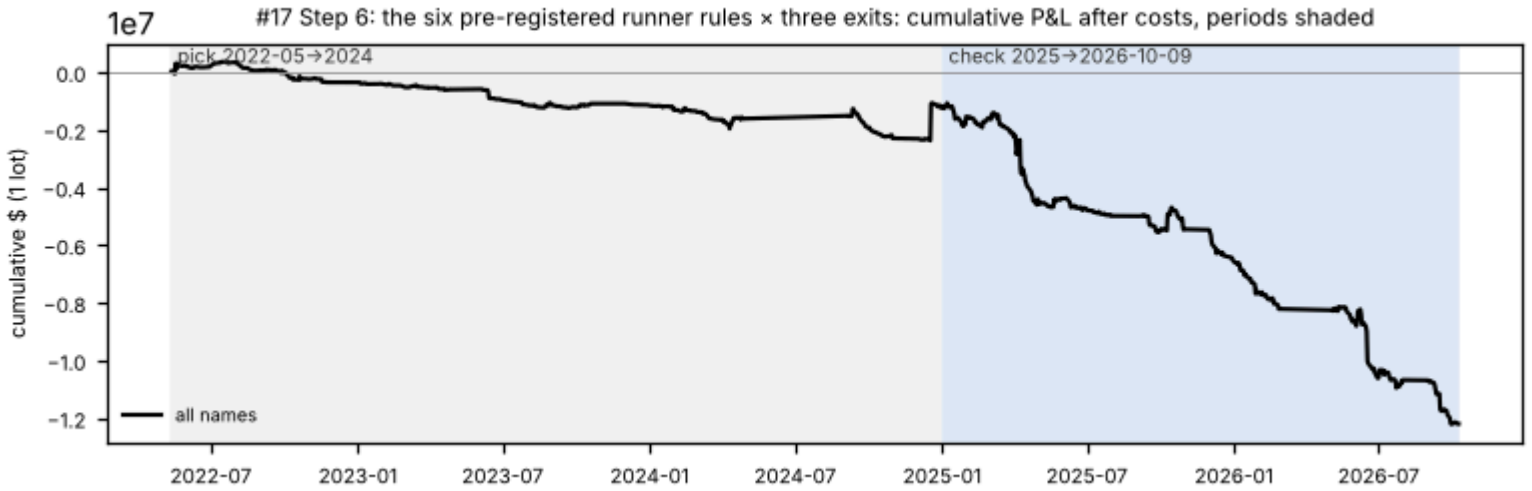
- pick 2022-05→2024: half at +40%, trail the rest 50%, out at 3:30 33.3%; time 20.4%; stop 15.9%; trail 11.9%; target 10.9%; time_armed 7.6%
- check 2025→2026-10-09: half at +40%, trail the rest 50%, out at 3:30 33.3%; time 21.0%; stop 17.5%; trail 10.7%; target 10.0%; time_armed 7.6%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	A1a	218	40.4	−1.16	−90.83	−19,801
2023	A1a	249	59.0	+6.29	+4.53	+1,127
2024	A1a	198	48.5	−3.03	−14.33	−2,837
2025	A1a	615	44.1	−1.95	+31.76	+19,534
2026	A1a	420	25.7	−15.71	−107.35	−45,087
2022	A1b	218	22.5	−9.14	−138.04	−30,094
2023	A1b	249	43.4	+5.73	−9.68	−2,411
2024	A1b	198	27.8	−27.44	−139.36	−27,594
2025	A1b	615	33.5	−3.87	−354.25	−217,863
2026	A1b	420	17.9	−47.38	−445.14	−186,959
2022	A1c	218	61.9	−10.50	−184.12	−40,138
2023	A1c	249	75.5	+15.00	−6.09	−1,516

year	name	trades	win %	mean %	mean $	total $
2024	A1c	198	59.1	−36.77	−162.76	−32,226
2025	A1c	615	63.9	−5.04	−364.75	−224,319
2026	A1c	420	38.1	−86.85	−770.39	−323,564
2022	A2a	341	37.5	−7.00	−59.61	−20,326
2023	A2a	403	46.9	−1.58	−10.09	−4,065
2024	A2a	337	44.2	−1.59	+25.18	+8,486
2025	A2a	1,058	30.9	−11.78	−123.28	−130,435
2026	A2a	444	35.4	−9.74	−23.72	−10,530
2022	A2b	341	27.3	−5.93	−72.18	−24,614
2023	A2b	403	32.5	+17.65	+48.07	+19,372
2024	A2b	337	38.6	+24.37	+116.92	+39,401
2025	A2b	1,058	35.4	−8.72	−69.17	−73,183
2026	A2b	444	36.7	−19.45	−60.99	−27,079
2022	A2c	341	62.8	−6.22	−36.77	−12,540
2023	A2c	403	68.5	+22.03	+56.14	+22,626
2024	A2c	337	59.9	+16.14	+58.87	+19,838
2025	A2c	1,058	56.6	−23.22	−214.00	−226,408
2026	A2c	444	52.3	−37.05	−105.79	−46,971
2022	A3a	493	48.5	+2.63	+48.55	+23,934
2023	A3a	440	48.2	−0.24	−24.33	−10,704
2024	A3a	306	44.4	−3.58	−37.49	−11,472
2025	A3a	1,516	36.7	−8.11	−74.40	−112,784
2026	A3a	919	37.8	−6.81	−28.46	−26,155
2022	A3b	493	27.2	−1.82	+56.60	+27,903
2023	A3b	440	29.8	−3.20	−4.90	−2,158
2024	A3b	306	29.1	−13.15	−75.28	−23,034
2025	A3b	1,516	27.6	−23.35	−402.33	−609,934
2026	A3b	919	30.6	−19.51	−120.87	−111,082
2022	A3c	493	65.9	−0.26	+88.84	+43,798
2023	A3c	440	74.3	+6.27	+113.37	+49,885
2024	A3c	306	54.2	−24.69	−113.46	−34,719
2025	A3c	1,516	52.0	−42.32	−642.72	−974,369
2026	A3c	919	57.3	−33.09	−197.72	−181,701
2023	B1a	911	43.4	−3.71	−72.13	−65,710
2024	B1a	1,515	41.7	−3.98	−81.80	−123,923
2025	B1a	3,411	42.0	−3.89	−85.43	−291,407
2026	B1a	2,939	41.3	−4.07	−94.18	−276,803
2023	B1b	911	36.2	−3.36	−78.60	−71,608
2024	B1b	1,515	33.4	+5.24	+118.21	+179,086
2025	B1b	3,411	36.1	−1.46	−51.14	−174,438

year	name	trades	win %	mean %	mean $	total $
2026	B1b	2,939	33.9	−5.53	−143.87	−422,822
2023	B1c	911	59.9	−8.62	−171.80	−156,506
2024	B1c	1,515	57.5	−2.32	−35.23	−53,378
2025	B1c	3,411	60.1	−7.60	−176.36	−601,549
2026	B1c	2,939	57.8	−15.01	−377.29	−1,108,868
2022	B2a	247	39.7	−5.52	−117.60	−29,047
2023	B2a	775	37.5	−5.44	−104.70	−81,144
2024	B2a	1,461	39.0	−5.27	−101.18	−147,825
2025	B2a	2,755	40.3	−4.73	−103.16	−284,195
2026	B2a	2,651	39.0	−5.42	−130.11	−344,915
2022	B2b	247	33.2	−17.67	−395.70	−97,738
2023	B2b	775	34.1	−10.28	−223.84	−173,474
2024	B2b	1,461	35.5	+3.36	+72.17	+105,442
2025	B2b	2,755	36.9	+1.04	+21.88	+60,290
2026	B2b	2,651	33.9	−9.84	−257.46	−682,525
2022	B2c	247	50.2	−33.36	−708.84	−175,082
2023	B2c	775	54.7	−20.25	−396.67	−307,420
2024	B2c	1,461	57.4	−5.17	−86.50	−126,381
2025	B2c	2,755	58.3	−7.52	−151.92	−418,538
2026	B2c	2,651	56.4	−20.87	−538.14	−1,426,613
2024	B3a	752	43.8	−2.80	+25.68	+19,313
2025	B3a	3,157	36.6	−6.90	−87.21	−275,316
2026	B3a	2,297	38.4	−5.60	−30.18	−69,326
2024	B3b	752	32.3	−0.53	+73.13	+54,994
2025	B3b	3,157	31.3	−7.37	−61.48	−194,107
2026	B3b	2,297	31.7	−8.05	−49.36	−113,389
2024	B3c	752	55.9	−8.30	+69.36	+52,157
2025	B3c	3,157	52.5	−20.30	−181.75	−573,791
2026	B3c	2,297	55.0	−19.28	−106.86	−245,456
2022	all names	3,897	43.9	−6.35	−90.77	−353,745
2023	all names	8,334	47.5	−2.26	−94.04	−783,707
2024	all names	13,707	44.3	−2.45	−7.64	−104,672
2025	all names	37,536	43.1	−9.30	−141.27	−5,302,813
2026	all names	29,010	42.2	−13.47	−194.76	−5,649,845



Amendments, deviations and integrity checks. The rules are chosen by the committed step6/trade.py choose() on the scratchpad population (39 MB, not committed); verify.py 0 differences in 480 figures; lookahead.py 0 differences in 104,748 values.

What it showed. No rule chosen on Pick made money on Check.

#18 Compass engine audit (six module groups)

Folder: `research_notes/Compass engine audit/, reports/Compass engine audit.md`. **Commits:** — / — / 9cf8110, 59e1a1a, d724ae4, 18cf86a, b84fa3c, 09799af; report 138ff0e.
Dates: 2026-10-21.

Hypothesis. The owner’s 16-module Compass specification picks direction and opens trades that pay after costs.

Rule. Each module tested under its own definition (exits and sizing, expected move/ATR/flow, clock and power hour, greeks and books, horizons, sweep and grade).

Fills and costs. Ask in, bid out, $0.65 each way.

Data. 0DTE SPY/QQQ/IWM/SPXW 980/980/621/1,106 sessions; daily bars for horizons.

Periods and holdouts. fit 2022–24 / check 2025–26.

Pass rule (as pre-registered). Picked on 2022–24, read on 2025–26, friction-adjusted.

Verdict. No directional rule survives after fees. What holds: the 15:58 flat, the −32% backstop, a breakeven lock at +0% with a target on, a trailed remainder, the r,q pricer from parity, Method A flips with three books, net/gross confidence, the MOC magnitude as a volatility gauge, an ATR > 1.5 block on counter-trend fades.

Metrics. not available — the audit committed row-level paths and events (paths_*.parquet, events_*.parquet) and per-cell tables; its exit stacks were scored by cell, and no single trade list per rule exists to recompute (spec stack A −4.3% / B −4.5% of debit a trade, exits_sizing.md line 14).

Amendments, deviations and integrity checks. Six blockers stated (no OPRA tape, no strike-level OI history before 2026-10, …).

What it showed. Frictions, not edges.

#19 Phase 2 hyp A: the low-moc_mag closing pin (SPXW short call spread, 15:30, held)

Folder: `research_notes/Compass phase 2/hyp_a_closing_pin`. **Commits:** prereg in-doc before the OOS read / — / 5deb793. **Dates:** 2026-10-21.

Hypothesis. On quiet closes (low moc_mag) the 15:30 short credit spread pins and pays.

Rule. SPXW short call spread, short strike at 1.00 × the move rule, 25 wide, 15:30 at the NBBO, held to settlement, on low-moc_mag sessions (the pre-registered cell).

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPXW 2022-05-13 → 2026-10-09.

Periods and holdouts. fit 2022–24 / OOS 2025–26 (read once).

Pass rule (as pre-registered). The pick fixed on fit; read OOS for a positive mean with an interval clear of zero.

Verdict. NO EDGE: fit +$26 a spread, OOS −$33 [−84, +13].

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	fit 2022–24	252	252 / 656	91.7	+2.3, +55.87	−12.2, −298.87	+1.9, +46.20	**+1.10**, +26.31	[+0.45, +1.66]; [+10.45, +39.79]
SPXW	OOS 2025–26	177	177 / 456	87.6	+2.6, +62.88	−29.5, −708.75	+2.0, +48.70	**−1.39**, −33.03	[−3.53, +0.30]; [−84.64, +7.73]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	fit 2022–24	+198.70, +8.6%, 2022-09-16, SPXW 15:30 3880.0/3905.0C	−687.30, −28.1%, 2023-03-08, SPXW 15:30 3985.0/4010.0C	2.06	908	93	13%
SPXW	OOS 2025–26	+173.70, +7.5%, 2025-04-14, SPXW 15:30 5450.0/5475.0C	−2,311.30, −100.1%, 2025-04-30, SPXW 15:30 5535.0/5560.0C	0.63	8,053	426	total ≤ 0 (best 5 days +794; without them −6,640)

Random baseline: not available — no random baseline was computed by this study

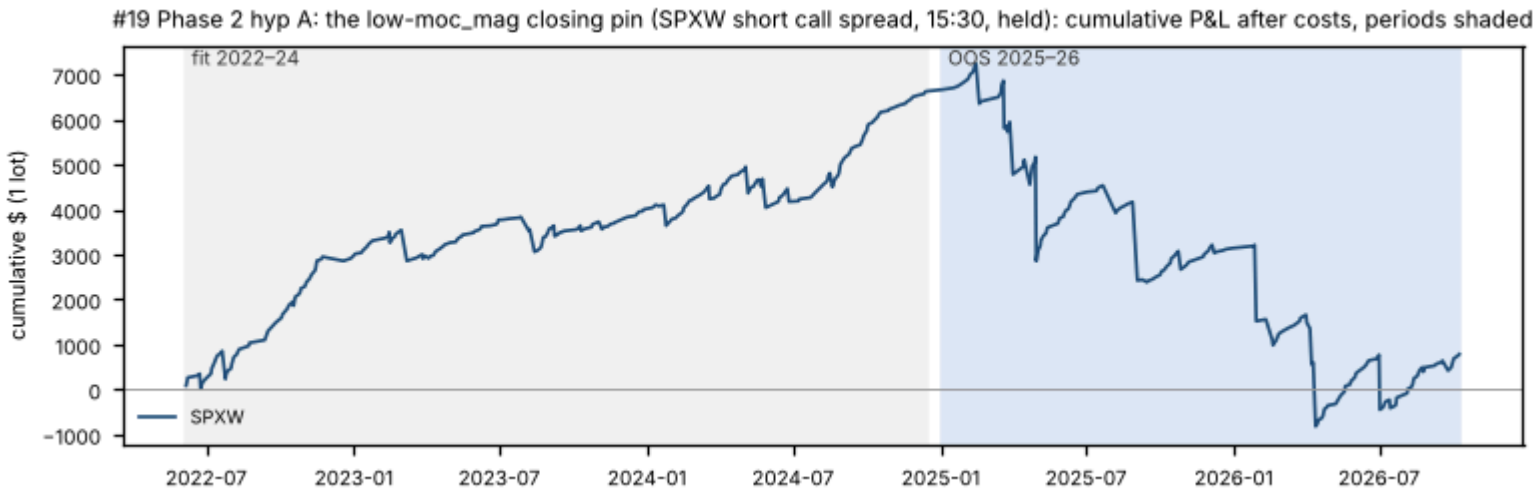
Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	55	92.7	+2.23	+53.19	+2,925
2023	SPXW	95	90.5	+0.47	+11.34	+1,078
2024	SPXW	102	92.2	+1.08	+25.76	+2,627
2025	SPXW	94	88.3	−1.59	−37.12	−3,489

year	name	trades	win %	mean %	mean $	total $
2026	SPXW	83	86.7	−1.17	−28.40	−2,357



Amendments, deviations and integrity checks. Cells and the pick written before the OOS run (s3_report.py fit, then oos).

What it showed. The pin made money on fit and lost out of sample.

#20a Phase 2 hyp B: boundary touch — SPY 2-wide 25-delta credit spread, held (the touch pick)

Folder: `research_notes/Compass phase 2/hyp_b_boundary_theta`. **Commits:** out/prereg.txt before the OOS read / — / 2c241f4. **Dates:** 2026-10-21.

Hypothesis. When price touches the straddle boundary, the far side's credit spread decays faster.

Rule. At a touch: sell the 2-wide spread on the touched side, short strike at 25 delta, held to settlement.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). The highest fit lower bound (N ≥ 300, ≥ 150 sessions, positive without the best 5); its lower bound was negative ("least bad").

Verdict. No edge: the touch adds nothing over the unconditional sell at the same minute (the increment flips sign between periods).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	fit 2022–24	2,890	512 / 553	83.5	+10.6, +18.66	−47.4, −83.65	+8.7, +15.70	**+1.07**, +1.81	[−0.80, +2.84]; [−1.55, +4.95]
SPY	OOS 2025–26	2,600	432 / 457	88.1	+11.9, +20.68	−55.2, −95.74	+10.4, +18.70	**+3.91**, +6.84	[+1.56, +5.68]; [+2.71, +9.95]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	fit 2022–24	+43.70, +28.2%, 2024-08-28, SPY 554.0/552.0 down	−195.30, −100.7%, 2024-05-31, SPY 525.0/527.0 up	1.13	4,005	172	26%
SPY	OOS 2025–26	+44.70, +29.0%, 2025-11-14, SPY 673.0/675.0 up	−191.30, −100.2%, 2026-04-13, SPY 684.0/686.0 up	1.60	4,526	64	9%

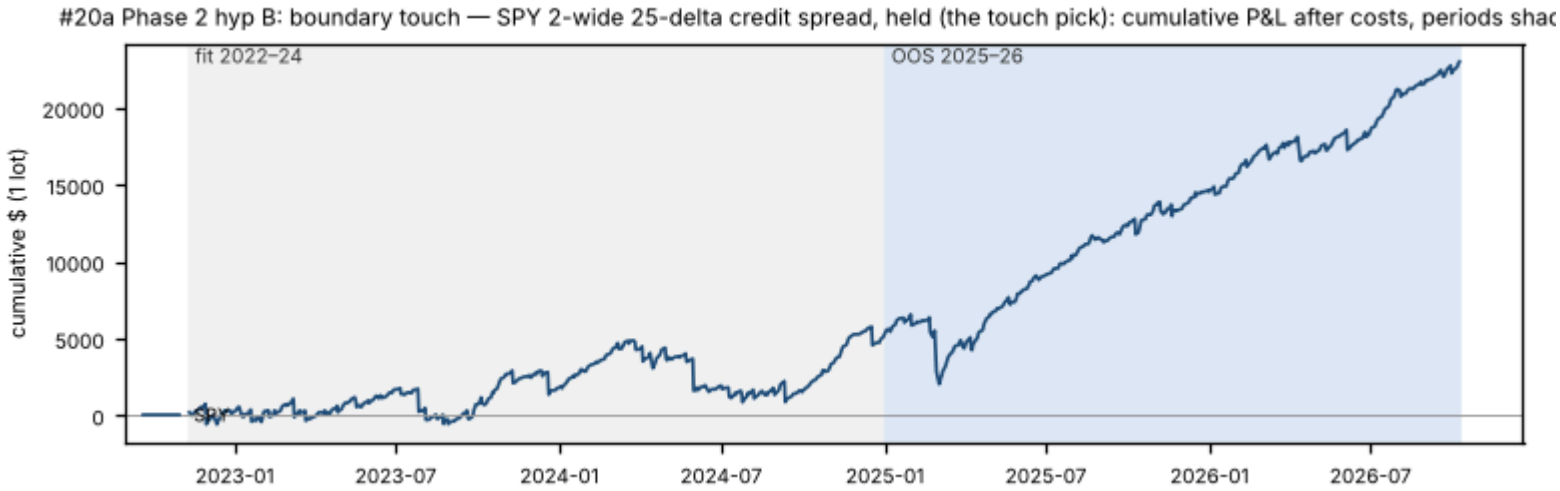
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	188	80.9	+0.75	+1.06	+200
2023	SPY	1,366	82.7	+0.71	+1.15	+1,567
2024	SPY	1,336	84.7	+1.48	+2.59	+3,465
2025	SPY	1,457	87.6	+3.68	+6.40	+9,319
2026	SPY	1,143	88.7	+4.21	+7.41	+8,465



Amendments, deviations and integrity checks. Deduplicated as analyze.py (one touch a session, side, e, mult, rule, width).

What it showed. Positive in both periods but no better than selling unconditionally.

#20b Phase 2 hyp B: the midday SPY iron condor (em1, mid-VIX), the midday pick

Folder: `research_notes/Compass phase 2/hyp_b_boundary_theta`. **Commits:** out/prereg.txt / — / 2c241f4. **Dates:** 2026-10-21.

Hypothesis. Midday iron condors on SPY are unconditional short premium that pays.

Rule. 11:30–13:30 every half hour: SPY 2-wide iron condor, shorts at the first listed strikes outside spot ± the ATM straddle, held to settlement, VIX in the fit's middle tercile.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). As 20a.

Verdict. "Needs X" (a forward paper run): +4.6% of risk fit, +3.3% OOS (CI $0.03–$10.69). Became the tracker's condor_em1 (all-VIX parent, #26).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	fit 2022–24	748	178 / 510	80.5	+15.6, +26.03	−40.9, −69.48	+12.8, +21.90	**+4.61**, +7.39	[+1.27, +7.77]; [+1.83, +12.66]
SPY	OOS 2025–26	1,202	280 / 456	81.6	+15.5, +25.89	−50.9, −84.73	+12.8, +22.40	**+3.25**, +5.55	[+0.21, +6.04]; [+0.58, +10.01]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	fit 2022–24	+62.40, +46.2%, 2024-09-18, None	−175.60, −101.5%, 2024-09-11, None	1.54	1,438	145	23%
SPY	OOS 2025–26	+56.40, +40.0%, 2026-04-29, None	−178.60, −101.5%, 2026-07-24, None	1.36	1,825	104	18%

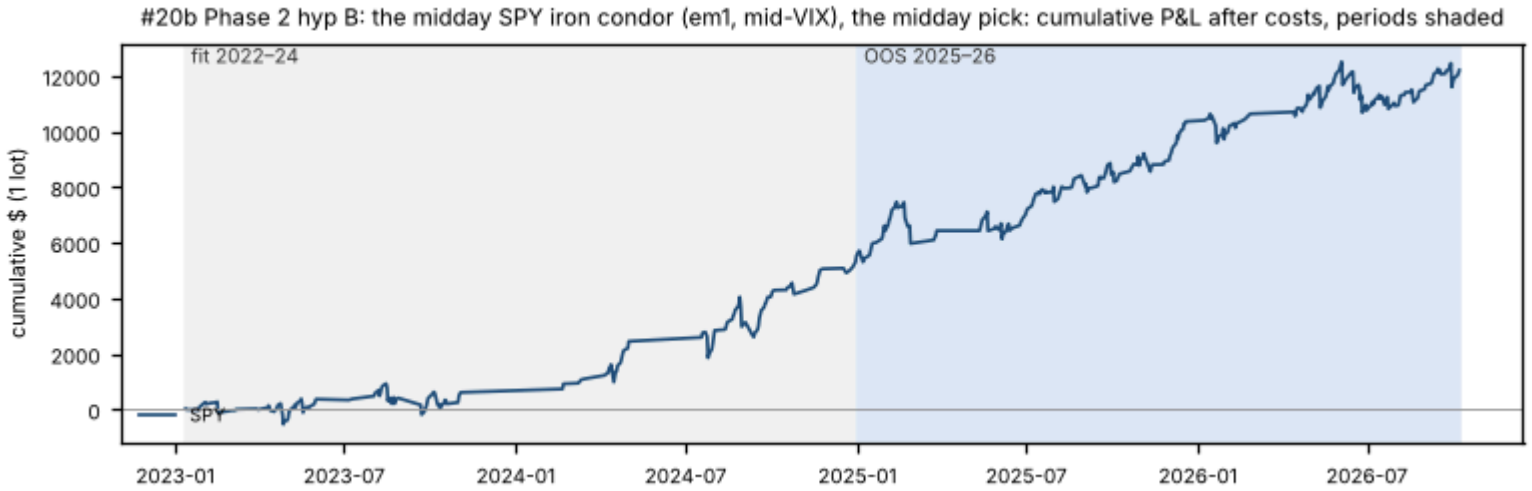
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2023	SPY	388	74.2	+1.05	+1.60	+619
2024	SPY	360	87.2	+8.44	+13.63	+4,905
2025	SPY	631	83.2	+4.65	+7.66	+4,836
2026	SPY	571	79.9	+1.72	+3.22	+1,840



Amendments, deviations and integrity checks. The VIX terciles recomputed from the fit d15 condors exactly as analyze.py.

What it showed. Small positive in both periods; $5 a condor on a four-leg structure.

#21a Phase 2 hyp C: 15:00 SPY/QQQ P1 iron ±2 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P1 iron ±2 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	521	521 / 553	63.1	+53.1, +49.60	−52.4, −50.42	+21.3, +23.40	**+14.26**, +12.74	[+9.21, +20.69]; [+8.42, +18.01]
SPY	fit 2022–24	525	525 / 553	62.1	+49.6, +47.81	−52.3, −51.56	+16.3, +19.40	**+10.95**, +10.14	[+6.16, +15.93]; [+5.96, +14.38]
all names	fit 2022–24	1,046	529 / 553	62.6	+51.4, +48.71	−52.4, −51.00	+19.3, +21.40	**+12.60**, +11.44	[+8.22, +17.89]; [+7.66, +15.67]
QQQ	OOS 2025–26	440	440 / 457	50.0	+75.0, +56.25	−70.7, −53.49	+0.1, +0.15	**+2.16**, +1.38	[−5.77, +10.53]; [−4.72, +7.83]
SPY	OOS 2025–26	438	438 / 457	54.6	+63.7, +52.13	−61.4, −50.60	+8.3, +8.65	**+6.84**, +5.45	[−0.71, +15.31]; [−0.57, +12.65]
all names	OOS 2025–26	878	440 / 457	52.3	+69.1, +54.10	−66.3, −52.12	+4.5, +4.40	**+4.49**, +3.41	[−2.96, +12.33]; [−1.96, +9.60]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+136.40, +257.4%, 2024-02-01, QQQ 1500 iron K0 422.0 ±2.0	−144.60, −101.8%, 2023-07-07, QQQ 1500 iron K0 369.0 ±2.0	1.69	683	92	10%
SPY	fit 2022–24	+130.40, +197.6%, 2024-12-31, SPY 1500 iron K0 586.0 ±2.0	−146.60, −101.8%, 2024-03-26, SPY 1500 iron K0 521.0 ±2.0	1.52	462	62	12%
all names	fit 2022–24	+136.40, +257.4%, 2024-02-01, QQQ 1500 iron K0 422.0 ±2.0	−146.60, −101.8%, 2024-03-26, SPY 1500 iron K0 521.0 ±2.0	1.60	870	37	10%
QQQ	OOS 2025–26	+153.40, +374.1%, 2026-01-28, QQQ 1500 iron K0 633.0 ±2.0	−120.60, −102.2%, 2026-09-14, QQQ 1500 iron K0 712.0 ±2.0	1.05	1,143	197	114%
SPY	OOS 2025–26	+146.90, +358.3%, 2026-09-16, SPY 1500 iron K0 754.0 ±2.0	−132.60, −102.0%, 2026-01-07, SPY 1500 iron K0 692.0 ±2.0	1.24	889	166	29%
all names	OOS 2025–26	+153.40, +374.1%, 2026-01-28, QQQ 1500 iron K0 633.0 ±2.0	−132.60, −102.0%, 2026-01-07, SPY 1500 iron K0 692.0 ±2.0	1.14	1,817	142	43%

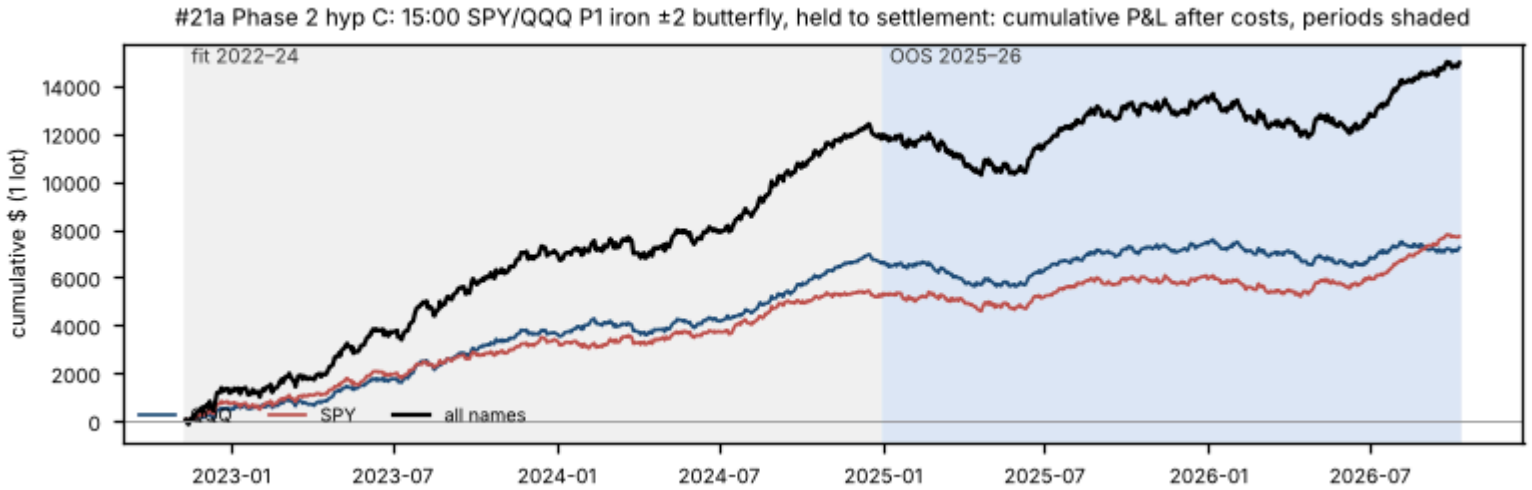
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	75.8	+18.94	+13.72	+453
2023	QQQ	247	63.6	+13.88	+13.15	+3,249
2024	QQQ	241	61.0	+14.00	+12.19	+2,937
2025	QQQ	246	52.0	+1.03	+3.07	+755
2026	QQQ	194	47.4	+3.58	−0.76	−148
2022	SPY	32	65.6	+30.49	+22.49	+720
2023	SPY	246	63.8	+10.46	+10.96	+2,696
2024	SPY	247	59.9	+8.92	+7.73	+1,909
2025	SPY	244	52.9	+1.88	+2.47	+604
2026	SPY	194	56.7	+13.06	+9.20	+1,785
2022	all names	65	70.8	+24.62	+18.04	+1,172
2023	all names	493	63.7	+12.17	+12.06	+5,946
2024	all names	488	60.5	+11.43	+9.93	+4,846
2025	all names	490	52.4	+1.46	+2.77	+1,359
2026	all names	388	52.1	+8.32	+4.22	+1,637



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21b Phase 2 hyp C: 15:00 SPY/QQQ P2 call ±2 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P2 call ±2 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	522	522 / 553	61.5	+49.9, +47.80	−51.6, −51.68	+18.2, +20.40	**+10.81**, +9.49	[+5.90, +17.14]; [+5.11, +14.83]
SPY	fit 2022–24	525	525 / 553	59.6	+47.6, +46.99	−50.3, −51.23	+14.3, +16.40	**+8.07**, +7.33	[+3.41, +13.02]; [+3.01, +11.65]
all names	fit 2022–24	1,047	530 / 553	60.6	+48.8, +47.40	−50.9, −51.45	+16.3, +18.40	**+9.44**, +8.41	[+5.18, +14.71]; [+4.54, +12.71]
QQQ	OOS 2025–26	441	441 / 457	48.3	+72.5, +55.44	−69.5, −54.36	−3.0, −2.60	**−0.92**, −1.33	[−8.71, +7.43]; [−7.44, +5.21]
SPY	OOS 2025–26	439	439 / 457	54.2	+61.5, +51.12	−61.7, −51.77	+6.2, +6.40	**+5.11**, +4.01	[−2.63, +13.75]; [−2.10, +11.26]
all names	OOS 2025–26	880	441 / 457	51.2	+66.7, +53.16	−65.8, −53.14	+2.1, +2.40	**+2.09**, +1.34	[−5.35, +10.06]; [−4.09, +7.53]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+135.40, +250.7%, 2024-02-01, QQQ 1500 call K0 422.0 ±2.0	−148.60, −101.8%, 2023-07-07, QQQ 1500 call K0 369.0 ±2.0	1.48	788	119	13%
SPY	fit 2022–24	+129.40, +369.7%, 2024-08-28, SPY 1500 call K0 558.0 ±2.0	−151.60, −101.7%, 2024-03-26, SPY 1500 call K0 521.0 ±2.0	1.35	577	152	16%
all names	fit 2022–24	+135.40, +250.7%, 2024-02-01, QQQ 1500 call K0 422.0 ±2.0	−151.60, −101.7%, 2024-03-26, SPY 1500 call K0 521.0 ±2.0	1.41	934	51	13%
QQQ	OOS 2025–26	+151.40, +352.1%, 2026-01-28, QQQ 1500 call K0 633.0 ±2.0	−132.60, −102.0%, 2025-11-05, QQQ 1500 call K0 626.0 ±2.0	0.95	1,413	197	total ≤ 0 (best 5 days +684; without them −1,268)
SPY	OOS 2025–26	+146.90, +358.3%, 2026-09-16, SPY 1500 call K0 754.0 ±2.0	−133.60, −102.0%, 2026-01-07, SPY 1500 call K0 692.0 ±2.0	1.17	1,081	205	39%
all names	OOS 2025–26	+151.40, +352.1%, 2026-01-28, QQQ 1500 call K0 633.0 ±2.0	−133.60, −102.0%, 2026-01-07, SPY 1500 call K0 692.0 ±2.0	1.05	2,167	212	109%

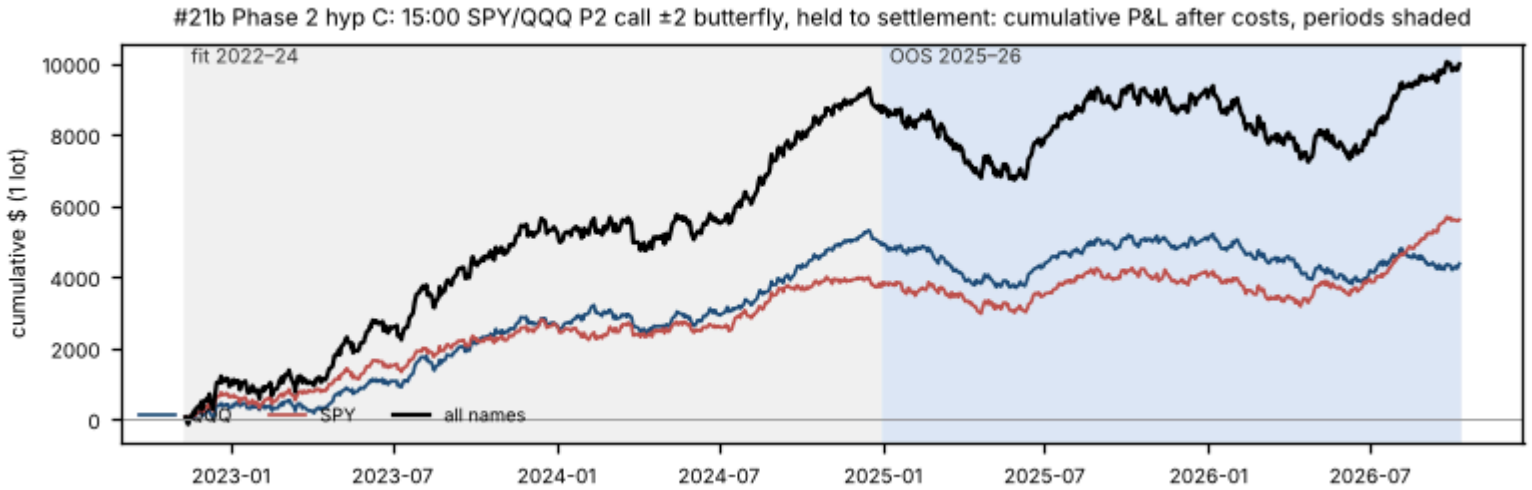
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	69.7	+14.87	+9.29	+307
2023	QQQ	247	61.9	+10.42	+9.74	+2,406
2024	QQQ	242	59.9	+10.66	+9.27	+2,243
2025	QQQ	247	50.2	−2.12	+0.31	+77
2026	QQQ	194	45.9	+0.60	−3.41	−662
2022	SPY	32	65.6	+26.62	+19.37	+620
2023	SPY	246	61.8	+8.01	+8.44	+2,076
2024	SPY	247	56.7	+5.73	+4.65	+1,150
2025	SPY	245	51.8	−0.32	+0.66	+161
2026	SPY	194	57.2	+11.96	+8.24	+1,599
2022	all names	65	67.7	+20.65	+14.25	+926
2023	all names	493	61.9	+9.22	+9.09	+4,483
2024	all names	489	58.3	+8.17	+6.94	+3,392
2025	all names	492	51.0	−1.22	+0.48	+238
2026	all names	388	51.5	+6.28	+2.42	+937



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21c Phase 2 hyp C: 15:00 SPY/QQQ P3 call ±5 butterfly, held to settlement

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md before the OOS run / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. Late-day ATM butterflies on quiet closes pin.

Rule. 15:00, SPY and QQQ, P3 call ±5 at the 4-leg NBBO, held to settlement, unconditional.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry (four legs); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY and QQQ 2022-11 → 2026-10.

Periods and holdouts. fit 2022–24 / OOS 2025–26.

Pass rule (as pre-registered). Edge only if the OOS mean is positive with its interval clear of zero AND positive without the best five sessions AND both OOS years positive.

Verdict. Not confirmed: the picks fell to +2 to +4.5% of risk OOS with intervals through zero and die under $0.01 a leg; the un-picked iron ±5 read +5.0/+6.0% (→ the tracker's fly5).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022–24	523	523 / 553	66.3	+16.0, +57.61	−19.9, −74.43	+6.9, +28.40	**+3.91**, +13.17	[+2.23, +6.26]; [+7.84, +20.38]
SPY	fit 2022–24	524	524 / 553	64.9	+14.7, +54.53	−19.2, −74.21	+5.4, +22.40	**+2.81**, +9.32	[+1.17, +4.83]; [+3.34, +16.31]
all names	fit 2022–24	1,047	530 / 553	65.6	+15.4, +56.08	−19.6, −74.32	+6.4, +25.40	**+3.36**, +11.25	[+1.85, +5.44]; [+6.15, +17.83]
QQQ	OOS 2025–26	438	438 / 457	55.5	+25.0, +76.31	−29.1, −98.48	+4.5, +17.40	**+0.90**, −1.51	[−2.31, +4.05]; [−12.39, +8.80]
SPY	OOS 2025–26	440	440 / 457	61.6	+21.4, +69.68	−25.5, −84.32	+6.6, +23.40	**+3.40**, +10.53	[+0.80, +5.95]; [+2.30, +19.47]
all names	OOS 2025–26	878	441 / 457	58.5	+23.1, +72.82	−27.4, −91.91	+5.0, +19.40	**+2.15**, +4.52	[−0.59, +4.78]; [−4.48, +13.82]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022–24	+286.40, +206.0%, 2024-08-28, QQQ 1500 call K0 472.0 ±5.0	−369.60, −100.7%, 2024-05-31, QQQ 1500 call K0 446.0 ±5.0	1.53	820	66	16%
SPY	fit 2022–24	+268.40, +136.9%, 2024-08-28, SPY 1500 call K0 558.0 ±5.0	−407.60, −100.6%, 2024-05-31, SPY 1500 call K0 521.0 ±5.0	1.36	945	169	21%
all names	fit 2022–24	+286.40, +206.0%, 2024-08-28, QQQ 1500 call K0 472.0 ±5.0	−407.60, −100.6%, 2024-05-31, SPY 1500 call K0 521.0 ±5.0	1.44	1,339	62	17%
QQQ	OOS 2025–26	+274.40, +169.4%, 2026-04-29, QQQ 1500 call K0 661.0 ±5.0	−470.60, −100.6%, 2026-05-12, QQQ 1500 call K0 702.0 ±5.0	0.97	2,306	457	total ≤ 0 (best 5 days +1,256; without them −1,916)
SPY	OOS 2025–26	+264.90, +118.8%, 2026-09-16, SPY 1500 call K0 754.0 ±5.0	−344.60, −100.8%, 2026-03-09, SPY 1500 call K0 670.0 ±5.0	1.33	1,260	118	25%
all names	OOS 2025–26	+274.40, +169.4%, 2026-04-29, QQQ 1500 call K0 661.0 ±5.0	−470.60, −100.6%, 2026-05-12, QQQ 1500 call K0 702.0 ±5.0	1.12	3,295	176	58%

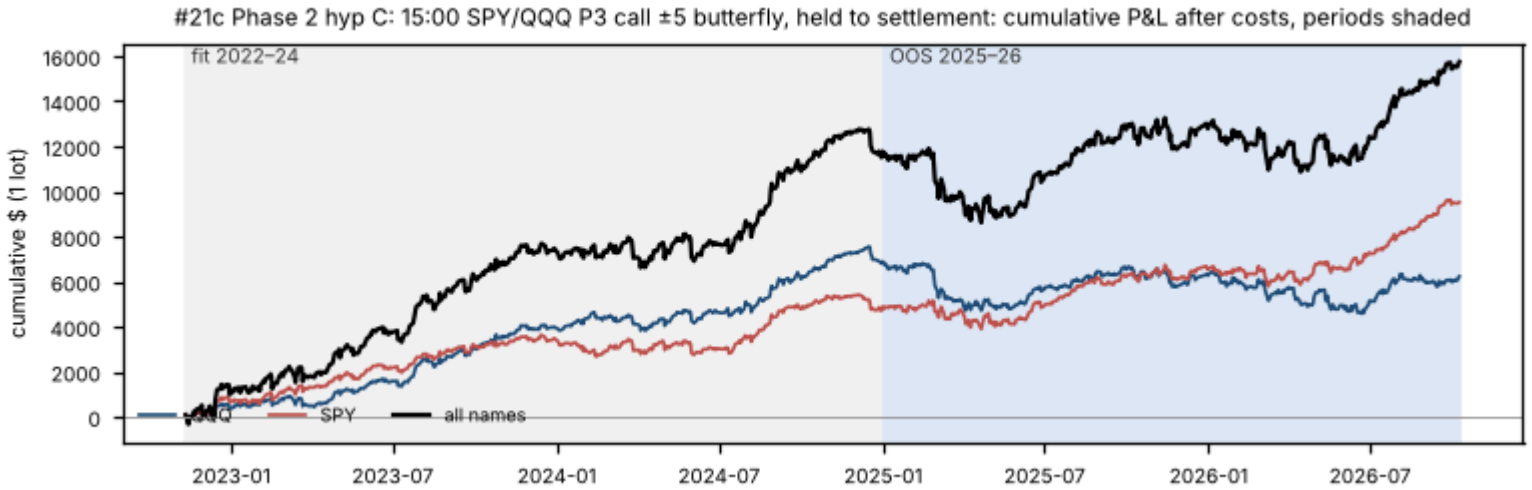
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022–24: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	32	71.9	+4.79	+10.67	+341
2023	QQQ	247	65.2	+3.82	+14.77	+3,648
2024	QQQ	244	66.8	+3.88	+11.89	+2,901
2025	QQQ	246	56.5	+0.26	−2.56	−629
2026	QQQ	192	54.2	+1.71	−0.16	−31
2022	SPY	33	72.7	+7.11	+19.84	+655
2023	SPY	247	66.0	+2.94	+11.47	+2,832
2024	SPY	244	62.7	+2.11	+5.74	+1,399
2025	SPY	247	59.5	+2.09	+6.62	+1,635
2026	SPY	193	64.2	+5.07	+15.53	+2,998
2022	all names	65	72.3	+5.97	+15.32	+996
2023	all names	494	65.6	+3.38	+13.12	+6,480
2024	all names	488	64.8	+2.99	+8.81	+4,300
2025	all names	493	58.0	+1.18	+2.04	+1,005
2026	all names	385	59.2	+3.40	+7.71	+2,967



Amendments, deviations and integrity checks. Fill stress ($0.01 a leg) read on both periods as a robustness check.

What it showed. Positive on fit, shrunk to noise out of sample.

#21d Phase 2 hyp C: debit vertical 0.70/0.30, universe B, held (the vertical pick)

Folder: `research_notes/Compass phase 2/hyp_c_debit_flies`. **Commits:** preregistration.md / — / faea67c. **Dates:** 2026-10-21.

Hypothesis. A debit vertical keeps the directional premium at lower cost.

Rule. The audit's universe B entries, 0.70Δ/0.30Δ debit vertical, held to settlement.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry; bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData)

Data. SPY, QQQ, IWM, SPXW 2022–2026.

Periods and holdouts. fit / OOS.

Pass rule (as pre-registered). Pre-registered expectation: negative OOS.

Verdict. Loses less than the naked contract but still loses (−6.7% of debit fit).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	fit 2022–24	173	173 / 182	54.9	+58.9, +43.45	−71.9, −51.06	+8.8, +9.70	**−0.05**, +0.84	[−12.07, +7.72]; [−8.88, +7.66]
QQQ	fit 2022–24	530	530 / 553	50.2	+69.1, +92.73	−71.0, −90.58	+0.4, +0.70	**−0.66**, +1.42	[−7.85, +7.82]; [−9.48, +14.58]
SPXW	fit 2022–24	656	656 / 683	45.4	+74.8, +884.62	−77.6, −952.87	−14.4, −155.80	**−8.38**, −118.16	[−13.19, −1.84]; [−182.50, −31.49]
SPY	fit 2022–24	530	530 / 553	44.0	+68.5, +80.70	−76.8, −91.93	−22.4, −25.30	**−12.94**, −16.04	[−19.06, −5.52]; [−24.17, −5.21]
all names	fit 2022–24	1,889	656 / 683	47.2	+69.8, +348.89	−75.2, −397.52	−9.3, −15.30	**−6.73**, −45.06	[−9.96, −3.11]; [−66.53, −16.04]
IWM	OOS 2025–26	439	439 / 457	49.7	+66.6, +60.84	−73.5, −61.89	−0.3, −0.30	**−3.93**, −0.94	[−10.90, +2.84]; [−8.20, +6.38]
QQQ	OOS 2025–26	441	441 / 457	51.5	+71.4, +135.51	−71.1, −133.23	+8.8, +13.70	**+2.26**, +5.10	[−5.49, +9.09]; [−10.73, +20.79]
SPXW	OOS 2025–26	441	441 / 457	51.7	+71.8, +1,075.79	−78.0, −1,139.59	+5.5, +70.70	**−0.55**, +5.78	[−8.93, +7.80]; [−102.60, +118.36]
SPY	OOS 2025–26	440	440 / 457	53.6	+72.9, +114.50	−68.8, −111.22	+11.9, +17.70	**+7.22**, +9.85	[−0.41, +14.64]; [−3.89, +23.09]
all names	OOS 2025–26	1,761	441 / 457	51.6	+70.7, +347.99	−72.9, −361.04	+5.7, +9.70	**+1.25**, +4.95	[−2.84, +4.65]; [−22.31, +32.57]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	fit 2022–24	+136.70, +92.4%, 2024-08-08, IWM CALL 0.70/0.30 m56	−213.30, −100.6%, 2024-06-12, IWM CALL 0.70/0.30 m146	1.04	717	78	405%
QQQ	fit 2022–24	+369.70, +95.5%, 2022-12-14, QQQ PUT 0.70/0.30 m54	−322.30, −100.4%, 2024-11-20, QQQ PUT 0.70/0.30 m66	1.03	3,066	231	202%
SPXW	fit 2022–24	+4,408.70, +107.8%, 2022-09-21, SPXW PUT 0.70/0.30 m59	−4,491.30, −100.0%, 2022-12-14, SPXW CALL 0.70/0.30 m90	0.77	79,805	574	total ≤ 0 (best 5 days +14,719; without them −92,231)
SPY	fit 2022–24	+337.70, +93.5%, 2023-03-22, SPY PUT 0.70/0.30 m96	−455.30, −100.3%, 2022-12-14, SPY CALL 0.70/0.30 m160	0.69	8,874	553	total ≤ 0 (best 5 days +1,312; without them −9,812)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	fit 2022–24	+4,408.70, +107.8%, 2022-09-21, SPXW PUT 0.70/0.30 m59	−4,491.30, −100.0%, 2022-12-14, SPXW CALL 0.70/0.30 m90	0.79	87,407	574	total ≤ 0 (best 5 days +13,894; without them −99,008)
IWM	OOS 2025–26	+217.70, +120.3%, 2026-09-16, IWM PUT 0.70/0.30 m108	−395.30, −94.6%, 2025-06-23, IWM PUT 0.70/0.30 m30	0.97	2,283	351	total ≤ 0 (best 5 days +1,048; without them −1,461)
QQQ	OOS 2025–26	+593.70, +117.6%, 2026-06-10, QQQ PUT 0.70/0.30 m62	−620.30, −67.1%, 2025-04-07, QQQ PUT 0.70/0.30 m66	1.08	2,773	186	113%
SPXW	OOS 2025–26	+6,048.70, +87.0%, 2025-04-11, SPXW CALL 0.70/0.30 m99	−7,091.30, −100.0%, 2025-04-10, SPXW PUT 0.70/0.30 m176	1.01	34,557	286	958%
SPY	OOS 2025–26	+689.70, +85.3%, 2025-04-11, SPY CALL 0.70/0.30 m52	−559.30, −100.2%, 2025-04-08, SPY CALL 0.70/0.30 m197	1.19	2,288	229	57%
all names	OOS 2025–26	+6,048.70, +87.0%, 2025-04-11, SPXW CALL 0.70/0.30 m99	−7,091.30, −100.0%, 2025-04-10, SPXW PUT 0.70/0.30 m176	1.03	32,600	264	304%

Random baseline: not available — no random baseline was computed by this study

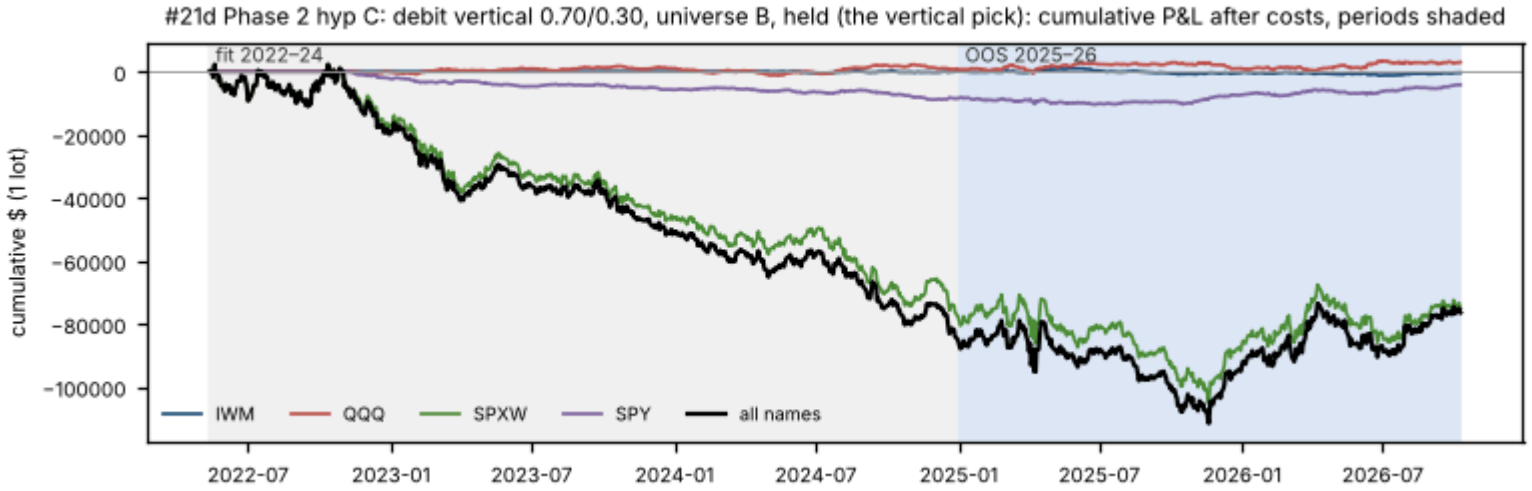
Exit reasons (share of trades):

- fit 2022–24: settle 100.0%
- OOS 2025–26: settle 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	173	54.9	−0.05	+0.84	+145
2025	IWM	245	47.8	−8.11	−1.75	−428
2026	IWM	194	52.1	+1.35	+0.08	+15
2022	QQQ	33	42.4	−2.60	−4.38	−144
2023	QQQ	248	51.6	+0.41	+4.88	+1,209
2024	QQQ	249	49.8	−1.46	−1.25	−311
2025	QQQ	247	50.6	−0.78	+0.91	+224
2026	QQQ	194	52.6	+6.14	+10.44	+2,025
2022	SPXW	159	47.2	−6.17	−110.29	−17,536
2023	SPXW	248	46.0	−8.83	−118.30	−29,338
2024	SPXW	249	43.8	−9.34	−123.04	−30,638
2025	SPXW	247	49.8	−2.42	−38.62	−9,539
2026	SPXW	194	54.1	+1.84	+62.30	+12,087
2022	SPY	33	30.3	−31.13	−65.75	−2,170
2023	SPY	248	42.7	−14.59	−14.39	−3,569
2024	SPY	249	47.0	−8.88	−11.09	−2,761
2025	SPY	246	51.6	+2.65	+4.53	+1,115
2026	SPY	194	56.2	+13.01	+16.59	+3,219
2022	all names	225	44.0	−9.31	−88.22	−19,850

year	name	trades	win %	mean %	mean $	total $
2023	all names	744	46.8	−7.67	−42.61	−31,699
2024	all names	920	48.4	−5.34	−36.48	−33,565
2025	all names	985	49.9	−2.16	−8.76	−8,629
2026	all names	776	53.7	+5.59	+22.35	+17,346



Amendments, deviations and integrity checks. —

What it showed. Verticals lose less than calls and puts, and still lose.

#22 Phase 2 hyp D: volatility breakout (straddles on the model’s top decile)

Folder: `research_notes/Compass phase 2/hyp_d_vol_breakout`. **Commits:** out/preregistered.md / — / da02507. **Dates:** 2026-10-21.

Hypothesis. The volatility model’s top decile marks straddles worth buying.

Rule. Buy straddles/strangles on the top-decile scan minutes.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry

Data. SPY/QQQ/IWM/SPXW 2022–2026.

Periods and holdouts. fit / cut / OOS (read once).

Pass rule (as pre-registered). Best-on-fit cell positive OOS.

Verdict. No edge: the best-on-fit cell flipped from +4.6% to −7.0%; the model forecasts volatility (AUC 0.71–0.73) but the straddle is already priced richer.

Metrics. not available — committed as cells by scan minute and decile (cells_*.csv, minutes_*.parquet); no trade list of the pre-registered cell.

Amendments, deviations and integrity checks. —

What it showed. Forecasting volatility is not trading it.

#24 Phase 3 hyp E: short-premium lift of the vol model’s quiet deciles

Folder: `research_notes/Compass phase 3/hyp_e_short_premium_lift`. **Commits:** out/preregistered.md / — / 02d0dcd. **Dates:** 2026-10-21.

Hypothesis. Quiet deciles lift the SPY short-premium spread over selling unconditionally.

Rule. Sell the SPY spread on the bottom deciles vs unconditionally, same minute.

Fills and costs. $0.65 a contract (leg) to open, nothing at expiry

Data. SPY 2022-11 → 2026-10.

Periods and holdouts. fit / OOS.

Pass rule (as pre-registered). A pre-registrable lift on fit.

Verdict. No edge established — not pre-registrable (no fit cell clear of zero); the D1 condor read later became condor_d20 (#26).

Metrics. not available — committed as by-scan cells and pooled tables; the candidate was set after the OOS read (see #26 condor_d20 for its trades).

Amendments, deviations and integrity checks. —

What it showed. No lift on fit.

#25 Phase 3 hyp F: the 15:30 SPY/QQQ long call ±5 butterfly (cash account)

Folder: `research_notes/Compass phase 3/hyp_f_cash_butterfly`. **Commits:** out/preregistered.md (20:44:54 UTC) / — / f2883c0; report b7ffa46. **Dates:** 2026-10-21.

Hypothesis. The cash-account call fly at 15:30 keeps hyp C's OOS edge under realistic complex-order fills.

Rule. 15:30, SPY and QQQ, long call butterfly ±5 around the ATM strike at the 4-leg NBBO package, held to settlement.

Fills and costs. $0.65 a leg, four legs to open, none at expiry; package at the NBBO (k = 0).

Data. SPY, QQQ 2022-11-11 → 2026-10-09.

Periods and holdouts. fit 2022-11 → 2024 / OOS 2025 → 2026-10-09 (the same OOS C read).

Pass rule (as pre-registered). Four parts: OOS mean > 0 with its interval clear of zero at k = 0; also under $0.01 a leg worse; positive without the best five sessions; both names positive.

Verdict. Passes all four, but the cell was named after C's OOS was printed: a conditionally justified rule, to be confirmed forward (≈ 450 sessions); dies at $0.02 a leg.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	fit 2022-11→2024	526	526 / 553	65.2	+13.3, +48.94	−16.6, −67.69	+5.1, +22.40	**+2.93**, +8.36	[+1.37, +5.42]; [+3.16, +16.58]
SPY	fit 2022-11→2024	522	522 / 553	63.8	+11.5, +44.54	−15.9, −64.33	+4.3, +18.40	**+1.56**, +5.12	[+0.11, +3.53]; [−0.53, +12.46]
all names	fit 2022-11→2024	1,048	529 / 553	64.5	+12.4, +46.77	−16.2, −65.98	+4.6, +19.40	**+2.25**, +6.75	[+0.83, +4.35]; [+1.80, +13.97]
QQQ	OOS 2025–26	439	439 / 457	60.1	+21.1, +66.53	−23.0, −83.58	+4.7, +18.40	**+3.48**, +6.69	[+0.65, +6.32]; [−2.40, +15.17]
SPY	OOS 2025–26	440	440 / 457	64.3	+17.1, +58.78	−18.9, −69.29	+5.9, +24.40	**+4.22**, +13.08	[+2.11, +6.41]; [+6.21, +19.91]
all names	OOS 2025–26	879	441 / 457	62.2	+19.0, +62.52	−21.1, −76.83	+5.2, +21.90	**+3.85**, +9.89	[+1.49, +6.20]; [+2.36, +16.75]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	fit 2022-11→2024	+287.40, +208.3%, 2024-08-28, QQQ 1530 call fly K0 472.0 ±5	−407.60, −100.6%, 2024-12-18, QQQ 1530 call fly K0 522.0 ±5	1.35	1,323	48	24%
SPY	fit 2022-11→2024	+226.40, +111.0%, 2024-08-28, SPY 1530 call fly K0 559.0 ±5	−422.60, −100.6%, 2024-05-31, SPY 1530 call fly K0 522.0 ±5	1.22	1,569	214	33%
all names	fit 2022-11→2024	+287.40, +208.3%, 2024-08-28, QQQ 1530 call fly K0 472.0 ±5	−422.60, −100.6%, 2024-05-31, SPY 1530 call fly K0 522.0 ±5	1.29	2,196	97	26%
QQQ	OOS 2025–26	+259.40, +146.6%, 2026-04-29, QQQ 1530 call fly K0 661.0 ±5	−334.60, −100.8%, 2025-03-04, QQQ 1530 call fly K0 502.0 ±5	1.20	1,654	161	41%
SPY	OOS 2025–26	+235.40, +103.7%, 2025-11-19, SPY 1530 call fly K0 663.0 ±5	−292.60, −74.5%, 2025-05-22, SPY 1530 call fly K0 587.0 ±5	1.53	1,121	82	19%
all names	OOS 2025–26	+259.40, +146.6%, 2026-04-29, QQQ 1530 call fly K0 661.0 ±5	−334.60, −100.8%, 2025-03-04, QQQ 1530 call fly K0 502.0 ±5	1.34	2,519	90	25%

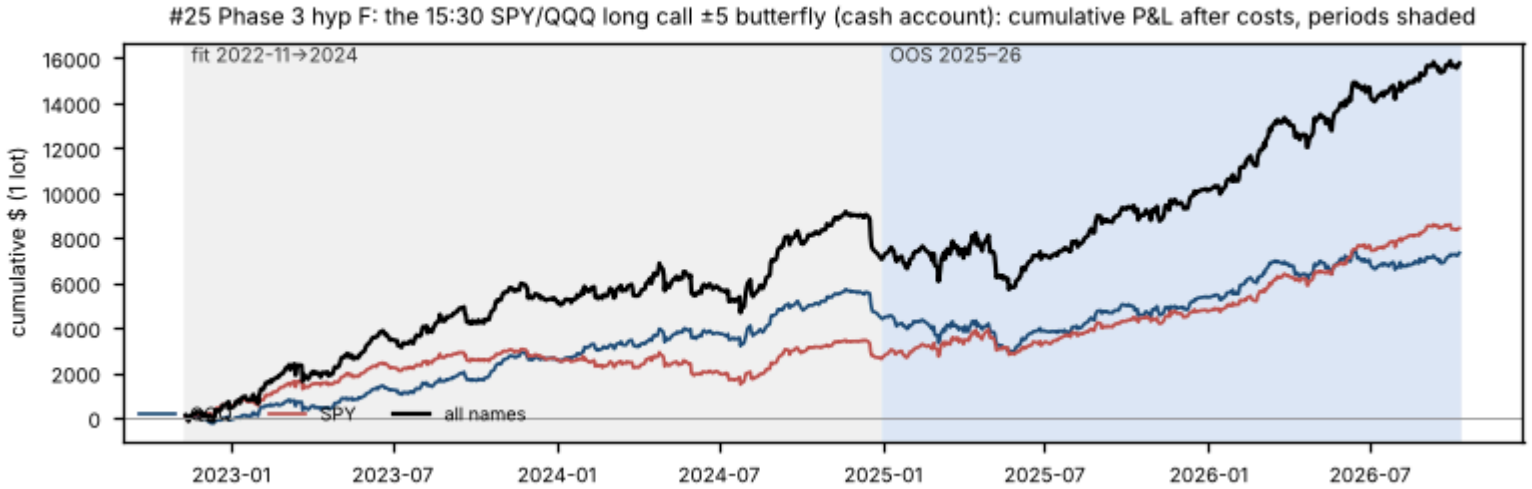
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- fit 2022-11→2024: held to settlement 100.0%
- OOS 2025–26: held to settlement 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ	33	60.6	−0.39	−3.25	−107
2023	QQQ	248	66.1	+3.12	+11.20	+2,778
2024	QQQ	245	64.9	+3.18	+7.05	+1,727
2025	QQQ	247	58.7	+2.84	+3.99	+986
2026	QQQ	192	62.0	+4.30	+10.15	+1,950
2022	SPY	32	71.9	+4.66	+16.35	+523
2023	SPY	248	66.1	+2.33	+8.65	+2,144
2024	SPY	242	60.3	+0.37	+0.02	+6
2025	SPY	246	63.0	+2.96	+8.36	+2,056
2026	SPY	194	66.0	+5.83	+19.07	+3,700
2022	all names	65	66.2	+2.10	+6.40	+416
2023	all names	496	66.1	+2.72	+9.92	+4,922
2024	all names	487	62.6	+1.78	+3.56	+1,733
2025	all names	493	60.9	+2.90	+6.17	+3,043
2026	all names	386	64.0	+5.07	+14.64	+5,649



Amendments, deviations and integrity checks. The selection is not independent of the OOS it was read on (stated in the report).

What it showed. +3.85% of risk a fly OOS, carried by the same data that chose it.

#26a Paper tracker: condor_em1 (SPY 2-wide iron condor at ± the ATM straddle, 11:30–13:30)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b. **Dates:** 2026-10-21 → 22.

Hypothesis. The midday SPY condor of hyp B (#20b), all-VIX form, pays forward.

Rule. 11:30, 12:00, 12:30, 13:00, 13:30: SPY 2-wide iron condor, shorts the first listed strikes outside spot ± the ATM straddle, held to the official close.

Fills and costs. Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i development 2022-11-11 → 2024-12-31 / ii selection 2025-01-02 → 2026-10-09 (read) / iii forward 2026-10-12 → (untouched). No historical period is untouched for any candidate (AUDIT.md §5).

Pass rule (as pre-registered). Confirmation: 60 → 120 forward sessions (the candidate's threshold).

Verdict. Historical periods through zero (i +3.47 [−0.04, +6.68], ii +4.10 [−0.51, +8.15]); the 15:58 close-out turns it negative; forward pending.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	i	2,358	525 / 553	80.2	+14.2, +23.96	−46.7, −79.50	+10.9, +19.40	**+2.16**, +3.47	[+0.08, +4.18]; [−0.04, +6.89]
SPY	ii	2,068	439 / 457	80.4	+17.5, +28.47	−59.1, −95.95	+13.4, +23.40	**+2.46**, +4.10	[−0.54, +5.02]; [−0.68, +8.15]
SPY	iii	44	9 / 9	86.4	+15.5, +25.93	−55.1, −90.93	+13.1, +22.90	**+5.84**, +9.99	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	i	+66.40, +50.7%, 2022-12-14, SPY 12:30 −C411/−C413/−P396/−P394	−191.60, −101.4%, 2023-07-07, SPY 13:30 −C444/−C446/−P441/−P439	1.22	2,849	165	17%
SPY	ii	+63.40, +47.3%, 2026-07-29, SPY 13:00 −C741/−C743/−P729/−P727	−187.60, −101.4%, 2025-04-09, SPY 13:30 −C536/−C538/−P512/−P510	1.22	2,910	81	17%
SPY	iii	+44.40, +29.0%, 2026-10-22, SPY 11:30 −C767/−C769/−P760/−P758	−170.60, −101.5%, 2026-10-16, SPY 11:30 −C772/−C774/−P766/−P764	1.81	455	4	170%

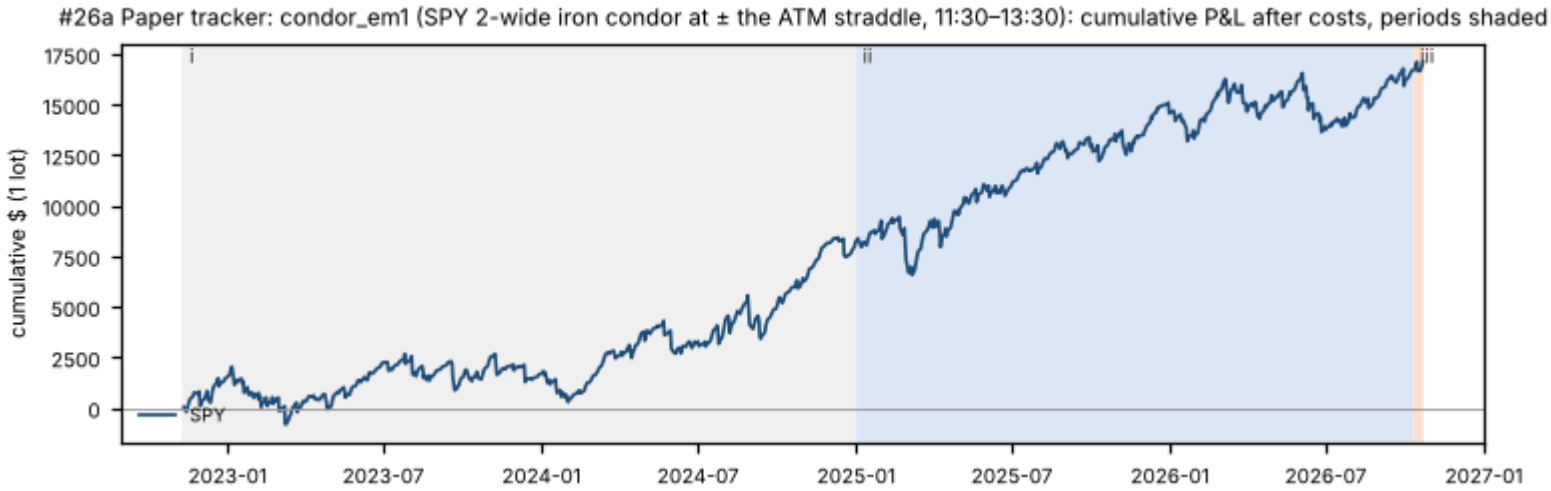
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- i: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%
- iii: held to settlement (official close) 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	159	81.8	+5.54	+9.38	+1,492
2023	SPY	1,080	77.9	+0.20	+0.16	+170
2024	SPY	1,119	82.2	+3.57	+5.84	+6,532
2025	SPY	1,129	81.4	+3.53	+5.65	+6,378
2026	SPY	983	79.6	+1.39	+2.59	+2,549



Amendments, deviations and integrity checks. Deterministic (a cold replay equals the committed CSV byte for byte); 100 trades re-derived, 0 differences; re-settled on the official close (12,338 of 20,469 trades changed).

What it showed. A few dollars a condor in both historical periods, inside the noise.

#26b Paper tracker: fly5 (SPY/QQQ ±5 butterflies at 15:00 and 15:30: iron, call, put)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b. **Dates:** 2026-10-21 → 22.

Hypothesis. The late-day ±5 fly of hyp C/F pays forward.

Rule. 15:00 and 15:30: SPY and QQQ ±5 iron, call and put butterflies (one trade seen three ways), held to the official close.

Fills and costs. Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i development 2022-11-11 → 2024-12-31 / ii selection 2025-01-02 → 2026-10-09 (read) / iii forward 2026-10-12 → (untouched). No historical period is untouched for any candidate (AUDIT.md §5).

Pass rule (as pre-registered). Confirmation: 250 forward sessions a name.

Verdict. Iron clear of zero in i and ii (+16.59, +16.20 a trade) but chosen on ii; hold-to-expiry leaves 100 shares a fly; the after-hours rule (#27) DROPS the iron as a SPY/QQQ hold-to-expiry strategy.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ call	i	1,049	528 / 553	66.7	+14.6, +52.97	−18.2, −70.84	+6.0, +25.40	**+3.69**, +11.78	[+2.25, +6.00]; [+6.99, +19.03]
QQQ iron	i	1,049	528 / 553	69.5	+15.8, +56.79	−18.3, −69.14	+7.6, +31.40	**+5.39**, +18.38	[+3.81, +7.82]; [+13.64, +25.55]
QQQ put	i	1,051	529 / 553	65.8	+14.5, +52.59	−17.8, −69.16	+6.0, +24.40	**+3.45**, +11.00	[+2.03, +5.76]; [+6.42, +18.23]
SPY call	i	1,046	525 / 553	64.1	+13.2, +50.06	−17.4, −68.57	+4.9, +20.40	**+2.24**, +7.42	[+0.85, +4.08]; [+2.07, +13.97]
SPY iron	i	1,046	525 / 553	69.0	+14.1, +52.81	−18.3, −69.92	+6.9, +28.40	**+4.08**, +14.80	[+2.77, +5.88]; [+10.06, +20.73]
SPY put	i	1,046	525 / 553	60.7	+13.1, +49.26	−16.6, −66.31	+3.8, +15.40	**+1.41**, +3.85	[−0.16, +3.37]; [−2.21, +11.04]
all names	i	6,287	530 / 553	66.0	+14.2, +52.51	−17.7, −68.89	+5.8, +24.40	**+3.38**, +11.21	[+2.05, +5.40]; [+6.67, +17.68]
QQQ call	ii	877	439 / 457	57.5	+23.1, +71.82	−26.1, −90.64	+4.3, +18.40	**+2.20**, +2.72	[−0.17, +4.54]; [−5.48, +10.66]
QQQ iron	ii	877	439 / 457	62.7	+25.0, +76.77	−26.7, −87.04	+7.8, +31.40	**+5.71**, +15.69	[+3.47, +7.94]; [+7.95, +22.93]
QQQ put	ii	877	439 / 457	56.4	+22.9, +71.07	−25.9, −90.24	+3.6, +15.40	**+1.68**, +0.81	[−0.72, +3.99]; [−7.46, +8.43]
SPY call	ii	880	441 / 457	61.6	+19.6, +65.60	−22.0, −75.18	+5.9, +24.40	**+3.65**, +11.52	[+1.91, +5.44]; [+5.61, +17.89]
SPY iron	ii	880	441 / 457	64.2	+20.4, +67.58	−22.3, −74.53	+7.2, +28.40	**+5.12**, +16.71	[+3.41, +6.91]; [+10.63, +23.00]
SPY put	ii	880	441 / 457	60.8	+19.6, +65.46	−21.8, −75.16	+5.6, +22.40	**+3.37**, +10.33	[+1.63, +5.21]; [+4.17, +16.78]
all names	ii	5,271	441 / 457	60.5	+21.8, +69.68	−24.2, −82.48	+5.7, +22.40	**+3.62**, +9.64	[+1.77, +5.47]; [+3.12, +15.94]
QQQ call	iii	18	9 / 9	77.8	+36.0, +108.26	−20.5, −77.10	+27.0, +79.90	**+23.42**, +67.07	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ iron	iii	18	9 / 9	77.8	+38.0, +112.76	−19.0, −69.60	+29.0, +86.90	**+25.29**, +72.23	n/a (< 60 sessions); n/a (< 60 sessions)
QQQ put	iii	18	9 / 9	77.8	+36.2, +108.40	−19.7, −73.10	+27.5, +78.40	**+23.75**, +68.07	n/a (< 60 sessions); n/a (< 60 sessions)
SPY call	iii	18	9 / 9	61.1	+22.0, +79.49	−9.8, −39.89	+13.8, +46.90	**+9.63**, +33.07	n/a (< 60 sessions); n/a (< 60 sessions)
SPY iron	iii	18	9 / 9	61.1	+22.2, +80.04	−9.6, −39.31	+13.9, +51.90	**+9.83**, +33.62	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY put	iii	18	9 / 9	61.1	+22.1, +79.67	−9.7, −39.60	+13.8, +51.40	**+9.72**, +33.29	n/a (< 60 sessions); n/a (< 60 sessions)
all names	iii	108	9 / 9	69.4	+30.3, +96.57	−13.4, −51.84	+17.5, +64.90	**+16.94**, +51.22	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ call	i	+294.40, +213.3%, 2024-08-28, QQQ 15:30 −C467/−C472/−C477	−407.60, −100.6%, 2024-12-18, QQQ 15:30 −C517/−C522/−C527	1.50	1,866	38	16%
QQQ iron	i	+296.40, +217.9%, 2024-08-28, QQQ 15:00 −C472/−P472/−P467/−C477	−351.60, −100.7%, 2024-07-25, QQQ 15:30 −C464/−P464/−P459/−C469	1.87	1,406	36	11%
QQQ put	i	+293.40, +211.1%, 2024-08-28, QQQ 15:00 −P467/−P472/−P477	−374.60, −100.7%, 2024-07-25, QQQ 15:30 −P459/−P464/−P469	1.47	1,702	38	18%
SPY call	i	+271.40, +138.5%, 2024-08-28, SPY 15:00 −C553/−C558/−C563	−422.60, −100.6%, 2024-05-31, SPY 15:30 −C517/−C522/−C527	1.30	2,046	195	23%
SPY iron	i	+273.40, +140.9%, 2024-08-28, SPY 15:00 −C558/−P558/−P553/−C563	−401.60, −100.7%, 2024-05-31, SPY 15:30 −C522/−P522/−P517/−C527	1.68	1,070	39	12%
SPY put	i	+271.40, +138.5%, 2024-08-28, SPY 15:00 −P553/−P558/−P563	−407.60, −100.6%, 2024-05-31, SPY 15:00 −P516/−P521/−P526	1.15	4,148	284	44%
all names	i	+296.40, +217.9%, 2024-08-28, QQQ 15:00 −C472/−P472/−P467/−C477	−422.60, −100.6%, 2024-05-31, SPY 15:30 −C517/−C522/−C527	1.48	8,394	53	16%
QQQ call	ii	+278.40, +171.9%, 2026-04-29, QQQ 15:00 −C656/−C661/−C666	−470.60, −100.6%, 2026-05-12, QQQ 15:00 −C697/−C702/−C707	1.07	3,590	184	95%
QQQ iron	ii	+280.40, +175.2%, 2026-04-29, QQQ 15:00 −C661/−P661/−P656/−C666	−349.60, −100.7%, 2026-03-09, QQQ 15:00 −C600/−P600/−P595/−C605	1.48	1,767	81	17%
QQQ put	ii	+278.40, +171.9%, 2026-04-29, QQQ 15:00 −P656/−P661/−P666	−365.60, −100.7%, 2026-03-09, QQQ 15:00 −P595/−P600/−P605	1.02	3,670	296	323%
SPY call	ii	+269.40, +120.8%, 2026-09-16, SPY 15:00 −C749/−C754/−C759	−344.60, −100.8%, 2026-03-09, SPY 15:00 −C665/−C670/−C675	1.40	1,765	99	18%
SPY iron	ii	+269.40, +120.8%, 2026-09-16, SPY 15:00 −C754/−P754/−P749/−C759	−343.60, −100.8%, 2026-03-09, SPY 15:00 −C670/−P670/−P665/−C675	1.63	1,299	35	13%
SPY put	ii	+268.40, +119.8%, 2026-09-16, SPY 15:00 −P749/−P754/−P759	−348.60, −100.8%, 2026-03-09, SPY 15:00 −P665/−P670/−P675	1.35	1,961	135	21%
all names	ii	+280.40, +175.2%, 2026-04-29, QQQ 15:00 −C661/−P661/−P656/−C666	−470.60, −100.6%, 2026-05-12, QQQ 15:00 −C697/−C702/−C707	1.30	12,048	107	24%
QQQ call	iii	+175.40, +72.5%, 2026-10-21, QQQ 15:30 −C721/−C726/−C731	−153.60, −43.8%, 2026-10-15, QQQ 15:00 −C720/−C725/−C730	4.91	43	1	100%
QQQ iron	iii	+179.40, +75.4%, 2026-10-21, QQQ 15:30 −C726/−P726/−P721/−C731	−143.60, −42.1%, 2026-10-15, QQQ 15:00 −C725/−P725/−P720/−C730	5.67	32	1	97%
QQQ put	iii	+176.40, +73.2%, 2026-10-21, QQQ 15:30 −P721/−P726/−P731	−147.60, −42.8%, 2026-10-15, QQQ 15:00 −P720/−P725/−P730	5.19	42	1	99%
SPY call	iii	+156.40, +47.1%, 2026-10-21, SPY 15:30 −C768/−C773/−C778	−71.60, −17.2%, 2026-10-19, SPY 15:30 −C763/−C768/−C773	3.13	121	2	125%
SPY iron	iii	+156.40, +47.1%, 2026-10-21, SPY 15:30 −C773/−P773/−P768/−C778	−73.60, −17.6%, 2026-10-19, SPY 15:30 −C768/−P768/−P763/−C773	3.20	124	2	125%

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY put	iii	+154.40, +46.2%, 2026-10-21, SPY 15:30 −P768/−P773/−P778	−74.60, −17.8%, 2026-10-19, SPY 15:30 −P763/−P768/−P773	3.16	125	2	126%
all names	iii	+179.40, +75.4%, 2026-10-21, QQQ 15:30 −C726/−P726/−P721/−C731	−153.60, −43.8%, 2026-10-15, QQQ 15:00 −C720/−C725/−C730	4.23	324	1	98%

Random baseline: not available — no random baseline was computed by this study

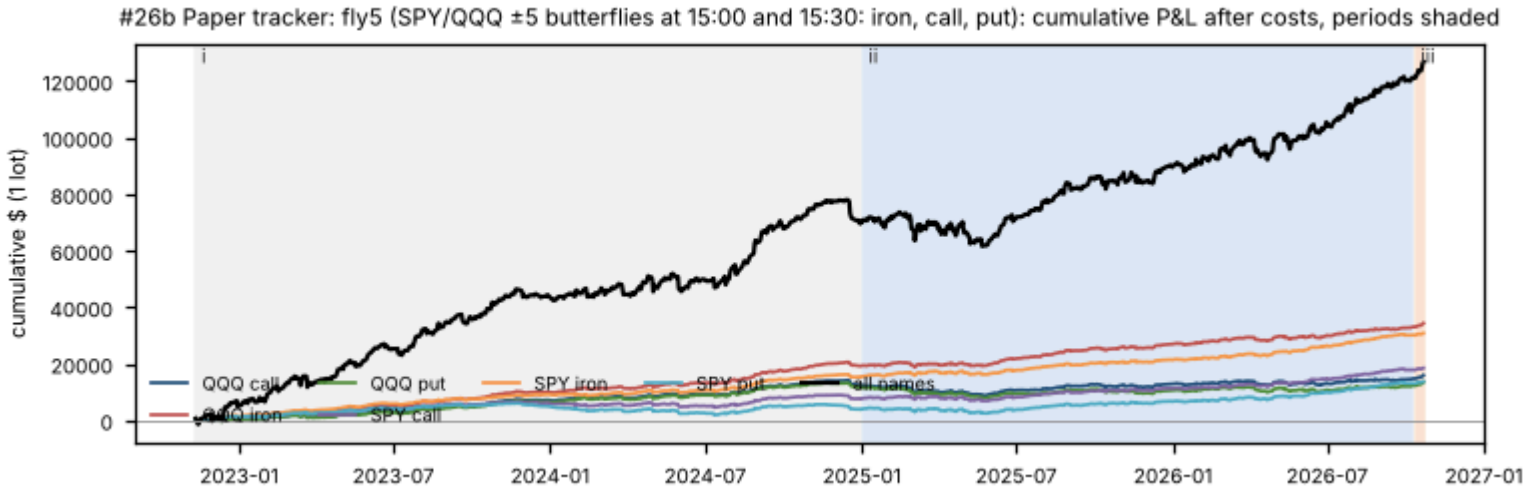
Exit reasons (share of trades):

- i: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%
- iii: held to settlement (official close) 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	QQQ call	65	66.2	+2.16	+3.75	+244
2023	QQQ call	495	67.7	+3.74	+14.07	+6,965
2024	QQQ call	489	65.8	+3.83	+10.53	+5,150
2025	QQQ call	493	57.2	+1.57	+0.95	+470
2026	QQQ call	402	58.7	+3.93	+7.78	+3,126
2022	QQQ iron	65	72.3	+3.99	+11.29	+734
2023	QQQ iron	495	69.5	+4.98	+18.83	+9,320
2024	QQQ iron	489	69.1	+6.00	+18.86	+9,224
2025	QQQ iron	493	63.7	+5.45	+15.54	+7,659
2026	QQQ iron	402	62.2	+6.91	+18.40	+7,399
2022	QQQ put	65	67.7	+2.01	+3.08	+200
2023	QQQ put	495	67.3	+3.51	+13.12	+6,492
2024	QQQ put	491	64.2	+3.58	+9.92	+4,872
2025	QQQ put	493	54.4	+0.76	−2.05	−1,013
2026	QQQ put	402	60.0	+3.80	+7.33	+2,948
2022	SPY call	65	69.2	+5.70	+17.31	+1,125
2023	SPY call	495	65.9	+2.68	+10.23	+5,065
2024	SPY call	486	61.5	+1.33	+3.23	+1,567
2025	SPY call	493	59.8	+2.22	+7.00	+3,453
2026	SPY call	405	63.7	+5.65	+17.99	+7,284
2022	SPY iron	65	72.3	+7.07	+21.86	+1,421
2023	SPY iron	495	70.5	+4.06	+15.77	+7,806
2024	SPY iron	486	67.1	+3.69	+12.86	+6,250
2025	SPY iron	493	62.1	+3.65	+11.92	+5,877
2026	SPY iron	405	66.7	+7.12	+23.29	+9,433
2022	SPY put	65	69.2	+5.59	+16.82	+1,093

year	name	trades	win %	mean %	mean $	total $
2023	SPY put	495	64.0	+2.14	+7.92	+3,918
2024	SPY put	486	56.2	+0.10	−2.03	−985
2025	SPY put	493	58.4	+1.82	+5.30	+2,612
2026	SPY put	405	63.7	+5.54	+17.47	+7,077
2022	all names	390	69.5	+4.42	+12.35	+4,817
2023	all names	2,970	67.5	+3.52	+13.32	+39,566
2024	all names	2,927	64.0	+3.09	+8.91	+26,079
2025	all names	2,958	59.3	+2.58	+6.44	+19,059
2026	all names	2,421	62.5	+5.49	+15.39	+37,266



Amendments, deviations and integrity checks. As 26a.

What it showed. Positive historically, but on the period that chose it, and not operationally clean.

#26c Paper tracker: condor_d20 (SPY 2-wide 20-delta iron condor, 13:00–15:00)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** defaults ba117a7, 6826b98 / official-close settlement a4ebe90, 99a3d4e, 16eb3c0 / replay 94af538, 30bb73e; audit 1ca1f0e; forward d7115b6, 2a7bd3b; candidate 3 99a3d4e. **Dates:** 2026-10-21 → 22.

Hypothesis. The 20-delta SPY condor (from hyp E's D1 read) pays forward.

Rule. 13:00, 13:30, 14:00, 14:30, 15:00: SPY 2-wide iron condor at 20 delta, held to the official close.

Fills and costs. Shorts at the bid, longs at the ask; $0.65 a contract (leg) to open, nothing at expiry.

Data. The replay of every stored session through the live tracker's code, settled on the official close (data_recorder/paper_backtest_official, git-ignored, every trade).

Periods and holdouts. i-a model training 2022-11 → 2024-06 (in-sample) / i-b decile cuts 2024-07 → 12 / ii selection 2025 → 2026-10-09 / iii forward.

Pass rule (as pre-registered). Confirmation: 150 forward sessions for the D1 re-test.

Verdict. Positive in ii (+7.67 [+3.87, +10.89]) but set after ii was read; the after-hours rule DROPS it as a hold-to-expiry strategy.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	i-a	1,823	403 / 422	76.9	+14.9, +24.87	−42.8, −72.11	+11.5, +20.40	**+1.53**, +2.42	[−0.37, +3.11]; [−0.79, +5.00]
SPY	i-b	572	125 / 131	83.9	+16.5, +27.06	−53.3, −87.05	+13.1, +22.40	**+5.25**, +8.71	[+2.06, +9.54]; [+3.44, +15.69]
SPY	ii	2,090	440 / 457	82.1	+17.0, +27.70	−51.5, −83.92	+13.4, +23.40	**+4.68**, +7.67	[+2.43, +6.59]; [+3.97, +10.78]
SPY	iii	41	9 / 9	95.1	+16.2, +26.94	−27.7, −46.10	+16.1, +27.40	**+14.02**, +23.38	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	i-a	+69.40, +54.2%, 2022-12-14, SPY 13:30 −C411/−C413/−P397/−P395	−191.60, −101.4%, 2023-07-07, SPY 13:30 −C444/−C446/−P441/−P439	1.14	1,769	75	28%
SPY	i-b	+63.40, +47.3%, 2024-09-18, SPY 13:00 −C569/−C571/−P557/−P555	−185.60, −101.4%, 2024-07-12, SPY 15:00 −C564/−C566/−P562/−P560	1.62	1,618	29	26%
SPY	ii	+60.40, +44.1%, 2025-04-10, SPY 13:30 −C530/−C532/−P509/−P507	−189.60, −101.4%, 2026-04-13, SPY 14:00 −C684/−C686/−P680/−P678	1.51	2,292	39	8%
SPY	iii	+41.40, +26.5%, 2026-10-22, SPY 13:00 −C767/−C769/−P761/−P759	−47.60, −28.7%, 2026-10-15, SPY 14:30 −C771/−C773/−P768/−P766	11.39	0	0	73%

Random baseline: not available — no random baseline was computed by this study

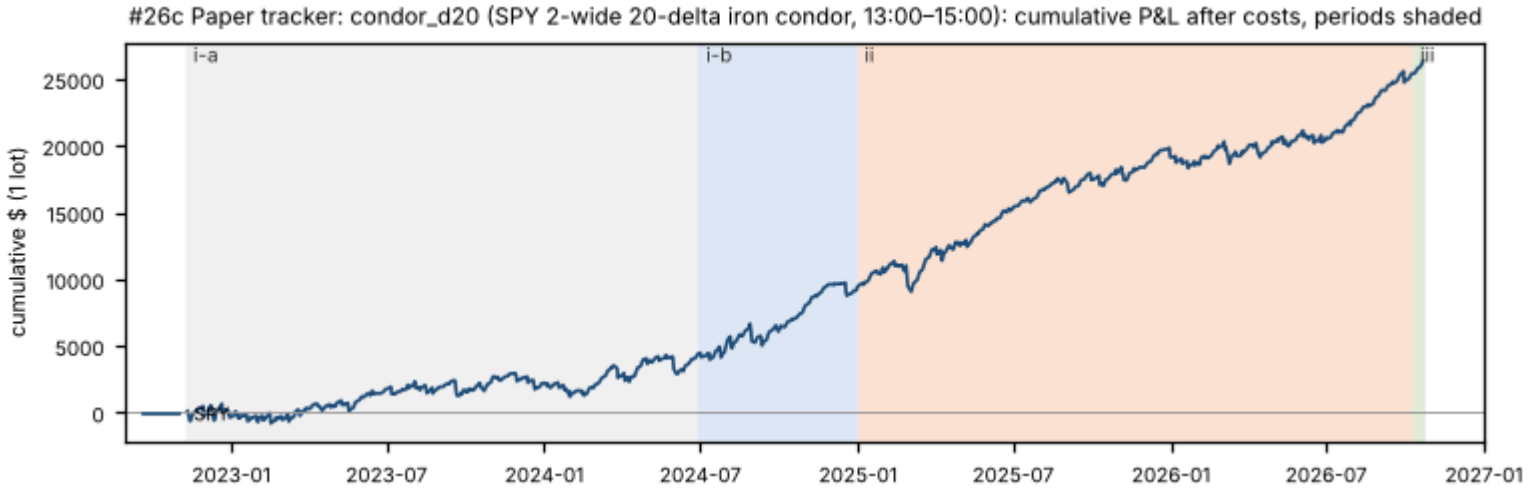
Exit reasons (share of trades):

- i-a: held to settlement (official close) 100.0%
- i-b: held to settlement (official close) 100.0%
- ii: held to settlement (official close) 100.0%

- iii: held to settlement (official close) 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPY	163	77.9	−1.08	−2.26	−369
2023	SPY	1,102	76.0	+1.43	+2.33	+2,569
2024	SPY	1,130	81.2	+3.89	+6.36	+7,190
2025	SPY	1,156	82.1	+5.26	+8.46	+9,784
2026	SPY	975	82.6	+4.38	+7.39	+7,208



Amendments, deviations and integrity checks. As 26a.

What it showed. Positive on the period it was chosen on.

#26d Paper tracker: fly_spxw (SPX ±5 / ±25 flies)

Folder: `scripts/paper_tracker.py; research_notes/Compass phase 3/backtest_replay, backtest_audit, forward_*`. **Commits:** scripts/paper_backtest.py (off unless listed). **Dates:** 2026-10-21.

Hypothesis. The SPXW analogue of fly5.

Rule. 15:00 and 15:30 SPXW ±5 and ±25 flies.

Fills and costs. SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC)

Data. One week, 2026-10-05 → 09.

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Run only as a one-week sanity run outside the repository; replaced by the pre-registered SPXW test (#28).

Metrics. not available — a one-week sanity run outside the repository; no rows were kept.

Amendments, deviations and integrity checks. —

What it showed. No result.

#27a After-hours exercise study: fly5 iron ±5, case (b) the post-market decision

Folder: `research_notes/Compass phase 3/backtest_audit/AFTERHOURS.md`. **Commits:** owner's rule fixed before the read / — / 633fbff. **Dates:** 2026-10-22.

Hypothesis. The SPY/QQQ hold-to-expiry candidates survive after-hours exercise.

Rule. fly5 iron held to expiry; case (b): the counterparty exercises on the 17:30 post-market price, the position re-marked at the next 09:30 open.

Fills and costs. As #26 plus the exercise/assignment fee.

Data. SPY/QQQ, period ii (2025-10-13 → 2026-10-09 overlap) and iii (2026-10-12 → 21).

Periods and holdouts. ii / iii (7 sessions with a next open).

Pass rule (as pre-registered). "If a candidate's mean under (b) is ≤ 0 or its block-20 interval includes zero, drop it as a SPY/QQQ hold-to-expiry strategy."

Verdict. DROP: (b) +22.52 a trade [−20.07, +52.89] in ii.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	ii	492	246 / 258	55.5	+119.5, +393.38	−131.7, −428.79	+15.8, +58.90	**+7.69**, +27.42	[−7.67, +18.13]; [−22.91, +61.48]
SPY	ii	495	248 / 258	56.0	+72.0, +257.24	−80.6, −286.78	+8.9, +36.40	**+4.82**, +17.65	[−6.54, +13.87]; [−20.04, +48.90]
all names	ii	987	248 / 258	55.7	+95.6, +324.81	−106.2, −357.95	+10.5, +41.40	**+6.25**, +22.52	[−5.84, +15.27]; [−18.92, +51.90]
QQQ	iii	14	7 / 7	42.9	+142.0, +504.15	−157.0, −513.29	−52.0, −170.60	**−28.86**, −77.24	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	iii	14	7 / 7	42.9	+55.9, +224.73	−49.7, −179.73	−28.1, −109.10	**−4.44**, −6.39	n/a (< 60 sessions); n/a (< 60 sessions)
all names	iii	28	7 / 7	42.9	+99.0, +364.44	−103.4, −346.51	−28.1, −109.10	**−16.65**, −41.81	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	ii	+2,196.40, +671.7%, 2026-06-22, QQQ 15:00 [["C", 736.0, "short", 1], ["P", 736.0, "short", 1], ["P", 731.0, "long", 1], ["C", 741.0, "long", 1]]	−2,195.60, −681.9%, 2026-04-07, QQQ 15:30 [["C", 585.0, "short", 1], ["P", 585.0, "short", 1], ["P", 580.0, "long", 1], ["C", 590.0, "long", 1]]	1.14	10,408	69	93%
SPY	ii	+1,153.40, +317.7%, 2026-06-22, SPY 15:00 [["C", 744.0, "short", 1], ["P", 744.0, "short", 1], ["P", 739.0, "long", 1], ["C", 749.0, "long", 1]]	−1,761.60, −550.5%, 2026-04-07, SPY 15:30 [["C", 657.0, "short", 1], ["P", 657.0, "short", 1], ["P", 652.0, "long", 1], ["C", 662.0, "long", 1]]	1.14	7,209	88	91%
all names	ii	+2,196.40, +671.7%, 2026-06-22, QQQ 15:00 [["C", 736.0, "short", 1], ["P", 736.0, "short", 1], ["P", 731.0, "long", 1], ["C", 741.0, "long", 1]]	−2,195.60, −681.9%, 2026-04-07, QQQ 15:30 [["C", 585.0, "short", 1], ["P", 585.0, "short", 1], ["P", 580.0, "long", 1], ["C", 590.0, "long", 1]]	1.14	15,715	84	89%
QQQ	iii	+1,066.40, +365.2%, 2026-10-16, QQQ 15:30 [["C", 711.0, "short", 1], ["P", 711.0, "short", 1], ["P", 706.0, "long", 1], ["C", 716.0, "long", 1]]	−875.60, −267.0%, 2026-10-14, QQQ 15:30 [["C", 729.0, "short", 1], ["P", 729.0, "short", 1], ["P", 724.0, "long", 1], ["C", 734.0, "long", 1]]	0.74	2,219	5	total ≤ 0 (best 5 days +1,152; without them −2,233)
SPY	iii	+373.40, +86.4%, 2026-10-13, SPY 15:00 [["C", 778.0, "short", 1], ["P", 778.0, "short", 1], ["P", 773.0, "long", 1], ["C", 783.0, "long", 1]]	−342.60, −108.1%, 2026-10-16, SPY 15:00 [["C", 762.0, "short", 1], ["P", 762.0, "short", 1], ["P", 757.0, "long", 1], ["C", 767.0, "long", 1]]	0.94	1,183	3	total ≤ 0 (best 5 days +845; without them −934)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
all names	iii	+1,066.40, +365.2%, 2026-10-16, QQQ 15:30 [["C", 711.0, "short", 1], ["P", 711.0, "short", 1], ["P", 706.0, "long", 1], ["C", 716.0, "long", 1]]	−875.60, −267.0%, 2026-10-14, QQQ 15:30 [["C", 729.0, "short", 1], ["P", 729.0, "short", 1], ["P", 724.0, "long", 1], ["C", 734.0, "long", 1]]	0.79	3,097	5	total ≤ 0 (best 5 days +1,222; without them −2,393)

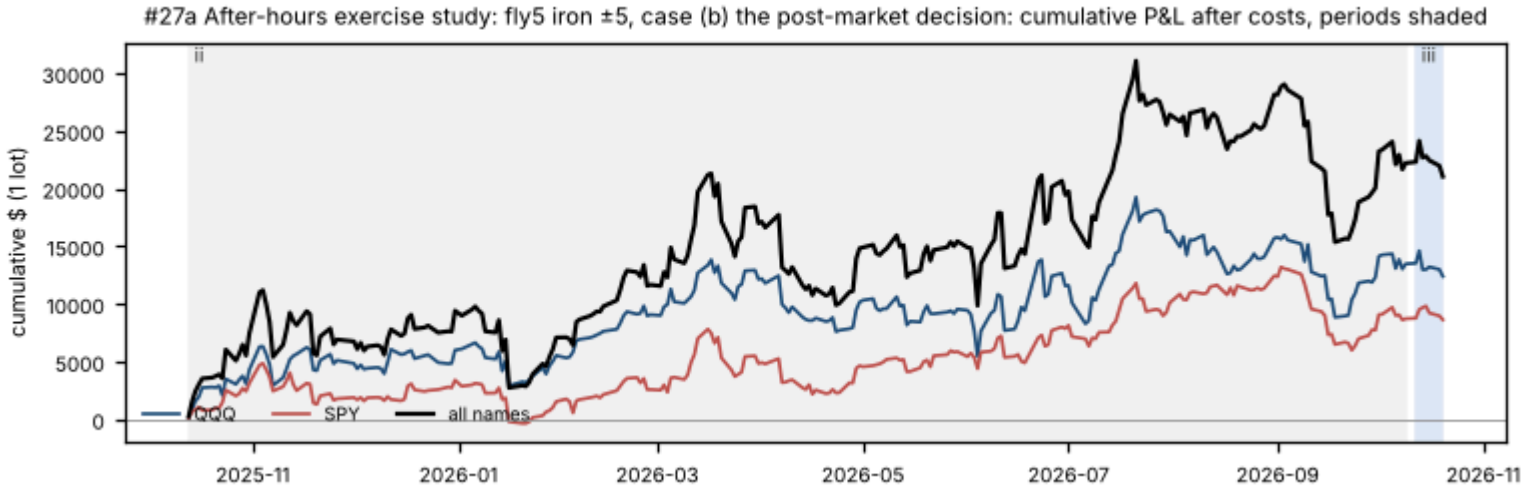
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ii: held; after-hours decision (b) on the 17:30 price 100.0%
- iii: held; after-hours decision (b) on the 17:30 price 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	QQQ	108	56.5	+10.86	+55.66	+6,011
2026	QQQ	398	54.8	+5.54	+16.07	+6,396
2025	SPY	108	59.3	+5.25	+31.63	+3,416
2026	SPY	401	54.6	+4.38	+13.05	+5,232
2025	all names	216	57.9	+8.05	+43.65	+9,427
2026	all names	799	54.7	+4.96	+14.55	+11,627



Amendments, deviations and integrity checks. 16 name-sessions of UW 1-minute post-market bars fetched once with the owner's leave; SPY 2025-04-09 official close flagged.

What it showed. Counting after-hours exercise, the iron fly's mean is inside the noise.

#27b After-hours exercise study: condor_d20, case (b)

Folder: `research_notes/Compass phase 3/backtest_audit/AFTERHOURS.md`. **Commits:** — / — / 633fbff. **Dates:** 2026-10-22.

Hypothesis. As 27a.

Rule. condor_d20 held to expiry, case (b).

Fills and costs. As 27a.

Data. As 27a.

Periods and holdouts. ii / iii.

Pass rule (as pre-registered). As 27a.

Verdict. DROP: (b) +3.84 [−19.94, +22.74] in ii; iii undecided (7 sessions).

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPY	ii	1,176	247 / 258	83.3	+37.0, +60.03	−170.6, −277.09	+14.8, +25.40	**+2.37**, +3.84	[−12.63, +14.57]; [−20.21, +22.96]
SPY	iii	36	8 / 8	94.4	+30.2, +51.93	−85.4, −146.10	+16.1, +27.40	**+23.77**, +40.93	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPY	ii	+867.40, +598.2%, 2026-09-16, SPY 14:00 [["C", 765.0, "short", 1], ["C", 767.0, "long", 1], ["P", 755.0, "short", 1], ["P", 753.0, "long", 1]]	−1,798.60, −1,145.6%, 2026-04-07, SPY 14:30 [["C", 658.0, "short", 1], ["C", 660.0, "long", 1], ["P", 654.0, "short", 1], ["P", 652.0, "long", 1]]	1.08	11,988	132	246%
SPY	iii	+353.40, +211.6%, 2026-10-16, SPY 13:00 [["C", 769.0, "short", 1], ["C", 771.0, "long", 1], ["P", 764.0, "short", 1], ["P", 762.0, "long", 1]]	−192.60, −116.0%, 2026-10-15, SPY 14:30 [["C", 771.0, "short", 1], ["C", 773.0, "long", 1], ["P", 768.0, "short", 1], ["P", 766.0, "long", 1]]	6.04	71	1	108%

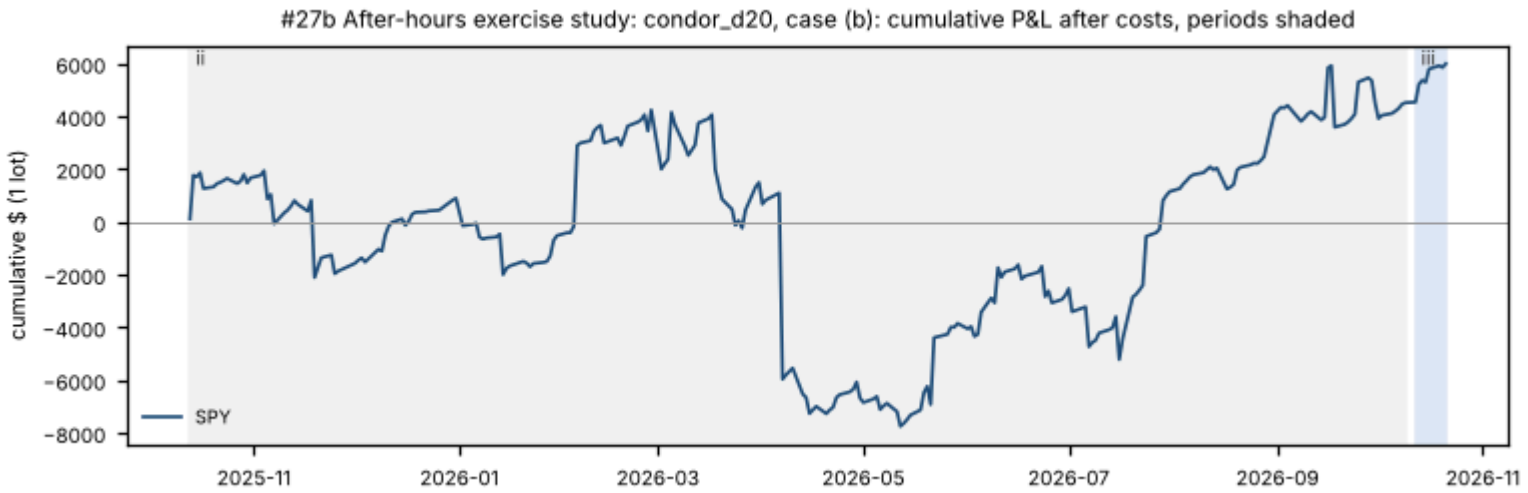
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- ii: held; after-hours decision (b) on the 17:30 price 100.0%
- iii: held; after-hours decision (b) on the 17:30 price 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	SPY	242	84.3	+1.60	+3.72	+900
2026	SPY	970	83.5	+3.36	+5.25	+5,095



Amendments, deviations and integrity checks. As 27a.

What it showed. Inside the noise once exercise is counted.

#28a SPXW analogue A: iron fly ±5×ratio at 15:00 and 15:30 (fly5 iron)

Folder: `research_notes/Compass phase 3/spxw_prereg`. **Commits:** 2820bb4 / — / 6bb230e. **Dates:** 2026-10-22.

Hypothesis. The ETF fly5 iron result transfers to cash-settled SPXW.

Rule. SPXW iron fly, wings ±5 × the SPX/SPY ratio, 15:00 and 15:30, package fill at the NBBO rounded against us to $0.05, held to cash settlement on the official SPX close.

Fills and costs. SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); about $4.93 to open a four-leg structure.

Data. SPXW 1,105 full sessions 2022-05-13 → 2026-10-21.

Periods and holdouts. development 2022-05 → 2024 / selection 2025 → 2026-10-09 / forward 8 sessions.

Pass rule (as pre-registered). (1) selection mean > 0 and block-20 lower bound > 0; (2) selection mean > 0 at +$0.05 package stress; (3) development mean > 0.

Verdict. FAIL: selection −$39.64 a session [−173.06, +87.48].

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	development	1,312	656 / 683	62.6	+12.7, +487.74	−18.3, −708.95	+4.6, +194.57	**+1.07**, +39.89	[−0.13, +2.35]; [−4.35, +88.04]
SPXW	selection	882	441 / 457	58.8	+15.0, +534.43	−22.5, −812.26	+3.5, +149.57	**−0.41**, −19.82	[−2.28, +1.41]; [−87.30, +45.60]
SPXW	forward	16	8 / 8	62.5	+10.9, +432.97	−7.7, −325.26	+1.4, +54.07	**+3.91**, +148.63	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	development	+2,462.89, +123.1%, 2022-12-14, None	−4,034.93, −100.1%, 2024-05-31, None	1.15	30,003	272	29%
SPXW	selection	+2,568.89, +114.4%, 2026-09-16, None	−3,549.93, −100.1%, 2026-03-09, None	0.94	39,380	422	total ≤ 0 (best 5 days +15,018; without them −32,499)
SPXW	forward	+921.07, +25.4%, 2026-10-16, None	−746.93, −17.3%, 2026-10-19, None	2.22	814	2	141%

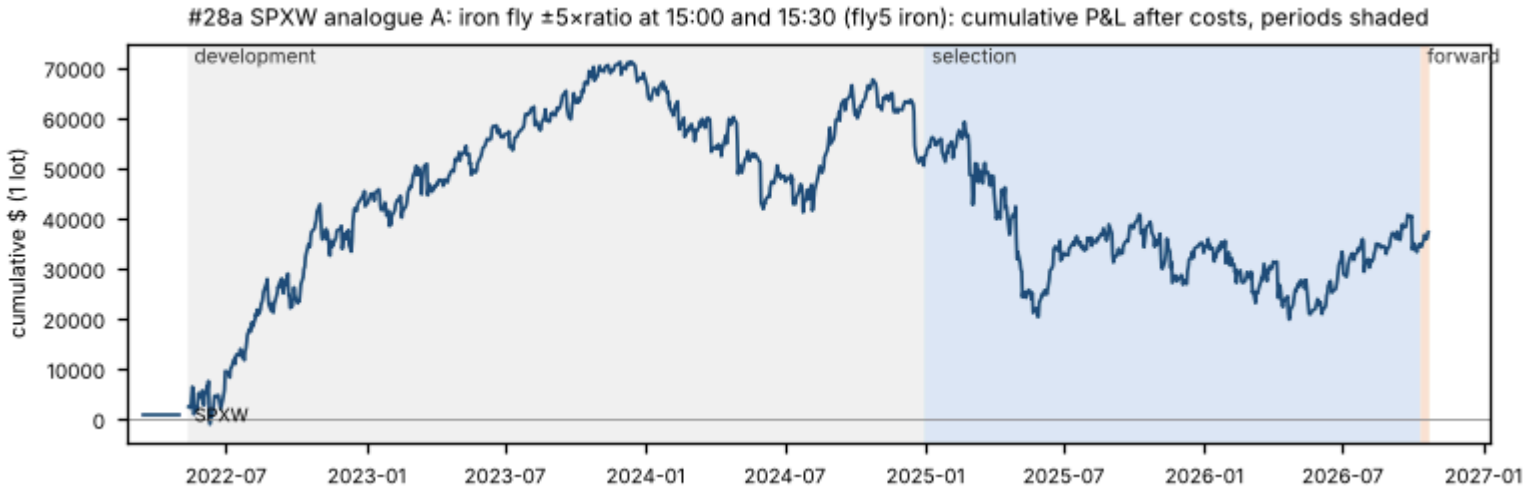
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: held to cash settlement (official SPX close) 100.0%
- selection: held to cash settlement (official SPX close) 100.0%
- forward: held to cash settlement (official SPX close) 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	318	63.5	+3.77	+133.26	+42,378
2023	SPXW	496	64.5	+1.27	+53.62	+26,596
2024	SPXW	498	60.0	−0.86	−33.41	−16,637
2025	SPXW	494	58.9	−0.89	−38.92	−19,225
2026	SPXW	404	58.9	+0.34	+10.20	+4,122



Amendments, deviations and integrity checks. The package rule cost nothing: the summed leg NBBO is always a multiple of $0.05.

What it showed. SPXW did not reproduce the ETF fly.

#28b SPXW analogue B: 20-delta iron condor, wings 2×ratio, 13:00–15:00 (condor_d20)

Folder: `research_notes/Compass phase 3/spxw_prereg`. **Commits:** 2820bb4 / — / 6bb230e. **Dates:** 2026-10-22.

Hypothesis. As 28a for condor_d20.

Rule. SPXW 20-delta iron condor, wings 2 × ratio, five entries 13:00–15:00, package fill, held to cash settlement.

Fills and costs. SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC)

Data. As 28a.

Periods and holdouts. As 28a.

Pass rule (as pre-registered). As 28a.

Verdict. FAIL: selection +$120.76 a session [−95.03, +286.78] — positive but the interval crosses zero.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	development	3,193	656 / 683	74.4	+16.9, +280.66	−47.0, −785.28	+13.0, +230.07	**+0.49**, +7.25	[−1.33, +2.18]; [−22.51, +35.23]
SPXW	selection	2,157	441 / 457	76.8	+17.5, +289.08	−51.4, −849.22	+13.4, +235.07	**+1.49**, +24.69	[−1.12, +3.46]; [−18.86, +57.64]
SPXW	forward	38	8 / 8	89.5	+15.4, +261.83	−15.0, −260.91	+14.3, +250.07	**+12.19**, +206.80	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	development	+704.89, +54.6%, 2022-12-14, SPXW 13:00 3945.0/3965.0P 4105.0/4125.0C	−1,879.75, −100.3%, 2023-12-20, SPXW 13:30 4745.0/4765.0P 4785.0/4805.0C	1.04	56,703	560	55%
SPXW	selection	+604.89, +43.5%, 2026-07-29, SPXW 13:00 7290.0/7310.0P 7430.0/7450.0C	−1,849.93, −100.3%, 2026-04-13, SPXW 14:30 6815.0/6835.0P 6860.0/6880.0C	1.13	35,113	171	24%
SPXW	forward	+434.89, +27.9%, 2026-10-16, SPXW 13:30 7620.0/7640.0P 7695.0/7715.0C	−645.93, −37.2%, 2026-10-15, SPXW 14:30 7685.0/7705.0P 7735.0/7755.0C	8.53	0	0	81%

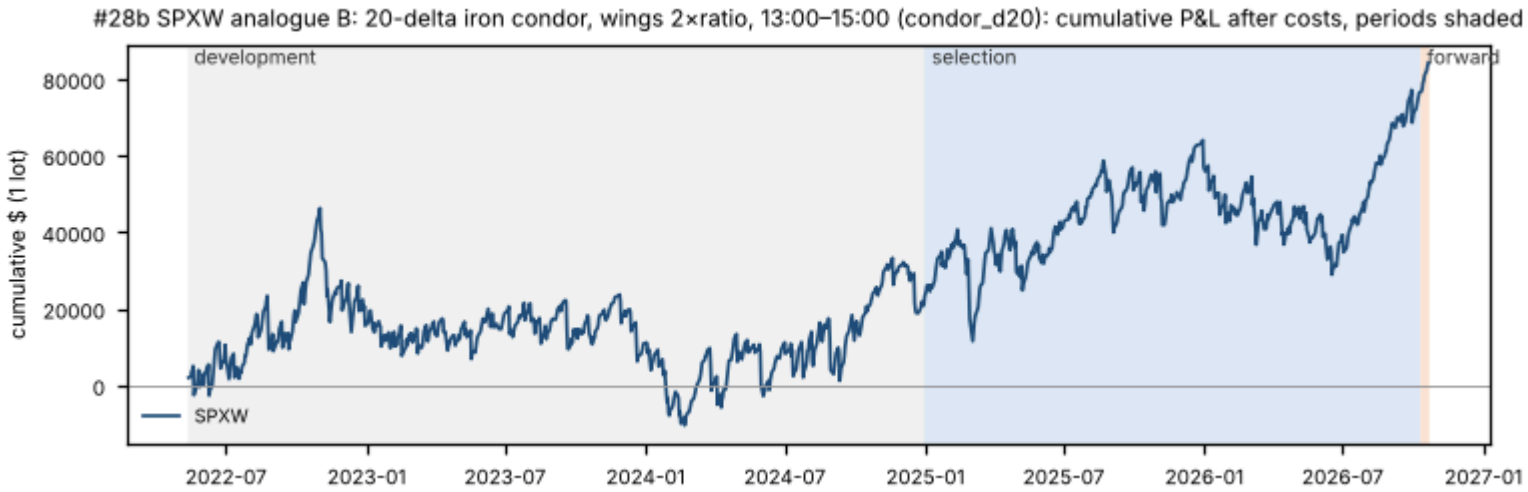
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- development: held to cash settlement (official SPX close) 100.0%
- selection: held to cash settlement (official SPX close) 100.0%
- forward: held to cash settlement (official SPX close) 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	794	73.8	+1.30	+19.65	+15,600
2023	SPXW	1,190	73.4	−0.27	−3.94	−4,694
2024	SPXW	1,209	75.6	+0.69	+10.13	+12,244
2025	SPXW	1,203	77.1	+1.80	+28.21	+33,937
2026	SPXW	992	76.9	+1.54	+27.40	+27,177



Amendments, deviations and integrity checks. As 28a; the ETF−SPXW late-day extrinsic gap measured beside it.

What it showed. Positive in both periods, inside the noise.

#29a Picker (GeorgeFi rule), PRIMARY: opening-range fade, stop 0.68, target 1.25, 5-minute stop, flat 15:58

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The opening-range sweep fade the GeorgeFi cards describe makes money as written.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exits: stop at 0.68 × ask, target 1.25 × ask (resting), out at 5 minutes unless +6% reached, flat 15:58.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). “Pass: mean > 0 with the interval clear of zero on each name, judged only on the span with the excluded dates removed” (PRIMARY only; mean % of premium, 20-session block interval; every name must pass).

Verdict. FAIL on every name: SPY −6.28% [−8.18, −4.41], QQQ −5.21%, IWM −9.47%, SPXW −4.96%.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	34.9	+21.8, +13.07	−26.3, −14.21	−14.1, −6.38	**−9.47**, −4.68	[−12.02, −7.26]; [−6.52, −3.22]
QQQ	whole span (card dates out)	814	698 / 1,005	36.4	+22.2, +28.41	−21.1, −26.89	−8.4, −8.88	**−5.34**, −6.78	[−7.05, −3.69]; [−9.68, −3.98]
SPXW	whole span (card dates out)	902	765 / 1,135	36.9	+21.8, +269.22	−20.8, −253.15	−7.9, −82.56	**−5.03**, −60.30	[−6.69, −3.51]; [−84.97, −37.22]
SPY	whole span (card dates out)	771	660 / 1,005	34.4	+21.7, +24.93	−21.1, −23.41	−9.3, −7.38	**−6.37**, −6.79	[−8.12, −4.67]; [−9.77, −4.19]
all names	whole span (card dates out)	2,925	898 / 1,135	35.8	+21.9, +101.88	−21.8, −92.58	−9.1, −10.38	**−6.14**, −22.97	[−7.36, −4.81]; [−31.17, −15.21]
QQQ	never-read 2026-10-14→21	6	5 / 6	66.7	+24.5, +49.12	−10.6, −20.38	+24.4, +44.62	**+12.77**, +25.96	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	50.0	+24.9, +290.78	−12.5, −145.89	+10.0, +87.44	**+6.21**, +72.44	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	50.0	+24.2, +25.29	−12.5, −14.38	+9.1, +4.62	**+5.82**, +5.46	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	55.6	+24.5, +114.47	−12.0, −65.19	+24.0, +23.12	**+8.27**, +34.62	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	34.9	+21.8, +13.07	−26.3, −14.21	−14.1, −6.38	**−9.47**, −4.68	[−12.02, −7.26]; [−6.52, −3.22]
QQQ	whole span (verdict)	820	703 / 1,018	36.6	+22.2, +28.68	−21.0, −26.86	−8.4, −8.38	**−5.21**, −6.54	[−7.13, −3.67]; [−9.75, −4.19]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (verdict)	908	770 / 1,148	37.0	+21.9, +269.41	−20.7, −252.59	−7.8, −77.56	**−4.96**, −59.43	[−6.73, −3.49]; [−85.73, −36.98]
SPY	whole span (verdict)	777	665 / 1,018	34.5	+21.8, +24.93	−21.0, −23.35	−9.1, −7.38	**−6.28**, −6.70	[−8.12, −4.61]; [−9.79, −4.25]
all names	whole span (verdict)	2,943	903 / 1,148	35.9	+21.9, +101.99	−21.7, −92.46	−9.1, −10.38	**−6.05**, −22.62	[−7.41, −4.82]; [−31.23, −14.52]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+59.62, +24.5%, 2025-04-09, IWM 2025-04-09 176.0P	−120.38, −58.4%, 2025-04-07, IWM 2025-04-07 176.0P	0.49	2,077	632	total ≤ 0 (best 5 days +281; without them −2,331)
QQQ	whole span (card dates out)	+140.62, +24.9%, 2025-04-09, QQQ 2025-04-09 423.0P	−300.38, −51.3%, 2025-04-07, QQQ 2025-04-07 416.0P	0.60	5,572	990	total ≤ 0 (best 5 days +680; without them −6,198)
SPXW	whole span (card dates out)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.62	54,720	1,135	total ≤ 0 (best 5 days +6,117; without them −60,512)
SPY	whole span (card dates out)	+127.62, +24.8%, 2025-04-09, SPY 2025-04-09 495.0C	−295.38, −45.4%, 2025-04-07, SPY 2025-04-07 495.0P	0.56	5,270	1,005	total ≤ 0 (best 5 days +561; without them −5,799)
all names	whole span (card dates out)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.61	67,530	1,135	total ≤ 0 (best 5 days +7,296; without them −74,497)
QQQ	never-read 2026-10-14→21	+53.62, +24.5%, 2026-10-15, QQQ 2026-10-15 720.0P	−35.38, −18.5%, 2026-10-19, QQQ 2026-10-19 720.0P	4.82	41	2	100%
SPXW	never-read 2026-10-14→21	+347.44, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.56, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	1.99	295	3	100%
SPY	never-read 2026-10-14→21	+29.62, +23.9%, 2026-10-21, SPY 2026-10-21 770.0C	−18.38, −19.1%, 2026-10-14, SPY 2026-10-14 774.0C	1.76	28	3	100%
all names	never-read 2026-10-14→21	+347.44, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.56, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	2.19	224	2	100%
IWM	whole span (verdict)	+59.62, +24.5%, 2025-04-09, IWM 2025-04-09 176.0P	−120.38, −58.4%, 2025-04-07, IWM 2025-04-07 176.0P	0.49	2,077	632	total ≤ 0 (best 5 days +281; without them −2,331)
QQQ	whole span (verdict)	+140.62, +24.9%, 2025-04-09, QQQ 2025-04-09 423.0P	−300.38, −51.3%, 2025-04-07, QQQ 2025-04-07 416.0P	0.62	5,572	1,003	total ≤ 0 (best 5 days +680; without them −6,042)
SPXW	whole span (verdict)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	54,720	1,148	total ≤ 0 (best 5 days +6,117; without them −60,077)
SPY	whole span (verdict)	+127.62, +24.8%, 2025-04-09, SPY 2025-04-09 495.0C	−295.38, −45.4%, 2025-04-07, SPY 2025-04-07 495.0P	0.56	5,270	1,018	total ≤ 0 (best 5 days +561; without them −5,766)
all names	whole span (verdict)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.62	67,530	1,148	total ≤ 0 (best 5 days +7,296; without them −73,874)

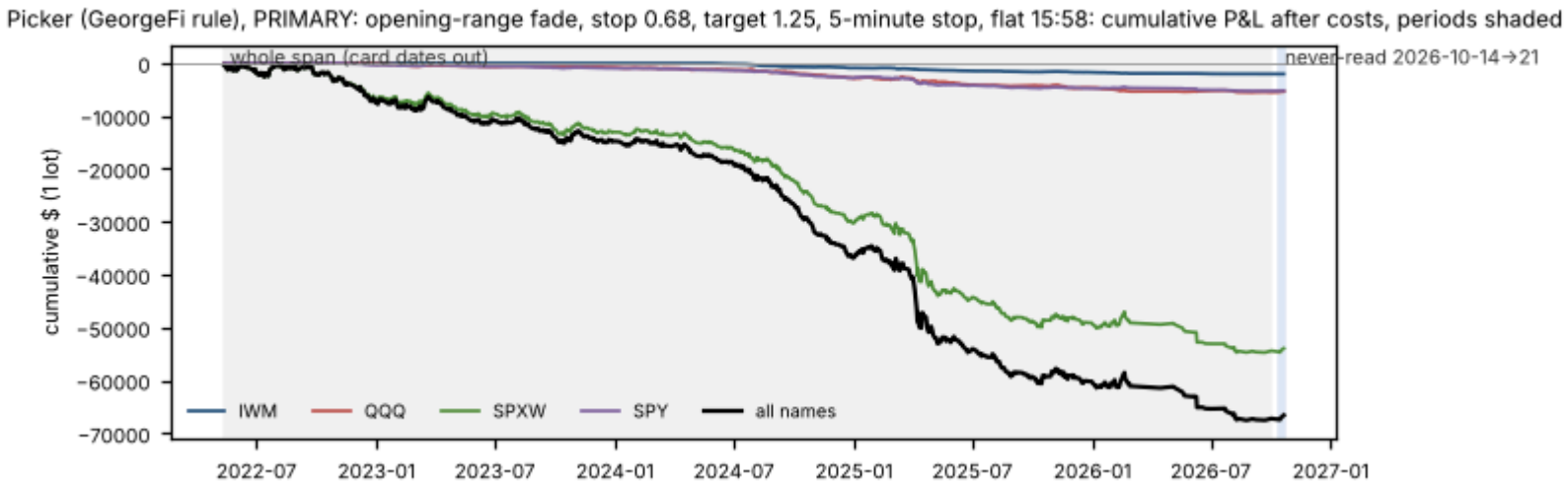
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): time5 47.2%; target 32.1%; stop 20.7%
- never-read 2026-10-14→21: target 55.6%; time5 44.4%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	28.9	−12.08	−6.01	−913
2025	IWM	232	37.9	−7.97	−3.56	−827
2026	IWM	54	38.9	−8.57	−5.75	−310
2022	QQQ	30	36.7	−4.19	−3.21	−96
2023	QQQ	248	38.7	−3.64	−2.77	−687
2024	QQQ	244	33.2	−6.55	−8.47	−2,066
2025	QQQ	233	38.2	−4.78	−7.64	−1,780
2026	QQQ	65	35.4	−8.19	−11.27	−732
2022	SPXW	164	40.9	−3.28	−41.70	−6,839
2023	SPXW	240	41.7	−3.18	−25.76	−6,183
2024	SPXW	228	28.5	−7.33	−75.63	−17,243
2025	SPXW	213	36.2	−5.64	−88.47	−18,844
2026	SPXW	63	42.9	−5.25	−77.00	−4,851
2022	SPY	33	30.3	−11.01	−11.59	−382
2023	SPY	240	39.2	−3.96	−2.35	−565
2024	SPY	228	26.8	−8.69	−8.13	−1,854
2025	SPY	213	35.7	−5.99	−9.06	−1,929
2026	SPY	63	42.9	−4.85	−7.53	−475
2022	all names	227	38.8	−4.52	−32.24	−7,318
2023	all names	728	39.8	−3.59	−10.21	−7,435
2024	all names	852	29.5	−8.32	−25.91	−22,077
2025	all names	891	37.0	−6.10	−26.24	−23,380
2026	all names	245	40.0	−6.66	−25.99	−6,368



Amendments, deviations and integrity checks. verify_picker.py re-derives trades from the raw files; the flow gate had one session of data (0 trades gated).

What it showed. The fade loses 5–9% of premium a trade on every name.

#29b Picker, SECONDARY: the narrative’s +6% breakeven lock and VIX trail

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The vendor’s narrative exits rescue the rule.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exits: stop, target, breakeven lock at +6%, VIX-scaled trail, flat 15:58.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). No verdict of its own (labelled).

Verdict. Labelled: SPY −6.29%, QQQ −5.44%, IWM −9.91%, SPXW −5.01%.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	33.8	+17.8, +10.60	−24.1, −12.28	−6.5, −3.38	**−9.91**, −4.55	[−12.27, −7.73]; [−5.94, −3.29]
QQQ	whole span (card dates out)	814	698 / 1,005	42.3	+16.8, +21.58	−22.0, −26.71	−2.0, −2.38	**−5.59**, −6.30	[−6.92, −4.02]; [−8.45, −3.87]
SPXW	whole span (card dates out)	902	765 / 1,135	41.4	+16.5, +210.61	−20.1, −243.10	−1.6, −12.55	**−4.98**, −55.48	[−6.45, −3.38]; [−72.59, −35.73]
SPY	whole span (card dates out)	771	660 / 1,005	38.4	+16.6, +19.60	−20.5, −22.36	−3.0, −2.38	**−6.25**, −6.25	[−8.05, −4.65]; [−8.21, −4.43]
all names	whole span (card dates out)	2,925	898 / 1,135	39.7	+16.8, +80.41	−21.4, −88.06	−2.7, −3.38	**−6.22**, −21.19	[−7.32, −4.97]; [−26.53, −14.47]
QQQ	never-read 2026-10-14→21	6	5 / 6	83.3	+24.5, +49.02	−33.2, −63.38	+24.5, +47.12	**+14.87**, +30.29	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	33.3	+25.0, +312.44	−26.9, −335.05	−17.7, −187.56	**−9.60**, −119.22	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	33.3	+24.2, +28.62	−28.5, −35.62	−19.7, −18.38	**−10.95**, −14.21	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	50.0	+24.5, +103.03	−28.3, −171.79	+10.3, +11.62	**−1.89**, −34.38	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	33.8	+17.8, +10.60	−24.1, −12.28	−6.5, −3.38	**−9.91**, −4.55	[−12.27, −7.73]; [−5.94, −3.29]
QQQ	whole span (verdict)	820	703 / 1,018	42.6	+16.9, +21.97	−22.0, −26.79	−2.0, −2.38	**−5.44**, −6.03	[−7.06, −4.10]; [−8.63, −4.07]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (verdict)	908	770 / 1,148	41.3	+16.6, +211.15	−20.2, −243.79	−1.7, −17.55	**−5.01**, −55.90	[−6.58, −3.47]; [−73.81, −37.26]
SPY	whole span (verdict)	777	665 / 1,018	38.4	+16.6, +19.66	−20.6, −22.47	−3.0, −2.38	**−6.29**, −6.31	[−8.04, −4.75]; [−8.37, −4.54]
all names	whole span (verdict)	2,943	903 / 1,148	39.8	+16.8, +80.58	−21.4, −88.48	−2.7, −3.38	**−6.20**, −21.27	[−7.38, −5.02]; [−26.92, −14.94]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+59.62, +24.5%, 2025-04-09, IWM 2025-04-09 176.0P	−60.38, −40.0%, 2024-08-05, IWM 2024-08-05 199.0P	0.44	2,010	632	total ≤ 0 (best 5 days +214; without them −2,205)
QQQ	whole span (card dates out)	+140.62, +24.9%, 2025-04-09, QQQ 2025-04-09 423.0P	−140.38, −34.9%, 2025-04-08, QQQ 2025-04-08 440.0P	0.59	5,174	1,001	total ≤ 0 (best 5 days +606; without them −5,736)
SPXW	whole span (card dates out)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.56, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.61	50,045	1,135	total ≤ 0 (best 5 days +5,590; without them −55,634)
SPY	whole span (card dates out)	+127.62, +24.8%, 2025-04-09, SPY 2025-04-09 495.0C	−110.38, −35.2%, 2025-04-23, SPY 2025-04-23 542.0P	0.55	4,923	913	total ≤ 0 (best 5 days +522; without them −5,340)
all names	whole span (card dates out)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.56, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.60	61,983	1,135	total ≤ 0 (best 5 days +6,242; without them −68,225)
QQQ	never-read 2026-10-14→21	+53.62, +24.5%, 2026-10-15, QQQ 2026-10-15 720.0P	−63.38, −33.2%, 2026-10-19, QQQ 2026-10-19 720.0P	3.87	15	2	100%
SPXW	never-read 2026-10-14→21	+347.44, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−562.55, −36.1%, 2026-10-15, SPXW 2026-10-15 7695.0P	0.47	998	6	total ≤ 0 (best 5 days −715; without them +0)
SPY	never-read 2026-10-14→21	+29.62, +23.9%, 2026-10-21, SPY 2026-10-21 770.0C	−58.38, −35.2%, 2026-10-15, SPY 2026-10-15 767.0P	0.40	95	6	total ≤ 0 (best 5 days −85; without them +0)
all names	never-read 2026-10-14→21	+347.44, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−562.55, −36.1%, 2026-10-15, SPXW 2026-10-15 7695.0P	0.60	957	6	total ≤ 0 (best 5 days −619; without them −0)
IWM	whole span (verdict)	+59.62, +24.5%, 2025-04-09, IWM 2025-04-09 176.0P	−60.38, −40.0%, 2024-08-05, IWM 2024-08-05 199.0P	0.44	2,010	632	total ≤ 0 (best 5 days +214; without them −2,205)
QQQ	whole span (verdict)	+140.62, +24.9%, 2025-04-09, QQQ 2025-04-09 423.0P	−140.38, −34.9%, 2025-04-08, QQQ 2025-04-08 440.0P	0.61	5,174	1,014	total ≤ 0 (best 5 days +606; without them −5,554)
SPXW	whole span (verdict)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.56, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.61	51,042	1,148	total ≤ 0 (best 5 days +5,590; without them −56,350)
SPY	whole span (verdict)	+127.62, +24.8%, 2025-04-09, SPY 2025-04-09 495.0C	−110.38, −35.2%, 2025-04-23, SPY 2025-04-23 542.0P	0.54	5,018	926	total ≤ 0 (best 5 days +522; without them −5,425)
all names	whole span (verdict)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−1,182.56, −34.4%, 2025-04-23, SPXW 2025-04-23 5445.0P	0.60	62,940	1,148	total ≤ 0 (best 5 days +6,242; without them −68,844)

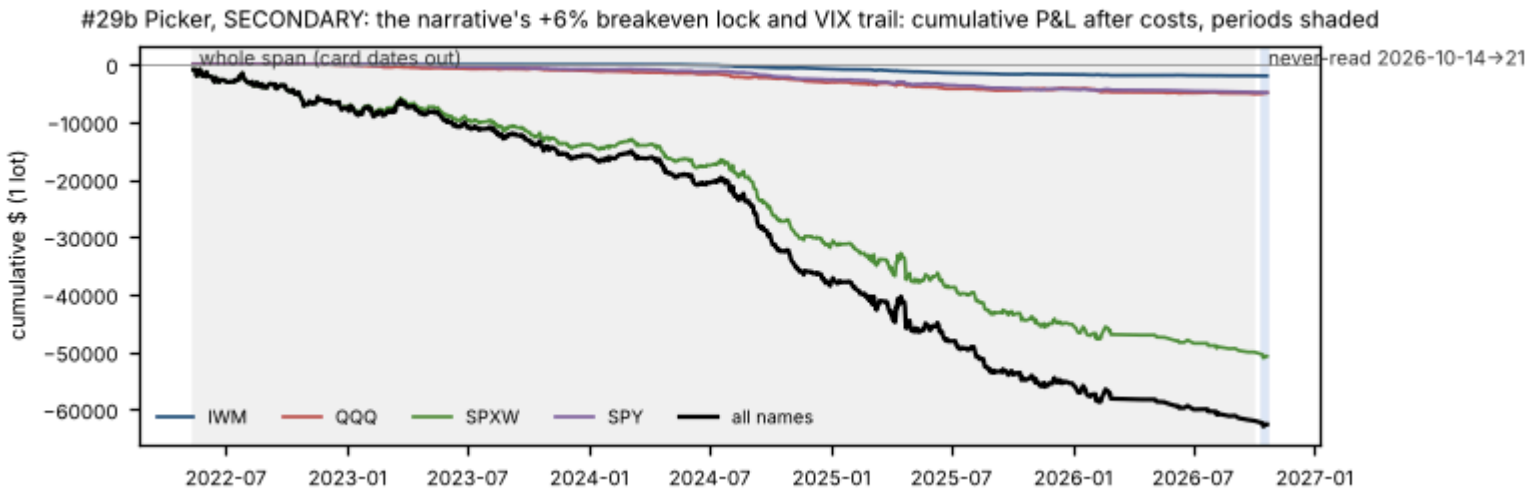
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): stop 30.8%; lock 27.4%; target 25.0%; trail 16.9%
- never-read 2026-10-14→21: target 50.0%; stop 38.9%; lock 11.1%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	32.2	−11.22	−5.28	−803
2025	IWM	232	33.2	−9.70	−4.18	−970
2026	IWM	54	40.7	−7.14	−4.04	−218
2022	QQQ	30	40.0	−4.56	−2.71	−81
2023	QQQ	248	46.8	−4.85	−3.95	−980
2024	QQQ	244	38.1	−7.23	−8.31	−2,026
2025	QQQ	233	44.6	−3.87	−4.93	−1,148
2026	QQQ	65	36.9	−6.99	−10.96	−712
2022	SPXW	164	36.6	−3.38	−44.99	−7,379
2023	SPXW	240	46.2	−3.54	−27.31	−6,553
2024	SPXW	228	41.2	−6.30	−76.68	−17,483
2025	SPXW	213	41.3	−5.52	−64.62	−13,764
2026	SPXW	63	34.9	−8.42	−88.59	−5,581
2022	SPY	33	36.4	−2.74	−2.41	−79
2023	SPY	240	39.6	−4.94	−3.16	−758
2024	SPY	228	37.3	−7.34	−7.90	−1,800
2025	SPY	213	40.4	−6.34	−7.71	−1,643
2026	SPY	63	31.7	−9.30	−9.87	−622
2022	all names	227	37.0	−3.44	−33.21	−7,540
2023	all names	728	44.2	−4.45	−11.39	−8,291
2024	all names	852	37.7	−7.72	−25.95	−22,113
2025	all names	891	39.8	−6.37	−19.67	−17,525
2026	all names	245	35.9	−7.98	−29.12	−7,133



Amendments, deviations and integrity checks. As 29a.

What it showed. The narrative exits change nothing.

#29c Picker, S2: “5 minutes” as a hard exit

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. Reading the 5-minute rule as a hard exit helps.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; exit at 5 minutes, or the stop/target first.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Sensitivity, no verdict.

Verdict. Labelled: SPY −4.20%, QQQ −3.64%, IWM −7.42%, SPXW −3.10%.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	438	394 / 634	35.8	+15.0, +8.80	−19.9, −10.28	−8.6, −3.38	**−7.42**, −3.44	[−9.45, −5.42]; [−4.94, −2.11]
QQQ	whole span (card dates out)	814	698 / 1,005	38.9	+14.5, +18.77	−15.4, −19.27	−4.8, −5.38	**−3.76**, −4.46	[−4.96, −2.44]; [−6.28, −2.37]
SPXW	whole span (card dates out)	902	765 / 1,135	40.4	+13.6, +172.58	−14.4, −172.74	−4.0, −42.55	**−3.11**, −33.39	[−4.33, −1.87]; [−47.88, −17.58]
SPY	whole span (card dates out)	771	660 / 1,005	36.4	+13.7, +16.05	−14.5, −15.63	−5.2, −4.38	**−4.21**, −4.08	[−5.64, −2.91]; [−5.65, −2.45]
all names	whole span (card dates out)	2,925	898 / 1,135	38.3	+14.1, +66.72	−15.6, −62.60	−4.8, −5.38	**−4.23**, −13.13	[−5.22, −3.18]; [−17.87, −7.66]
QQQ	never-read 2026-10-14→21	6	5 / 6	66.7	+24.5, +49.12	−10.6, −20.38	+24.4, +44.62	**+12.77**, +25.96	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	never-read 2026-10-14→21	6	5 / 6	33.3	+15.0, +177.44	−10.2, −117.56	−4.0, −52.56	**−1.78**, −19.22	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	6	5 / 6	33.3	+13.7, +16.62	−10.8, −11.88	−5.7, −6.88	**−2.63**, −2.38	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	18	5 / 6	44.4	+19.4, +73.08	−10.5, −55.85	−3.0, −4.88	**+2.79**, +1.45	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	438	394 / 634	35.8	+15.0, +8.80	−19.9, −10.28	−8.6, −3.38	**−7.42**, −3.44	[−9.45, −5.42]; [−4.94, −2.11]
QQQ	whole span (verdict)	820	703 / 1,018	39.1	+14.7, +19.15	−15.4, −19.27	−4.7, −5.38	**−3.64**, −4.23	[−4.99, −2.47]; [−6.40, −2.42]

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (verdict)	908	770 / 1,148	40.3	+13.6, +172.61	−14.4, −172.33	−4.0, −42.55	**−3.10**, −33.29	[−4.46, −1.80]; [−48.84, −17.23]
SPY	whole span (verdict)	777	665 / 1,018	36.4	+13.7, +16.05	−14.5, −15.60	−5.2, −4.38	**−4.20**, −4.07	[−5.61, −2.85]; [−5.69, −2.44]
all names	whole span (verdict)	2,943	903 / 1,148	38.3	+14.1, +66.77	−15.6, −62.56	−4.8, −5.38	**−4.18**, −13.04	[−5.32, −3.14]; [−18.40, −7.47]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+32.62, +13.4%, 2025-04-09, IWM 2025-04-09 176.0P	−48.38, −40.7%, 2026-02-13, IWM 2026-02-13 261.0P	0.48	1,520	632	total ≤ 0 (best 5 days +175; without them −1,683)
QQQ	whole span (card dates out)	+82.62, +24.8%, 2025-03-11, QQQ 2025-03-11 472.0C	−101.38, −39.9%, 2026-02-20, QQQ 2026-02-20 604.0P	0.62	3,627	1,005	total ≤ 0 (best 5 days +512; without them −4,139)
SPXW	whole span (card dates out)	+907.45, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.55, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	30,589	1,038	total ≤ 0 (best 5 days +4,437; without them −34,552)
SPY	whole span (card dates out)	+87.62, +17.0%, 2025-04-09, SPY 2025-04-09 495.0C	−98.38, −32.9%, 2026-02-05, SPY 2026-02-05 679.0C	0.59	3,274	914	total ≤ 0 (best 5 days +401; without them −3,549)
all names	whole span (card dates out)	+907.45, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.55, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.66	38,812	1,038	total ≤ 0 (best 5 days +4,920; without them −43,319)
QQQ	never-read 2026-10-14→21	+53.62, +24.5%, 2026-10-15, QQQ 2026-10-15 720.0P	−35.38, −18.5%, 2026-10-19, QQQ 2026-10-19 720.0P	4.82	41	2	100%
SPXW	never-read 2026-10-14→21	+247.44, +22.3%, 2026-10-21, SPXW 2026-10-21 7725.0C	−222.56, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	0.75	328	6	total ≤ 0 (best 5 days −115; without them +0)
SPY	never-read 2026-10-14→21	+24.62, +19.9%, 2026-10-21, SPY 2026-10-21 770.0C	−18.38, −19.1%, 2026-10-14, SPY 2026-10-14 774.0C	0.70	32	6	total ≤ 0 (best 5 days −14; without them +0)
all names	never-read 2026-10-14→21	+247.44, +22.3%, 2026-10-21, SPXW 2026-10-21 7725.0C	−222.56, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	1.05	224	5	100%
IWM	whole span (verdict)	+32.62, +13.4%, 2025-04-09, IWM 2025-04-09 176.0P	−48.38, −40.7%, 2026-02-13, IWM 2026-02-13 261.0P	0.48	1,520	632	total ≤ 0 (best 5 days +175; without them −1,683)
QQQ	whole span (verdict)	+82.62, +24.8%, 2025-03-11, QQQ 2025-03-11 472.0C	−101.38, −39.9%, 2026-02-20, QQQ 2026-02-20 604.0P	0.64	3,627	1,018	total ≤ 0 (best 5 days +512; without them −3,983)
SPXW	whole span (verdict)	+907.45, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.55, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.68	30,812	1,051	total ≤ 0 (best 5 days +4,437; without them −34,667)
SPY	whole span (verdict)	+87.62, +17.0%, 2025-04-09, SPY 2025-04-09 495.0C	−98.38, −32.9%, 2026-02-05, SPY 2026-02-05 679.0C	0.59	3,298	927	total ≤ 0 (best 5 days +401; without them −3,563)
all names	whole span (verdict)	+907.45, +16.4%, 2025-04-09, SPXW 2025-04-09 4960.0C	−992.55, −34.1%, 2026-02-05, SPXW 2026-02-05 6810.0C	0.66	38,991	1,051	total ≤ 0 (best 5 days +4,920; without them −43,292)

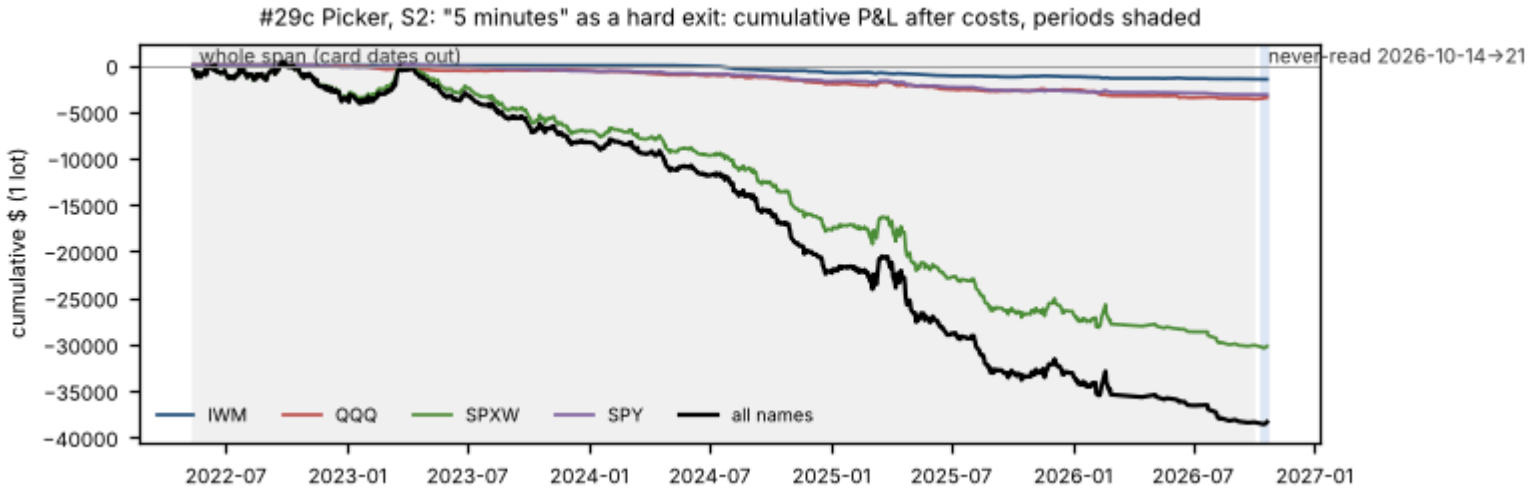
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): time5 80.8%; target 13.2%; stop 6.0%
- never-read 2026-10-14→21: time5 77.8%; target 22.2%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	152	29.6	−10.15	−5.22	−794
2025	IWM	232	40.1	−5.41	−1.88	−436
2026	IWM	54	35.2	−8.37	−5.15	−278
2022	QQQ	30	36.7	−2.27	−1.41	−42
2023	QQQ	248	41.9	−2.69	−2.10	−521
2024	QQQ	244	36.1	−4.88	−5.92	−1,446
2025	QQQ	233	42.1	−2.49	−2.73	−636
2026	QQQ	65	30.8	−7.32	−12.71	−826
2022	SPXW	164	45.7	−1.49	−20.18	−3,309
2023	SPXW	240	42.9	−2.73	−15.81	−3,793
2024	SPXW	228	34.2	−3.91	−46.37	−10,573
2025	SPXW	213	39.0	−3.45	−41.29	−8,794
2026	SPXW	63	42.9	−4.62	−59.70	−3,761
2022	SPY	33	39.4	−5.32	−5.77	−190
2023	SPY	240	38.8	−3.46	−1.79	−430
2024	SPY	228	31.6	−4.76	−4.87	−1,110
2025	SPY	213	37.6	−3.95	−4.82	−1,027
2026	SPY	63	39.7	−5.24	−6.42	−405
2022	all names	227	43.6	−2.15	−15.60	−3,542
2023	all names	728	41.2	−2.96	−6.52	−4,744
2024	all names	852	33.2	−5.53	−16.34	−13,923
2025	all names	891	39.7	−3.83	−12.23	−10,893
2026	all names	245	37.1	−6.32	−21.51	−5,270



Amendments, deviations and integrity checks. As 29a.

What it showed. Shorter holds lose less, still lose.

#29d Picker, S3: SPXW rides to the official close when in the money at 15:58

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. SPXW's cash settlement adds value.

Rule. Opening range = the 09:30–09:59 bars; 10:00–11:59: the first bar with high ≥ ORH × (1 + 2 bp) and close < ORH → PUT (the low side → CALL); buy the nearest OTM 0DTE contract at the ask one bar later; SPXW only; ITM at 15:58 → settle.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Sensitivity, no verdict.

Verdict. Labelled: identical to PRIMARY (no trade was open at 15:58).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
SPXW	whole span (card dates out)	902	765 / 1,135	36.9	+21.8, +269.22	−20.8, −253.15	−7.9, −82.56	**−5.03**, −60.30	[−6.69, −3.51]; [−84.97, −37.22]
SPXW	never-read 2026-10-14→21	6	5 / 6	50.0	+24.9, +290.78	−12.5, −145.89	+10.0, +87.44	**+6.21**, +72.44	n/a (< 60 sessions); n/a (< 60 sessions)
SPXW	whole span (verdict)	908	770 / 1,148	37.0	+21.9, +269.41	−20.7, −252.59	−7.8, −77.56	**−4.96**, −59.43	[−6.73, −3.49]; [−85.73, −36.98]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
SPXW	whole span (card dates out)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.62	54,720	1,135	total ≤ 0 (best 5 days +6,117; without them −60,512)
SPXW	never-read 2026-10-14→21	+347.44, +25.0%, 2026-10-19, SPXW 2026-10-19 7690.0C	−222.56, −17.8%, 2026-10-14, SPXW 2026-10-14 7760.0C	1.99	295	3	100%
SPXW	whole span (verdict)	+1,387.44, +25.1%, 2025-04-09, SPXW 2025-04-09 4960.0C	−2,952.55, −45.6%, 2025-04-07, SPXW 2025-04-07 4970.0P	0.63	54,720	1,148	total ≤ 0 (best 5 days +6,117; without them −60,077)

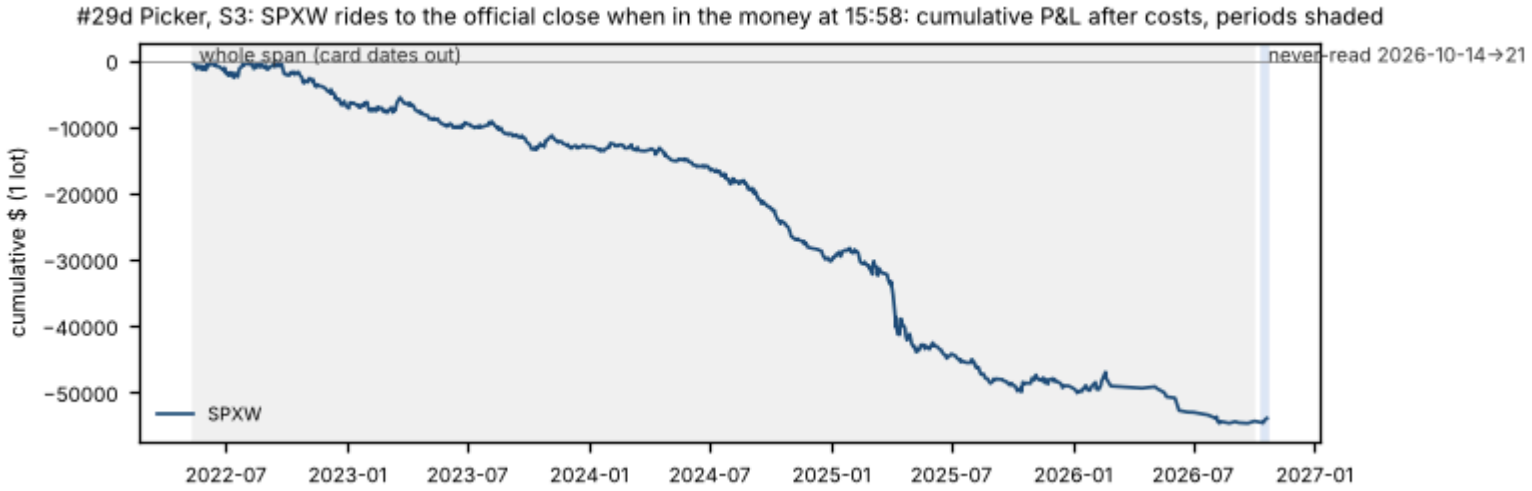
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- whole span (card dates out): time5 48.9%; target 31.5%; stop 19.6%
- never-read 2026-10-14→21: time5 50.0%; target 50.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	SPXW	164	40.9	−3.28	−41.70	−6,839
2023	SPXW	240	41.7	−3.18	−25.76	−6,183
2024	SPXW	228	28.5	−7.33	−75.63	−17,243
2025	SPXW	213	36.2	−5.64	−88.47	−18,844
2026	SPXW	63	42.9	−5.25	−77.00	−4,851



Amendments, deviations and integrity checks. As 29a.

What it showed. No trade reached 15:58.

#29e Picker, IMPROVED (picked on the cards): PDH sweep → put, 10:00–12:00, +25% or 5 minutes

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. The owner's "improved" parameters, traced to the cards, work.

Rule. SPXW/SPY/IWM: prior-day-high sweep → put, 10:00–12:00, sell at +25% or at 5 minutes.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData); a target at its resting limit

Data. SPY, QQQ (846 sessions), IWM (484), SPXW (972), 2022 → 2026-10-21, the 134 GeorgeFi card dates excluded.

Periods and holdouts. The whole tested span (verdict) and the never-read line 2026-10-14 → 21 (no verdict). The 2022–2026 NBBO had been read by the engine audit for the same universe (disclosed).

Pass rule (as pre-registered). Labelled, never pooled; the owner's rule shown for reading only.

Verdict. FAIL on every name (read only): SPXW −3.81% [−6.76, −0.95], SPY −4.77%, IWM −10.38%.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
IWM	whole span (card dates out)	106	106 / 615	34.9	+13.8, +7.95	−23.4, −12.32	−11.2, −5.38	**−10.38**, −5.24	[−14.45, −6.32]; [−8.72, −2.14]
SPXW	whole span (card dates out)	217	217 / 1,105	40.1	+16.9, +196.64	−17.9, −198.02	−6.3, −62.55	**−3.95**, −39.79	[−6.85, −1.03]; [−77.49, −3.10]
SPY	whole span (card dates out)	191	191 / 977	38.7	+16.7, +17.98	−18.6, −18.62	−8.2, −6.38	**−4.92**, −4.44	[−7.55, −1.17]; [−7.75, −0.03]
all names	whole span (card dates out)	514	275 / 1,125	38.5	+16.2, +94.61	−19.3, −91.05	−8.1, −8.38	**−5.64**, −19.53	[−8.26, −3.06]; [−37.02, −2.94]
SPXW	never-read 2026-10-14→21	1	1 / 1	100.0	+25.2, +337.44	—, —	+25.2, +337.44	**+25.18**, +337.44	n/a (< 60 sessions); n/a (< 60 sessions)
SPY	never-read 2026-10-14→21	1	1 / 1	100.0	+24.0, +34.62	—, —	+24.0, +34.62	**+24.05**, +34.62	n/a (< 60 sessions); n/a (< 60 sessions)
all names	never-read 2026-10-14→21	2	1 / 1	100.0	+24.6, +186.03	—, —	+24.6, +186.03	**+24.61**, +186.03	n/a (< 60 sessions); n/a (< 60 sessions)
IWM	whole span (verdict)	106	106 / 615	34.9	+13.8, +7.95	−23.4, −12.32	−11.2, −5.38	**−10.38**, −5.24	[−14.45, −6.32]; [−8.72, −2.14]
SPXW	whole span (verdict)	218	218 / 1,145	40.4	+17.0, +198.24	−17.9, −198.02	−6.2, −62.55	**−3.81**, −38.06	[−6.93, −1.10]; [−80.79, −3.24]
SPY	whole span (verdict)	192	192 / 1,017	39.1	+16.8, +18.20	−18.6, −18.62	−8.0, −6.38	**−4.77**, −4.24	[−7.69, −1.37]; [−7.82, −0.14]
all names	whole span (verdict)	516	276 / 1,145	38.8	+16.3, +95.52	−19.3, −91.05	−8.0, −8.38	**−5.52**, −18.73	[−8.38, −2.95]; [−38.68, −2.53]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
IWM	whole span (card dates out)	+27.62, +23.8%, 2024-07-17, IWM 2024-07-17 224.0P	−130.38, −60.9%, 2025-04-07, IWM 2025-04-07 182.0P	0.35	625	512	total ≤ 0 (best 5 days +111; without them −667)
SPXW	whole span (card dates out)	+477.44, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.56, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.66	9,157	1,084	total ≤ 0 (best 5 days +2,257; without them −10,892)
SPY	whole span (card dates out)	+49.62, +22.0%, 2025-11-20, SPY 2025-11-20 667.0P	−109.38, −49.5%, 2026-02-20, SPY 2026-02-20 685.0P	0.61	880	977	total ≤ 0 (best 5 days +218; without them −1,067)
all names	whole span (card dates out)	+477.44, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.56, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.65	10,551	1,104	total ≤ 0 (best 5 days +2,479; without them −12,518)
SPXW	never-read 2026-10-14→21	+337.44, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	+337.44, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	∞	0	0	100%
SPY	never-read 2026-10-14→21	+34.62, +24.0%, 2026-10-20, SPY 2026-10-20 769.0P	+34.62, +24.0%, 2026-10-20, SPY 2026-10-20 769.0P	∞	0	0	100%
all names	never-read 2026-10-14→21	+337.44, +25.2%, 2026-10-20, SPXW 2026-10-20 7715.0P	+34.62, +24.0%, 2026-10-20, SPY 2026-10-20 769.0P	∞	0	0	100%
IWM	whole span (verdict)	+27.62, +23.8%, 2024-07-17, IWM 2024-07-17 224.0P	−130.38, −60.9%, 2025-04-07, IWM 2025-04-07 182.0P	0.35	625	512	total ≤ 0 (best 5 days +111; without them −667)
SPXW	whole span (verdict)	+477.44, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.56, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.68	9,157	1,124	total ≤ 0 (best 5 days +2,257; without them −10,554)
SPY	whole span (verdict)	+49.62, +22.0%, 2025-11-20, SPY 2025-11-20 667.0P	−109.38, −49.5%, 2026-02-20, SPY 2026-02-20 685.0P	0.63	880	1,017	total ≤ 0 (best 5 days +218; without them −1,032)
all names	whole span (verdict)	+477.44, +25.0%, 2024-08-07, SPXW 2024-08-07 5310.0P	−1,172.56, −49.7%, 2026-02-20, SPXW 2026-02-20 6870.0P	0.66	10,551	1,124	total ≤ 0 (best 5 days +2,479; without them −12,146)

Random baseline: not available — no random baseline was computed by this study

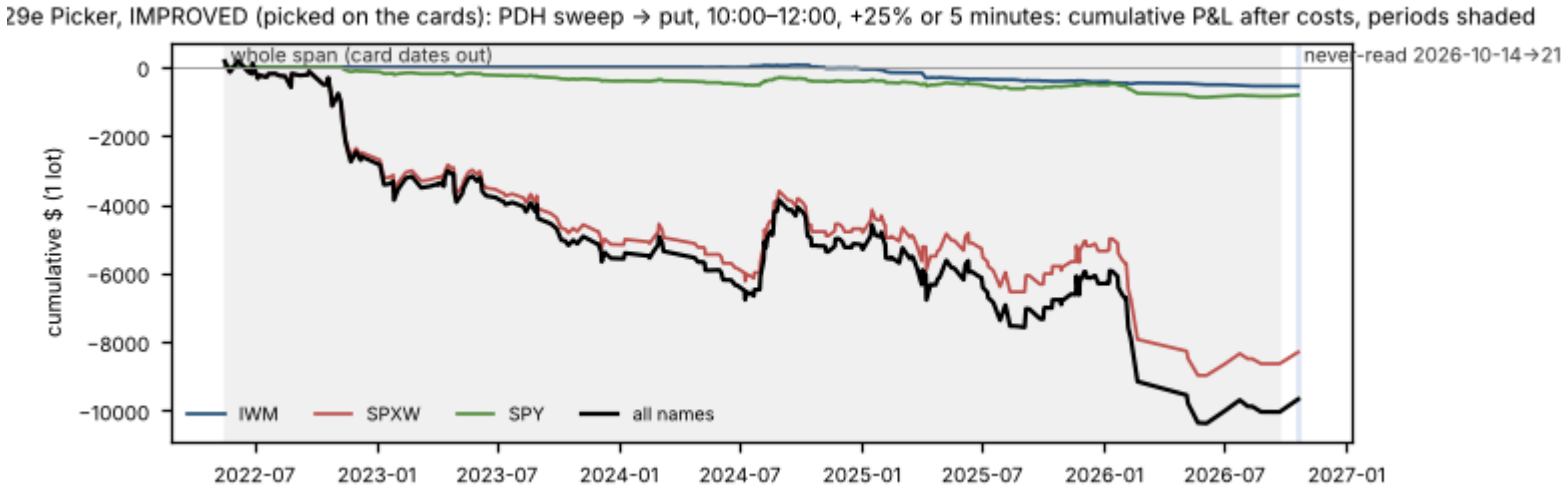
Exit reasons (share of trades):

- whole span (card dates out): time5 81.9%; target 18.1%
- never-read 2026-10-14→21: target 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2024	IWM	37	45.9	−6.18	−1.51	−56
2025	IWM	52	32.7	−12.05	−7.16	−372
2026	IWM	17	17.6	−14.41	−7.49	−127
2022	SPXW	33	42.4	−5.31	−75.28	−2,484
2023	SPXW	61	37.7	−5.07	−43.87	−2,676
2024	SPXW	50	38.0	−1.12	+9.04	+452
2025	SPXW	57	47.4	−2.05	−11.33	−646
2026	SPXW	17	29.4	−10.24	−173.14	−2,943

year	name	trades	win %	mean %	mean $	total $
2022	SPY	7	28.6	−15.74	−16.66	−117
2023	SPY	61	36.1	−6.37	−4.74	−289
2024	SPY	50	38.0	−1.93	−0.14	−7
2025	SPY	57	47.4	−2.37	−1.88	−107
2026	SPY	17	29.4	−10.94	−17.32	−294
2022	all names	40	40.0	−7.13	−65.02	−2,601
2023	all names	122	36.9	−5.72	−24.30	−2,965
2024	all names	137	40.1	−2.78	+2.84	+390
2025	all names	166	42.8	−5.29	−6.78	−1,126
2026	all names	51	25.5	−11.87	−65.98	−3,365



Amendments, deviations and integrity checks. Parameters picked from the cards, whose dates are excluded.

What it showed. Negative on every name.

#29f Picker: flow gate and grade check

Folder: `research_notes/Compass phase 3/picker_prereg`. **Commits:** c502122 / — / 1e10053. **Dates:** 2026-10-22.

Hypothesis. UW sweep flow on the trade’s side, and the vendor’s grades, separate winners.

Rule. Gate: trade-side UW sweep ask-premium over the 5 minutes before entry > the other side. Grade: A+/A/B+ on the vendor cards.

Fills and costs. —

Data. Flow: one session (2026-10-21). Grades: the D1 ledger (489 cards).

Periods and holdouts. —

Pass rule (as pre-registered). Descriptive only.

Verdict. Flow gate: 0 of 5 trades gated (one session). Grades (vendor-recorded, not re-marked): A+ +8.09% [+1.81, +19.44], A +3.12%, B+ −3.23%; A+ − A +4.97 [−3.79, +15.09].

Metrics. not available — one session of flow (5 trades) and the vendor’s own recorded returns (see #1, vendor-recorded).

Amendments, deviations and integrity checks. —

What it showed. No usable gate; the grades rank the vendor’s own record.

#30a Compass features: the base trades (every feature is judged against them)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. (The reference set.)

Rule. Every base trade, fixed exit: 0DTE sold at the 15:58 bid (SPXW held to settlement), multi-day at a fixed horizon.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). —

Verdict. Descriptive: −13.35% a trade (exploration), −11.03% (holdout).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	651,232	2,596 / 3,671	26.9	+168.2, +744.76	−80.0, −330.73	−75.1, −57.37	**−13.35**, −41.86	[−16.39, −10.41]; [−58.11, −25.11]
0DTE→60 DTE, 15 names	holdout (opened)	580,234	3,577 / 3,686	28.2	+158.6, +672.15	−77.5, −306.91	−67.3, −48.37	**−11.03**, −31.26	[−13.95, −7.83]; [−44.18, −10.15]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.45, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.55, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.83	27,484,233	3,671	total ≤ 0 (best 5 days +4,027,391; without them −31,288,704)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.45, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−32,491.28, −100.0%, 2026-10-19, SPXW 1-7 7740.0P	0.86	18,135,830	3,686	total ≤ 0 (best 5 days +2,431,046; without them −20,566,876)

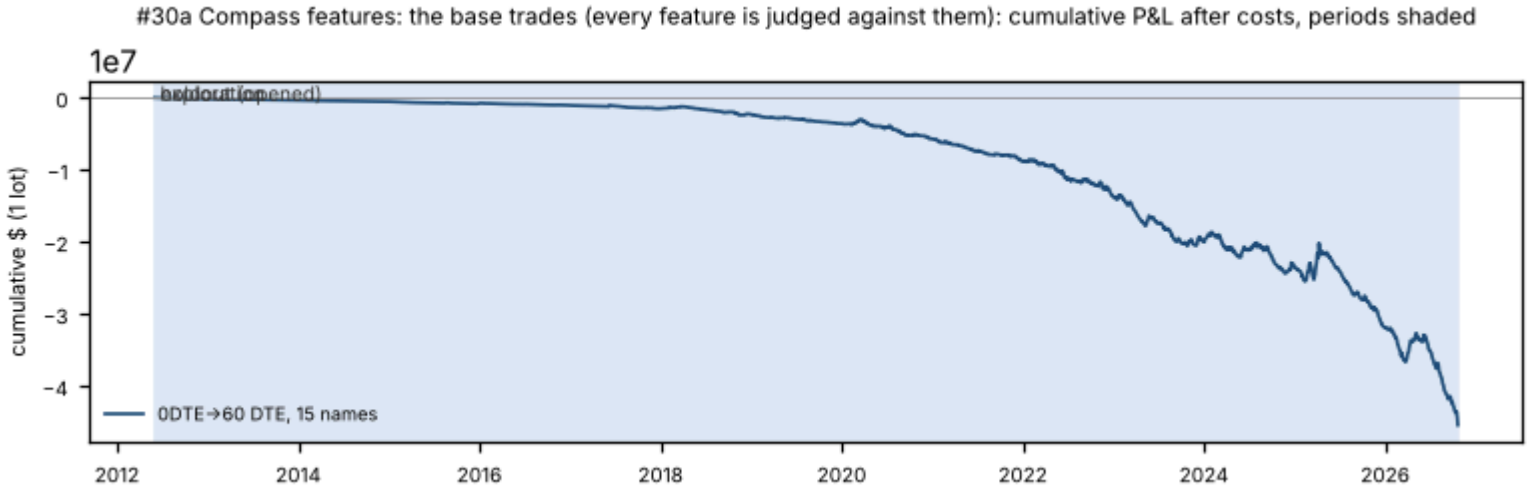
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	7,483	33.0	−16.21	−20.40	−152,617
2013	0DTE→60 DTE, 15 names	13,829	33.4	−10.08	−13.43	−185,703
2014	0DTE→60 DTE, 15 names	14,053	34.9	−8.57	−12.10	−170,008
2015	0DTE→60 DTE, 15 names	14,318	31.9	−10.04	−17.20	−246,310
2016	0DTE→60 DTE, 15 names	14,750	32.1	−10.00	−19.08	−281,487
2017	0DTE→60 DTE, 15 names	26,450	26.0	−15.28	−17.11	−452,492
2018	0DTE→60 DTE, 15 names	45,006	26.7	−18.90	−18.17	−817,830
2019	0DTE→60 DTE, 15 names	41,854	25.4	−26.84	−29.98	−1,254,812
2020	0DTE→60 DTE, 15 names	52,798	26.2	−16.34	−41.08	−2,169,063
2021	0DTE→60 DTE, 15 names	50,593	24.4	−26.27	−60.05	−3,038,340
2022	0DTE→60 DTE, 15 names	90,557	27.0	−14.06	−54.23	−4,910,848
2023	0DTE→60 DTE, 15 names	194,773	28.2	−9.72	−32.29	−6,288,933
2024	0DTE→60 DTE, 15 names	210,416	27.7	−8.39	−16.66	−3,504,658
2025	0DTE→60 DTE, 15 names	223,997	27.6	−10.48	−37.20	−8,331,655
2026	0DTE→60 DTE, 15 names	230,589	26.8	−11.16	−58.95	−13,592,388



Amendments, deviations and integrity checks. The holdout join duplicated rows on the first read (777,571 rows for 580,274 trades); fixed with no feature, threshold or rule changed.

What it showed. The base trades lose 11–13% of the debit a trade.

#30b Compass features V1: veto a spread over 10% of mid (PASSED)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. Skipping wide spreads makes a losing set lose less.

Rule. The base trades with (ask − bid) ÷ mid ≤ 10% at the entry quote.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). Holdout Δ (with − without, % of debit) > 0 with its 20-session block 95% interval clear of zero.

Verdict. PASS: holdout Δ +2.61 [+1.76, +3.46] (−8.43 with vs −11.03 without).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	538,468	2,594 / 3,671	28.0	+163.3, +776.73	−78.6, −355.88	−70.2, −61.69	**−10.89**, −38.87	[−14.09, −7.65]; [−57.55, −19.77]
0DTE→60 DTE, 15 names	holdout (opened)	474,526	3,572 / 3,686	29.5	+153.1, +676.26	−76.1, −321.72	−62.5, −49.69	**−8.43**, −27.16	[−11.25, −5.25]; [−40.74, −5.56]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.45, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.55, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.85	21,147,853	3,671	total ≤ 0 (best 5 days +4,129,699; without them −25,057,444)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.45, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−29,000.69, −100.0%, 2026-10-19, MU 1-7 1000.0C	0.88	12,888,643	3,686	total ≤ 0 (best 5 days +2,364,189; without them −15,252,832)

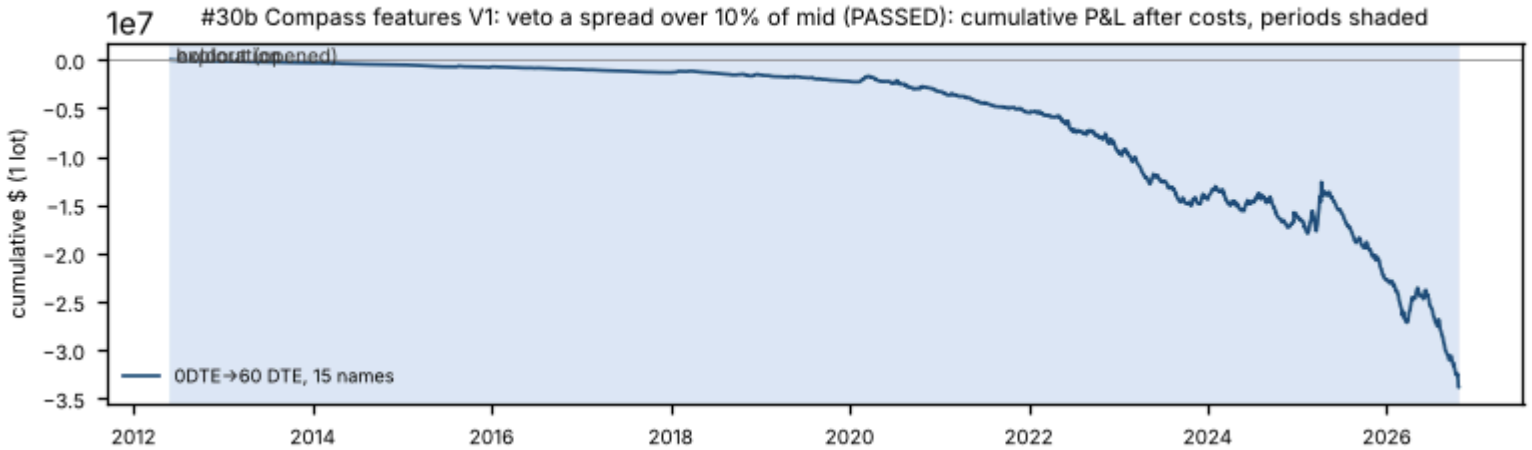
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	6,826	33.0	−16.46	−21.97	−149,981
2013	0DTE→60 DTE, 15 names	12,329	33.5	−10.84	−14.71	−181,409
2014	0DTE→60 DTE, 15 names	12,822	35.8	−7.77	−12.46	−159,705
2015	0DTE→60 DTE, 15 names	13,158	32.9	−7.88	−17.12	−225,237
2016	0DTE→60 DTE, 15 names	14,153	32.2	−10.33	−19.94	−282,182
2017	0DTE→60 DTE, 15 names	15,517	30.0	−12.71	−20.74	−321,751
2018	0DTE→60 DTE, 15 names	23,379	32.4	−8.45	−9.78	−228,634
2019	0DTE→60 DTE, 15 names	25,570	31.1	−15.74	−26.36	−673,917
2020	0DTE→60 DTE, 15 names	32,249	29.7	−8.71	−32.88	−1,060,327
2021	0DTE→60 DTE, 15 names	35,164	26.9	−22.75	−61.28	−2,154,972
2022	0DTE→60 DTE, 15 names	76,404	28.0	−12.52	−52.47	−4,009,075
2023	0DTE→60 DTE, 15 names	169,119	29.1	−8.26	−29.21	−4,940,725
2024	0DTE→60 DTE, 15 names	179,301	28.5	−6.45	−10.20	−1,829,470
2025	0DTE→60 DTE, 15 names	201,814	28.0	−9.12	−32.15	−6,488,858
2026	0DTE→60 DTE, 15 names	195,189	27.4	−10.44	−56.92	−11,110,146



Amendments, deviations and integrity checks. As 30a.

What it showed. A cost filter: the with-set still loses 8.4% a trade.

#30c Compass features X3: sell a same-day long option at the 14:00 bid (PASSED)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. Selling 0DTE longs at 14:00 beats holding to 15:58/settlement.

Rule. 0DTE base trades entered before 14:00, sold at the 14:00 bid.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b.

Verdict. PASS: holdout Δ +7.78 [+4.14, +10.95] (−9.51 with vs −17.29 without).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	291,090	964 / 2,388	28.3	+109.4, +381.24	−62.6, −184.44	−45.1, −26.37	**−13.88**, −24.21	[−16.20, −11.58]; [−33.58, −12.55]
0DTE→60 DTE, 15 names	holdout (opened)	233,454	1,290 / 2,428	29.0	+122.0, +271.22	−63.3, −127.51	−44.8, −22.37	**−9.51**, −11.72	[−13.87, −7.42]; [−19.67, −5.30]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+36,117.45, +538.3%, 2025-04-09, SPXW 0DTE 4940.0C	−9,452.55, −96.9%, 2025-04-09, SPXW 0DTE 5040.0P	0.82	7,048,089	2,388	total ≤ 0 (best 5 days +1,948,354; without them −8,996,442)
0DTE→60 DTE, 15 names	holdout (opened)	+19,167.45, +891.5%, 2025-11-20, SPXW 0DTE 6780.0P	−4,252.55, −93.3%, 2025-11-21, SPXW 0DTE 6580.0P	0.87	3,066,916	2,418	total ≤ 0 (best 5 days +1,176,529; without them −3,911,950)

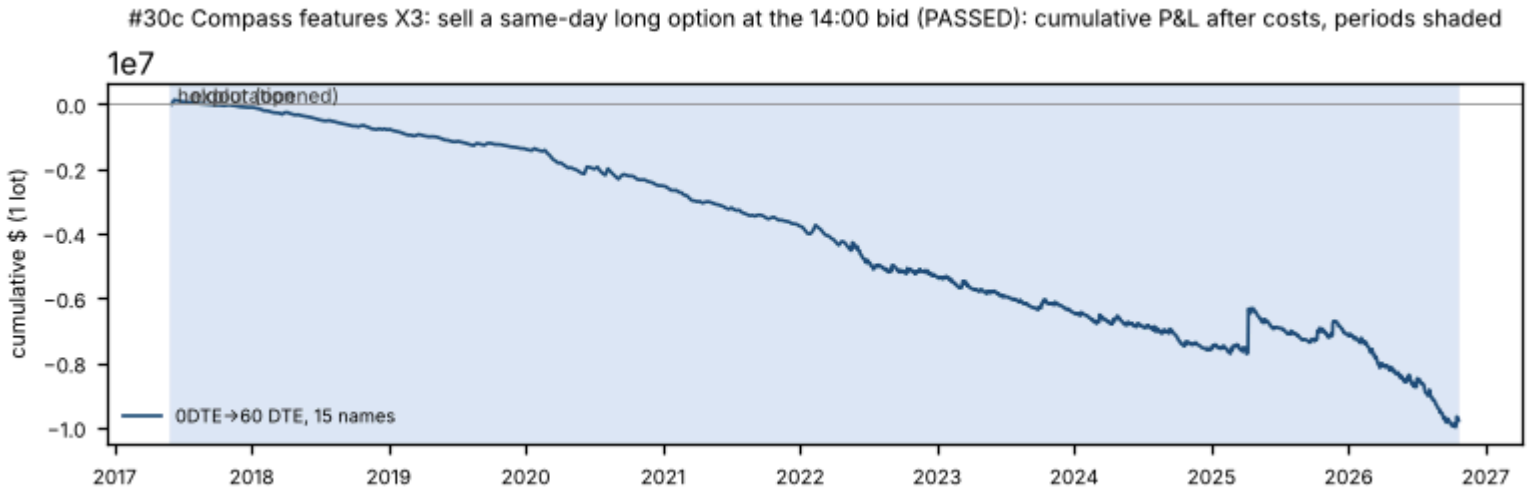
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: sold at the 14:00 bid 100.0%
- holdout (opened): sold at the 14:00 bid 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	0DTE→60 DTE, 15 names	9,741	20.5	−13.11	−10.64	−103,674
2018	0DTE→60 DTE, 15 names	23,860	24.5	−23.82	−29.00	−691,889
2019	0DTE→60 DTE, 15 names	21,472	23.1	−24.02	−26.75	−574,307
2020	0DTE→60 DTE, 15 names	29,501	26.7	−15.98	−39.20	−1,156,410
2021	0DTE→60 DTE, 15 names	27,934	25.9	−18.56	−44.60	−1,245,834
2022	0DTE→60 DTE, 15 names	47,279	29.8	−11.71	−33.06	−1,562,925
2023	0DTE→60 DTE, 15 names	72,946	30.8	−8.99	−15.24	−1,111,973
2024	0DTE→60 DTE, 15 names	84,765	29.3	−11.39	−13.45	−1,139,864
2025	0DTE→60 DTE, 15 names	93,284	30.7	−6.49	+4.56	+425,798
2026	0DTE→60 DTE, 15 names	113,762	28.4	−11.23	−23.05	−2,622,433



Amendments, deviations and integrity checks. As 30a.

What it showed. Selling at 14:00 halves the loss; the with-set still loses 9.5% a trade.

#30d Compass features BV1b: the V1 veto at 3% (PASSED, Phase B)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. A stricter spread veto.

Rule. The base trades with spread ≤ 3% of mid.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b (Phase B, read once).

Verdict. PASS: holdout Δ +4.30 [+2.66, +5.85] (−6.73 with vs −11.03) — the holdout was not virgin for it.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	265,416	2,576 / 3,671	30.8	+140.1, +924.53	−74.0, −478.80	−55.6, −82.37	**−7.95**, −46.00	[−11.44, −4.19]; [−75.14, −14.27]
0DTE→60 DTE, 15 names	holdout (opened)	251,808	3,437 / 3,686	31.9	+134.9, +692.20	−73.0, −363.11	−53.1, −65.37	**−6.73**, −26.77	[−9.38, −3.86]; [−40.12, −9.68]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+73,267.45, +453.4%, 2025-03-25, SPXW 31-60 5900.0P	−27,252.55, −77.6%, 2025-04-08, SPXW 31-60 5175.0P	0.86	12,394,931	3,671	total ≤ 0 (best 5 days +3,637,140; without them −15,846,316)
0DTE→60 DTE, 15 names	holdout (opened)	+45,477.45, +143.5%, 2026-03-30, SPXW 31-60 6220.0C	−26,112.55, −90.4%, 2026-03-30, SPXW 31-60 6510.0P	0.89	6,740,309	3,686	total ≤ 0 (best 5 days +1,733,658; without them −8,473,967)

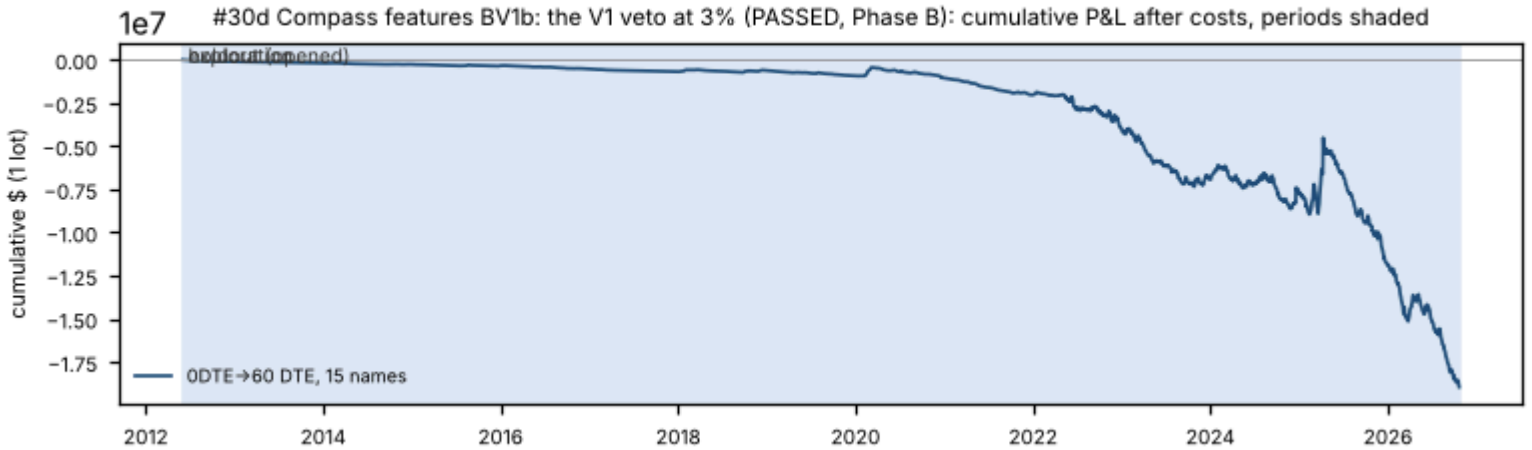
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: fixed exit 100.0%
- holdout (opened): fixed exit 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	0DTE→60 DTE, 15 names	3,539	34.4	−15.15	−28.74	−101,727
2013	0DTE→60 DTE, 15 names	5,476	35.3	−10.01	−16.44	−90,048
2014	0DTE→60 DTE, 15 names	5,572	37.0	−5.91	−11.37	−63,337
2015	0DTE→60 DTE, 15 names	4,958	35.1	−4.82	−16.56	−82,108
2016	0DTE→60 DTE, 15 names	8,797	33.2	−10.98	−22.50	−197,896
2017	0DTE→60 DTE, 15 names	5,822	31.2	−13.88	−27.01	−157,225
2018	0DTE→60 DTE, 15 names	8,205	38.0	+3.72	+8.55	+70,128
2019	0DTE→60 DTE, 15 names	11,814	36.1	−9.84	−26.10	−308,398
2020	0DTE→60 DTE, 15 names	12,459	35.2	+1.78	−9.92	−123,574
2021	0DTE→60 DTE, 15 names	14,837	31.7	−15.12	−65.73	−975,297
2022	0DTE→60 DTE, 15 names	36,317	32.2	−7.40	−54.28	−1,971,370
2023	0DTE→60 DTE, 15 names	85,208	32.3	−6.90	−34.23	−2,916,704
2024	0DTE→60 DTE, 15 names	93,592	31.3	−4.85	−8.80	−823,985
2025	0DTE→60 DTE, 15 names	120,615	30.1	−7.12	−34.04	−4,105,168
2026	0DTE→60 DTE, 15 names	100,013	29.2	−10.06	−71.02	−7,102,777



Amendments, deviations and integrity checks. 33 variants tried for 3 candidates.

What it showed. Still −6.7% a trade.

#30e Compass features BX3e: the 14:00 exit on the physically settled names only (PASSED, Phase B)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. X3 without SPXW.

Rule. X3 on SPY/QQQ/singles (physical settlement).

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30d.

Verdict. PASS: holdout Δ +7.67 [+4.23, +10.69] (−11.03 with vs −18.71).

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
0DTE→60 DTE, 15 names	exploration	225,023	897 / 2,388	27.3	+110.7, +206.30	−63.5, −107.62	−47.5, −20.37	**−15.96**, −21.91	[−18.40, −13.67]; [−28.33, −16.59]
0DTE→60 DTE, 15 names	holdout (opened)	203,289	1,257 / 2,428	28.4	+122.1, +152.15	−63.8, −80.05	−46.4, −20.37	**−11.03**, −14.09	[−15.23, −9.12]; [−19.41, −10.34]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
0DTE→60 DTE, 15 names	exploration	+10,633.63, +181.8%, 2022-02-04, AMZN 0DTE 3045.0C	−6,109.37, −97.1%, 2020-09-04, AMZN 0DTE 3150.0P	0.72	4,930,273	2,388	total ≤ 0 (best 5 days +867,361; without them −5,797,634)
0DTE→60 DTE, 15 names	holdout (opened)	+7,778.63, +421.6%, 2026-07-02, MU 0DTE 1055.0P	−3,458.37, −99.4%, 2021-12-17, AMZN 0DTE 3330.0P	0.75	3,023,529	2,418	total ≤ 0 (best 5 days +482,248; without them −3,346,742)

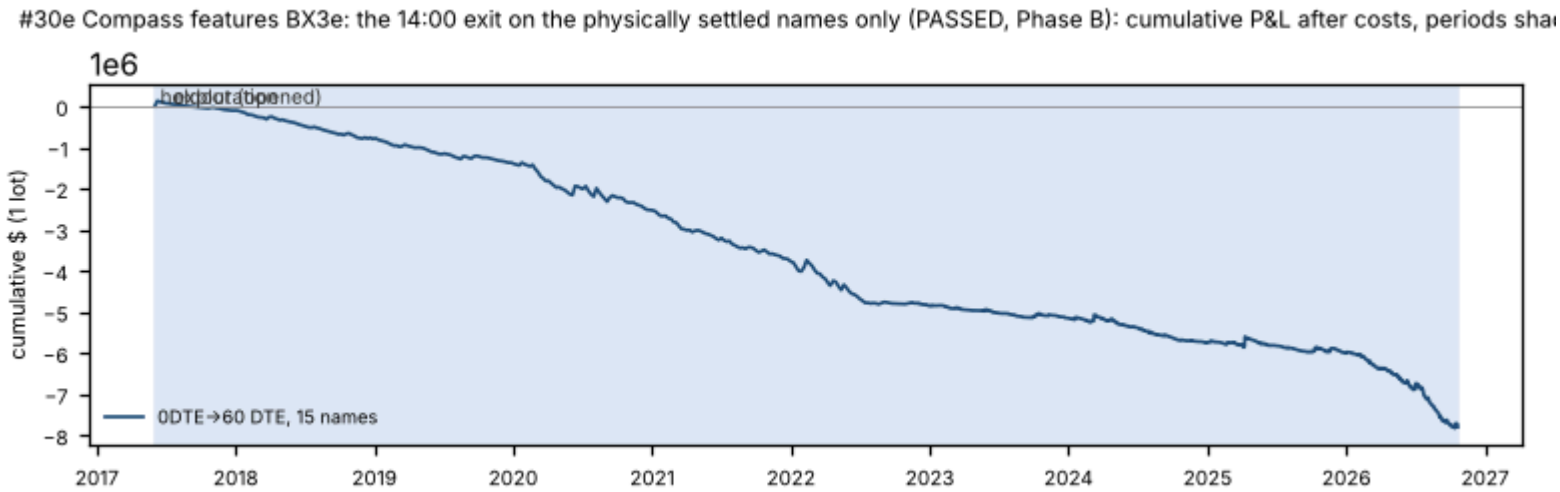
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration: sold at the 14:00 bid 100.0%
- holdout (opened): sold at the 14:00 bid 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2017	0DTE→60 DTE, 15 names	9,741	20.5	−13.11	−10.64	−103,674
2018	0DTE→60 DTE, 15 names	23,860	24.5	−23.82	−29.00	−691,889
2019	0DTE→60 DTE, 15 names	21,472	23.1	−24.02	−26.75	−574,307
2020	0DTE→60 DTE, 15 names	29,501	26.7	−15.98	−39.20	−1,156,410
2021	0DTE→60 DTE, 15 names	27,934	25.9	−18.56	−44.60	−1,245,834
2022	0DTE→60 DTE, 15 names	33,013	28.0	−15.44	−32.63	−1,077,365
2023	0DTE→60 DTE, 15 names	51,821	29.9	−10.03	−5.80	−300,459
2024	0DTE→60 DTE, 15 names	63,319	28.4	−13.63	−9.52	−602,667
2025	0DTE→60 DTE, 15 names	71,666	30.1	−7.83	−3.51	−251,270
2026	0DTE→60 DTE, 15 names	95,985	28.1	−12.29	−18.66	−1,790,893



Amendments, deviations and integrity checks. As 30d.

What it showed. Still −11% a trade.

#30f Compass features VN1 (venue SPY vs SPXW) and BC3 (V1 + X3 + D6)

Folder: `research_notes/Compass features/`. **Commits:** inventory 04ce685; prereg a2e14e4 / Phase B prereg b6737c7; forward rule b101ae1 / Phase A 2a49049; Phase B e5dd8c6; forward wired 850268e. **Dates:** 2026-10-22.

Hypothesis. The cheaper venue, and the stacked model, lose less.

Rule. VN1: keep the SPXW/SPY venue with the lower ((ask − mid) + fee ÷ 100) ÷ mid. BC3: V1 veto + X3 exit + D6 direction.

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); bought at the ask, sold at the bid of the 1-minute NBBO (ThetaData) (0DTE at 13 fixed minutes; 1–60 DTE at the EOD closing ask)

Data. The base trades: 0DTE at 13 fixed minutes (SPY/QQQ/SPXW and the singles' expiry days) and 1–7, 8–30, 31–60 DTE at the EOD close, 15 names, five strike buckets, 2012-06 → 2026-10 (by span).

Periods and holdouts. exploration / holdout — the holdout (43 months + AVGO, MU, QQQ) was OPENED: read twice in Phase A (the first on a table with duplicated rows) and once in Phase B; forward from 2026-10-22 (0 sessions).

Pass rule (as pre-registered). As 30b.

Verdict. PASS: VN1 holdout Δ +1.18 [+0.50, +2.20] (failed on the duplicated table); BC3 +6.77 [+4.82, +8.19] (−4.26 with vs −11.03).

Metrics. not available — VN1 pairs venues and BC3 needs the D6 direction feature; both are computed inside the study's screen code, and their per-trade arms are not on disk.

Amendments, deviations and integrity checks. VN1 failed on the first (duplicated) holdout read and passes on the corrected one.

What it showed. Cost and timing filters; no direction feature passed.

#31 Compass direction: post-earnings drift in 31–60 DTE options (rerun)

Folder: `research_notes/Compass direction/`. **Commits:** 38b11bf / ledger code 043d723 / 94fb69d. **Dates:** 2026-10-22.

Hypothesis. After a report, options on the side of the earnings-day move drift in its direction (PEAD).

Rule. At the close after the report: the 31–60 DTE expiry nearest 45 days; buy five strike buckets on the picked side (the sign of the report-day return) at the closing ask; sell at the closing bid 10 full sessions later ($0 if no bid).

Fills and costs. $0.68748 a contract each way (SPY, QQQ, IWM: $0.65 commission + $0.03748 pass-throughs), nothing at expiry; the closing NBBO (EOD).

Data. Exploration set X: 1,601 names, 3,129 events (12,114 picked-side trades) 2019-04 → 2026-09.

Periods and holdouts. Exploration screen only; the 1,600 holdout names (2019-01 → 2022-12) UNOPENED.

Pass rule (as pre-registered). Screen: Δ (picked − both sides) > 0 with its block-20 interval clear of zero; then the holdout once.

Verdict. FAIL: Δ −0.90 pp [−3.53, +1.71]; the holdout is not fetched and not read.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	exploration (holdout unopened)	12,114	893 / 1,920	26.1	+79.5, +471.43	−60.9, −313.71	−47.1, −86.37	**−24.29**, −108.97	[−28.19, −19.82]; [−136.77, −79.20]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	exploration (holdout unopened)	+18,028.63, +195.5%, 2023-10-30, CMG 2023-12-15 1880.0C	−13,866.37, −72.0%, 2026-06-26, MU 2026-08-07 1040.0C	0.53	1,326,491	1,840	total ≤ 0 (best 5 days +160,543; without them −1,480,599)

Random baseline: not available — no random baseline was computed by this study

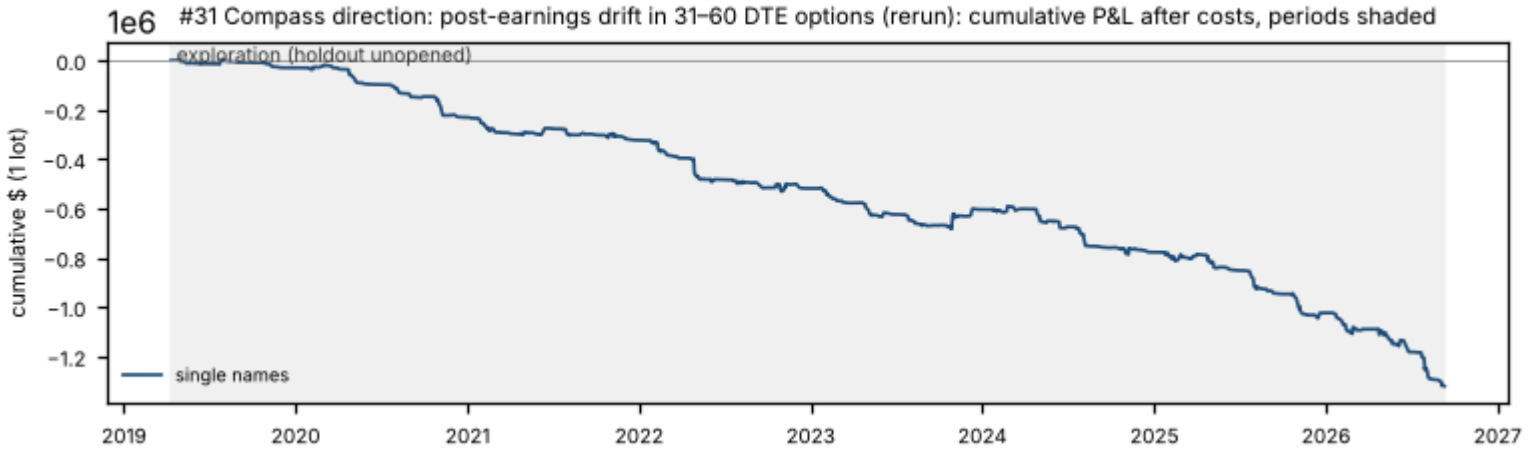
Exit reasons (share of trades):

- exploration (holdout unopened): closing bid 10 sessions later 94.5%; no bid at exit ($0) 5.5%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2019	single names	702	29.8	−10.35	−41.98	−29,472
2020	single names	1,187	19.4	−27.94	−169.27	−200,926
2021	single names	1,485	28.6	−20.78	−62.05	−92,150
2022	single names	1,549	28.3	−19.35	−126.64	−196,162
2023	single names	1,520	25.1	−25.15	−56.19	−85,408

year	name	trades	win %	mean %	mean $	total $
2024	single names	1,604	28.3	−24.67	−107.80	−172,908
2025	single names	2,170	27.1	−24.51	−113.65	−246,622
2026	single names	1,897	22.9	−32.71	−156.25	−296,410



Amendments, deviations and integrity checks. Rerun for this report at 94fb69d from the stored files: screen.json equal to the committed one to the last float digit (summation order); the trade table expl_trades.parquet rewritten in the scratchpad.

What it showed. The picked side loses 24% of the debit a trade, no better than the other side.

#32 Compass H1 / edge H1: overnight reversal of the late-day move (SPY, QQQ stock)

Folder: `research_notes/Compass H1/`. **Commits:** 38aca66 / — / 4a02144. **Dates:** 2026-10-23.

Hypothesis. The 15:00 → close move reverses overnight.

Rule. Short 100 shares at the official close if the close > the 15:00 price, long if below; exit at the next 09:30 open.

Fills and costs. $0.01 a share each way ($2.00 a trade at 1×); 2× shown in the study.

Data. SPY 2017-01-03 → 2025-10-10, QQQ 2012-06-01 → 2025-10-10.

Periods and holdouts. Single test; no holdout.

Pass rule (as pre-registered). In each name: mean $ a trade after costs > 0 and the 20-session block interval clear of zero (both names).

Verdict. FAIL: SPY +$11.09 [−4.42, +28.72], QQQ +$9.42 [−0.06, +18.61].

Metrics (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	3,299	3,299 / 3,420	50.5	+0.535, +133.53	−0.491, −117.20	+0.007, +0.92	**+0.027**, +9.42	[−0.006, +0.064]; [−0.19, +18.84]
SPY	single test	2,177	2,177 / 2,266	50.8	+0.479, +181.49	−0.431, −164.89	+0.010, +4.00	**+0.032**, +11.09	[−0.013, +0.084]; [−4.56, +27.78]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+2,400.88, +5.4%, 2024-08-02, 100 QQQ short	−1,449.35, −3.4%, 2025-04-07, 100 QQQ short	1.16	8,311	1,157	32%
SPY	single test	+3,026.86, +11.1%, 2020-03-13, 100 SPY short	−1,750.42, −3.5%, 2025-04-07, 100 SPY short	1.14	10,434	565	46%

Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- single test: next 09:30 open 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	142	48.6	−0.03	−1.78	−252
2013	QQQ	247	52.6	+0.04	+3.03	+747
2014	QQQ	244	48.4	−0.03	−3.18	−776
2015	QQQ	246	49.6	−0.02	−1.75	−430
2016	QQQ	248	46.8	+0.02	+1.43	+355

year	name	trades	win %	mean %	mean $	total $
2017	QQQ	248	52.4	+0.01	+0.93	+231
2018	QQQ	247	45.7	−0.05	−8.40	−2,074
2019	QQQ	248	43.5	−0.08	−14.19	−3,520
2020	QQQ	247	54.7	+0.19	+38.01	+9,388
2021	QQQ	247	55.9	+0.06	+17.11	+4,227
2022	QQQ	249	52.6	+0.12	+36.64	+9,122
2023	QQQ	246	52.0	+0.03	+11.10	+2,731
2024	QQQ	248	54.0	+0.09	+41.73	+10,349
2025	QQQ	192	49.0	−0.00	+5.13	+984
2017	SPY	245	52.2	+0.03	+6.27	+1,537
2018	SPY	246	47.2	−0.03	−8.82	−2,171
2019	SPY	248	42.3	−0.05	−15.09	−3,743
2020	SPY	251	57.4	+0.21	+60.54	+15,197
2021	SPY	251	53.8	+0.02	+8.11	+2,036
2022	SPY	247	53.8	+0.08	+31.70	+7,830
2023	SPY	247	49.0	+0.01	+6.02	+1,486
2024	SPY	249	52.6	+0.04	+22.58	+5,622
2025	SPY	193	48.2	−0.04	−18.96	−3,659
2012	all names	142	48.6	−0.03	−1.78	−252
2013	all names	247	52.6	+0.04	+3.03	+747
2014	all names	244	48.4	−0.03	−3.18	−776
2015	all names	246	49.6	−0.02	−1.75	−430
2016	all names	248	46.8	+0.02	+1.43	+355
2017	all names	493	52.3	+0.02	+3.59	+1,769
2018	all names	493	46.5	−0.04	−8.61	−4,245
2019	all names	496	42.9	−0.06	−14.64	−7,263
2020	all names	498	56.0	+0.20	+49.37	+24,585
2021	all names	498	54.8	+0.04	+12.58	+6,263
2022	all names	496	53.2	+0.10	+34.18	+16,952
2023	all names	493	50.5	+0.02	+8.55	+4,216
2024	all names	497	53.3	+0.07	+32.14	+15,971
2025	all names	385	48.6	−0.02	−6.95	−2,675



Amendments, deviations and integrity checks. verify_h1.py: 50 random trades re-derived, 0 differences.

What it showed. The longs carry the overnight drift; the shorts lose.

#34 Edge H3: earnings IV-crush short iron condor (single names)

Folder: `research_notes/Edge search/H3`. **Commits:** 001a136 / — / 34c8fd8. **Dates:** 2026-10-23.

Hypothesis. Implied volatility collapses across the report enough to pay a short condor.

Rule. Shorts at K0 ± 0.5 implied move, longs ± 1.5, first expiry after R within 14 days, sold at the closing quote before the report, bought back at the reaction session's close.

Fills and costs. Bid/ask both ways, $0.68748 a contract each way.

Data. Exploration: 1,601 X names, 2,209 condors 2019-04 → 2022-12.

Periods and holdouts. Exploration screen; holdout (1,600 H names) not fetched, UNOPENED.

Pass rule (as pre-registered). Screen: mean > 0 with the block interval clear of zero.

Verdict. FAIL: −$161.54 a condor [−180.68, −142.46] (−30.2% of max risk); at the mid −$20.

Metrics (return = % of max risk; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
single names	exploration (holdout unopened)	2,209	554 / 957	30.3	+21.0, +118.26	−52.5, −283.09	−22.6, −44.81	**−30.24**, −161.54	[−32.38, −27.80]; [−180.79, −142.84]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
single names	exploration (holdout unopened)	+1,560.19, +19.3%, 2022-04-26, CMG 2022-04-29 1270.0/1385.0P 1495.0/1605.0C	−6,539.81, −111.2%, 2022-07-26, CMG 2022-07-29 1180.0/1270.0P 1365.0/1455.0C	0.18	356,842	957	total ≤ 0 (best 5 days +4,919; without them −361,761)

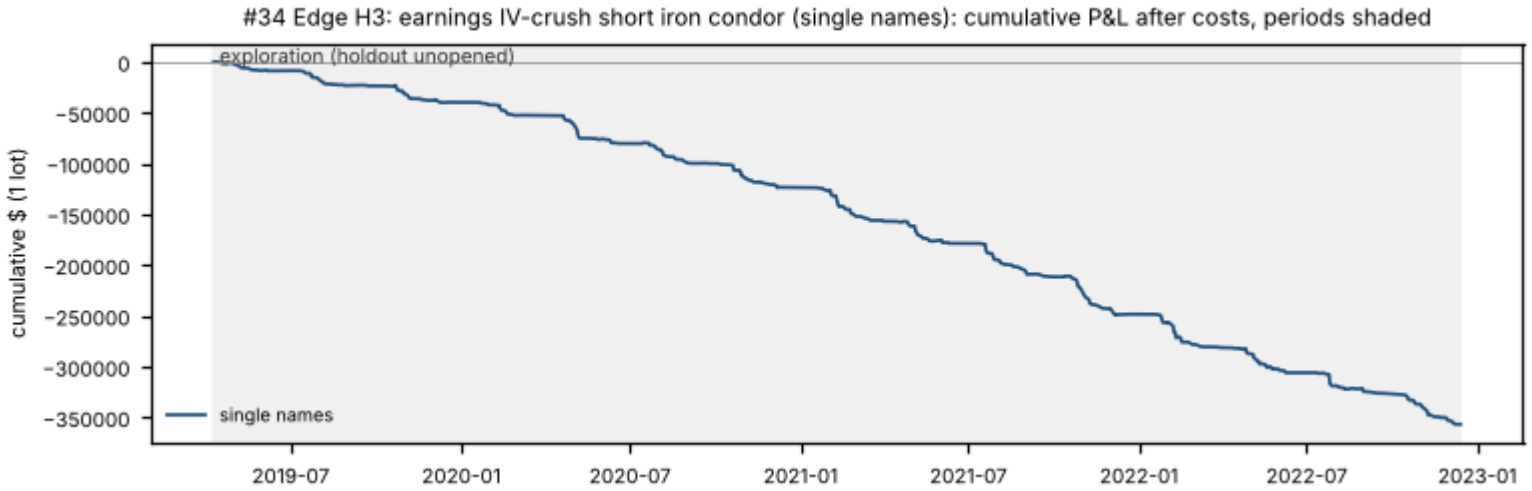
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- exploration (holdout unopened): quotes 100.0%; split-sized move: booked at max loss 0.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2019	single names	403	35.0	−28.00	−99.06	−39,920
2020	single names	504	26.0	−32.37	−166.49	−83,909
2021	single names	649	30.0	−32.72	−192.07	−124,656
2022	single names	653	30.9	−27.50	−165.94	−108,357



Amendments, deviations and integrity checks. —

What it showed. The spread and fees take the crush.

#35 Edge H4: market intraday momentum (SPY, QQQ stock)

Folder: `research_notes/Edge search/H4`. **Commits:** b14b640 / — / dfa575d. **Dates:** 2026-10-23.

Hypothesis. The first half hour's return predicts the last half hour (Gao, Han, Li & Zhou 2018).

Rule. r1 = the 09:59 bar close ÷ the previous official close − 1; long 100 at the 15:30 bar open if r1 > 0, short if < 0; exit at the official close.

Fills and costs. $0.005 a share entry + $0.01 auction ($1.50 a trade).

Data. Test 2025-10-13 → 2026-10-22 (257 sessions a name); exploration (context) SPY 2017 → 2025-10-10, QQQ 2012 → .

Periods and holdouts. test (the verdict) / exploration (context only, spent by S12 and others).

Pass rule (as pre-registered). Each name: mean > 0 with the block-20 interval clear of zero.

Verdict. FAIL: SPY −$2.95 [−19.01, +15.37], QQQ +$10.32 [−8.56, +35.77].

Metrics (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	exploration (context)	3,319	3,319 / 3,419	48.1	+0.194, +45.75	−0.193, −47.21	−0.008, −1.50	**−0.007**, −2.53	[−0.017, +0.005]; [−5.13, +0.56]
SPY	exploration (context)	2,184	2,184 / 2,265	48.9	+0.192, +72.19	−0.190, −72.03	−0.003, −1.25	**−0.003**, −1.51	[−0.018, +0.013]; [−6.94, +4.41]
QQQ	test 2025-10-13→2026-10-22	257	257 / 267	49.8	+0.180, +120.02	−0.152, −98.53	−0.001, −0.50	**+0.014**, +10.32	[−0.016, +0.048]; [−9.41, +35.26]
SPY	test 2025-10-13→2026-10-22	257	257 / 267	45.5	+0.126, +89.37	−0.114, −80.10	−0.012, −8.50	**−0.004**, −2.95	[−0.028, +0.020]; [−19.50, +14.78]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	exploration (context)	+779.50, +4.2%, 2020-03-13, 100 QQQ long	−752.50, −4.5%, 2020-03-18, 100 QQQ short	0.90	12,532	1,938	total ≤ 0 (best 5 days +3,462; without them −11,870)
SPY	exploration (context)	+1,249.04, +4.8%, 2020-03-13, 100 SPY long	−744.51, −2.6%, 2020-02-28, 100 SPY short	0.96	9,267	1,417	total ≤ 0 (best 5 days +4,113; without them −7,402)
QQQ	test 2025-10-13→2026-10-22	+902.49, +1.3%, 2026-07-29, 100 QQQ short	−377.50, −0.6%, 2025-11-20, 100 QQQ long	1.21	2,278	142	92%
SPY	test 2025-10-13→2026-10-22	+573.49, +0.8%, 2026-07-29, 100 SPY short	−372.50, −0.5%, 2026-09-30, 100 SPY long	0.93	3,144	240	total ≤ 0 (best 5 days +1,737; without them −2,495)

Random baseline: not available — no random baseline was computed by this study

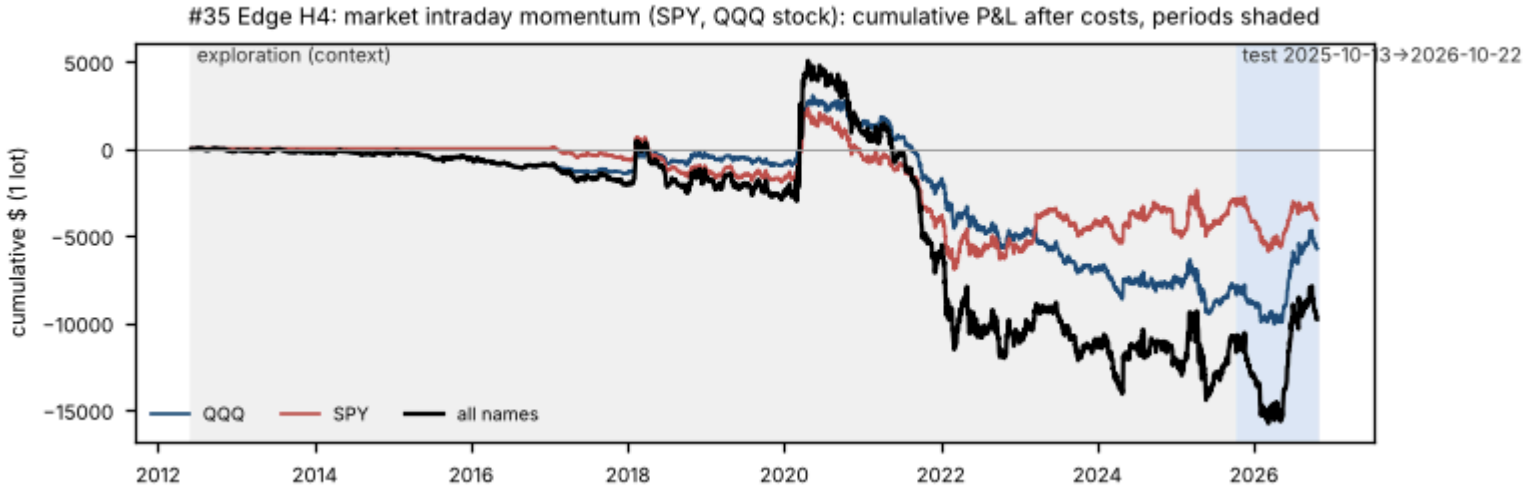
Exit reasons (share of trades):

- exploration (context): official close 100.0%
- test 2025-10-13→2026-10-22: official close 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	142	48.6	−0.01	−0.84	−120
2013	QQQ	245	46.1	−0.01	−0.40	−99
2014	QQQ	245	46.1	−0.00	−0.32	−79
2015	QQQ	250	45.6	−0.01	−0.86	−216
2016	QQQ	251	48.6	−0.02	−1.89	−475
2017	QQQ	249	45.0	−0.01	−1.55	−385
2018	QQQ	248	55.2	+0.03	+3.97	+985
2019	QQQ	249	48.6	−0.01	−2.06	−513
2020	QQQ	251	52.2	+0.05	+8.79	+2,206
2021	QQQ	250	44.4	−0.04	−13.21	−3,302
2022	QQQ	250	47.6	−0.04	−12.83	−3,209
2023	QQQ	248	48.0	−0.02	−6.82	−1,691
2024	QQQ	248	48.0	−0.01	−2.62	−649
2025	QQQ	247	47.8	−0.01	−5.26	−1,299
2026	QQQ	203	51.7	+0.02	+15.21	+3,088
2017	SPY	246	44.7	−0.01	−2.71	−666
2018	SPY	248	50.8	−0.01	−2.21	−548
2019	SPY	248	50.8	−0.01	−2.07	−513
2020	SPY	251	46.6	+0.03	+5.04	+1,265
2021	SPY	251	43.0	−0.03	−14.11	−3,543
2022	SPY	250	50.0	−0.02	−7.61	−1,903
2023	SPY	248	51.2	+0.02	+6.98	+1,731
2024	SPY	249	49.4	−0.00	−2.61	−649
2025	SPY	247	52.2	+0.00	+2.81	+695
2026	SPY	203	46.3	+0.00	+0.42	+84
2012	all names	142	48.6	−0.01	−0.84	−120
2013	all names	245	46.1	−0.01	−0.40	−99
2014	all names	245	46.1	−0.00	−0.32	−79
2015	all names	250	45.6	−0.01	−0.86	−216
2016	all names	251	48.6	−0.02	−1.89	−475
2017	all names	495	44.8	−0.01	−2.12	−1,051
2018	all names	496	53.0	+0.01	+0.88	+437
2019	all names	497	49.7	−0.01	−2.07	−1,026
2020	all names	502	49.4	+0.04	+6.91	+3,471
2021	all names	501	43.7	−0.03	−13.66	−6,845
2022	all names	500	48.8	−0.03	−10.22	−5,112

year	name	trades	win %	mean %	mean $	total $
2023	all names	496	49.6	−0.00	+0.08	+40
2024	all names	497	48.7	−0.01	−2.61	−1,298
2025	all names	494	50.0	−0.00	−1.22	−603
2026	all names	406	49.0	+0.01	+7.81	+3,172



Amendments, deviations and integrity checks. A-priori MDE $28–37 a trade: weak evidence either way.

What it showed. No momentum after costs.

#37a Edge H7: the pre-FOMC drift (SPY, QQQ stock) — PASSED

Folder: `research_notes/Edge search/H7_H8_H9`. **Commits:** edca472 / forward rule fcc1f4d / 6c94bb8; wired 712b0ad; baseline 827bd5e. **Dates:** 2026-10-23.

Hypothesis. Stocks drift up in the 24 hours before a scheduled FOMC statement (Lucca & Moench 2015).

Rule. Buy 100 at the official close of the session before each scheduled 2:00 pm statement (2013-03-20 on); sell at the 13:58 bar's close on the statement day.

Fills and costs. $0.01 a share auction entry + $0.005 exit ($1.50 a trade).

Data. SPY 77 statements 2017-02 → 2026-09; QQQ 108 2013-03 → 2026-09.

Periods and holdouts. Single test (no study had read the signed pre-FOMC return); forward from 2026-10-28 (N 366 SPY / 483 QQQ).

Pass rule (as pre-registered). Each name: mean > 0 with the block-20 interval (calendar sessions, 2,000 draws, seed 20261023) clear of zero at 1× and 2×; Bonferroni m = 13 for an EDGE.

Verdict. PASS both names: SPY +$70.43 [+14.64, +129.31], QQQ +$69.59 [+23.10, +122.14]; QQQ clears Bonferroni (EDGE under rule 5); not "works" before its forward N.

Metrics (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	108	108 / 3,462	61.1	+0.645, +195.18	−0.473, −127.77	+0.110, +17.50	**+0.210**, +69.59	[+0.049, +0.355]; [+17.74, +114.93]
SPY	single test	77	77 / 2,487	55.8	+0.562, +224.67	−0.281, −124.63	+0.060, +15.50	**+0.190**, +70.43	[+0.059, +0.355]; [+11.12, +131.07]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+1,161.50, +2.5%, 2024-07-30, 100 QQQ long, FOMC 2024-07-31	−542.50, −0.8%, 2026-07-28, 100 QQQ long, FOMC 2026-07-29	2.40	1,003	1,370	56%
SPY	single test	+823.10, +1.5%, 2024-07-30, 100 SPY long, FOMC 2024-07-31	−620.50, −1.6%, 2021-01-26, 100 SPY long, FOMC 2021-01-27	2.28	1,065	295	66%

Random baseline: descriptive baseline against every non-FOMC session over the same clock window (H7_BASELINE.md, 827bd5e), no new verdict.

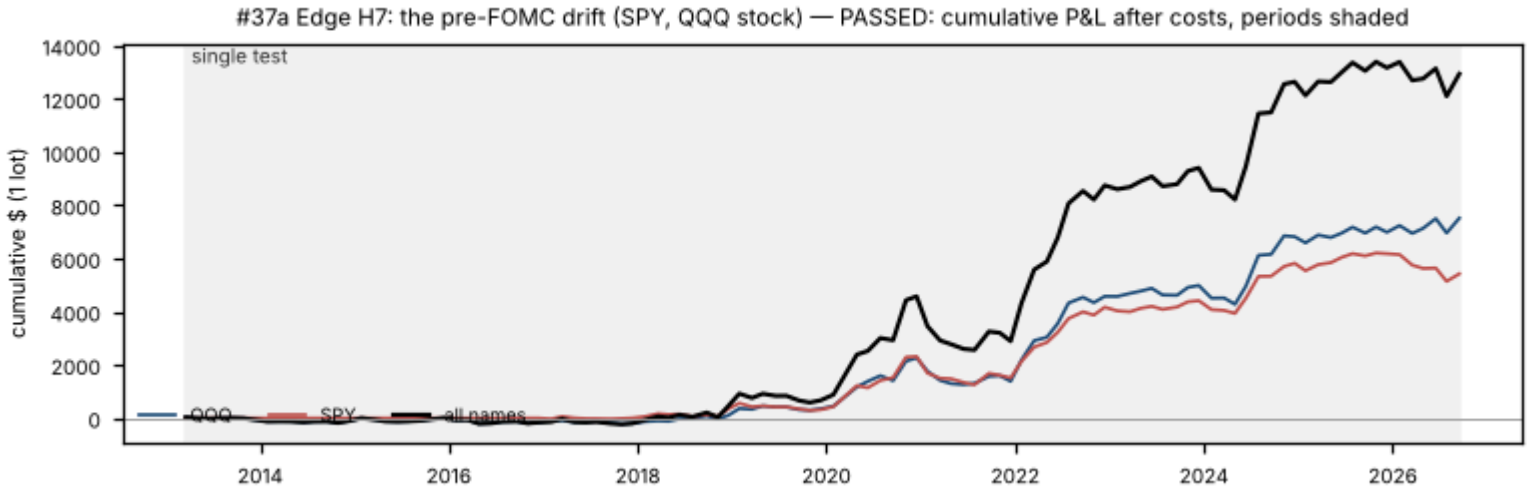
Exit reasons (share of trades):

- single test: 13:58 bar close on the statement day 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2013	QQQ	7	42.9	−0.12	−9.93	−70
2014	QQQ	8	50.0	−0.05	−3.44	−28
2015	QQQ	8	62.5	+0.12	+12.57	+101

year	name	trades	win %	mean %	mean $	total $
2016	QQQ	8	62.5	−0.20	−20.75	−166
2017	QQQ	8	62.5	−0.02	−4.56	−37
2018	QQQ	8	75.0	+0.23	+38.19	+305
2019	QQQ	8	50.0	+0.19	+30.71	+246
2020	QQQ	7	85.7	+1.12	+273.00	+1,911
2021	QQQ	8	37.5	−0.34	−109.81	−879
2022	QQQ	8	87.5	+1.28	+399.29	+3,194
2023	QQQ	8	62.5	+0.16	+51.19	+410
2024	QQQ	8	62.5	+0.46	+229.09	+1,833
2025	QQQ	8	50.0	+0.04	+20.79	+166
2026	QQQ	6	66.7	+0.11	+88.00	+528
2017	SPY	8	37.5	+0.00	+0.50	+4
2018	SPY	8	62.5	+0.17	+43.94	+352
2019	SPY	8	50.0	−0.01	−4.65	−37
2020	SPY	7	85.7	+0.90	+285.20	+1,996
2021	SPY	8	12.5	−0.26	−100.59	−805
2022	SPY	8	87.5	+0.82	+332.50	+2,660
2023	SPY	8	62.5	+0.07	+30.82	+247
2024	SPY	8	62.5	+0.30	+175.51	+1,404
2025	SPY	8	62.5	+0.08	+45.25	+362
2026	SPY	6	33.3	−0.18	−126.53	−759
2013	all names	7	42.9	−0.12	−9.93	−70
2014	all names	8	50.0	−0.05	−3.44	−28
2015	all names	8	62.5	+0.12	+12.57	+101
2016	all names	8	62.5	−0.20	−20.75	−166
2017	all names	16	50.0	−0.01	−2.03	−33
2018	all names	16	68.8	+0.20	+41.06	+657
2019	all names	16	50.0	+0.09	+13.03	+208
2020	all names	14	85.7	+1.01	+279.10	+3,907
2021	all names	16	25.0	−0.30	−105.20	−1,683
2022	all names	16	87.5	+1.05	+365.90	+5,854
2023	all names	16	62.5	+0.11	+41.00	+656
2024	all names	16	62.5	+0.38	+202.30	+3,237
2025	all names	16	56.2	+0.06	+33.02	+528
2026	all names	12	50.0	−0.03	−19.27	−231



Amendments, deviations and integrity checks. The gross is the night (close → 09:30 ≈ +$73–74); QQQ before March 2016 −$3.00.

What it showed. About +$70 a trade (0.2% of notional) in both names, carried overnight.

#37b Edge H8: turn of the month (SPY, QQQ stock)

Folder: `research_notes/Edge search/H7_H8_H9`. **Commits:** edca472 / — / 6c94bb8. **Dates:** 2026-10-23.

Hypothesis. Returns concentrate from the last trading day to the third of the new month.

Rule. Buy 100 at the official close of the second-to-last session of each month, sell at the official close of the third session of the next.

Fills and costs. $0.01 a share each way ($2.00 a trade).

Data. SPY 117 turns 2017-01 → 2026-09; QQQ 172 2012-06 → 2026-09.

Periods and holdouts. Single test.

Pass rule (as pre-registered). As H7.

Verdict. SPY FAIL; QQQ PASS at 1× only (+$110.22 [+0.84, +218.26]) — not an EDGE.

Metrics (return = % of notional ($ = per 100 sh); win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
QQQ	single test	172	172 / 3,651	59.3	+1.844, +503.91	−1.737, −463.44	+0.398, +80.50	**+0.387**, +110.22	[+0.006, +0.694]; [−10.71, +211.82]
SPY	single test	117	117 / 2,498	60.7	+1.479, +607.54	−1.491, −596.91	+0.414, +154.00	**+0.311**, +134.00	[−0.052, +0.642]; [−20.87, +264.39]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
QQQ	single test	+3,373.00, +4.9%, 2026-07-30, 100 QQQ long, turn 2026-07	−2,370.00, −3.8%, 2026-01-29, 100 QQQ long, turn 2026-01	1.58	4,243	1,015	64%
SPY	single test	+2,808.00, +3.8%, 2026-07-30, 100 SPY long, turn 2026-07	−2,464.00, −4.5%, 2024-07-30, 100 SPY long, turn 2024-07	1.57	4,630	498	66%

Random baseline: not available — no random baseline was computed by this study

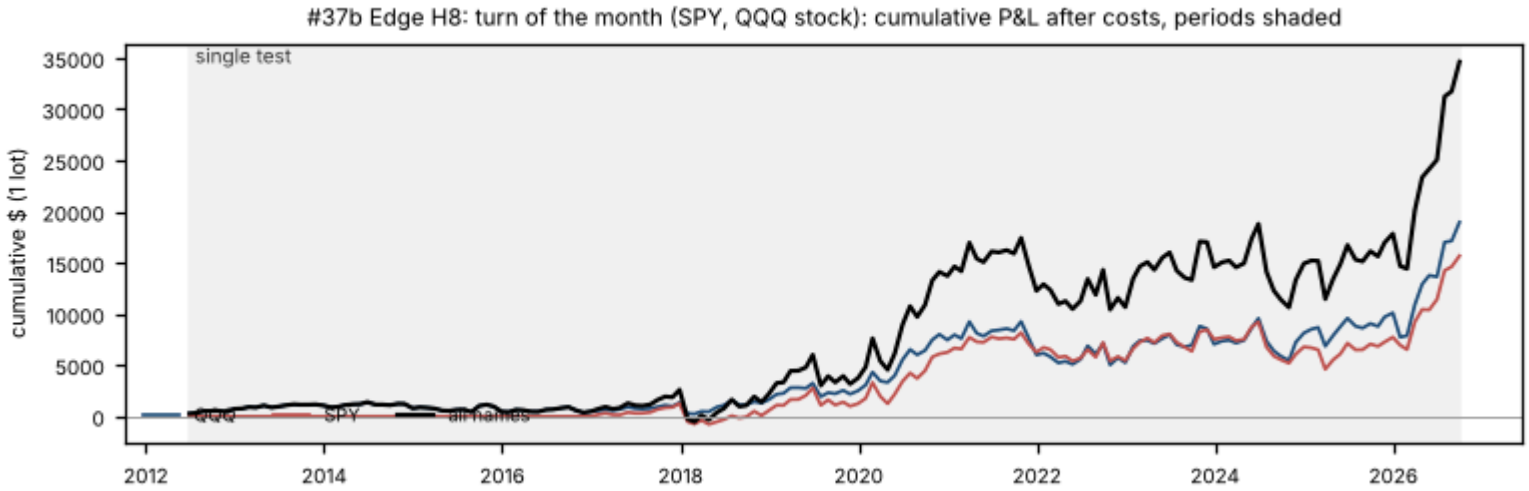
Exit reasons (share of trades):

- single test: official close, 3rd session 100.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2012	QQQ	7	71.4	+1.62	+103.62	+725
2013	QQQ	12	66.7	+0.42	+27.51	+330
2014	QQQ	12	50.0	−0.19	−22.33	−268
2015	QQQ	12	41.7	−0.19	−27.50	−330

year	name	trades	win %	mean %	mean $	total $
2016	QQQ	12	50.0	+0.03	+1.67	+20
2017	QQQ	12	66.7	+0.49	+71.83	+862
2018	QQQ	12	58.3	+0.18	+23.67	+284
2019	QQQ	12	66.7	+0.45	+73.50	+882
2020	QQQ	12	66.7	+1.68	+413.08	+4,957
2021	QQQ	12	50.0	−0.24	−121.75	−1,461
2022	QQQ	12	50.0	−0.19	−63.33	−760
2023	QQQ	12	58.3	+0.61	+150.50	+1,806
2024	QQQ	12	58.3	+0.17	+90.67	+1,088
2025	QQQ	12	66.7	+0.27	+164.25	+1,971
2026	QQQ	9	77.8	+1.50	+983.56	+8,852
2017	SPY	12	75.0	+0.42	+104.92	+1,259
2018	SPY	12	58.3	−0.18	−59.58	−715
2019	SPY	12	66.7	+0.24	+58.62	+703
2020	SPY	12	75.0	+1.24	+417.33	+5,008
2021	SPY	12	41.7	+0.05	+1.08	+13
2022	SPY	12	50.0	−0.14	−69.50	−834
2023	SPY	12	58.3	+0.48	+174.25	+2,091
2024	SPY	12	58.3	−0.12	−61.58	−739
2025	SPY	12	58.3	+0.10	+79.08	+949
2026	SPY	9	66.7	+1.25	+882.44	+7,942
2012	all names	7	71.4	+1.62	+103.62	+725
2013	all names	12	66.7	+0.42	+27.51	+330
2014	all names	12	50.0	−0.19	−22.33	−268
2015	all names	12	41.7	−0.19	−27.50	−330
2016	all names	12	50.0	+0.03	+1.67	+20
2017	all names	24	70.8	+0.45	+88.37	+2,121
2018	all names	24	58.3	−0.00	−17.96	−431
2019	all names	24	66.7	+0.34	+66.06	+1,585
2020	all names	24	70.8	+1.46	+415.21	+9,965
2021	all names	24	45.8	−0.09	−60.33	−1,448
2022	all names	24	50.0	−0.17	−66.42	−1,594
2023	all names	24	58.3	+0.54	+162.37	+3,897
2024	all names	24	58.3	+0.03	+14.54	+349
2025	all names	24	62.5	+0.19	+121.67	+2,920
2026	all names	18	72.2	+1.37	+933.00	+16,794



Amendments, deviations and integrity checks. The nine 2026 turns are about half of each total.

What it showed. Positive means, intervals at or through zero.

#36 Edge H5 (SPY − SPXW late straddle) and H6 (monthly put spreads), forward only

Folder: `research_notes/Edge search/`. **Commits:** e114e30 / — / wired 23c0cc3. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. (See the prereg.)

Fills and costs. —

Data. Forward only.

Periods and holdouts. Forward; no historical read.

Pass rule (as pre-registered). At N forward trades only.

Verdict. No verdict.

Metrics. not available — forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is “not available” because no forward session has been counted. H5: no forward session read yet (N = 1,326 sessions); H6: entries from 2026-11-02 (N = 449 months). Their historical reads are spent (the SPXW report, S17).

Amendments, deviations and integrity checks. —

What it showed. Nothing to report yet.

#38 Edge H9: overnight drift (forward only, withdrawn)

Folder: `research_notes/Edge search/`. **Commits:** edca472 / — / wired 712b0ad, removed c68f389. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. (See the prereg.)

Fills and costs. —

Data. Forward only.

Periods and holdouts. Forward; no historical read.

Pass rule (as pre-registered). At N forward trades only.

Verdict. No verdict.

Metrics. not available — forward-only: no trade exists. Every metric (trades, win rate, mean, interval, drawdown, by year) is “not available” because no forward session has been counted. Withdrawn by the owner 2026-10-23 (c68f389) before any forward row was written; the registered rule edca472 is unchanged.

Amendments, deviations and integrity checks. —

What it showed. Nothing to report yet.

#39 Compass scanner (score ≥ 80, SPY/QQQ/IWM 0DTE)

Folder: `research_notes/Compass scanner/`. **Commits:** 142fcf4 / — / 9e3888e. **Dates:** 2026-10-23.

Hypothesis. The owner’s scanner score (RSI/VWAP/RVOL/structure) gives a forward move worth an option trade.

Rule. Signal check first: the underlying’s forward return after a signal vs non-signal minutes; only then the option P&L.

Fills and costs. —

Data. SPY/QQQ/IWM bars, Pick 2017-07-05 → 2022-11-18 and the option layer’s Pick.

Periods and holdouts. Pick / Check; the 28 holdout months and AMD/AVGO UNOPENED.

Pass rule (as pre-registered). At 15 or 30 minutes: excess > 0 with its block-20 interval clear of zero and a positive signal mean.

Verdict. FAIL at the gate: 15 min −0.30 bp [−0.56, −0.07] (wrong side); 30 min −0.19 bp [−0.57, +0.21]. The study stopped.

Metrics. not available — signal-only: the gate failed before any option was priced, so trades, win rate, average win/loss, biggest win/loss, profit factor, drawdown, top-5 share and by-year are not available; no exit or option P&L was ever read.

Amendments, deviations and integrity checks. No held-out month read for an outcome.

What it showed. The scanner’s signals do not move the underlying their way.

#40a Compass EMA ladder: PREVIEW 2025-03, 13 versions (rerun)

Folder: `research_notes/Compass EMA ladder/`. **Commits:** 0dd8c5d / — / cb79c23. **Dates:** 2026-10-23.

Hypothesis. The owner's EMA ladder entries on 0DTE options make money.

Rule. EMA 9 / ladder L1 (V1, the preview's base) and V2–V13 (one change each); the best-R/R execution contract; exits per PREREG_EMA_LADDER §3.

Fills and costs. $0.68748 a contract each way; ask in, bid out; bid 0 → $0, no exit fee.

Data. SPY, QQQ, IWM 0DTE, 21 sessions 2025-03-03 → 31 (inside Pick, not held out).

Periods and holdouts. PREVIEW, no verdict; the full study was never run; the 28-month holdout UNOPENED for this rule.

Pass rule (as pre-registered). None for a preview (the study's rule waits for the full run).

Verdict. PREVIEW: every version loses.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
V1	preview 2025-03	2,781	21 / 21	15.5	+37.5, +13.57	−17.7, −5.74	−14.7, −4.37	**−9.20**, −2.76	n/a (< 60 sessions); n/a (< 60 sessions)
V10	preview 2025-03	1,996	21 / 21	17.0	+32.1, +12.46	−15.8, −5.82	−13.3, −4.37	**−7.62**, −2.71	n/a (< 60 sessions); n/a (< 60 sessions)
V11	preview 2025-03	2,765	21 / 21	15.6	+37.6, +13.58	−17.5, −5.78	−14.6, −4.37	**−8.88**, −2.76	n/a (< 60 sessions); n/a (< 60 sessions)
V12	preview 2025-03	2,350	21 / 21	18.5	+32.2, +15.79	−14.5, −7.10	−11.8, −5.37	**−5.91**, −2.87	n/a (< 60 sessions); n/a (< 60 sessions)
V13	preview 2025-03	2,794	21 / 21	22.3	+23.3, +32.57	−11.3, −13.12	−7.8, −8.37	**−3.61**, −2.91	n/a (< 60 sessions); n/a (< 60 sessions)
V2	preview 2025-03	768	21 / 21	13.2	+84.7, +31.46	−32.1, −11.21	−27.6, −8.37	**−16.78**, −5.60	n/a (< 60 sessions); n/a (< 60 sessions)
V3	preview 2025-03	253	21 / 21	10.7	+163.8, +60.14	−46.8, −18.55	−45.3, −14.37	**−24.32**, −10.15	n/a (< 60 sessions); n/a (< 60 sessions)
V4	preview 2025-03	2,214	21 / 21	12.5	+65.4, +23.59	−19.1, −6.24	−16.3, −5.37	**−8.50**, −2.51	n/a (< 60 sessions); n/a (< 60 sessions)
V5	preview 2025-03	511	21 / 21	11.2	+131.3, +43.08	−34.7, −12.58	−31.4, −10.37	**−16.15**, −6.37	n/a (< 60 sessions); n/a (< 60 sessions)
V6	preview 2025-03	192	21 / 21	8.3	+233.2, +95.00	−50.3, −20.88	−48.0, −17.37	**−26.68**, −11.22	n/a (< 60 sessions); n/a (< 60 sessions)
V7	preview 2025-03	2,749	21 / 21	15.5	+37.3, +13.58	−17.7, −5.76	−14.7, −4.37	**−9.15**, −2.75	n/a (< 60 sessions); n/a (< 60 sessions)
V8	preview 2025-03	2,101	21 / 21	14.8	+46.0, +17.13	−17.9, −5.76	−15.0, −4.37	**−8.50**, −2.38	n/a (< 60 sessions); n/a (< 60 sessions)
V9	preview 2025-03	2,076	21 / 21	14.8	+46.0, +17.19	−17.9, −5.77	−14.9, −4.37	**−8.40**, −2.37	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V1	preview 2025-03	+147.63, +328.1%, 2025-03-28, QQQ 473P m34	−31.37, −40.2%, 2025-03-05, QQQ 486P m24	0.43	7,668	21	total ≤ 0 (best 5 days −931; without them −6,737)
V10	preview 2025-03	+147.63, +328.1%, 2025-03-28, QQQ 473P m34	−31.37, −40.2%, 2025-03-05, QQQ 486P m24	0.44	5,407	21	total ≤ 0 (best 5 days −321; without them −5,087)
V11	preview 2025-03	+147.63, +328.1%, 2025-03-28, QQQ 473P m34	−31.37, −40.2%, 2025-03-05, QQQ 486P m24	0.44	7,618	21	total ≤ 0 (best 5 days −922; without them −6,696)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V12	preview 2025-03	+147.63, +328.1%, 2025-03-28, QQQ 473P m34	−41.37, −57.5%, 2025-03-07, QQQ 489P m346	0.50	6,741	21	total ≤ 0 (best 5 days −611; without them −6,131)
V13	preview 2025-03	+357.63, +211.6%, 2025-03-03, QQQ 509P m195	−98.37, −29.6%, 2025-03-05, QQQ 495P m24	0.71	8,143	21	total ≤ 0 (best 5 days +687; without them −8,830)
V2	preview 2025-03	+247.63, +604.0%, 2025-03-17, QQQ 482C m229	−47.37, −61.5%, 2025-03-11, QQQ 483C m18	0.42	4,301	21	total ≤ 0 (best 5 days +118; without them −4,419)
V3	preview 2025-03	+305.63, +679.2%, 2025-03-28, QQQ 473P m30	−146.37, −61.5%, 2025-03-03, QQQ 506P m30	0.39	3,072	21	total ≤ 0 (best 5 days +861; without them −3,429)
V4	preview 2025-03	+494.63, +1,545.7%, 2025-03-03, QQQ 501P m317	−35.37, −53.6%, 2025-03-07, QQQ 479P m209	0.54	6,344	19	total ≤ 0 (best 5 days +423; without them −5,980)
V5	preview 2025-03	+287.63, +653.7%, 2025-03-28, QQQ 473P m35	−46.37, −66.2%, 2025-03-11, QQQ 463P m90	0.43	3,791	19	total ≤ 0 (best 5 days +842; without them −4,095)
V6	preview 2025-03	+306.63, +696.9%, 2025-03-28, QQQ 473P m35	−146.37, −61.5%, 2025-03-03, QQQ 506P m30	0.41	3,106	21	total ≤ 0 (best 5 days +1,008; without them −3,162)
V7	preview 2025-03	+147.63, +328.1%, 2025-03-28, QQQ 473P m34	−31.37, −40.2%, 2025-03-05, QQQ 486P m24	0.43	7,571	21	total ≤ 0 (best 5 days −931; without them −6,640)
V8	preview 2025-03	+376.63, +941.6%, 2025-03-03, QQQ 502P m316	−28.37, −53.5%, 2025-03-19, QQQ 471P m269	0.52	5,580	20	total ≤ 0 (best 5 days +266; without them −5,265)
V9	preview 2025-03	+376.63, +941.6%, 2025-03-03, QQQ 502P m316	−28.37, −53.5%, 2025-03-19, QQQ 471P m269	0.52	5,494	20	total ≤ 0 (best 5 days +266; without them −5,180)

Random baseline: not available — no random baseline was computed by this study

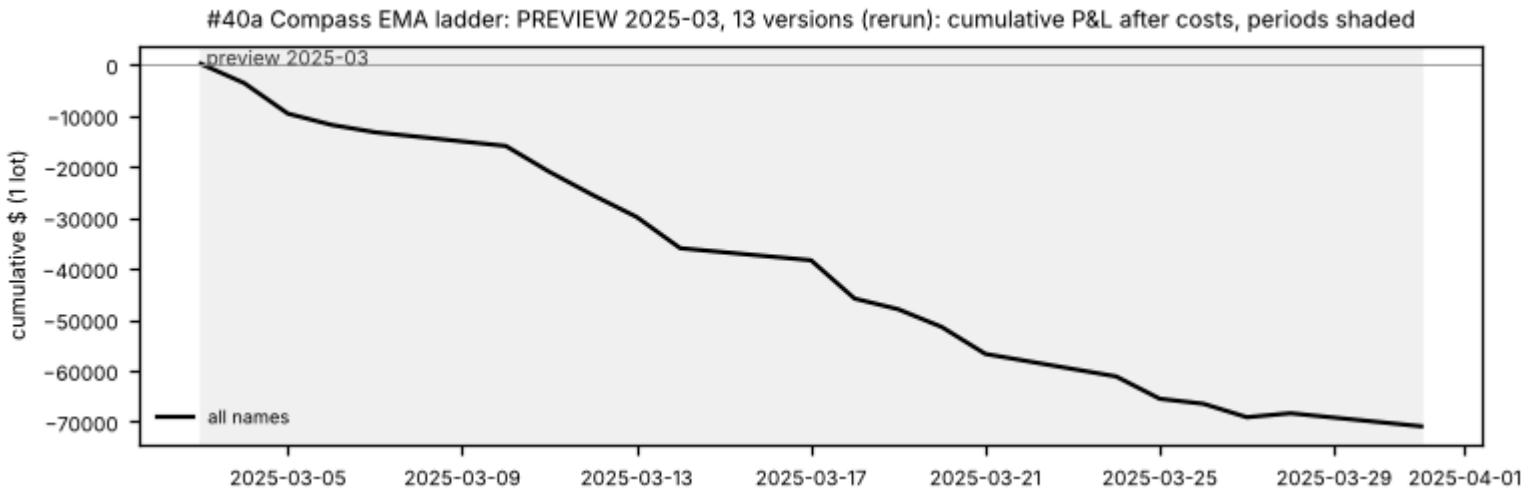
Exit reasons (share of trades):

- preview 2025-03: signal 99.6%; 15:58 0.2%; 14:00 0.2%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	V1	2,781	15.5	−9.20	−2.76	−7,668
2025	V10	1,996	17.0	−7.62	−2.71	−5,407
2025	V11	2,765	15.6	−8.88	−2.76	−7,618
2025	V12	2,350	18.5	−5.91	−2.87	−6,741
2025	V13	2,794	22.3	−3.61	−2.91	−8,143
2025	V2	768	13.2	−16.78	−5.60	−4,301
2025	V3	253	10.7	−24.32	−10.15	−2,568
2025	V4	2,214	12.5	−8.50	−2.51	−5,557
2025	V5	511	11.2	−16.15	−6.37	−3,254
2025	V6	192	8.3	−26.68	−11.22	−2,154
2025	V7	2,749	15.5	−9.15	−2.75	−7,571
2025	V8	2,101	14.8	−8.50	−2.38	−5,000
2025	V9	2,076	14.8	−8.40	−2.37	−4,914

year	name	trades	win %	mean %	mean $	total $
2025	all names	23,550	16.1	−8.54	−3.01	−70,896



Amendments, deviations and integrity checks. Rerun for this report at cb79c23/e9d13c0 from the stored files: preview_2025-03.json byte-identical to the original run (sha256 81edab0c934effe5…).

What it showed. All 13 versions lose 3.6–26.7% of premium a trade in the month.

#40b Compass EMA ladder: “take profits are the nearest EMAs” PREVIEW (E1 ATM, E2 best-R/R)

Folder: `research_notes/Compass EMA ladder/`. **Commits:** rule aad56cf / — / e9d13c0. **Dates:** 2026-10-23.

Hypothesis. Targets at the nearest EMAs rescue the entries.

Rule. E1: the ATM contract (3% spread veto); E2: the best-R/R contract (3% cap); targets at the nearest EMAs, exits on the ATM chart.

Fills and costs. As 40a.

Data. 2025-02 (19 sessions, fresh) and 2025-03 (21, seen before).

Periods and holdouts. PREVIEW, no verdict.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: both versions lose in both months.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib’s 20-session moving-block bootstrap over the period’s calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
E1	preview 2025-02 (fresh)	1,761	19 / 20	30.8	+15.1, +16.25	−10.7, −10.83	−6.9, −5.37	**−2.75**, −2.49	n/a (< 60 sessions); n/a (< 60 sessions)
E2	preview 2025-02 (fresh)	1,544	19 / 20	27.1	+18.5, +8.60	−14.0, −6.42	−10.3, −4.37	**−5.21**, −2.34	n/a (< 60 sessions); n/a (< 60 sessions)
E1	preview 2025-03 (seen before)	2,179	21 / 21	32.5	+15.5, +22.22	−10.8, −14.19	−6.6, −7.37	**−2.28**, −2.36	n/a (< 60 sessions); n/a (< 60 sessions)
E2	preview 2025-03 (seen before)	2,044	21 / 21	29.4	+18.5, +9.16	−14.5, −7.05	−10.4, −5.37	**−4.76**, −2.29	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days’ share of profit
E1	preview 2025-02 (fresh)	+175.63, +82.5%, 2025-02-03, SPY 592C m52	−86.37, −61.3%, 2025-02-07, QQQ 527.5C m77	0.67	4,393	20	total ≤ 0 (best 5 days +1; without them −4,395)
E2	preview 2025-02 (fresh)	+79.63, +118.8%, 2025-02-25, SPY 596P m359	−31.37, −64.0%, 2025-02-07, QQQ 530C m77	0.50	3,617	20	total ≤ 0 (best 5 days −393; without them −3,224)
E1	preview 2025-03 (seen before)	+139.63, +133.0%, 2025-03-19, QQQ 484P m348	−64.37, −21.9%, 2025-03-11, QQQ 473P m120	0.75	5,134	21	total ≤ 0 (best 5 days +357; without them −5,491)
E2	preview 2025-03 (seen before)	+108.63, +142.9%, 2025-03-05, QQQ 502P m346	−24.37, −32.1%, 2025-03-13, QQQ 466P m42	0.54	4,672	21	total ≤ 0 (best 5 days −557; without them −4,116)

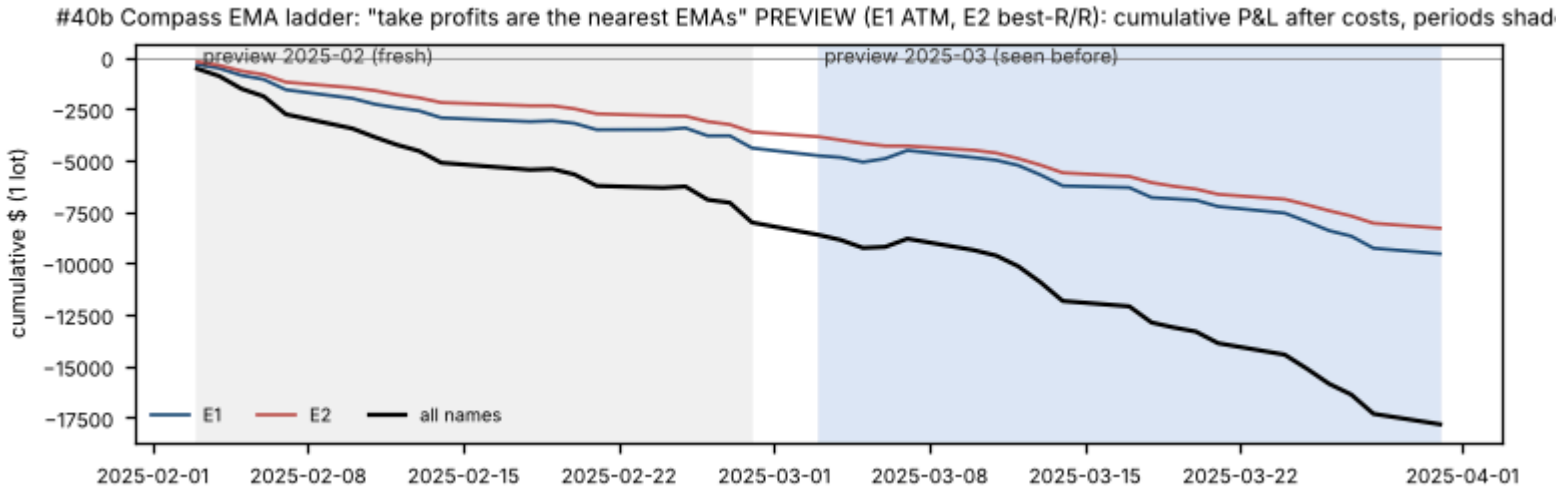
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- preview 2025-02 (fresh): stop 72.4%; target 27.6%
- preview 2025-03 (seen before): stop 70.3%; target 29.7%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	E1	3,940	31.7	−2.49	−2.42	−9,527
2025	E2	3,588	28.4	−4.95	−2.31	−8,289
2025	all names	7,528	30.2	−3.66	−2.37	−17,817



Amendments, deviations and integrity checks. Rerun byte-identical (targets_2025-02.json 6de804ea3ddfa9d5..., targets_2025-03.json 2b005e352e86e126...).

What it showed. E1 −2.3 to −2.8% and E2 −4.8 to −5.2% of premium a trade.

#41a EMA system v0: PREVIEW on 2026-10-12 → 16 (5 sessions)

Folder: `research_notes/EMA system/`. **Commits:** af32405 / — / 8765f71. **Dates:** 2026-10-23.

Hypothesis. The owner's EMA options system as specified (v0 defaults pinned).

Rule. RULEBOOK_v0.md: Config A/B entries, exits X1 (+1R), X2 (+2R), X3 (runner), X4; SPXW (SPX prints), SPY, QQQ, IWM.

Fills and costs. $0.68748 SPY/QQQ/IWM; SPXW $1.18748/$1.27748; ask in, bid out.

Data. 5 full sessions 2026-10-12 → 16 (a falling week: SPY −1.96%).

Periods and holdouts. PREVIEW, no verdict.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: every row net positive in a falling week, 86% of trades puts.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config A X1	preview 2026-10-12→16	57	5 / 5	61.4	+41.4, +134.99	−38.2, −103.27	+25.1, +32.63	**+10.67**, +43.03	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview 2026-10-12→16	53	5 / 5	49.1	+75.4, +232.16	−42.0, −103.07	−13.9, −11.37	**+15.56**, +61.38	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview 2026-10-12→16	48	5 / 5	29.2	+135.6, +560.15	−34.9, −101.33	−22.8, −28.87	**+14.82**, +91.60	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview 2026-10-12→16	48	5 / 5	33.3	+89.2, +3,182.94	−33.0, −920.11	−21.3, −233.75	**+7.71**, +447.58	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview 2026-10-12→16	55	5 / 5	63.6	+38.8, +125.30	−39.2, −106.01	+26.0, +32.63	**+10.41**, +41.19	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview 2026-10-12→16	49	5 / 5	46.9	+70.7, +208.98	−42.6, −102.36	−16.9, −12.37	**+10.54**, +43.78	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview 2026-10-12→16	46	5 / 5	28.3	+124.5, +596.57	−35.4, −100.87	−23.8, −30.37	**+9.80**, +96.23	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview 2026-10-12→16	46	5 / 5	32.6	+81.5, +3,352.92	−33.9, −915.35	−22.1, −248.75	**+3.72**, +476.48	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X1	preview 2026-10-12→16	+717.45, +69.7%, 2026-10-14, SPXW 7770.0P 09:53	−772.55, −51.2%, 2026-10-15, SPXW 7665.0P 09:55	2.08	624	2	100%
Config A X2	preview 2026-10-12→16	+1,427.45, +138.6%, 2026-10-14, SPXW 7770.0P 09:53	−527.55, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.17	728	2	100%
Config A X3	preview 2026-10-12→16	+3,277.45, +337.9%, 2026-10-16, SPXW 7715.0P 11:42	−527.55, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.28	609	2	100%
Config A X4	preview 2026-10-12→16	+21,704.45, +210.7%, 2026-10-14, SPXW 7770.0P 09:53	−3,605.55, −44.5%, 2026-10-16, SPXW 7640.0P 14:37	1.73	5,507	2	100%
Config B X1	preview 2026-10-12→16	+717.45, +69.7%, 2026-10-14, SPXW 7770.0P 09:53	−772.55, −51.2%, 2026-10-15, SPXW 7665.0P 09:55	2.07	624	2	100%
Config B X2	preview 2026-10-12→16	+1,427.45, +138.6%, 2026-10-14, SPXW 7770.0P 09:53	−527.55, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	1.81	606	2	100%
Config B X3	preview 2026-10-12→16	+3,277.45, +337.9%, 2026-10-16, SPXW 7715.0P 11:42	−527.55, −65.1%, 2026-10-16, SPXW 7640.0P 14:37	2.33	578	2	100%
Config B X4	preview 2026-10-12→16	+21,704.45, +210.7%, 2026-10-14, SPXW 7770.0P 09:53	−3,605.55, −44.5%, 2026-10-16, SPXW 7640.0P 14:37	1.77	5,507	2	100%

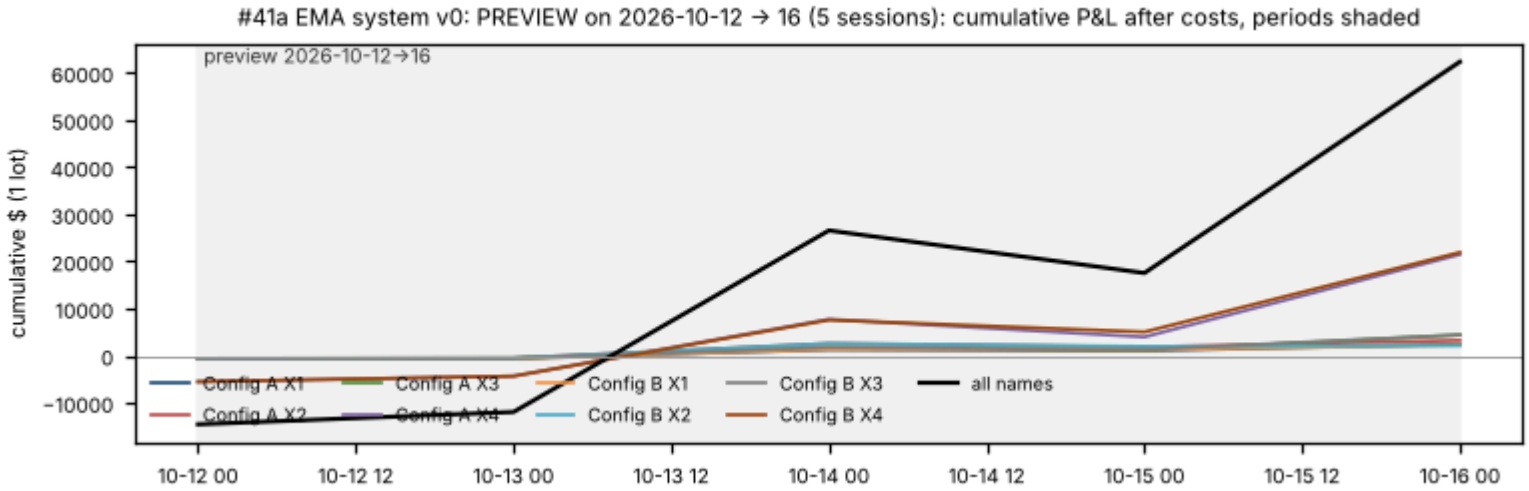
Random baseline: not available — no random baseline was computed by this study

Exit reasons (share of trades):

- preview 2026-10-12→16: stop 45.0%; runner 24.9%; +1R 17.4%; +2R 12.2%; flat 0.5%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2026	Config A X1	57	61.4	+10.67	+43.03	+2,452
2026	Config A X2	53	49.1	+15.56	+61.38	+3,253
2026	Config A X3	48	29.2	+14.82	+91.60	+4,397
2026	Config A X4	48	33.3	+7.71	+447.58	+21,484
2026	Config B X1	55	63.6	+10.41	+41.19	+2,265
2026	Config B X2	49	46.9	+10.54	+43.78	+2,145
2026	Config B X3	46	28.3	+9.80	+96.23	+4,427
2026	Config B X4	46	32.6	+3.72	+476.48	+21,918
2026	all names	402	44.0	+10.51	+155.08	+62,341



Amendments, deviations and integrity checks. —

What it showed. One falling week, mostly puts.

#41b EMA system v1: PREVIEW on three drawn weeks (up, flat, down)

Folder: `research_notes/EMA system/`. **Commits:** f171654 / — / f512d21. **Dates:** 2026-10-23.

Hypothesis. v0 with four changes (the three passed rules V1, X3, VN1 and a fix).

Rule. RULEBOOK_v1.md.

Fills and costs. As 41a.

Data. Weeks drawn by seed 2026102310: up 2025-07-21, flat 2026-02-02, down 2026-07-13.

Periods and holdouts. PREVIEW, no verdict; no day in a held-out month.

Pass rule (as pre-registered). None.

Verdict. PREVIEW: about zero pooled; the sign follows the week.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config A X1	preview week: up	20	5 / 5	45.0	+17.1, +27.12	−43.8, −70.77	−27.4, −18.37	**−16.37**, −26.72	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: up	20	5 / 5	20.0	+50.7, +158.04	−47.4, −74.72	−35.8, −23.37	**−27.76**, −28.17	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: up	21	5 / 5	9.5	+22.9, +9.63	−30.2, −58.90	−30.2, −21.37	**−25.11**, −52.37	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: up	21	5 / 5	9.5	+28.5, +121.25	−28.1, −564.96	−29.4, −213.75	**−22.71**, −499.61	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: up	18	5 / 5	50.0	+17.1, +27.12	−43.8, −79.64	−9.9, −6.37	**−13.33**, −26.26	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview week: up	18	5 / 5	22.2	+50.7, +158.04	−47.9, −80.98	−35.8, −23.37	**−25.99**, −27.87	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview week: up	19	5 / 5	10.5	+22.9, +9.63	−29.6, −62.90	−30.2, −18.37	**−24.09**, −55.26	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: up	19	5 / 5	10.5	+28.5, +121.25	−27.3, −602.16	−28.0, −183.75	**−21.43**, −526.01	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	preview week: flat	32	5 / 5	68.8	+43.9, +217.85	−40.6, −208.01	+28.7, +33.63	**+17.52**, +84.77	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: flat	27	5 / 5	55.6	+73.6, +367.64	−36.5, −245.84	+50.8, +46.63	**+24.68**, +94.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: flat	24	5 / 5	45.8	+87.5, +486.11	−31.2, −163.95	−15.7, −11.87	**+23.21**, +133.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: flat	24	5 / 5	45.8	+70.7, +3,677.41	−27.4, −1,418.16	−3.4, −59.75	**+17.55**, +917.31	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: flat	32	5 / 5	68.8	+43.9, +217.85	−40.6, −208.01	+28.7, +33.63	**+17.52**, +84.77	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview week: flat	27	5 / 5	55.6	+73.6, +367.64	−36.5, −245.84	+50.8, +46.63	**+24.68**, +94.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X3	preview week: flat	24	5 / 5	45.8	+87.5, +486.11	−31.2, −163.95	−15.7, −11.87	**+23.21**, +133.99	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: flat	24	5 / 5	45.8	+70.7, +3,677.41	−27.4, −1,418.16	−3.4, −59.75	**+17.55**, +917.31	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	preview week: down	12	5 / 5	33.3	+63.1, +201.08	−42.9, −349.01	−30.3, −94.46	**−7.58**, −165.65	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X2	preview week: down	12	5 / 5	16.7	+79.9, +530.54	−48.9, −295.58	−47.3, −138.96	**−27.42**, −157.90	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X3	preview week: down	13	5 / 5	23.1	+18.5, +60.57	−33.0, −245.88	−28.5, −83.37	**−21.10**, −175.16	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X4	preview week: down	13	5 / 5	30.8	+34.5, +1,459.30	−36.5, −2,730.51	−28.5, −833.75	**−14.66**, −1,441.33	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X1	preview week: down	12	5 / 5	33.3	+63.1, +201.08	−42.9, −349.01	−30.3, −94.46	**−7.58**, −165.65	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X2	preview week: down	12	5 / 5	16.7	+79.9, +530.54	−48.9, −295.58	−47.3, −138.96	**−27.42**, −157.90	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
Config B X3	preview week: down	13	5 / 5	23.1	+18.5, +60.57	−33.0, −245.88	−28.5, −83.37	**−21.10**, −175.16	n/a (< 60 sessions); n/a (< 60 sessions)
Config B X4	preview week: down	13	5 / 5	30.8	+34.5, +1,459.30	−36.5, −2,730.51	−28.5, −833.75	**−14.66**, −1,441.33	n/a (< 60 sessions); n/a (< 60 sessions)
Config A X1	three weeks pooled	64	15 / 257	54.7	+39.2, +166.89	−42.4, −194.85	+9.2, +9.63	**+2.23**, +2.98	[−14.91, +28.63]; [−71.71, +164.43]
Config A X2	three weeks pooled	59	15 / 257	35.6	+69.8, +343.23	−44.3, −186.88	−29.4, −23.37	**−3.69**, +1.80	[−21.32, +33.47]; [−60.09, +239.59]
Config A X3	three weeks pooled	58	15 / 257	27.6	+66.5, +346.76	−31.2, −135.93	−26.8, −18.37	**−4.22**, −2.78	[−23.21, +41.16]; [−96.79, +243.90]
Config A X4	three weeks pooled	58	15 / 257	29.3	+57.2, +2,737.13	−29.7, −1,310.85	−23.8, −173.75	**−4.24**, −124.37	[−20.19, +29.83]; [−552.41, +1,586.55]
Config B X1	three weeks pooled	62	15 / 257	56.5	+39.2, +166.89	−42.3, −207.00	+11.4, +11.13	**+3.71**, +4.07	[−13.33, +28.63]; [−71.71, +164.43]
Config B X2	three weeks pooled	57	15 / 257	36.8	+69.8, +343.23	−44.4, −195.55	−28.0, −23.37	**−2.29**, +2.95	[−21.32, +33.47]; [−60.09, +239.59]
Config B X3	three weeks pooled	56	15 / 257	28.6	+66.5, +346.76	−31.0, −141.49	−25.5, −17.37	**−3.12**, −1.99	[−22.39, +41.16]; [−96.79, +243.90]
Config B X4	three weeks pooled	56	15 / 257	30.4	+57.2, +2,737.13	−29.5, −1,365.32	−22.3, −158.75	**−3.15**, −119.93	[−19.76, +29.83]; [−572.95, +1,586.55]

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X1	preview week: up	+97.45, +19.1%, 2025-07-23, SPXW 6340.0C 12:43	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.31	552	4	total ≤ 0 (best 5 days −534; without them −0)
Config A X2	preview week: up	+387.45, +76.0%, 2025-07-23, SPXW 6340.0C 12:43	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.53	815	5	total ≤ 0 (best 5 days −563; without them −0)
Config A X3	preview week: up	+17.63, +43.0%, 2025-07-25, QQQ 567.0C 12:09	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	1,100	5	total ≤ 0 (best 5 days −1,100; without them −0)
Config A X4	preview week: up	+208.25, +50.8%, 2025-07-25, QQQ 567.0C 12:09	−3,225.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	10,492	5	total ≤ 0 (best 5 days −10,492; without them +0)
Config B X1	preview week: up	+97.45, +19.1%, 2025-07-23, SPXW 6340.0C 12:43	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.34	529	4	total ≤ 0 (best 5 days −473; without them −0)
Config B X2	preview week: up	+387.45, +76.0%, 2025-07-23, SPXW 6340.0C 12:43	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.56	776	5	total ≤ 0 (best 5 days −502; without them +0)
Config B X3	preview week: up	+17.63, +43.0%, 2025-07-25, QQQ 567.0C 12:09	−322.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	1,050	5	total ≤ 0 (best 5 days −1,050; without them −0)
Config B X4	preview week: up	+208.25, +50.8%, 2025-07-25, QQQ 567.0C 12:09	−3,225.55, −35.4%, 2025-07-22, SPXW 6285.0P 10:13	0.02	9,994	5	total ≤ 0 (best 5 days −9,994; without them +0)
Config A X1	preview week: flat	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.55, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	2.30	567	2	100%
Config A X2	preview week: flat	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.55, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.87	1,208	3	100%
Config A X3	preview week: flat	+4,287.45, +376.1%, 2026-02-03, SPXW 6925.0P 11:51	−782.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.51	920	3	100%
Config A X4	preview week: flat	+28,224.45, +247.6%, 2026-02-03, SPXW 6925.0P 11:51	−7,825.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.19	4,734	3	100%
Config B X1	preview week: flat	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.55, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	2.30	567	2	100%
Config B X2	preview week: flat	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.55, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.87	1,208	3	100%
Config B X3	preview week: flat	+4,287.45, +376.1%, 2026-02-03, SPXW 6925.0P 11:51	−782.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.51	920	3	100%
Config B X4	preview week: flat	+28,224.45, +247.6%, 2026-02-03, SPXW 6925.0P 11:51	−7,825.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	2.19	4,734	3	100%
Config A X1	preview week: down	+627.45, +90.9%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.29	2,615	4	total ≤ 0 (best 5 days −1,988; without them +0)
Config A X2	preview week: down	+1,047.45, +151.8%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.36	2,942	4	total ≤ 0 (best 5 days −1,895; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
Config A X3	preview week: down	+137.45, +19.9%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.07	2,415	4	total ≤ 0 (best 5 days −2,277; without them +0)
Config A X4	preview week: down	+5,084.45, +73.7%, 2026-07-13, SPXW 7530.0P 12:17	−6,325.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.24	23,822	4	total ≤ 0 (best 5 days −18,737; without them −0)
Config B X1	preview week: down	+627.45, +90.9%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.29	2,615	4	total ≤ 0 (best 5 days −1,988; without them +0)
Config B X2	preview week: down	+1,047.45, +151.8%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.36	2,942	4	total ≤ 0 (best 5 days −1,895; without them +0)
Config B X3	preview week: down	+137.45, +19.9%, 2026-07-13, SPXW 7530.0P 12:17	−632.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.07	2,415	4	total ≤ 0 (best 5 days −2,277; without them +0)
Config B X4	preview week: down	+5,084.45, +73.7%, 2026-07-13, SPXW 7530.0P 12:17	−6,325.55, −35.9%, 2026-07-17, SPXW 7430.0P 09:45	0.24	23,822	4	total ≤ 0 (best 5 days −18,737; without them −0)
Config A X1	three weeks pooled	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.55, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	1.03	2,615	136	2095%
Config A X2	three weeks pooled	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.55, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.01	2,942	137	4791%
Config A X3	three weeks pooled	+4,287.45, +376.1%, 2026-02-03, SPXW 6925.0P 11:51	−782.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.97	3,197	138	total ≤ 0 (best 5 days +4,184; without them −4,346)
Config A X4	three weeks pooled	+28,224.45, +247.6%, 2026-02-03, SPXW 6925.0P 11:51	−7,825.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.87	23,822	138	total ≤ 0 (best 5 days +31,822; without them −39,035)
Config B X1	three weeks pooled	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−872.55, −66.1%, 2026-02-03, SPXW 6925.0P 10:25	1.05	2,615	136	1591%
Config B X2	three weeks pooled	+1,357.45, +119.1%, 2026-02-03, SPXW 6925.0P 11:51	−1,052.55, −47.6%, 2026-02-05, SPXW 6785.0P 10:46	1.02	2,942	137	3045%
Config B X3	three weeks pooled	+4,287.45, +376.1%, 2026-02-03, SPXW 6925.0P 11:51	−782.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.98	3,197	138	total ≤ 0 (best 5 days +4,197; without them −4,308)
Config B X4	three weeks pooled	+28,224.45, +247.6%, 2026-02-03, SPXW 6925.0P 11:51	−7,825.55, −59.3%, 2026-02-03, SPXW 6925.0P 10:25	0.87	23,822	138	total ≤ 0 (best 5 days +32,095; without them −38,812)

Random baseline: not available — no random baseline was computed by this study

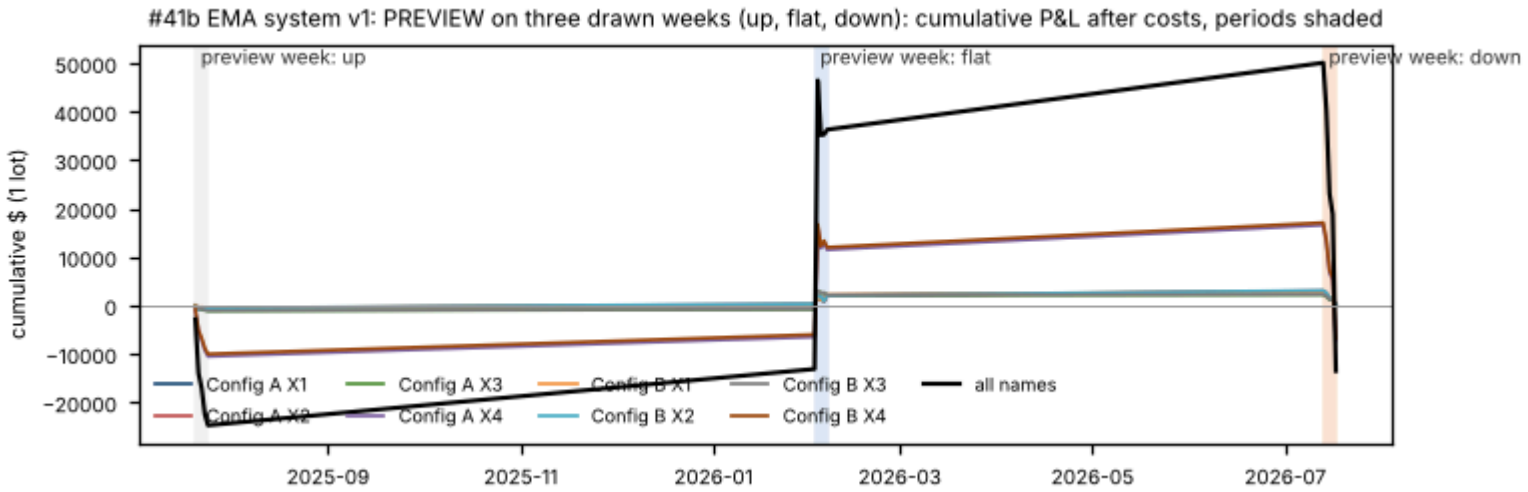
Exit reasons (share of trades):

- preview week: up: stop 46.2%; runner 34.6%; +1R 11.5%; +2R 5.1%; 14:00 2.6%
- preview week: flat: runner 31.8%; stop 25.2%; +1R 19.6%; +2R 12.1%; 14:00 11.2%
- preview week: down: stop 56.0%; runner 32.0%; +1R 6.0%; 14:00 4.0%; +2R 2.0%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2025	Config A X1	20	45.0	−16.37	−26.72	−534
2026	Config A X1	44	59.1	+10.68	+16.47	+725
2025	Config A X2	20	20.0	−27.76	−28.17	−563
2026	Config A X2	39	43.6	+8.65	+17.18	+670
2025	Config A X3	21	9.5	−25.11	−52.37	−1,100

year	name	trades	win %	mean %	mean $	total $
2026	Config A X3	37	37.8	+7.64	+25.37	+939
2025	Config A X4	21	9.5	−22.71	−499.61	−10,492
2026	Config A X4	37	40.5	+6.24	+88.60	+3,278
2025	Config B X1	18	50.0	−13.33	−26.26	−473
2026	Config B X1	44	59.1	+10.68	+16.47	+725
2025	Config B X2	18	22.2	−25.99	−27.87	−502
2026	Config B X2	39	43.6	+8.65	+17.18	+670
2025	Config B X3	19	10.5	−24.09	−55.26	−1,050
2026	Config B X3	37	37.8	+7.64	+25.37	+939
2025	Config B X4	19	10.5	−21.43	−526.01	−9,994
2026	Config B X4	37	40.5	+6.24	+88.60	+3,278
2025	all names	156	21.8	−22.17	−158.38	−24,708
2026	all names	314	45.9	+8.41	+35.74	+11,223



Amendments, deviations and integrity checks. —

What it showed. The sign follows the week.

#42 George A+ entry + EMA targets

Folder: `(research_notes/George A+ EMA targets/ was planned)`. **Commits:** none. **Dates:** 2026-10-23.

Hypothesis. The GeorgeFi “A+” entry with EMA price-target exits.

Rule. —

Fills and costs. —

Data. —

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Never run.

Metrics. not available — the run was stopped before its prereg was committed; no file exists on disk or in history, so every metric is not available.

Amendments, deviations and integrity checks. —

What it showed. No result.

#43 Compass ICT: sweep → displacement/MSS → FVG on SPXW 0DTE

Folder: `research_notes/Compass ICT/`. **Commits:** 9d6c37e / Amendment 1 9dbc0bf / preview 315fe8e; exploration 0943030. **Dates:** 2026-10-23.

Hypothesis. The owner's ICT rule (v0) and its variants trade SPXW 0DTE at a profit.

Rule. HTF bias, a sweep of a level in a killzone, displacement/MSS, entry in the FVG; 2 contracts; TP1 half, TP2, stop; V1 OTE, V2 out 14:00, V3 range gate, V4 5-min; four ablations.

Fills and costs. SPXW $1.18748 a contract each way under $1 of premium, $1.27748 at $1 and up (commission, Cboe fee, SPXW surcharge, ORF, OCC); ask in, bid out.

Data. SPXW 0DTE, 818 full sessions 2022-05-16 → 2026-10-09 (the 28 held-out months out); preview 2026-10-09 → 22.

Periods and holdouts. exploration / preview (10 sessions). Holdout A and holdout B UNOPENED; finalists named, never read.

Pass rule (as pre-registered). Finalists: v0 plus at most 2 candidates with exploration mean % > 0 and underlying return > 0; holdout B pass: options mean % > 0, interval clear of zero, ≥ 80 trades.

Verdict. Exploration only: v0 −0.8% [−7.7, +7.2], +$48 a trade; no holdout read.

Metrics (return = % of premium; win = P&L > 0 after fees; breakeven is a loss; the interval is audit_lib's 20-session moving-block bootstrap over the period's calendar sessions):

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
A no FVG	exploration 2022-05→2026-10-09	202	158 / 1,123	33.2	+66.1, +1,839.67	−29.7, −695.46	−4.9, −110.11	**+2.10**, +145.40	[−6.84, +12.24]; [−150.93, +476.28]
A no HTF bias	exploration 2022-05→2026-10-09	357	281 / 1,135	37.3	+37.0, +893.24	−26.4, −628.49	−4.7, −105.11	**−2.77**, −61.57	[−7.87, +2.08]; [−204.13, +87.84]
A no killzones	exploration 2022-05→2026-10-09	245	175 / 1,126	33.1	+43.7, +1,039.77	−22.7, −519.33	−4.3, −95.11	**−0.77**, −3.87	[−6.30, +6.13]; [−148.35, +185.23]
A no sweep	exploration 2022-05→2026-10-09	1,266	698 / 1,128	41.3	+30.8, +864.12	−28.7, −754.22	−2.7, −55.11	**−4.14**, −85.66	[−6.24, −1.67]; [−141.27, −24.78]
V1 OTE	exploration 2022-05→2026-10-09	94	79 / 1,117	33.0	+36.6, +832.95	−22.6, −526.30	−12.1, −215.11	**−3.07**, −78.04	[−11.17, +6.37]; [−264.98, +154.33]
V2 out 14:00	exploration 2022-05→2026-10-09	142	116 / 1,117	39.4	+34.0, +895.96	−21.2, −447.05	−3.5, −65.11	**+0.55**, +82.59	[−6.41, +8.90]; [−102.52, +310.29]
V3 range gate	exploration 2022-05→2026-10-09	84	71 / 1,105	36.9	+26.4, +713.28	−24.9, −499.21	−5.4, −85.11	**−5.98**, −51.74	[−14.28, +2.70]; [−259.58, +236.68]
V4 5-min	exploration 2022-05→2026-10-09	37	37 / 1,077	27.0	+77.6, +1,528.39	−41.1, −988.60	−5.2, −85.11	**−9.02**, −308.33	[−24.97, +17.97]; [−673.90, +278.78]
v0	exploration 2022-05→2026-10-09	142	116 / 1,117	37.3	+35.5, +954.70	−22.4, −491.54	−3.9, −75.11	**−0.76**, +48.26	[−7.77, +7.54]; [−154.18, +292.19]
A no FVG	preview 2026-10-09→22	2	2 / 6	50.0	+15.1, +174.89	−37.8, −1,005.11	−11.4, −415.11	**−11.35**, −415.11	n/a (< 60 sessions); n/a (< 60 sessions)
A no HTF bias	preview 2026-10-09→22	4	3 / 3	25.0	+98.7, +3,454.89	−32.1, −925.11	−25.5, −575.11	**+0.57**, +169.89	n/a (< 60 sessions); n/a (< 60 sessions)
A no killzones	preview 2026-10-09→22	4	3 / 6	0.0	—, —	−50.0, −722.56	−47.2, −825.11	**−50.00**, −722.56	n/a (< 60 sessions); n/a (< 60 sessions)
A no sweep	preview 2026-10-09→22	17	9 / 9	35.3	+68.8, +1,563.22	−42.0, −1,029.66	−8.9, −205.11	**−2.86**, −114.52	n/a (< 60 sessions); n/a (< 60 sessions)
V1 OTE	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−10.0, −265.11	−10.0, −265.11	**−10.04**, −265.11	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	trades	sessions (traded / span)	win %	avg win %, $	avg loss %, $	median %, $	mean %, $	20-session block 95% (%; $)
V2 out 14:00	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−40.6, −885.11	−40.6, −885.11	**−40.60**, −885.11	n/a (< 60 sessions); n/a (< 60 sessions)
V3 range gate	preview 2026-10-09→22	1	1 / 1	100.0	+6.7, +174.89	—, —	+6.7, +174.89	**+6.68**, +174.89	n/a (< 60 sessions); n/a (< 60 sessions)
V4 5-min	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−13.1, −105.11	−13.1, −105.11	**−13.14**, −105.11	n/a (< 60 sessions); n/a (< 60 sessions)
v0	preview 2026-10-09→22	1	1 / 1	0.0	—, —	−40.6, −885.11	−40.6, −885.11	**−40.60**, −885.11	n/a (< 60 sessions); n/a (< 60 sessions)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
A no FVG	exploration 2022-05→2026-10-09	+21,544.89, +282.7%, 2025-04-08, SPXW 5220.0- ×2 m84	−4,075.11, −91.8%, 2026-02-20, SPXW 6860.0- ×2 m47	1.31	22,178	337	194%
A no HTF bias	exploration 2022-05→2026-10-09	+9,734.89, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−10,805.11, −69.9%, 2025-04-07, SPXW 5075.01 ×2 m53	0.84	45,401	1,011	total ≤ 0 (best 5 days +35,064; without them −57,045)
A no killzones	exploration 2022-05→2026-10-09	+9,734.89, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,284.93, −97.8%, 2025-05-05, SPXW 5655.01 ×2 m33	0.99	16,929	518	total ≤ 0 (best 5 days +30,154; without them −31,103)
A no sweep	exploration 2022-05→2026-10-09	+24,994.89, +262.6%, 2025-04-04, SPXW 5245.0- ×2 m6	−13,625.11, −92.2%, 2025-04-07, SPXW 4895.0- ×2 m3	0.81	121,646	1,083	total ≤ 0 (best 5 days +58,764; without them −167,214)
V1 OTE	exploration 2022-05→2026-10-09	+4,614.89, +219.8%, 2023-08-10, SPXW 4510.0- ×2 m86	−2,125.11, −37.7%, 2026-06-12, SPXW 7405.0- ×2 m37	0.78	7,452	1,117	total ≤ 0 (best 5 days +16,419; without them −23,755)
V2 out 14:00	exploration 2022-05→2026-10-09	+9,734.89, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,045.11, −51.4%, 2026-06-12, SPXW 7395.0- ×2 m31	1.31	7,081	584	214%
V3 range gate	exploration 2022-05→2026-10-09	+7,704.89, +157.9%, 2022-10-14, SPXW 3675.0- ×2 m30	−3,045.11, −51.4%, 2026-06-12, SPXW 7395.0- ×2 m31	0.84	7,290	1,008	total ≤ 0 (best 5 days +16,389; without them −20,736)
V4 5-min	exploration 2022-05→2026-10-09	+4,024.89, +156.0%, 2026-06-04, SPXW 7550.01 ×2 m61	−3,765.11, −64.3%, 2026-06-12, SPXW 7380.0- ×2 m27	0.57	13,996	1,077	total ≤ 0 (best 5 days +13,134; without them −24,543)
v0	exploration 2022-05→2026-10-09	+9,734.89, +333.4%, 2026-07-27, SPXW 7465.0- ×2 m16	−3,284.93, −97.8%, 2025-05-05, SPXW 5655.01 ×2 m33	1.16	10,622	878	398%
A no FVG	preview 2026-10-09→22	+174.89, +15.1%, 2026-10-21, SPXW 7745.0- ×2 m330	−1,005.11, −37.8%, 2026-10-14, SPXW 7785.01 ×2 m18	0.17	1,005	6	total ≤ 0 (best 5 days −830; without them +0)
A no HTF bias	preview 2026-10-09→22	+3,454.89, +98.7%, 2026-10-15, SPXW 7695.01 ×2 m35	−1,625.11, −45.4%, 2026-10-15, SPXW 7695.01 ×2 m13	1.24	885	1	100%
A no killzones	preview 2026-10-09→22	−295.11, −11.0%, 2026-10-20, SPXW 7720.0- ×2 m106	−944.93, −94.5%, 2026-10-21, SPXW 7745.0- ×2 m347	0.00	2,890	6	total ≤ 0 (best 5 days −2,890; without them +0)
A no sweep	preview 2026-10-09→22	+6,544.89, +320.8%, 2026-10-16, SPXW 7740.0- ×2 m86	−2,225.11, −74.7%, 2026-10-21, SPXW 7720.0- ×2 m38	0.83	5,616	4	total ≤ 0 (best 5 days +5,134; without them −7,081)
V1 OTE	preview 2026-10-09→22	−265.11, −10.0%, 2026-10-14, SPXW 7775.01 ×2 m23	−265.11, −10.0%, 2026-10-14, SPXW 7775.01 ×2 m23	0.00	265	1	total ≤ 0 (best 5 days −265; without them +0)
V2 out 14:00	preview 2026-10-09→22	−885.11, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	−885.11, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	0.00	885	1	total ≤ 0 (best 5 days −885; without them +0)

name	period	biggest win ($, %, date, contract)	biggest loss ($, %, date, contract)	profit factor	max DD $	longest DD (sessions)	top-5 days' share of profit
V3 range gate	preview 2026-10-09→22	+174.89, +6.7%, 2026-10-14, SPXW 7760.01 ×2 m30	+174.89, +6.7%, 2026-10-14, SPXW 7760.01 ×2 m30	∞	0	0	100%
V4 5-min	preview 2026-10-09→22	−105.11, −13.1%, 2026-10-13, SPXW 7805.01 ×2 m286	−105.11, −13.1%, 2026-10-13, SPXW 7805.01 ×2 m286	0.00	105	1	total ≤ 0 (best 5 days −105; without them +0)
v0	preview 2026-10-09→22	−885.11, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	−885.11, −40.6%, 2026-10-14, SPXW 7790.01 ×2 m20	0.00	885	1	total ≤ 0 (best 5 days −885; without them +0)

Random baseline: not available — no random baseline was computed by this study

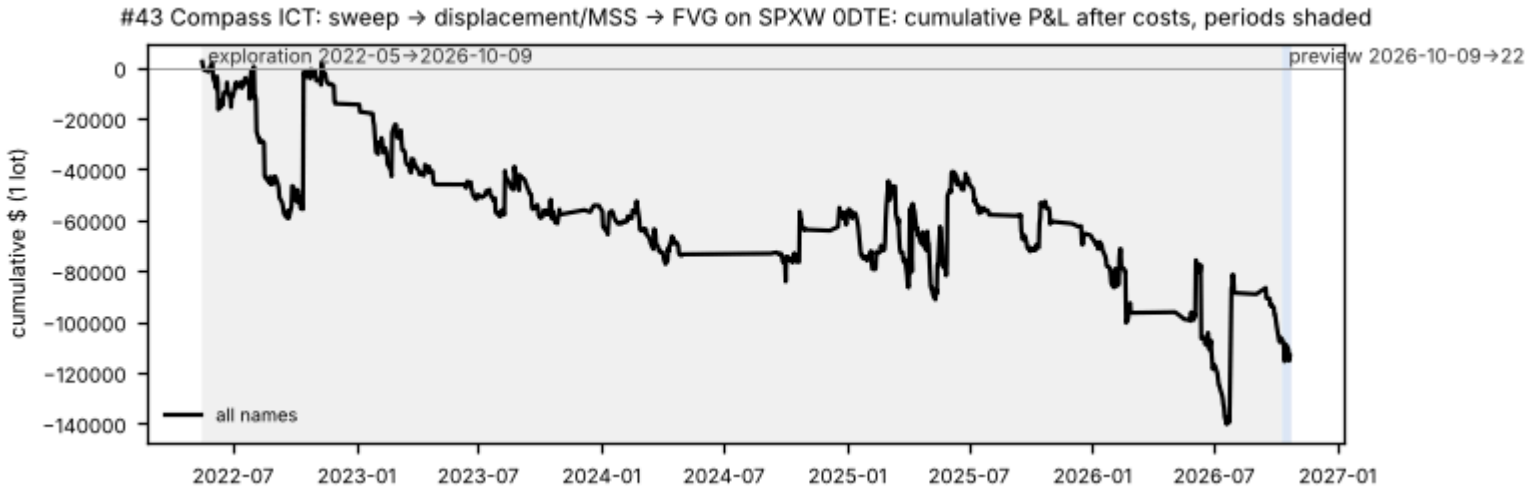
Exit reasons (share of trades):

- exploration 2022-05→2026-10-09: TP1+stop@entry 45.9%; stop 29.9%; TP1+TP2 7.3%; TP1 6.5%; TP1+flat 15:58 5.2%; flat 15:58 3.3%; TP2 1.5%; TP1+14:00 0.4%
- preview 2026-10-09→22: stop 56.2%; TP1+stop@entry 21.9%; TP1 12.5%; TP1+TP2 3.1%; TP1+flat 15:58 3.1%; TP2 3.1%

By year (all periods together, labelled by entry year):

year	name	trades	win %	mean %	mean $	total $
2022	A no FVG	40	25.0	−4.77	+69.70	+2,788
2023	A no FVG	61	39.3	−2.11	+6.70	+408
2024	A no FVG	29	37.9	+22.77	+384.20	+11,142
2025	A no FVG	43	34.9	+3.39	+483.73	+20,800
2026	A no FVG	31	25.8	−2.78	−212.85	−6,598
2022	A no HTF bias	68	39.7	−1.74	+21.47	+1,460
2023	A no HTF bias	102	37.3	−4.08	−93.24	−9,511
2024	A no HTF bias	51	31.4	−6.64	−81.48	−4,155
2025	A no HTF bias	90	36.7	−4.25	−203.99	−18,360
2026	A no HTF bias	50	40.0	+5.39	+185.29	+9,265
2022	A no killzones	52	36.5	−3.95	+35.04	+1,822
2023	A no killzones	70	31.4	−0.30	+46.32	+3,242
2024	A no killzones	35	25.7	−2.20	+11.47	+401
2025	A no killzones	60	36.7	−2.28	−137.85	−8,271
2026	A no killzones	32	28.1	+1.60	−32.29	−1,033
2022	A no sweep	233	43.8	−3.11	−43.09	−10,040
2023	A no sweep	307	41.4	−4.40	−93.48	−28,698
2024	A no sweep	201	40.3	−5.24	−70.07	−14,084
2025	A no sweep	338	40.5	−2.14	−27.23	−9,203
2026	A no sweep	204	40.2	−7.03	−237.12	−48,372
2022	V1 OTE	17	35.3	−12.64	−310.70	−5,282
2023	V1 OTE	26	38.5	+1.90	+93.54	+2,432
2024	V1 OTE	10	30.0	−5.04	−62.11	−621
2025	V1 OTE	25	32.0	+2.47	+29.89	+747
2026	V1 OTE	17	23.5	−8.48	−286.87	−4,877

year	name	trades	win %	mean %	mean $	total $
2022	V2 out 14:00	30	43.3	−6.77	+18.64	+559
2023	V2 out 14:00	40	32.5	−1.83	+8.14	+326
2024	V2 out 14:00	20	35.0	−4.67	+33.90	+678
2025	V2 out 14:00	35	48.6	+4.09	+106.89	+3,741
2026	V2 out 14:00	18	33.3	+14.67	+307.67	+5,538
2022	V3 range gate	18	44.4	−9.73	+90.59	+1,631
2023	V3 range gate	20	25.0	−9.41	−152.61	−3,052
2024	V3 range gate	11	36.4	−15.04	−146.00	−1,606
2025	V3 range gate	27	44.4	+1.27	+24.89	+672
2026	V3 range gate	9	33.3	−0.17	−201.78	−1,816
2022	V4 5-min	8	12.5	−42.24	−999.77	−7,998
2023	V4 5-min	11	18.2	−25.19	−391.21	−4,303
2024	V4 5-min	6	50.0	+24.11	+477.39	+2,864
2025	V4 5-min	5	40.0	+46.46	+618.89	+3,094
2026	V4 5-min	8	25.0	−13.62	−646.34	−5,171
2022	v0	30	40.0	−7.48	+30.81	+924
2023	v0	40	30.0	−3.82	−43.36	−1,734
2024	v0	20	35.0	−5.30	+15.90	+318
2025	v0	35	45.7	+0.73	−6.25	−219
2026	v0	18	33.3	+17.20	+371.00	+6,678
2022	all names	496	39.9	−4.83	−28.50	−14,136
2023	all names	677	37.4	−3.78	−60.40	−40,890
2024	all names	383	36.8	−2.82	−13.22	−5,063
2025	all names	658	39.8	−0.91	−10.63	−6,997
2026	all names	387	36.2	−2.28	−119.86	−46,386



Amendments, deviations and integrity checks. Amendment 1 (ablations become candidates) committed before any exploration result.

What it showed. v0 is about zero, and 398% of its profit comes from its five best days.

#44 Compass EMA: the owner’s three EMA / ORB models

Folder: `research_notes/Compass EMA/`. **Commits:** 9af93fc (prereg + HOLDOUT_LEDGER) / — / none. **Dates:** 2026-10-23.

Hypothesis. (See the prereg.)

Rule. PREREG_COMPASS_EMA.md v0.

Fills and costs. —

Data. SPXW (primary), QQQ (secondary).

Periods and holdouts. Holdouts A and B UNOPENED.

Pass rule (as pre-registered). (See the prereg.)

Verdict. Never computed.

Metrics. not available — the run was stopped before computing: a partial engine and a session list exist in the scratchpad (compass_ema_partial/), no signal, trade or P&L.

Amendments, deviations and integrity checks. —

What it showed. No result.

#45 DR-GCE gates 1 and 2 (Mode 1 Gamma Vacuum Squeeze)

Folder: `research_notes/DR-GCE/`. **Commits:** 01d5c8e, 294d4f2 / A1 041baba / b20eb9c, 00eb3d5. **Dates:** 2026-10-23.

Hypothesis. Mode 1 signals touch +18% within 4 minutes far more often than random (gate 1), and in their own direction (gate 2).

Rule. Signal rows on SPXW 0DTE; the contract under the gate's rules; a touch = +18% within 4 minutes before the stop.

Fills and costs. Real NBBO.

Data. 504 exploration sessions 2023-12-01 → 2026-10-09; 98 Mode 1 signals.

Periods and holdouts. Exploration; the 28 held-out months UNOPENED.

Pass rule (as pre-registered). Gate 1: touch-rate lower bound above the random baseline's 97.5%; gate 2: signal − opposite side touch difference clear of zero.

Verdict. Gate 1 PASS (narrow: 32.7% [23.3, 40.2] vs 16.0% [9.2, 22.4]); gate 2 FAIL (the lift is volatility, not direction). No backtest was written.

Metrics. not available — signal-only: touch outcomes (signals.csv, gate2_pairs.csv), never traded to an exit; no P&L metric exists.

Amendments, deviations and integrity checks. Guard selftest 5 of 5.

What it showed. The lift is volatility after the impulse, not direction.

#46 Trend-day test

Folder: `–`. **Commits:** none. **Dates:** —.

Hypothesis. —

Rule. —

Fills and costs. —

Data. —

Periods and holdouts. —

Pass rule (as pre-registered). —

Verdict. Never registered, never run.

Metrics. not available — never run (the trend-day share is only described in step 6).

Amendments, deviations and integrity checks. —

What it showed. No result.